



**WORLDWIDE INFLUENZA CENTRE
THE FRANCIS CRICK INSTITUTE
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LONDON**

**Report prepared for the WHO annual consultation
on the composition of influenza vaccine for the
Southern Hemisphere 2017**

26th – 28th September 2016

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The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and presented in Flu New Europe (<http://flunewseurope.org>), the joint ECDC-WHO Europe weekly/monthly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office:

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WHO Vaccine Recommendation Meeting for the Southern Hemisphere 2017 influenza season.

WHO CC, London - Final Report for the meeting: 26-28 September 2016.

Influenza Activity in the WHO European Region, February - August 2016

Weekly reporting on influenza activity started in week 40/2015. During the course of the 2015-2016 season transmission peaked in week 06/2016 (**Figure E1-A**) declining thereafter. The duration of the season was 18 weeks (weeks 51/2015-16/2016) with no distinct West to East pattern of spread across the Region; in fact the season would appear to have started earlier in Eastern European countries (**Figure E2**).

For the whole WHO European Region between weeks 40/2015 and 20/2016, over 139,000 influenza detections were reported to the European Surveillance System (TESSy) - with type A detections (70%) predominating over type B (30%) (**Table E1**). Within type A, the A(H1N1)pdm09 subtype (91%) predominated over A(H3N2) viruses (9%) (**Table E1**) and this pattern was evident for both sentinel and non-sentinel influenza detections (**Figure E1-B/C**). Within type B, 91% of the viruses for which lineage-determination was performed were of the Victoria lineage (**Table E1**). The majority of countries in the Region reported influenza A virus detections to have outnumbered those of influenza B; however, some countries in Western Europe reported the majority of detections as influenza B and some in Southern Europe and West Asia had similar proportions of influenza A and B detections. Influenza B detections predominated in the latter part of the season (**Figure E1-B/C**). Influenza A(H1N1)pdm09 detections greatly outnumbered A(H3N2) detections in most countries, with the exception of some in Southern Europe (**Figures E2 & E5**).

Between weeks 40/2015 and 20/2016, eight countries (Finland, France, Ireland, Romania, Slovakia, Spain, Sweden and the UK) reported a total of 4909 laboratory-confirmed hospitalized influenza cases, with type A detections (81%) predominating over type B (19%). Within type A, A(H1N1)pdm09 viruses predominated in proportions similar to that observed in ILI and ARI patients (**Table E1 & Figure E1B/C**).

Figures E3 and E4 show the distribution of viruses in genetic reporting categories for A(H3N2) viruses and influenza B viruses, respectively, since week 05/2016. Reporting of A(H1N1)pdm09 viruses was not sub-divided by genetic group. **Figure E3** shows for A(H3N2) that viruses in subclade 3C.2a predominated; in no week did the number of reported viruses of the 3C.3a subclade outnumber those reported for the 3C.2a subclade.

Figure E4 shows that viruses of the B/Victoria lineage have predominated since week 05/2016; in no week in which more than a single virus was reported did the number of viruses reported in the B/Yamagata lineage exceed the number in the B/Victoria lineage.

Figure E5 illustrates the pattern of circulation of viruses in various sectors of the European Region, with different patterns of circulation. For example, Ukraine and Kazakhstan showed predominantly A(H1N1)pdm09 viruses throughout the season, whilst in France influenza B viruses predominated throughout. Over the whole season (weeks 40/2015 to

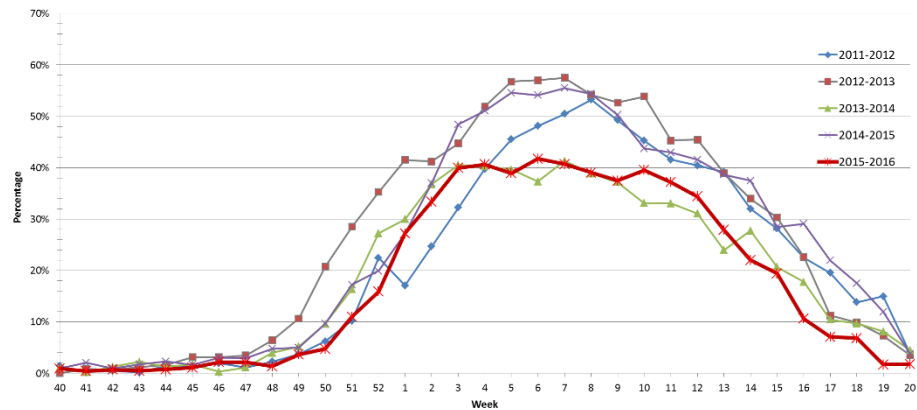
20/2016), for hospitalized patients where age was reported, the group aged 15-64 years accounted for the largest number of cases (**Figure E6**).

Between weeks 40/2015 and 20/2016, 2,700 A(H1N1)pdm09, 172 A(H3N2) and 523 type B viruses were tested for susceptibility to neuraminidase inhibitors (NAI). Of these, 31 viruses showed loss of susceptibility to at least one NAI due to NA amino acid substitution: 26 A(H1N1)pdm09 viruses showed highly reduced inhibition by oseltamivir associated with NA-H275Y, one A(H3N2) virus showed reduced inhibition by oseltamivir associated with NA-E119V and four B/Victoria lineage viruses showed highly reduced inhibition by oseltamivir and zanamivir associated with NA-R374K. All other viruses showed no molecular or phenotypic evidence of reduced inhibition by NAI.

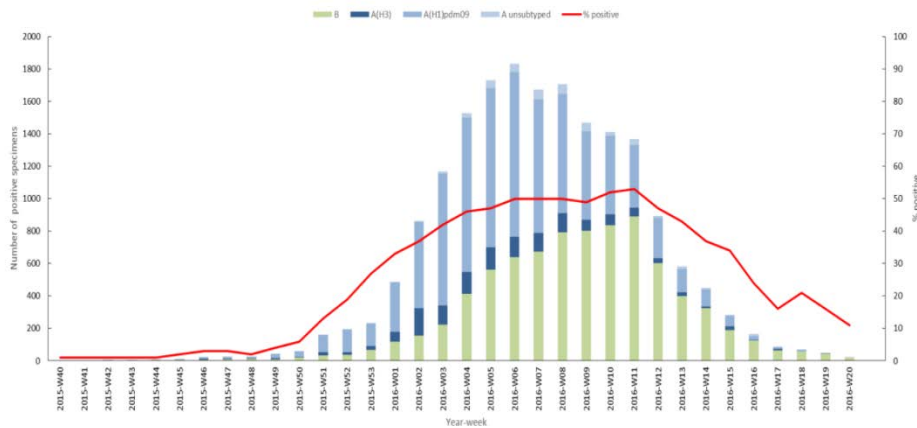
Data from the 18 countries or regions reporting to the European monitoring of excess mortality for public health action project (EuroMOMO - <http://www.euromomo.eu>) suggested a pattern of excess all-cause mortality among those aged 15–64 years between the end of 2015 and week 14/2016. This may have been associated with influenza, as well as other factors. The level of excess all-cause mortality was similar to that of the 2012-2013 winter season and slightly lower than that of the 2014–2015 winter season.

Figure E1. Influenza Activity in the WHO European Region 2015-2016 (weeks 40/2015-20/2016)

A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel Influenza Detections 2015-2016 season



C. Non-sentinel Influenza Detections 2015-2016 season

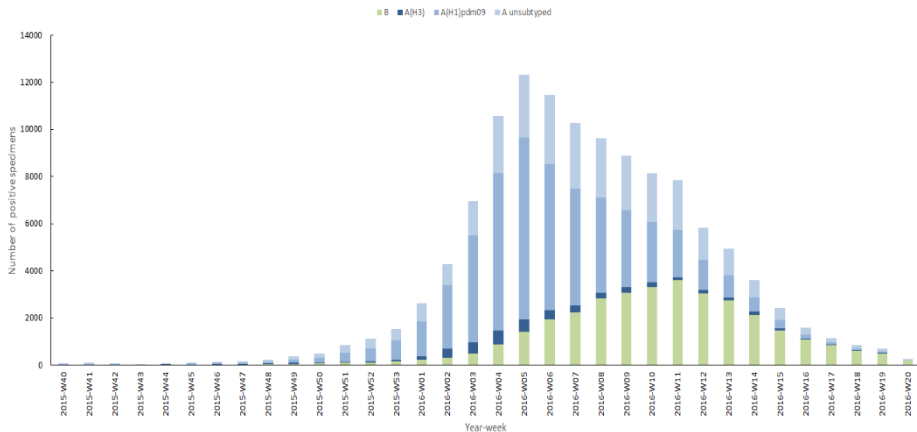


Figure E2. Development of the 2015-2016 Influenza Season in the WHO European Region

Intensity of Influenza Activity and Dominant Virus (sub)type

Country	Week	40	41	42	43	44	45	46	47	48	49	50	51	52	53	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
Iceland		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	-	AH1	AH1	AH1	AH1	AH1	-	-	-	-	-	-	-	-	-	-	-		
Ireland		-	-	-	-	-	-	-	-	-	-	-	-	-	-	B/AH1	B/AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	-	-	-	-	-	-	-		
Portugal		-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	A/B	-	-	-	-	-	-	-	-	-		
Spain		-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B	B	B	B	B	B		
United Kingdom (Northern Ireland)		-	B	AH3	A	A	-	-	AH1	AH1	A	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	A/B	B/AH1	B	A/B	A/B	B	B	A/B	
United Kingdom (Wales)		-	-	-	-	-	-	-	AH1	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B	B	B	B	B	B	B	B		
United Kingdom (Scotland)		-	-	-	-	-	A	-	-	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	B	B	B	B	B	B	B		
United Kingdom (England)		-	-	-	-	-	-	-	-	-	-	-	A	-	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	B	B	B	A/B	B	B		
France		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B	B	B	B	B	B	B	B	B	B	-	-	-	-	-	-	-		
Belgium		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B	B	B/AH1	B/AH1	B	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	A	-	-	-	-	-	-		
Netherlands		-	-	-	-	-	-	-	-	AH1	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B	B	B	B	B	B	A/B	-	
Luxembourg		-	-	-	-	-	-	-	-	-	-	-	B	-	-	B	-	-	B	B	B	B	B	B	B	B	B	B/AH1	B/AH1	B/AH1	-	-	-	-	-	-	
Switzerland		-	A	-	A	A/B	-	A	A	A/B	A	A/B	A/B	A/B	A	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B	B	B	B	B	B/AH1	A/B	A/B	-	-	-	-	-	
Germany		-	-	-	-	-	-	-	-	-	-	-	AH1	-	-	AH1	AH1	AH1	AH1	AH1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Denmark		-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	-	-	-	-	-	-	-		
Norway		-	-	-	-	A	A	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B	B	B	B	B	B	B	B	B	
Italy		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Austria		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Malta		-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B	B	B	B	B	-	-	-	-	
Slovenia		-	-	-	-	-	-	-	-	-	-	-	A	A	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	A	A/B	A/B	A/B	B	B	B	B	-	-	-	
Sweden		-	-	-	-	-	-	-	-	-	A	A	A	A	A	A	A	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B	B	B	B	B	B	B	B	B	B	B	
Croatia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	-	AH1	AH1	AH1	AH1	AH1	AH1	A	AH1	AH1	AH3	AH3	AH3	-	-	-	-	-	-	-	
Czech Republic		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	B/AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	A/B	-	-	-	-	-	-	
Bosnia and Herzegovina		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Slovakia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Albania		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	-	-	-	B/AH1	B/AH1	B/AH1	B/AH1	B	B	-	-	-	-	-	-	-	-	-	
Hungary		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B	B	B	B	B	B	-	-	-	-	-	-	-	-	-	
Poland		-	-	AH1	-	-	A	A	-	A	A	A	-	A	A	A	A	A	AH1	AH1	AH1	AH1	AH1	AH1	B	B	B	B	A	A	B	B	AH1	-	-	-	-
Montenegro		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	-	AH1	AH1	-	AH1	-	-	-	-	-	
Serbia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
The former Yugoslav Republic of Macedonia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	A	A	A	A	A	A	A	A/B	A/B	A/B	A/B	A/B	A/B	-	-	-	-
Greece		-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	A	-	AH1	-	B/AH1	B	-	B	B	B	B	B	B	-	-	-
Lithuania		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	-	-	-	-	-	-	-	-	-	
Bulgaria		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	-	B	B/AH1	-	-	-	-	-	-	-	-	
Latvia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	A	AH1	A	AH1	A	AH1	AH1	AH1	AH1	A	A	A	A/B	A/B	B	A/B	B	B	B	B	B	
Romania		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B	-	-	-	-	-	-	-	
Estonia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B/AH1	-	B/AH1	B/AH1	B/AH1	-	AH1	-	B/AH1	AH1	-	A	-	-	-	-	-	-	
Finland		-	-	-	-	-	-	-	-	-	AH1H3	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B	-	B	AH1	B	B	-	B	-	-	-	-	
Belarus		-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B	B	B
Republic of Moldova		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	-	-	-	-	-	-	-	-	-	-	-	
Ukraine		-	-	-	-	-	-	-	-	-	-	-	-	-	-	B/AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1H3	AH1	B/AH1H3	-	-	-	-	-	
Cyprus		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Israel		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Turkey		-	-	-	-	-	-	-	-	-	-	-	-	-	-	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	
Georgia		-	-	-	-	-	-	-	-	-	AH1	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1
Armenia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	-	-	AH1	AH1	-	AH1	AH1	-	AH1	-	-	-	-	-	
Azerbaijan		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B
Uzbekistan		-	-	-	-	-	-	-	-	-	-	-	AH1	-	-	AH1	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B	-	B	B	B	B	B	B	B	B	B	
Kazakhstan		-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	-	AH1	AH1	AH1	AH1	AH1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Russian Federation		-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	A/B	A/B	B	B	B	B	B	
Tajikistan		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Kyrgyzstan		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	

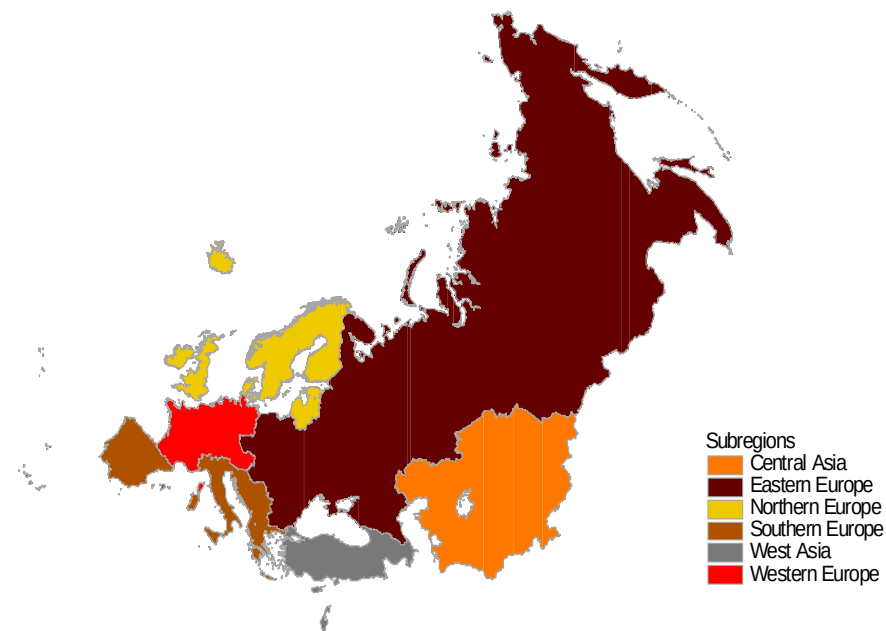
WEST

EAST

Very High = exceptionally high level of activity
 High = levels of activity higher than usual
 Medium = usual levels of activity
 Low = no activity or activity at baseline* level

* Baseline influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.

Table E1. Distribution of Influenza Detections in the WHO European Region¹: 2015-2016 Season (weeks 40/2015-20/2016)



Influenza Detections*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	14018	71%	10729	42%	27580	90%	34940	68%	8874	79%	1173	91%	97314	70%
Influenza A subtyped	10981		5396		25446		15938		8803		1170		67734	
<i>A (H1N1)pdm09</i>	9501	87%	5113	95%	24409	96%	15324	96%	6382	72%	1004	86%	61733	91%
<i>A (H3N2)</i>	1480	13%	283	5%	1037	4%	614	4%	2421	28%	166	14%	6001	9%
Influenza B	5701	29%	14636	58%	3067	10%	16390	32%	2335	21%	117	9%	42246	30%
Influenza B lineage determined	1033		3698		614		2556		156		1		8058	
<i>B/Yamagata lineage</i>	84	8%	116	3%	69	11%	460	18%	0		0		729	9%
<i>B/Victoria lineage</i>	949	92%	3582	97%	545	89%	2096	82%	156	100%	1	100%	7329	91%
Total	19719		25365		30647		51330		11209		1290		139560	

*combined ILI, ARI sentinel and non-sentinel respiratory specimens positive for influenza

¹ WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: <http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>

Figure E3. Genetically characterised influenza A(H3N2) viruses reported to TESSy from week 5/2016

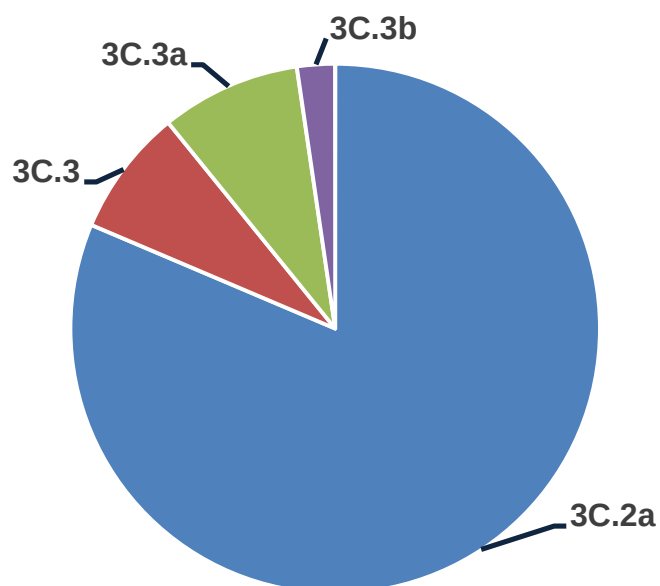


Figure E4. Genetically characterised influenza B viruses reported to TESSy from week 5/2016

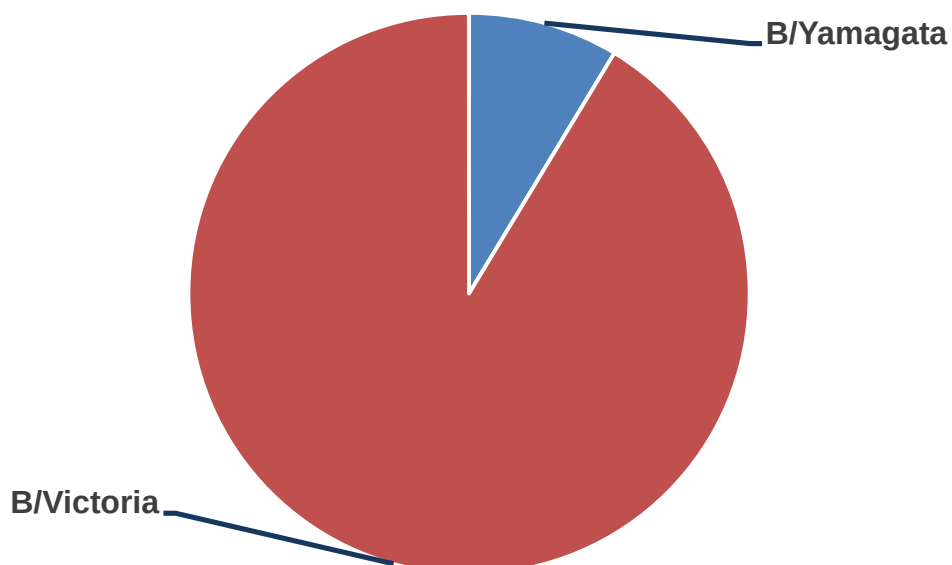
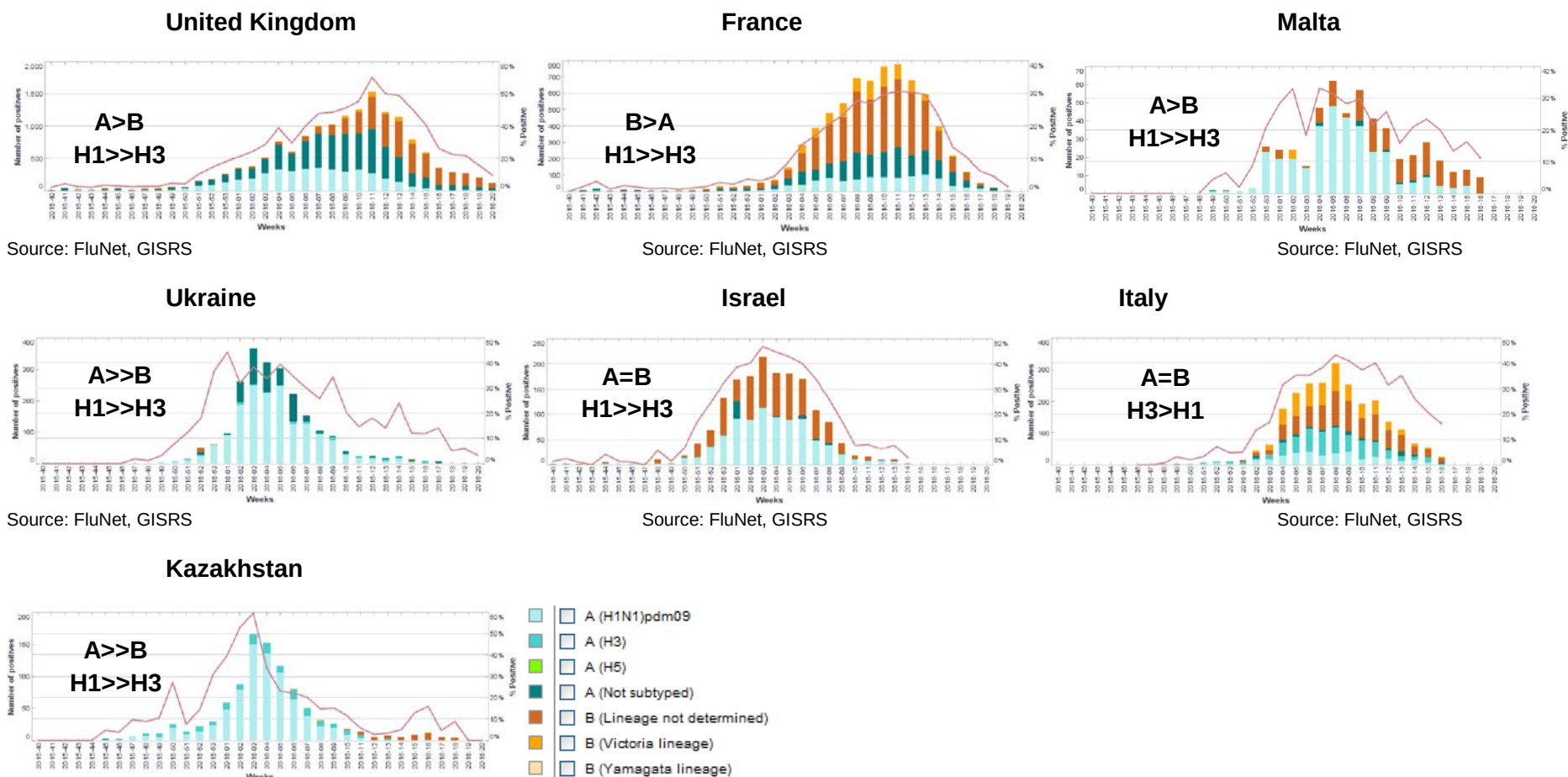


Figure E5. Patterns of Influenza Type and Subtype Circulation in the WHO European Region, combined sentinel and non-sentinel reporting



Source: FluNet, GISRS

Source: FluNet, GISRS

Source: FluNet, GISRS

Source: FluNet, GISRS

Source: FluNet, GISRS

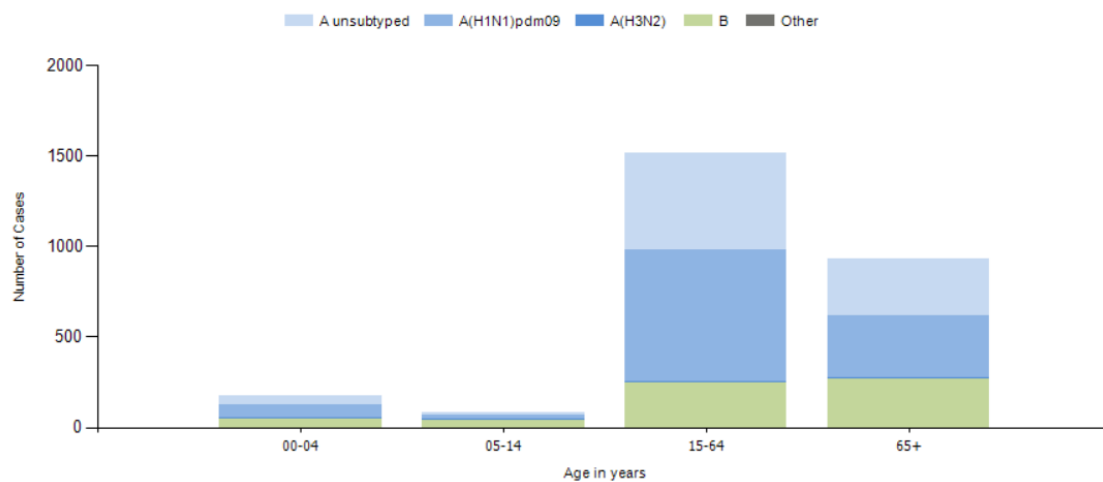
Source: FluNet, GISRS

Source: FluNet, GISRS

= represents similar ratio; > represents a ratio of at least 1.5:1; >> represents a ratio of ≥ 5

Figure E6. Hospitalised influenza cases

Distribution of virus (sub)type by age group in laboratory-confirmed hospitalised cases (ICU)*



*Reporting countries: Finland, France, Ireland, Romania, Slovakia, Spain, Sweden, and the United Kingdom

Influenza specimens received at WHO CC, London: the Francis Crick Institute, Mill Hill Laboratory.

Samples, viruses or clinical samples, with collection dates after 2016-01-31 have been received from 48 countries in Europe, Africa, the Middle East, Central Asia and the Far East. Seventy *per cent* of the samples received were type A viruses, with A(H1N1)pdm09 viruses outnumbering A(H3N2) viruses at a ratio of 1.9:1 (**Table 1**). Of the type B viruses, viruses of the B/Victoria lineage predominated over those of the B/Yamagata lineage at a ratio of over 6:1. These proportions are different from those reported within the WHO European region with a higher sampling rate of A(H3N2) viruses (**Figures E1B & E1C, Table E1**). **Table 2** shows the recovery/isolation rates from specimens received with collection dates after 2016-01-31. Both viruses and clinical samples were recovered efficiently. For A(H3N2) viruses, efficient recovery was obtained as assessed by detection of virus neuraminidase activity using the MUNANA sialidase assay, but approximately two thirds (67%) of these could not be assessed by HI assay due to HA titres being low (<4 HA units in the presence of 20nM oseltamivir). **Table 3** summarises the reactivity of A(H1N1)pdm09, A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages in HI assays with post-infection ferret antisera raised against vaccine viruses and representative reference viruses. For A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PRNA) is also summarised. Numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera for their respective homologous reference/vaccine viruses are shown. The antiserum raised against the cell culture-propagated cultivar of A/Switzerland/9715293/2013 had a low homologous titre and relative titres for the test viruses could only be estimated as ≤ 2 -fold or ≥ 4 -fold reduced compared with the titre of this antiserum with its homologous virus.

Table 1. Summary of clinical samples and virus isolates received, with collection dates after 2016-01-31, by country*

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2			B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
2016														
FEBRUARY														
Albania	16			10	10						6	6		
Algeria	7			4	0	3	0	1						
Belarus	3			3	3									
Bosnia & Herzegovina	4	1	0	2	1	1	0	1						
Bulgaria	47			34	32	1	0	1			12	11		
Cyprus	9			8	8						1	1		
Czech Republic	8			8	8									
Egypt	7			6	0	1	0	0						
Estonia	8			6	6						2	2		
Finland	5			4	4	1	1	0						
France	2			1	1	1	0							
Germany	13			6	6	2	2	0					5	5
Georgia	5			5	4									
Ghana	12			10	4	2	0	2						
Greece	4			4	2									
Iceland	5			4	4								1	1
Italy	22			7	7	10	7	3			4	4	1	1
Jordan	13			12	11	1	0	1						
Kazakhstan	13			11	8	2	1	1						
Kyrgyzstan	4			4	3									
Latvia	2			2	2									
Lithuania	12			10	10						2	2		
Macedonia	2			1	0	1	0							
Montenegro	18	1	0	14	12	1	0	0			2	2		
Morocco	5			3	3	1	0	1	1	0				
Netherlands	1			1	1									
Niger	16			10	in progress								6	in progress
Oman	20			6	6	8	1	5			5	5	1	1
Palestine	2			2	1									
Portugal	1			1	1									
Romania	6			3	3	2	2	0			1	1		
Russia	18			11	10	1	1	0			5	5	1	1
Senegal	1	1	in progress											
Serbia	30			10	6	9	1	7	2	0	9	9		
Slovakia	9			3	3	1	1	0			5	5		
Slovenia	17			5	5	9	4	4			3	3		
South Africa	1			1	1									
Spain	17			15	14						2	2		
Sweden	5			1	1	2	2	0			2	2		
Tajikistan	3			2	0	1	0	0						
Turkey	62			26	21	21	11	8	6	0	9	9		
Ukraine	48			41	36	7	2	5						
United Kingdom	1												1	1
MARCH														
Albania	2			2	2									
Algeria	2			1	0	1	0	1						
Bulgaria	16			6	5	2	0	2			8	8		
Cameroon	3								3	0				
Cote d'Ivoire	2			2	2									
Czech Republic	11			2	2	1	1	0			8	8		
Egypt	4										2	2	2	2
Estonia	10			7	7						3	3		
Finland	5			2	2						3	3		
France	3			1	1	1	0						1	1
Georgia	6			2	2	1	1	0			3	3		
Germany	9			1	0	1	0	1			7	7	1	1
Ghana	18			16	12	2	0	2						
Iceland	5			3	3	1	0	1			1	1		
Italy	10			1	1	5	4	1			4	4		
Jordan	23			7	6	7	3		2	0	6	6	1	1
Kazakhstan	10			6	4	1	0	1			3	3		
Macedonia	1			1	0									
Madagascar	1					1	0	1						
Montenegro	9	1	0	5	4	1	0	1	2	0				
Niger	5			4	in progress								1	1
Norway	2			2	2									
Oman	3			1	1	2	0	2						
Palestine	2			1	0						1	1		
Portugal	10			6	6						4	3		
Romania	9			6	6	1	1	0			1	1	1	1
Russia	4			1	1						3	3		
Senegal	5	4	in progress	1	1									
Serbia	15			3	0	4	0	3	8	0				
Slovakia	10			3	3						5	5	2	2
Slovenia	14			3	3	6	3	3			5	5		
Sweden	5			1	1	1	1	0			2	2	1	1
Turkey	10			1	0	4	3	1	2	0	3	3		
Ukraine	17			10	9	7	0	7						
United Kingdom	1					1	0	1						
APRIL														
Cameroon	3								3	0				
Cote d'Ivoire	25	1	0	16	16	8	4	4						
France	24			11	11	1	0	1			12	12		
Ghana	3			3	2									
Georgia	8					5	1	2			3	2		
Iceland	8			2	2	2	0	1			3	3	1	1
Italy	3			1	1	1	0	1			1	1		
Jordan	18			1	1	1	0	1	1	0	13	13	2	2
Kazakhstan	4								2	0	2	2		
Niger	1												1	1
Portugal	6			1	1						5	5		
Romania	8			3	3						5	5		
Russia	6			1	1	3	0	3			2	2		
Senegal	1			1	1									
Serbia	3			1	0				2	0				
Slovakia	2			1	1								1	1
Slovenia	4			1	1						2	2	1	1
Sweden	1					1	0	1						

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2			B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
Tajikistan	4	1	0			1	0	1	2	0				
Turkey	1										1	1		
Ukraine	4			3	2	1	0	1						
United Kingdom	8			2	2	1	1				3	3	2	2
MAY														
Cameroon	6	1	0			3	0	1	2	0				
Cote d'Ivoire	4			1	1	3	1	2						
France	2										2	2		
Ghana	2					2	0	2						
Iceland	4					2	1	1			2	2		
Jordan	8					1	0	1			7	7		
Madagascar	8					7	1	6			1	1		
Norway	6			1	1	2	2	0			2	2	1	1
Senegal	3			3	3									
Slovenia	2										2	2		
South Africa	9			2	2	2	2	0			5	5		
Ukraine	1			1	1									
United Kingdom	4			1	1						3	3		
JUNE														
Cameroon	8					7	0	3	1	0				
Egypt	2					1	0	0	1	0				
Ghana	16			5	3	9	0	7					2	2
Hong Kong	1										1	1		
Iceland	1										1	1		
Kazakhstan	1								1	0				
Madagascar	1										1	1		
Niger	2					2	0	2						
Norway	3					3	1	2						
Senegal	2	2	in progress											
Serbia	1								1	0				
South Africa	19			1	1	11	4	6			7	7		
United Kingdom	3					3	0	3						
JULY														
Cameroon	4					4	1							
Egypt	4					2	0	2	1	0	1	1		
France	2			1	1	1	in progress							
Ghana	8			1	1	5	0	1			2	2		
Hong Kong	4					2	2	0			1	1	1	1
Niger	4					4	0	0						
South Africa	17			1	1	11	1	10			5	4		
United Kingdom	1					1	0	1						
AUGUST														
Hong Kong	5			3	3	1	0	1					1	1
United Kingdom	6					6	in progress							
	1085	13	0	493	408	260	75	135	43	0	237	233	39	33
48 Countries				70.6%					29.4%					

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

* As of 2016-0-9-16

Table 2. Isolation rates for specimens collected after 2016-01-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated ¹	Percent recovered	Number processed	Number propagated ¹	Percent recovered
H1	137	77	56%	356	336	94%
H3	115	12 ²	10%	147	62 ²	42%
		74 ³	74% (14%) ⁴ 64%		79 ³	96% (44%) ⁴ 54%
B Victoria	55	53	96%	182	180	99%
B Yamagata	14	12	86%	25	25	100%

1. Propagated to sufficient HA titre to perform HI assay

2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay

3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay

4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2016-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction microneutralisation assays

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ¹	413	393 (95.2%)	10 (2.4%)	10 (2.4%)
H3 ^{2a}	74	0	1 (1.4%)	73 (98.6%)
H3 ^{2b}	74	48 (64.9%)	26 (35.1%) ⁶	
H3 ^{2c}	74	51 (68.9%)	18 (24.34%)	5 (6.8%)
H3 ^{2d}	74	26 (35.1%)	35 (47.3%)	13 (17.6%)
H3 ^{2e}	74	65(87.8%)	9 (12.2%)	0
H3 ^{3a}	88	83 (94.3%)	5 (5.7%)	0
H3 ^{3b}	88	40 (45.5%)	19 (21.6%)	29 (32.9%)
H3 ^{3c}	88	72 (81.8%)	15 (17.0%)	1 (1.2%)
B/Victoria ⁴	234	203 (86.8%)	22 (9.4%)	9 (3.8%)
B/Yamagata ⁵	37	30 (81.1%)	7 (18.9%)	0

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009

2a. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)

2b. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)

2c. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)

2d. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a)

2e. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)

3a. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a) in plaque-reduction neutralisation assays

3b. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a) in plaque-reduction neutralisation assays

3c. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/7295/2014 (genetic subgroup 3C.2a) in plaque-reduction neutralisation assays

4. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016)

5. Compared to homologous titres for ferret antisera raised against vaccine virus B/Phuket/3073/2013

6. Unable to differentiate fold drop in titre for most viruses as it falls outside the dynamic range of the antiserum

Influenza A(H1N1)pdm09 virus analysis.

Influenza A(H1N1)pdm09 viruses were received from NICs in 46 countries or administrative regions in four WHO Regions: African (5: Algeria, Cote d'Ivoire, Ghana, Niger and Senegal), Eastern Mediterranean (4: Egypt, Jordan, Oman and Palestine), European (36 countries) and Western Pacific (1: Hong Kong SAR). **Table 4** summarises the antigenic (HI) results for viruses, recovered from clinical samples with collection dates after 2016-01-31 that were propagated successfully at the Francis Crick Institute.

The vast majority of H1N1 viruses collected after 2016-01-31 (97%) were antigenically similar to the vaccine virus (**Table 4**). **Tables 5-1 to 5-22** show the results of HI assays of H1N1 viruses carried out since our February 2016 report. Test viruses in each table are sorted by date of collection and viruses with collection dates after 2016-01-31 are shown below red lines in **Tables 5-4** and **5-6 to 5-22**. A summary of the results for viruses with collection dates after 2016-01-31 is shown in **Table 5-23**.

Over 97% of the test viruses were recognised by the antisera raised against the A/California/7/2009 vaccine virus at titres within 4-fold of the titres for the homologous virus, and 95% within 2-fold of the homologous titre. With the exception of the antiserum raised against A/Christchurch/16/2010 the other antisera also recognised the test viruses very well, with 90 to 100% of viruses being recognised at titres within 4-fold of the homologous titres for the antisera. With the additional exception of the antiserum raised against A/Lviv/N6/2009, 87% to 100% of the test viruses were recognised at titres within 2-fold of the homologous titres of the antisera (**Table 5-23**). Antisera raised against two viruses with HA genes in genetic subgroup 6B.2, against A/Ukraine/6907/2015 used only in **Table 5-14** and against A/Hong Kong/90/2016 used only in **Table 5-16**, recognised all the viruses tested at titres within 2-fold of the titres of the antisera with the homologous viruses. The antiserum raised against A/Lviv/N6/2009 recognised only 41% of test viruses at titres within 2-fold of the titre of the antiserum for the homologous virus (90% at titres within 4-fold of the homologous titre) while the antiserum raised against A/Christchurch/16/2010 recognised only 20% of the test viruses collected since 2016-01-31 at titres within 4-fold of the titre for the homologous virus. Of the test viruses for which the genetic group or subgroup were known (89% of the total examined) two test viruses fell into genetic group 6B (<1%) and three in genetic subgroup 6B.2 (<1%) with the vast majority (>98%) falling into genetic subgroup 6B.1.

Figure 1 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. Over the last seven years the HA genes have evolved compared to A/California/7/2009 and eight genetic groups have been designated with viruses in group 6 forming clusters designated groups 6A, 6B and 6C. Viruses collected after 2016-01-31 fell into genetic group 6B and in two subgroups, 6B.1 and 6B.2 (**Figure 1**). Subgroup 6B and subgroups 6B.1 and 6B.2 are defined by the following amino acid substitutions in **HA1** and **HA2**:

- **P83S, D97N, S185T, S203T K163Q, A256T, K283E and I321V** in **HA1**, and **E47K, S124N** and **E172K** in **HA2** compared with A/California/7/2009 define **genetic group 6B**, e.g. A/South Africa/3626/2013.
- **S84N, S162N (+CHO)** and **I216T** in **HA1** define **subgroup 6B.1** within group 6B, e.g. A/Slovenia/2903/2015
- **V152T** and **V173I** in **HA1** and **E164G** and **D174E** in **HA2** define **subgroup 6B.2** within group 6B, e.g. A/Israel/Q-504/2015.

Another minor grouping is presently detected within group 6B viruses. This group is represented by viruses from Ghana, Honduras, Kyrgyzstan and Latvia, with the amino acid substitutions **A48P, S69P** in **HA1** and **N43S** in **HA2**.

An amino acid sequence alignment of the HAs for representative viruses is shown in **Figure 2**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3 illustrates the location of amino acid residues on an H1-HA structure defining genetic group 6B and the two major subgroups 6B.1 and 6B.2.

Figure 4 shows a phylogenetic tree for NA genes of representative H1N1 viruses. The HA and NA phylogenetic trees are generally congruent.

Figure 5 shows a NA amino acid alignment for a selected group of these viruses, annotated as **Figure 2** but lacking a proteolytic-processing site.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 678 viruses, with 415 having collection dates after 2016-01-31. All but five viruses tested showed normal inhibition (NI) to both neuraminidase inhibitors (**Table 14**). Four showed highly reduced inhibition (HRI) by oseltamivir due to NA H275Y amino acid substitution and one showed reduced inhibition (RI) by zanamivir with the virus, A/Odessa/699/2016, being polymorphic at two NA positions, Q136X and Y274X.

Conclusion.

Although new genetic subgroups have emerged in recent months, the vast majority of A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus A/California/7/2009 and retained susceptibility to oseltamivir and zanamivir.

Table 4. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2016-01-31

MONTH	H1N1pdm09				
2016	Fold reduction in HI titre				
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³
FEBRUARY					
Albania	10	10	10		
Algeria	4	0	0		
Belarus	3	3	3		
Bosnia & Herzegovina	2	1	1		
Bulgaria	34	32	31	1	
Cyprus	8	8	8		
Czech Republic	8	8	8		
Egypt	6	0	0		
Estonia	6	6	6		
Finland	4	4	4		
France	1	1	1		1
Georgia	5	4	3		
Germany	6	6	6		
Ghana	10	4	4		
Greece	4	2	2		
Iceland	4	4	4		
Italy	7	7	6	1	
Jordan	12	11	11		
Kazakhstan	11	8	7	1	
Kyrgyzstan	4	3	3		
Latvia	2	2	2		
Lithuania	10	10	10		
Macedonia	1	0	0		
Montenegro	14	12	12		
Morocco	3	3	3		
Netherlands	1	1	1		
Niger	10	in progress	4		
Oman	6	6	6		
Palestine	2	1	1		
Portugal	1	1	1		
Romania	3	3	3		
Russia	11	10	9	1	
Serbia	10	6	6		
Slovakia	3	3	3		
Slovenia	5	5	4		1
South Africa	1	1	1		
Spain	15	14	14		
Sweden	1	1	1		
Tajikistan	2	0	0		
Turkey	26	21	21		
Ukraine	41	36	31	3	2
MARCH					
Albania	2	2	2		
Algeria	1	0	0		
Bulgaria	6	5	5		
Cote d'Ivoire	2	2	2		
Czech Republic	2	2	2		
Estonia	7	7	7		
Finland	2	2	2		
France	1	1	1		
Georgia	2	2	2		
Ghana	16	12	12		
Iceland	3	3	3		
Italy	1	1	1		
Jordan	7	6	6		
Kazakhstan	6	4	4		
Macedonia	1	0	0		
Montenegro	5	4	4		
Niger	4	in progress	1		
Norway	2	2	2		
Oman	1	1	1		
Palestine	1	0	0		
Portugal	6	6	6	1	
Romania	6	6	5		
Russia	1	1	1		
Senegal	1	1	1		
Serbia	3	0	0		
Slovakia	3	3	2	1	
Slovenia	3	3	3		
Sweden	1	1	1		
Turkey	1	0	0		
Ukraine	10	9	7		2
APRIL					
Cote d'Ivoire	16	16	16		
France	11	11	11		
Ghana	3	2	2		
Iceland	2	2			2
Italy	1	1	1		
Jordan	1	1		1	
Portugal	1	1	1		
Romania	3	3	3		
Russia	1	1	1		
Senegal	1	1	1		
Serbia	1	0	0		
Slovakia	1	1			1
Slovenia	1	1	1		
Ukraine	3	2	1		1
United Kingdom	2	2	2		
MAY					
Cote d'Ivoire	1	1	1		
Norway	1	1	1		
Senegal	3	3	3		
South Africa	2	2	2		
Ukraine	1	1	1		
United Kingdom	1	1	1		
JUNE					
Ghana	5	3	3		
South Africa	1	1	1		
JULY					
Ghana	1	1	1		
France	1	1	1		
South Africa	1	1	1		
AUGUST					
Hong Kong	3	3	3		
46 Countries	493	408	393 95.2%	10 2.4%	10 2.4%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assays

3. Reduction in HI titre with post-infection ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-02-16)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/15
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg
	Ferret number			F05/14 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²
	Genetic group						4	5	6	7	6A	6B	6B.1
REFERENCE VIRUSES													
A/California/7/2009	NIBSC	2009-04-09	E4/E3	2560	1280	1280	640	2560	1280	5120	2560	2560	2560
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	320	80	80	40	80	80	80	80
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	1280	320	160	160	160	320	160	320
A/Christchurch/16/2010		2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	2560	1280	2560
A/Astrakhan/1/2011		2011-02-28	MDCK1/MDCK5	1280	1280	640	1280	2560	1280	5120	2560	1280	2560
A/St. Petersburg/27/2011		2011-02-14	E1/E4	1280	640	640	640	1280	640	2560	1280	1280	1280
A/St. Petersburg/100/2011		2011-03-14	E1/E3	2560	1280	1280	1280	2560	1280	5120	5120	2560	5120
A/Hong Kong/5659/2012		2012-05-21	MDCK4/MDCK1	1280	320	320	320	1280	640	2560	1280	1280	640
A/South Africa/3626/2013		2013-06-06	E1/E3	12580	640	1280	640	1280	640	2560	1280	1280	1280
A/Slovenia/2903/2015	clone 37	2015-10-26	E4	1280	640	640	640	1280	640	2560	2560	1280	1280
TEST VIRUSES													
A/Ghana/DILI-15-0386/2015		2015-04-27	Cx/MDCK1	2560	640	1280	1280	1280	1280	5120	2560	1280	5120
A/Ghana/DILI-15-0455/2015		2015-05-07	Cx/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120
A/Ghana/DILI-15-0544/2015		2015-06-10	Cx/MDCK1	1280	640	640	640	1280	1280	2560	2560	1280	2560
A/Ghana/DILI-15-0661/2015		2015-07-08	Cx/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120
A/Aragon/1615/2015		2015-09-29	SIAT3/SIAT1	1280	1280	640	1280	1280	1280	2560	2560	1280	2560
A/Madrid/SO13656/2015		2015-10-29	SIAT1/SIAT1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Navarra/1736/2015		2015-11-14	SIAT1/SIAT1	2560	640	640	640	2560	1280	5120	2560	1280	5120
A/Egypt/7185/2015		2015-12-19	MDCK1	2560	1280	640	1280	2560	1280	5120	5120	1280	5120
A/Ukraine/6959/2015		2015-12-21	C2/SIAT1	1280	640	640	640	1280	640	1280	1280	1280	2560
A/Ukraine/6958/2015		2015-12-29	C1/SIAT1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120
A/Castilla La Mancha/16013/2015		2015-12-30	SIAT1/SIAT1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Turkey/TR-09/2016		2016-01-14	MDCK1	1280	640	320	1280	1280	640	2560	2560	1280	2560
A/Turkey/TR-10/2016		2016-01-14	MDCK1	1280	640	320	1280	1280	640	2560	2560	1280	5120
A/Turkey/TR-12/2016		2016-01-14	MDCK1	640	320	320	640	640	320	2560	1280	640	2560
A/Greece/39/2016		2016-01-15	MDCK2	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Greece/41/2016		2016-01-16	MDCK2	1280	640	320	1280	1280	640	2560	2560	1280	2560
A/Egypt/1565/2016		2016-01-23	MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	5120
A/Egypt/1602/2016		2016-01-23	MDCK1	2560	1280	640	1280	2560	1280	5120	5120	1280	5120

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-02-17)

Haemagglutination inhibition titre													
Post-infection ferret antisera													
Viruses	Other	Collection	Passage	A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov
	information	date	history	7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/15
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg
	Ferret number			F05/14 ⁻¹	F09/15 ⁻¹	F14/13 ⁻¹	F15/14 ⁻¹	F22/13 ⁻¹	F26/14 ⁻¹	F24/11 ⁻¹	F30/12 ⁻¹	F03/14 ⁻¹	F02/16 ⁻²
	Genetic group						4	5	6	7	6A	6B	6B.1
REFERENCE VIRUSES													
A/California/7/2009	NIBSC	2009-04-09	E4/E3	1280	320	320	320	640	320	1280	1280	640	640
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	<	320	320	160	80	40	80	80	80	<
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	1280	320	80	80	80	160	80	80
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	2560	2560	640	2560	2560	1280	2560
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	1280	1280	640	2560	2560	1280	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	640	640	1280	640	2560	1280	1280	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	640	1280	1280	640	2560	2560	2560	2560
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	320	320	320	1280	640	640	640
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	640	640	640	2560	1280	1280	1280
A/Slovenia/2903/2015	clone 37	2015-10-26	E4	1280	640	640	2560	2560	1280	5120	2560	1280	2560
TEST VIRUSES													
A/Oman/6329/2015	6B.1	2015-11-17	SIAT2/MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	2560
A/Oman/6337/2015	6B	2015-11-18	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Oman/6338/2015	6B	2015-11-18	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Oman/6268/2015	6B.1	2015-12-11	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Istanbul/201/2015		2015-12-31	MDCK1/MDCK1	1280	640	320	640	640	320	2560	1280	1280	2560
A/Jordan/20012/2016	6B.1	2016-01-05	Cx/MDCK1	1280	640	320	640	640	640	2560	2560	1280	5120
A/Jordan/10094/2016	6B.1	2016-01-06	Cx/MDCK1	1280	640	320	640	640	640	2560	1280	1280	2560
A/Jordan/10090/2016	6B.1	2016-01-06	Cx/MDCK1	640	640	320	640	640	640	2560	1280	1280	2560
A/Jordan/20024/2016	6B.1	2016-01-07	Cx/MDCK1	1280	1280	640	640	640	640	2560	2560	2560	2560
A/Jordan/10119/2016	6B.1	2016-01-07	Cx/MDCK1	1280	1280	640	640	640	640	2560	2560	2560	2560
A/Jordan/10118/2016	6B.1	2016-01-07	Cx/MDCK1	1280	640	320	640	640	640	2560	1280	1280	2560
A/Jordan/10228/2016	6B.1	2016-01-10	Cx/MDCK1	1280	1280	640	640	640	640	2560	2560	1280	2560
A/Jordan/10227/2016	6B.1	2016-01-10	Cx/MDCK1	1280	640	320	640	640	640	2560	2560	1280	2560
A/Jordan/10209/2016	6B.1	2016-01-10	Cx/MDCK1	1280	1280	320	640	640	640	2560	2560	1280	2560
A/Jordan/20081/2016	6B.1	2016-01-12	Cx/MDCK1	1280	640	640	640	640	640	2560	1280	1280	2560
A/Turkey/TR-18/2016	6B.1	2016-01-15	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Turkey/TR-34/2016	6B.1	2016-01-15	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Turkey/TR-36/2016	6B.1	2016-01-15	MDCK1	1280	640	320	1280	640	640	2560	2560	2560	2560
A/Greece/50/2016	6B.1	2016-01-18	MDCK1	640	320	320	640	640	320	1280	1280	1280	2560
A/Moldova/45.03/2016	6B.1	2016-01-20	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Moldova/47.03/2016	6B.1	2016-01-20	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560
A/Moldova/48.03/2016	6B.1	2016-01-20	MDCK1	640	320	160	320	640	640	2560	1280	1280	2560
A/Moldova/59.03/2016	6B.1	2016-01-21	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Moldova/62.03/2016	6B.1	2016-01-22	MDCK1	1280	1280	320	640	1280	640	2560	2560	1280	2560
A/Greece/65/2016	6B.1	2016-01-22	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-02-23)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/15
	Passage history			NIBSC	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg
	Ferret number			F05/14 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²
	Genetic group						4	5	6	7	6A	6B	6B.1
REFERENCE VIRUSES													
A/California/7/2009	NIBSC	2009-04-09	E4/E3	1280	640	640	640	2560	640	5120	2560	1280	2560
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	320	160	80	80	80	80	80	80
A/Lviv/6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	1280	320	160	160	160	320	160	320
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	2560	1280	2560	2560	1280	2560
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	1280	2560	1280	2560	2560	1280	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	640	1280	1280	2560	2560	1280	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	2560	1280	2560
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	320	640	320	1280	1280	640	640
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	640	1280	640	1280	640	2560	1280	1280	1280
A/Slovenia/2903/2015	clone 37	2015-10-26	E4	1280	640	640	640	1280	640	2560	2560	1280	2560
TEST VIRUSES													
A/Michigan/45/2015	6B.1	2015-09-07	E3/E1	1280	640	640	1280	1280	1280	2560	2560	1280	2560
A/Stockholm/41/2015	6B.1	2015-09-22	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560
A/Italy-FVG/104/2015	6B.1	2015-10-13	SIAT3/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Norway/2634/2015	6B	2015-10-27	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Paris/2415/2015	6B.1	2015-11-09	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Croatia/2662/2015	6B.1	2015-11-14	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560
A/Belgium/2015G0778/2015	6B.1	2015-11-24	MDCK2	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Odessa/7/1/2016	6B.2	2015-11-27	SIAT2/MDCK1	640	640	320	320	640	640	1280	1280	1280	2560
A/Ukraine/6903/2015	6B.1	2015-12-01	C1/MDCK1	640	640	320	640	1280	640	1280	1280	1280	2560
A/Switzerland/15818956/2015	6B.1	2015-12-03	MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	2560
A/Ukraine/587/2015	6B.1	2015-12-09	SIAT1/MDCK1	1280	1280	640	1280	2560	1280	2560	2560	1280	2560
A/Kazakhstan/4235/2015	6B.1	2015-12-11	MDCK1	640	320	320	640	640	640	2560	1280	1280	1280
A/Odessa/69/2016	6B.1	2015-12-11	SIAT2/MDCK1	640	320	160	320	640	320	1280	640	640	1280
A/Kharkiv/595/2015	6B.1	2015-12-11	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Kazakhstan/4222/2015	6B.1	2015-12-12	MDCK2	640	640	320	640	1280	640	2560	1280	1280	2560
A/Algeria/G0164-15/2015	6B.1	2015-12-13	MDCK1	1280	640	320	1280	1280	640	2560	2560	1280	2560
A/Odessa/64/2016	6B.2	2015-12-14	SIAT2/MDCK1	320	320	320	320	640	320	640	640	640	1280
A/Odessa/66/2016	6B.2	2015-12-14	SIAT2/MDCK1	640	640	640	640	1280	640	2560	1280	1280	2560
A/Rennes/2549/2015	6B.1	2015-12-14	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Khmelnitsky/80/2016	6B.1	2015-12-15	SIAT2/MDCK1	1280	640	640	1280	2560	640	2560	2560	1280	2560
A/Parma/293/2015	6B	2015-12-15	MDCK2/MDCK1	1280	640	640	640	1280	1280	2560	2560	1280	2560
A/Bretagne/2522/2015	6B.1	2015-12-16	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Switzerland/15925910/2015	6B.1	2015-12-17	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Georgia/1473/2015	6B	2015-12-23	MDCK1	2560	1280	640	1280	2560	1280	2560	2560	2560	2560
A/Israel/Q-636/2015	6B.2	2015-12-29	C1/MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	2560
A/Israel/Q-638/2015	6B.2	2015-12-29	C1/MDCK1	320	320	320	320	320	320	640	320	320	640
A/Hungary/1/2016	6B.1	2016-01-04	MDCK3/MDCK1	320	640	640	640	2560	320	640	640	640	1280
A/Hungary/2/2016	6B.1	2016-01-11	MDCK3/MDCK1	640	320	320	320	320	640	1280	1280	640	1280
A/Hungary/13/2016	6B.1	2016-01-12	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280
A/Hungary/12/2016	6B	2016-01-14	MDCK1/MDCK1	40	160	320	40	80	80	80	80	160	160
A/Thuringen/4/2016	6B.1	2016-01-18	C2/MDCK1	640	320	160	320	640	320	1280	640	640	640
A/Berlin/6/2016	6B.1	2016-01-19	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280
A/Brandenburg/3/2016	6B.1	2016-01-19	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Egypt/1612/2016	6B.1	2016-01-23	MDCK2	640	640	640	640	1280	640	2560	1280	640	1280
A/Rheinland-Pfalz/8/2016	6B.1	2016-01-25	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-03-01)

Haemagglutination inhibition titre																		
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera														
				A/Cal	A/Cal	A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel		
				7/09	7/09	7/09	69/09	N6/09	16/10	1/11	27/11	100/11	100/11	5659/12	3626/13	2903/15	Q-504/15	
				NIBSC	Egg	Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg	MDCK	
				F05/14 ¹	F06/16 ¹	F07/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F05/16 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
				New	New													
REFERENCE VIRUSES																		
A/California/7/2009	NIBSC	2009-04-09	E4/E3	1280	1280	2560	640	640	1280	1280	640	5120	2560	2560	1280	2560	1280	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	<	40	80	320	320	160	40	80	80	80	80	80	80	<	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	80	160	160	1280	1280	320	160	160	160	320	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	1280	1280	5120	2560	1280	2560	2560	2560	1280	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	1280	2560	1280	640	2560	2560	1280	2560	2560	2560	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	2560	1280	640	640	1280	1280	2560	2560	2560	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	2560	640	640	640	1280	640	2560	2560	2560	1280	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	320	640	160	160	320	640	320	640	1280	640	640	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	640	1280	640	640	640	1280	640	1280	2560	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4	1280	1280	2560	640	640	640	1280	640	2560	2560	2560	1280	2560	1280	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	1280	2560	640	320	640	1280	640	2560	2560	1280	ND	2560	2560	
TEST VIRUSES																		
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	1280	2560	640	640	640	1280	1280	2560	2560	2560	ND	2560	2560	
A/Dagestan/1328/2015		2015-01-01	MDCK1/MDCK1	640	1280	1280	320	320	640	640	640	2560	2560	1280	1280	2560	1280	
A/Zambia/06-00038/2015	6B	2015-06-22	MDCK2/SIAT1	1280	1280	1280	640	640	1280	1280	640	2560	2560	2560	1280	2560	2560	
A/Zambia/06-00044/2015	6B	2015-06-23	MDCK2/MDCK1	1280	1280	1280	320	320	640	1280	640	2560	1280	1280	1280	2560	1280	
A/Zambia/06-00039/2015	6B	2015-06-23	MDCK2/SIAT1	1280	1280	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	5120	2560	
A/Zambia/06-00043/2015	6B	2015-06-24	MDCK2/MDCK1	1280	1280	2560	1280	640	1280	1280	640	5120	5120	2560	1280	2560	2560	
A/Zambia/04-00302/2015	6B	2015-07-11	MDCK2/MDCK1	1280	1280	1280	640	640	1280	1280	640	2560	2560	2560	1280	2560	1280	
A/Zambia/04-00308/2015	6B	2015-07-20	MDCK2/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Zambia/01-00118/2015	6B	2015-07-28	MDCK2/MDCK1	640	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	1280	1280	
A/Zambia/04-00339/2015	6B	2015-08-11	MDCK2/MDCK1	640	640	1280	320	160	320	640	320	1280	1280	1280	640	1280	1280	
A/Ghana/ARI-15-0134/2015	6B.1	2015-09-22	MDCK1	1280	1280	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	2560	2560	
A/Oman/5446/2015	6B.1	2015-10-17	SIAT2/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	2560	1280	2560	1280	
A/Oman/5450/2015	6B.1	2015-10-17	SIAT2/MDCK1	1280	1280	2560	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Oman/5437/2015	6B.1	2015-10-18	SIAT2/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Netherlands/2916/2015	6B.1	2015-11-06	SIAT3/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	2560	
A/Oman/6272/2015	6B.1	2015-11-15	SIAT2/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Oman/6227/2015	6B	2015-11-16	SIAT2/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Oman/6285/2015	6B.1	2015-11-16	SIAT2/MDCK1	1280	1280	2560	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Oman/6328/2015	6B	2015-11-16	SIAT2/MDCK1	640	1280	1280	320	320	320	1280	320	2560	2560	1280	1280	2560	1280	
A/Ukraine/6863/2015	6B.1	2015-11-30	MDCK1	1280	2560	2560	1280	640	1280	2560	1280	5120	5120	2560	2560	2560	2560	
A/Netherlands/2941/2015	6B.1	2015-11-30	SIAT3/MDCK1	1280	1280	2560	640	320	640	1280	640	2560	5120	2560	1280	5120	2560	
A/Netherlands/2942/2015	6B.1	2015-12-02	SIAT3/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	2560	
A/Urengoy/2/2015		2015-12-10	MDCK1/MDCK1	640	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Ukraine/6865/2015	6B.1	2015-12-11	MDCK1	1280	1280	2560	640	640	640	1280	640	2560	2560	2560	1280	2560	2560	
A/Netherlands/3078/2015	6B.1	2015-12-14	MDCK/MDCK1	640	640	1280	320	320	320	640	320	1280	1280	1280	640	1280	1280	
A/Netherlands/3027/2015	6B.1	2015-12-15	MDCK/MDCK1	1280	1280	2560	640	640	640	1280	640	2560	2560	1280	1280	2560	2560	
A/Netherlands/2996/2015	6B.1	2015-12-16	MDCK12/MDCK1/MDCK1	160	80	160	320	320	320	640	320	640	1280	640	320	1280	1280	
A/Netherlands/3057/2015	6B.1	2015-12-25	MDCK2/MDCK1	320	160	320	640	320	640	1280	160	1280	2560	1280	640	2560	1280	
A/Kazakhstan/4232/2015	6B.1		MDCK1	640	640	1280	320	160	320	640	320	1280	1280	1280	640	1280	1280	
A/Kazakhstan/4147/2015	6B.1		MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Kazakhstan/4148/2015	6B.1		MDCK1	640	1280	1280	320	320	640	640	640	2560	2560	1280	640	1280	1280	
A/Kazakhstan/4230/2015	6B.1		MDCK1	1280	1280	1280	640	640	640	1280	640	2560	2560	2560	1280	2560	2560	
A/Kazakhstan/4231/2015	6B.1		MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	2560	1280	2560	2560	
A/Netherlands/051/2016	6B.1	2016-01-05	MDCK2/MDCK1	1280	1280	1280	320	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Netherlands/050/2016	6B.1	2016-01-06	MDCK2/MDCK1	1280	1280	1280	640	320	640	1280	640	2560	2560	1280	1280	2560	1280	
A/Netherlands/056/2016	6B.1	2016-02-01	MDCK2/MDCK1	640	1280	1280	320	320	640	1280	640	2560	2560	1280	1280	2560	1280	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Sequences in phylogenetic trees

Vaccine

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-03-08)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/15	Q-504/15
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK
	Ferret number			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2
REFERENCE VIRUSES														
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	1280	640	640	1280	1280	5120	2560	2560	2560	1280
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	320	80	80	40	80	80	80	80	<
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT3/MDCK3	160	1280	1280	320	160	160	160	320	160	320	160
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	2560	640	5120	2560	2560	2560	2560
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	1280	640	1280	2560	1280	5120	2560	2560	2560	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	640	640	2560	640	2560	2560	2560	2560	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	2560	1280	1280	1280
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	320	640	320	1280	1280	640	640	640
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	640	640	640	1280	640	2560	1280	2560	1280	1280
A/Slovenia/2903/2015	clone 37	2015-10-26	E4	1280	640	1280	640	2560	640	2560	2560	2560	1280	2560
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	640	1280	2560	640	5120	2560	2560	2560	2560
TEST VIRUSES														
A/Dakar/36/2014	6B.1	2014-02-04	C1/SIAT3	1280	640	640	640	1280	640	2560	2560	2560	2560	2560
A/Dakar/18/2015	6B.1	2015-01-15	C1/SIAT3	2560	640	640	1280	2560	1280	5120	2560	2560	5120	5120
A/Jordan/30397/2015	6B	2015-07-29	Cx/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Jordan/20766/2015	6B.2	2015-10-07	Cx/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Jordan/30455/2015	6B.1	2015-10-08	Cx/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560
A/Antsirabel/4350/2015	6B	2015-10-15	MDCK1	1280	640	640	1280	2560	1280	2560	2560	1280	5120	2560
A/Antsirabel/4351/2015	6B	2015-10-15	MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	2560
A/Dakar/12/2015	6B	2015-10-20	C1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Antananarivo/4502/2015	6B	2015-10-22	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	1280
A/Jordan/30495/2015	6B.1	2015-11-01	Cx/MDCK1	80	640	640	320	640	160	640	320	320	1280	1280
A/Jordan/30498/2015	6B.1	2015-11-09	Cx/MDCK1	640	320	320	320	640	320	1280	1280	1280	1280	1280
A/Antananarivo/4723/2015	6B	2015-11-10	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280
A/Antananarivo/4787/2015	6B	2015-11-17	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280
A/Antsirabel/4798/2015	6B	2015-11-17	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280
A/Maevatanana/4905/2015	6B	2015-11-17	MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560
A/Jordan/30527/2015	6B.1	2015-11-19	Cx/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Egypt/52/2015	6B.1	2015-11-22	MDCK1	2560	1280	640	1280	2560	1280	2560	2560	2560	2560	2560
A/Antsirabel/4887/2015	6B	2015-11-23	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Antananarivo/4845/2015	6B	2015-11-24	MDCK1	1280	640	640	1280	1280	1280	5120	2560	1280	2560	2560
A/Antananarivo/4937/2015	6B	2015-11-27	MDCK1	640	320	320	640	1280	320	1280	1280	640	1280	1280
A/Antsirabel/5054/2015	6B	2015-12-07	MDCK1	1280	640	640	1280	2560	1280	2560	2560	1280	2560	1280
A/Kazakhstan/4217/2015	6B.1	2015-12-08	MDCK2	1280	640	640	1280	1280	640	2560	2560	1280	2560	2560
A/Jordan/20887/2015	6B.1	2015-12-19	Cx/MDCK1	640	320	320	320	1280	320	1280	1280	1280	1280	1280
A/Egypt/62/2015	6B.1	2015-12-21	MDCK1	1280	640	640	1280	1280	640	2560	2560	2560	2560	2560
A/Jordan/13989/2015	6B.1	2015-12-22	Cx/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Athens/2407/2015	6B.1	2015-12-28	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Jordan/14065/2015	6B.1	2015-12-29	Cx/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560
A/Jordan/20915/2015	6B.1	2015-12-29	Cx/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560
A/Athens/2421/2015	6B.1	2015-12-30	MDCK1	1280	320	320	640	1280	640	1280	1280	1280	2560	1280
A/Athens/19/2016	6B.1	2016-01-04	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280
A/Jordan/10030/2016	6B.1	2016-01-04	Cx/MDCK1	2560	1280	640	1280	2560	640	2560	2560	1280	2560	2560
A/Jordan/10026/2016	6B.1	2016-01-04	Cx/MDCK1	1280	640	320	640	1280	320	1280	1280	1280	2560	1280
A/Athens/76/2016	6B.1	2016-01-08	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Athens/71/2016	6B.1	2016-01-09	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Athens/75/2016	6B.1	2016-01-10	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Moldova/16.02/2016	6B.1	2016-01-11	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560
A/Moldova/26.02/2016	6B.1	2016-01-11	MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560
A/Athens/141/2016	6B.1	2016-01-12	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280
A/Moldova/7.02/2016	6B.1	2016-01-12	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Moldova/32.02/2016	6B.1	2016-01-14	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Moldova/34.02/2016	6B.1	2016-01-15	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Moldova/55.03/2016	6B.1	2016-01-18	MDCK1	1280	1280	640	640	2560	1280	2560	2560	1280	2560	2560
A/Moldova/57.03/2016	6B.1	2016-01-18	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-03-16)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/SL P	A/SL P	A/HK	A/Sth Afr	A/Slov	A/Israel	
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3628/13	2903/15	Q-504/15	
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Ferret number			F06/16 ^a	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/13 ¹	F03/14 ¹	F02/16 ^a	F08/16 ^a	
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/Columbia/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	1280	640	1280	1280	5120	2560	2560	2560	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	320	80	40	40	80	40	40	80	80	<
A/Lviv/16/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	80	160	160	320	160	320	160	
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	1280	2560	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	1280	1280	2560	2560	2560	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	640	640	1280	1280	2560	2560	2560	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	640	1280	1280	2560	2560	2560	1280	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	320	320	1280	640	640	640	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	1280	640	1280	1280	2560	1280	2560	1280	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/JE1	2560	640	1280	1280	1280	1280	5120	2560	2560	1280	2560	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	320	320	640	640	640	1280	1280	1280	2560	2560	
TEST VIRUSES															
A/Yokohama/50/2015	6B	2015-02-02	MDCK2/MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Yokohama/50/2015	6B	2015-02-02	E3/E1	640	640	640	640	640	640	1280	640	1280	1280	1280	
A/Jordan/30338/2015	6B	2015-05-02	Cx/MDCK1	1280	320	320	640	640	320	1280	1280	640	1280	1280	
A/Jordan/20342/2015	6B	2015-05-23	Cx/MDCK1	1280	320	320	640	1280	640	2560	1280	1280	2560	1280	
A/Central African Republic/674/2015	6B	2015-06-26	MDCK1/MDCK2	1280	320	320	640	640	320	1280	1280	1280	1280	1280	
A/Jordan/20636/2015	6B.2	2015-07-24	MDCK1	1280	320	320	640	640	320	1280	1280	640	1280	1280	
A/Central African Republic/809/2015	6B	2015-08-11	MDCK1/MDCK2	1280	320	320	640	640	320	1280	1280	1280	1280	1280	
A/Central African Republic/854/2015	6B	2015-08-18	MDCK1/MDCK1	640	320	640	320	640	320	1280	640	640	1280	1280	
A/Central African Republic/848/2015	6B	2015-08-19	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	
A/Central African Republic/926/2015	6B	2015-09-08	MDCK1/MDCK1	640	320	320	320	640	320	1280	640	640	1280	1280	
A/Fukushima/74/2015	6B.1	2015-09-16	MDCK2/MDCK2/MDCK1	160	320	160	320	640	320	640	640	640	1280	1280	
A/Yokohama/90/2015	6B	2015-10-04	MDCK1/MDCK2/MDCK1	640	320	160	320	640	320	1280	640	640	1280	1280	
A/Shanghai-Putuo/SWL1860/2015	6B.2	2015-10-29	E3/E1	1280	320	320	640	640	320	1280	1280	640	1280	1280	
A/Egypt/7176/2015	6B.1	2015-12-16	MDCK2	320	80	40	80	640	160	640	640	640	1280	640	
A/Egypt/7177/2015	6B.1	2015-12-17	MDCK1	1280	320	320	320	640	320	1280	640	640	1280	1280	
A/Egypt/7187/2015	6B.1	2015-12-18	MDCK1	1280	320	160	320	640	320	1280	1280	640	1280	1280	
A/Athens/2395/2015	6B.1	2015-12-22	MDCK2	640	320	160	320	640	320	1280	640	640	1280	1280	
A/Jordan/20906/2015	6B.1	2015-12-23	MDCK1	1280	320	320	640	640	320	1280	640	640	1280	1280	
A/Jordan/10004/2016	6B.1	2016-01-03	Cx/MDCK2	640	320	320	320	320	320	640	640	640	1280	640	
A/Jordan/10029/2016	6B	2016-01-04	Cx/MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Jordan/10054/2016	6B.1	2016-01-05	MDCK1	640	160	160	320	640	320	1280	640	320	1280	640	
A/Jordan/10045/2016	6B.1	2016-01-05	MDCK2	640	320	160	320	640	320	1280	640	640	1280	1280	
A/Jordan/10101/2016	6B.1	2016-01-06	MDCK1	640	320	160	320	640	320	1280	640	640	1280	1280	
A/Jordan/10214/2016	6B.1	2016-01-10	MDCK1	640	320	160	320	640	320	1280	640	640	1280	640	
A/Athens/822/2016	6B.1	2016-01-11	MDCK2	1280	320	320	640	1280	320	1280	1280	1280	1280	1280	
A/Jordan/10347/2016	6B.1	2016-01-12	MDCK1	1280	320	320	640	1280	320	1280	1280	640	1280	1280	
A/Jordan/10309/2016	6B.1	2016-01-12	MDCK2	1280	640	640	640	1280	640	1280	1280	1280	2560	1280	
A/Jordan/10353/2016	6B.1	2016-01-12	MDCK2	1280	640	640	640	1280	640	1280	1280	1280	2560	1280	
A/Jordan/10371/2016	6B.1	2016-01-13	MDCK1	640	320	160	320	640	320	1280	640	640	640	640	
A/Jordan/10352/2016	6B.1	2016-01-13	MDCK1	640	320	160	320	640	320	1280	640	640	640	640	
A/Armenia/611/2016	6B.1	2016-01-13	MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Armenia/774/2016	6B.1	2016-01-15	MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Fes/231/2016	6B.1	2016-01-15	MDCK1/MDCK1	160	80	80	80	160	160	320	160	320	320	320	
A/Armenia/816/2016	6B.1	2016-01-16	MDCK1	1280	640	640	1280	1280	2560	2560	1280	2560	2560	2560	
A/Armenia/851/2016	6B.1	2016-01-16	MDCK1	320	160	160	160	320	160	640	640	320	640	640	
A/Armenia/893/2016	6B.1	2016-01-17	MDCK1	640	160	160	320	640	320	1280	640	640	1280	1280	
A/Moldova/58.03/2016	6B.1	2016-01-19	MDCK2	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Meknes/344/2016	6B.1	2016-01-21	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560	1280	
A/Catalonia/8590S/2016	6B.1	2016-01-25	C0/MDCK1	1280	320	320	640	640	320	1280	1280	1280	2560	1280	
A/Greece/97/2016	6B.1	2016-01-26	C2/MDCK1	640	320	320	320	640	320	640	640	640	1280	640	
A/Catalonia/8610S/2016	6B.1	2016-01-27	C0/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Catalonia/8608S/2016	6B.1	2016-01-27	C0/MDCK1	1280	320	160	320	640	320	2560	1280	1280	2560	1280	
A/Catalonia/8607S/2016	6B.1	2016-01-27	C0/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Catalonia/8621S/2016	6B.1	2016-01-27	C0/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Catalonia/160626710S/2016	6B.1	2016-02-02	C0/MDCK1	1280	320	160	320	640	320	2560	1280	1280	2560	1280	
A/Catalonia/160626718S/2016	6B.1	2016-02-02	C0/MDCK1	1280	320	320	640	640	640	2560	1280	1280	2560	1280	
A/Catalonia/8664S/2016	6B.1	2016-02-03	C0/MDCK1	1280	320	320	640	1280	640	2560	1280	1280	2560	1280	
A/Catalonia/8687S/2016	6B.1	2016-02-04	C0/MDCK1	1280	320	160	320	640	320	1280	1280	1280	1280	1280	
A/Catalonia/163529394NS/2016	6B.1	2016-02-05	C0/MDCK1	640	320	160	320	640	320	1280	1280	1280	1280	1280	
A/Catalonia/163529493NS/2016	6B.1	2016-02-07	C0/MDCK1	1280	320	160	320	640	320	1280	1280	640	1280	1280	
A/Catalonia/163529479NS/2016	6B.1	2016-02-07	C0/MDCK1	1280	320	160	320	640	320	2560	1280	1280	2560	1280	
A/Catalonia/163529538NS/2016	6B.1	2016-02-09	C0/MDCK1	640	320	160	320	640	320	1280	640	1280	1280	1280	
A/Temara/602/2016	6B.1	2016-02-17	MDCK1/MDCK1	640	320	320	320	640	320	1280	640	640	1280	1280	
A/Benimellal/615/16	6B.1	2016-02-18	MDCK1/MDCK1	1280	320	160	320	640	320	1280	1280	640	1280	1280	
A/Fes/644/2016	6B.1	2016-02-22	MDCK1/MDCK1	1280	320	320	320	640	320	1280	640	640	1280	1280	

^a Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

G155E
G155E>G, D222G

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-03-23)

				Haemagglutination inhibition titre											
				Post-infection ferret antisera											
Viruses	Other information	Collection date	Passage history	A/Cal 7/09 Egg F06/16 ¹	A/Bayern 69/09 MDCK F09/15 ¹	A/Lviv N6/09 MDCK F14/13 ¹	A/Chch 16/10 Egg F15/14 ¹	A/Astrak 1/11 MDCK F22/13 ¹	A/St. P 27/11 Egg F26/14 ¹	A/St. P 100/11 Egg F24/11 ¹	A/HK 5659/12 MDCK F30/12 ¹	A/Sth Afr 3626/13 Egg F03/14 ¹	A/Slov 2903/15 Egg F02/16 ²	A/Israel Q-504/15 MDCK F08/16 ²	
							4	5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	640	320	1280	640	2560	2560	1280	2560	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	160	40	40	40	40	40	40	80	<	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	80	640	640	160	80	80	80	160	160	160	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	640	1280	1280	2560	1280	640	2560	1280	1280	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	320	320	1280	640	2560	1280	1280	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	640	320	1280	640	1280	1280	1280	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	640	80	40	80	320	160	640	640	320	320	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	320	640	320	1280	640	1280	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	320	320	320	1280	640	2560	1280	1280	2560	2560	
TEST VIRUSES															
A/Central African Republic/1282/2015	6B	2015-11-26	MDCK1/MDCK2	640	320	160	320	640	320	1280	1280	1280	2560	1280	
A/Central African Republic/851/2015	6B.1	2015-08-19	MDCK1/MDCK3	640	320	160	320	640	640	1280	1280	1280	2560	1280	
A/Catalonia/16352886NS/2016	6B.1	2016-01-18	C0/MDCK1	640	320	160	320	1280	640	1280	1280	1280	2560	1280	
A/Greece/188/2016	6B.1	2016-01-25	Cs/MDCK1	640	320	320	640	1280	640	1280	1280	1280	2560	1280	
A/Bosnia & Herzegovina/94-G/2016	6B.1	2016-02-06	Cs/MDCK2	640	320	160	320	640	640	1280	1280	1280	1280	1280	
A/Greece/239/2016	6B.1	2016-02-06	Cs/MDCK1	1280	320	320	640	1280	640	1280	1280	1280	2560	1280	
A/Greece/249/2016	6B.1	2016-02-09	Cs/MDCK2	640	160	160	320	640	320	1280	640	640	1280	640	
* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)				Vaccine											
¹ < = <40; ² < = <80															

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-04-12)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/15	Q-504/15	
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Ferret number			F06/16 ¹	F09/15 ¹	F14/13 ²	F15/14 ²	F22/13 ²	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	640	320	1280	640	2560	1280	640	1280	1280	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	640	320	160	80	80	80	80	80	80	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	160	160	160	320	160	320	160	
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	2560	1280	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK6	1280	640	320	640	1280	640	2560	1280	640	1280	1280	
A/St. Petersburg/2/2011	6	2011-02-14	E1/E3	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	2560	1280	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	640	320	640	640	640	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	640	640	640	1280	640	2560	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	1280	640	640	640	1280	640	2560	2560	1280	2560	1280	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	640	640	1280	1280	2560	2560	1280	2560	2560	
TEST VIRUSES															
A/Cameroon/15V-5175/2015	6B	2015-07-03	MDCK2	2560	1280	1280	1280	2560	1280	5120	2560	2560	2560	2560	
A/Cameroon/NGHP 68/2015	6B	2015-08-12	MDCK1	1280	1280	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Cameroon/GARO 67/2015	6B	2015-09-28	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Cameroon/15V-7628/2015	6B	2015-10-22	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Cameroon/15V-8162/2015	6B	2015-11-04	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Cameroon/15V-8177/2015	6B	2015-11-09	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Cameroon/15V-8406/2015	6B	2015-11-16	MDCK1	1280	640	320	640	1280	640	1280	1280	1280	1280	1280	
A/Cameroon/15V-8802/2015	6B	2015-11-17	SIAT1	640	640	320	640	640	640	1280	1280	1280	1280	1280	
A/Ramallah/5R/2015	6B.1	2015-11-29	MDCK1	1280	640	640	640	1280	640	2560	1280	640	1280	1280	
A/Gaziantep/5206/2015	6B.1	2015-12-30	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Bayburt/5202/2015	6B.1	2015-12-30	MDCK1/MDCK1	1280	640	320	320	640	640	1280	1280	1280	1280	1280	
A/Hatay/5204/2015	6B.1	2015-12-30	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Ankara/2449/2015	6B.1	2015-12-31	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/S. Urfa/14/2016	6B.1	2015-12-31	SIAT1/MDCK2	1280	640	320	320	640	320	1280	1280	640	1280	1280	
A/S. Urfa/16/2016	6B.1	2015-12-31	SIAT1/MDCK1	1280	640	320	320	1280	640	2560	1280	1280	2560	1280	
A/Adana/20/2016	6B.1	2015-12-31	SIAT1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	
A/Adana/23/2016	6B.1	2015-12-31	SIAT1/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	
A/Adana/24/2016	6B.1	2015-12-31	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Batman/2/2016	6B.1	2016-01-02	MDCK1/MDCK1	1280	640	320	320	1280	640	1280	1280	1280	1280	1280	
A/Siirt/3/2016	6B.1	2016-01-02	MDCK1/MDCK1	1280	640	320	320	640	640	1280	1280	1280	1280	1280	
A/Mardin/13/2016	6B.1	2016-01-02	MDCK1/MDCK1	1280	640	320	320	1280	640	2560	1280	640	2560	1280	
A/Bitlis/14/2016	6B.1	2016-01-02	MDCK1/MDCK1	1280	640	320	320	1280	320	1280	1280	640	1280	1280	
A/Ankara/45/2016	6B.1	2016-01-04	SIAT1/MDCK2	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	
A/Ankara/17/2016	6B.1	2016-01-04	SIAT1/MDCK1	640	320	320	320	640	320	1280	640	640	1280	640	
A/Ankara/18/2016	6B.1	2016-01-04	SIAT1/MDCK1	2560	1280	1280	1280	2560	1280	5120	2560	2560	2560	2560	
A/Ankara/20/2016	6B.1	2016-01-04	SIAT1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	
A/Ankara/36/2016	6B.1	2016-01-04	SIAT1/MDCK1	1280	640	320	640	640	640	2560	1280	640	1280	1280	
A/Ankara/43/2016	6B.1	2016-01-04	SIAT1/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Adana/8/2016	6B.1	2016-01-04	MDCK1/MDCK1	1280	640	320	320	1280	640	2560	1280	1280	1280	1280	
A/Istanbul/2/2016	6B.1	2016-01-04	SIAT1/MDCK1	1280	640	640	640	2560	1280	5120	2560	1280	2560	1280	
A/Adana/27/2016	6B.1	2016-01-04	MDCK1/MDCK1	2560	1280	640	640	1280	640	5120	1280	1280	2560	2560	
A/Adana/28/2016	6B.1	2016-01-04	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	
A/Adana/29/2016	6B.1	2016-01-04	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	1280	1280	
A/Adana/30/2016	6B.1	2016-01-04	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	
A/Adana/31/2016	6B.1	2016-01-04	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	
A/Latvia/01-006305/2016	6B.1	2016-01-04	C1/MDCK1	640	320	320	320	640	320	1280	640	640	1280	640	
A/Nigde/11/2016	6B.1	2016-01-05	SIAT1/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	
A/Aksaray/50/2016	6B.1	2016-01-05	SIAT1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	
A/Mersin/58/2016	6B.1	2016-01-05	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	1280	2560	2560	
A/Manisa/92/2016	6B.1	2016-01-06	MDCK1/MDCK1	1280	640	320	640	1280	640	5120	2560	1280	2560	1280	
A/Nigde/141/2016	6B.1	2016-01-06	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	1280	2560	2560	
A/Latvia/01-019811/2016		2016-01-10	Cx/MDCK1	640	320	160	320	640	320	640	640	640	1280	640	
A/Turkey/TR-03/2016	6B.1	2016-01-13	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Turkey/TR-05/2016	6B.1	2016-01-13	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Turkey/TR-07/2016	6B.1	2016-01-13	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Turkey/TR-22/2016	6B.1	2016-01-13	MDCK1	1280	1280	640	1280	1280	1280	5120	2560	1280	2560	2560	
A/Turkey/TR-23/2016	6B.1	2016-01-13	MDCK1	640	320	160	320	640	320	1280	640	640	640	640	
A/Turkey/TR-24/2016	6B.1	2016-01-13	MDCK1	1280	320	320	320	640	320	1280	1280	640	1280	1280	
A/Turkey/TR-13/2016	6B.1	2016-01-14	MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280	
A/Turkey/TR-14/2016	6B.1	2016-01-14	MDCK1	1280	640	640	1280	1280	1280	5120	2560	1280	2560	2560	
A/Turkey/TR-15/2016	6B.1	2016-01-14	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Turkey/TR-16/2016	6B.1	2016-01-14	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Latvia/01-069507/2016	6B.1	2016-01-18	C1/MDCK1	160	320	320	160	320	160	640	320	320	640	640	
A/Latvia/01-045064/2016	6B.1	2016-01-19	C2/MDCK1	640	640	320	320	640	640	1280	1280	640	1280	1280	
A/Latvia/01-049333/2016	6B.1	2016-01-21	C2/MDCK1	640	640	320	320	640	320	1280	1280	640	1280	1280	
A/Latvia/01-074070/2016	6B.1	2016-01-30	C1/MDCK1	640	640	320	320	640	320	1280	1280	640	1280	1280	
A/Hebron/1H/2016	6B.1	2016-02-02	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Latvia/03-011553/2016	6B.1	2016-02-29	Cx/MDCK1	640	320	160	320	640	320	1280	640	640	1280	640	
A/Latvia/02-072980/2016	6B	2016-02-29	Cx/MDCK1	1280	1280	640	640	1280	640	2560	1280	1280	2560	1280	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ <= <40; ² <= <80

Vaccine

Sequences in phylogenetic trees

G155E
G155E>G, D222G

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-04-19)

Haemagglutination inhibition titre																	
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera													
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	Yoko	Yoko	
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/15	Q-504/15	94/15	94/15	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	MDCK	Egg	MDCK	Egg	MDCK	MDCK	Egg	
Ferret number				F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	JPN	JPN	
Genetic group							4	5	6	7	6A	6B	6B.1	6B.2	6B.1	6B.1	
REFERENCE VIRUSES																	
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	640	640	1280	640	2560	2560	1280	2560	1280	2560	5120	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	640	320	80	80	40	80	80	80	80	80	80	80	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	160	160	160	320	160	320	160	320	160	
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	2560	640	2560	2560	1280	2560	2560	2560	5120	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	1280	640	640	1280	640	2560	2560	1280	1280	2560	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	1280	640	640	1280	640	2560	2560	1280	2560	1280	2560	5120	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	1280	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	320	640	320	1280	1280	640	1280	640	640	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	1280	640	640	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	1280	640	640	640	1280	640	1280	2560	1280	2560	2560	2560	5120	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	640	640	1280	1280	640	2560	1280	2560	2560	2560	2560	
TEST VIRUSES																	
A/Yokohama/94/2015	6B.1	2015-10-21	MDCK1/MDCK2/MDCK1	80	320	640	320	320	160	320	320	320	640	640	1280	1280	
A/Yokohama/94/2015	6B.1	2015-10-21	E4/E1	1280	640	640	640	1280	640	2560	2560	1280	2560	1280	2560	2560	
A/Spain/8740/2016	6B.1	2016-01-25	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	2560	1280	
A/Serbia-NS/1439/2016	6B.1	2016-01-25	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	ND	ND	
A/Spain/9134/2016	6B.1	2016-01-26	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560	2560	2560	
A/Spain/9333/2016	6B.1	2016-01-26	MDCK1	640	640	320	640	640	640	1280	1280	1280	1280	1280	2560	1280	
A/Cyprus/F66/2016	6B.1	2016-01-26	MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	ND	ND	
A/Spain/9779/2016	6B.1	2016-01-27	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	2560	2560	
A/Cyprus/F75/2016	6B.1	2016-01-27	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	ND	ND	
A/Spain/10181/2016	6B.1	2016-01-28	MDCK1	640	640	320	640	640	320	1280	1280	640	1280	1280	2560	1280	
A/Serbia-NS/1447/2016	6B.1	2016-01-28	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	ND	ND	
A/Spain/12301/2016	6B.1	2016-01-29	MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560	1280	2560	2560	
A/Serbia-NS/1451/2016	6B.1	2016-01-29	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560	ND	ND	
A/Serbia-NS/1454/2016	6B.1	2016-01-29	MDCK1	1280	640	320	640	1280	640	2560	1280	2560	2560	2560	ND	ND	
A/Cyprus/F116/2016	6B.1	2016-02-01	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	ND	ND	
A/Serbia-NS/1460/2016	6B.1	2016-02-01	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	ND	ND	
A/Serbia-NS/1465/2016	6B.1	2016-02-01	MDCK1	1280	640	640	640	1280	640	2560	2560	2560	2560	2560	ND	ND	
A/Spain/12109/2016	6B.1	2016-02-03	MDCK1	640	640	320	320	640	320	1280	1280	1280	2560	1280	2560	1280	
A/Spain/12112/2016	6B.1	2016-02-03	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	5120	2560	
A/Spain/12113/2016	6B.1	2016-02-03	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	2560	2560	
A/Spain/12293/2016	6B.1	2016-02-03	MDCK1	1280	640	320	640	640	640	1280	1280	640	1280	1280	2560	1280	
A/Cyprus/F142/2016	6B.1	2016-02-03	MDCK1	2560	640	640	640	1280	640	2560	2560	1280	2560	2560	ND	ND	
A/Cyprus/F138/2016	6B.1	2016-02-03	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	ND	ND	
A/Cyprus/F147/2016	6B.1	2016-02-04	MDCK1	1280	640	320	640	640	640	1280	1280	1280	1280	1280	ND	ND	
A/Niedersachsen/6/2016	6B.1	2016-02-04	C2/MDCK1	2560	1280	640	1280	2560	1280	2560	2560	1280	2560	2560	ND	ND	
A/Cyprus/F152/2016	6B.1	2016-02-05	MDCK1	1280	640	320	640	1280	640	1280	1280	1280	1280	1280	ND	ND	
A/Cyprus/F179/2016	6B.1	2016-02-07	MDCK1	640	640	320	640	640	640	1280	1280	640	1280	1280	2560	2560	
A/Cyprus/F176/2016	6B.1	2016-02-07	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	ND	ND	
A/Sachsen/20/2016	6B.1	2016-02-08	C2/MDCK1	1280	640	320	320	640	640	1280	1280	1280	2560	1280	ND	ND	
A/Hessen/6/2016	6B.1	2016-02-10	C2/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560	ND	ND	
A/Baden-Wuerttemberg/18/2016	6B.1	2016-02-10	C1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	ND	ND	
A/Cyprus/F220/2016	6B.1	2016-02-12	MDCK1	640	640	320	640	1280	640	1280	1280	640	1280	1280	2560	2560	
A/Serbia-NS/1541/2016	6B.1	2016-02-13	MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	ND	ND	
A/Thuringen/17/2016	6B.1	2016-02-22	C2/MDCK1	2560	640	640	1280	1280	1280	2560	2560	2560	2560	2560	ND	ND	
A/Serbia-NS/1583/2016	6B.1	2016-02-23	MDCK1	1280	1280	640	640	1280	1280	2560	2560	1280	2560	2560	ND	ND	
A/Bayern/28/2016	6B.1	2016-02-26	C1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	ND	ND	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80; ND = Not done

Sequences in phylogenetic trees

Vaccine

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-04-26)

Haemagglutination inhibition titre															
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/SL P	A/SL P	A/HK	A/Slh Afr	A/Slh Afr	A/Slav	A/Israel
				70/9	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	3626/13	2903/15	Q-504/15
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	Egg	MDCK
				F06/16 ¹	F09/15 ³	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F13/16 ¹	F02/16 ¹	F08/16 ¹
				4	5	6	7	6A	6B	6B	6B.1	6B.2			
REFERENCE VIRUSES															
A/California/1/2009	clone 38-32	2009-04-09	E3/E2	640	640	640	320	1280	640	1280	1280	640	1280	640	
A/Bayern/69/2009		2009-07-01	MDCK/5MDCK1	40	320	320	80	40	40	80	40	40	80	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/5IAT1/MDCK3	80	1280	640	160	80	80	80	160	160	160	160	
A/Christchurch/16/2010	4	2010-07-12	EU/E3	1280	1280	1280	2560	1280	640	2560	1280	1280	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/5MDCK5	1280	1280	640	640	1280	640	2560	1280	1280	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	EU/E4	640	640	320	320	1280	320	1280	1280	640	1280	640	
A/St. Petersburg/100/2011	7	2011-03-14	EU/E4	640	640	320	320	640	320	1280	1280	640	1280	640	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK-4/MDCK2	320	160	80	160	320	160	640	640	320	640	320	
A/South Africa/3626/2013	6B	2013-06-06	EU/E3	640	640	320	320	640	320	640	640	1280	640	640	
A/Slovenia/2903/2015	clone 37	6B.1	EU/E1	1280	640	640	640	1280	640	2560	1280	2560	1280	1280	
A/Israel/Q-504/2015	6B.2		C1/MDCK2	1280	640	320	640	1280	640	1280	1280	1280	1280	1280	
TEST VIRUSES															
A/Ayon/1884/2015	6B.1	2015-10-07	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Hatay/5028/2015	6B.1	2015-12-09	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Adana/5047/2015	6B.1	2015-12-11	SIAT1/MDCK1	1280	640	320	320	640	320	1280	1280	1280	1280	1280	
A/Van/5063/2015	6B.1	2015-12-11	SIAT1/MDCK1	640	640	320	640	640	320	1280	1280	1280	1280	1280	
A/Ankara/2193/2015	6B.1	2015-12-11	SIAT1/MDCK1	640	320	160	320	640	320	1280	640	1280	1280	1280	
A/Van/5084/2015	6B.1	2015-12-14	SIAT1/MDCK1	640	1280	320	320	1280	320	1280	640	ND	1280	1280	
A/Ankara/5091/2015	6B.1	2015-12-15	SIAT1/MDCK1	640	640	320	640	1280	640	2560	1280	1280	ND	1280	
A/Ankara/2217/2015	6B.1	2015-12-15	MDCK1/MDCK1	640	640	320	320	1280	320	1280	1280	640	ND	1280	
A/Ankara/5099/2015	6B.1	2015-12-16	SIAT1/MDCK1	640	320	320	320	640	320	1280	1280	640	ND	1280	
A/Ankara/5108/2015	6B.1	2015-12-18	SIAT1/MDCK1	640	320	320	320	640	320	1280	1280	640	ND	1280	
A/Ankara/5129/2015	6B.1	2015-12-21	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	ND	1280	
A/Ankara/5136/2015	6B.1	2015-12-22	MDCK1/MDCK1	640	640	320	640	1280	640	1280	1280	1280	ND	1280	
A/Ankara/5138/2015	6B.1	2015-12-22	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	ND	2560	
A/Ankara/5141/2015	6B.1	2015-12-22	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	ND	1280	
A/Ankara/2269/2015	6B.1	2015-12-22	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	ND	1280	
A/Van/5146/2015	6B.1	2015-12-23	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	640	1280	2560	
A/Ankara/5147/2015	6B.1	2015-12-23	MDCK1/MDCK1	640	640	320	320	640	320	1280	1280	640	ND	1280	
A/Ankara/2275/2015	6B.1	2015-12-23	MDCK1/MDCK1	640	320	320	320	640	320	1280	640	640	ND	1280	
A/Hatay/5162/2015	6B.1	2015-12-24	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	640	1280	1280	
A/Cankiri/2306/2015	6B.1	2015-12-24	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	640	1280	1280	
A/Adana/9/2016	6B.1	2015-12-24	SIAT1/MDCK1	1280	640	320	640	640	640	1280	1280	640	1280	2560	
Altmir/5174/2015	6B.1	2015-12-25	MDCK1/MDCK1	1280	640	320	320	640	640	1280	1280	640	640	1280	
A/Ankara/2309/2015	6B.1	2015-12-25	MDCK1/MDCK1	1280	640	320	320	640	320	1280	1280	640	640	1280	
A/Ankara/2315/2015	6B.1	2015-12-25	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	640	1280	2560	
A/Van/2323/2015	6B.1	2015-12-25	MDCK1/MDCK1	1280	640	320	320	640	640	1280	1280	640	1280	2560	
A/Ankara/2324/2015	6B.1	2015-12-25	MDCK1/MDCK1	1280	640	320	320	640	640	2560	1280	640	1280	2560	
A/Adana/2/2016	6B.1	2015-12-25	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	640	1280	1280	
A/Ankara/2339/2015	6B.1	2015-12-28	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	2560	
A/Ankara/2342/2015	6B.1	2015-12-28	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	640	1280	2560	
A/Adana/3/2016	6B.1	2015-12-28	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	640	1280	2560	
A/Gaziantep/5188/2015	6B.1	2015-12-29	MDCK1/MDCK1	1280	640	320	640	320	1280	1280	1280	1280	1280	1280	
A/Ankara/2366/2015	6B.1	2015-12-29	SIAT1/MDCK2	640	320	160	320	640	320	1280	640	640	640	1280	
A/Ankara/2388/2015	6B.1	2015-12-29	SIAT1/MDCK1	1280	640	320	640	1280	320	1280	1280	640	1280	1280	
A/Kyrgyzstan/95/2015	6B.1	2015-12-29	MDCK1	1280	640	320	640	1280	320	2560	1280	1280	ND	2560	
A/Ankara/2404/2015	6B.1	2015-12-30	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Mersin/2149/2015	6B.1	2015-12-30	SIAT1/MDCK2	1280	640	320	640	1280	640	2560	1280	1280	1280	2560	
A/Adana/7/2016	6B.1	2015-12-30	MDCK1/MDCK2	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Adana/12/2016	6B.1	2015-12-30	SIAT1/MDCK1	1280	320	320	320	640	320	1280	1280	640	1280	1280	
A/Adana/18/2016	6B.1	2015-12-30	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Adana/19/2016	6B.1	2015-12-30	SIAT1/MDCK1	640	640	320	640	1280	640	1280	1280	1280	1280	1280	
A/Adana/32/2016	6B.1	2015-12-30	MDCK1/MDCK1	1280	640	320	640	1280	320	1280	1280	640	1280	1280	
A/Kyrgyzstan/161/2016	6B.1	2016-01-06	MDCK2	1280	640	320	640	1280	640	2560	1280	1280	ND	1280	
A/Kyrgyzstan/182/2016	6B.1	2016-01-14	MDCK2	1280	640	320	640	1280	640	1280	1280	1280	2560	2560	
A/Kyrgyzstan/183/2016	6B	2016-01-14	MDCK2	1280	640	320	640	1280	640	2560	1280	640	ND	1280	
A/Cyprus/F40/2016	6B.1	2016-01-15	MDCK2	640	640	320	320	640	640	1280	1280	640	1280	1280	
A/Kyrgyzstan/109/2016	6B.1	2016-01-17	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Kyrgyzstan/119/2016	6B	2016-01-18	MDCK2	640	320	160	320	640	320	1280	640	640	1280	1280	
A/Kyrgyzstan/165/2016	6B.1	2016-01-19	MDCK2	1280	640	320	640	1280	640	2560	1280	1280	1280	2560	
A/Kyrgyzstan/190/2016	6B.1	2016-01-21	MDCK1	640	640	320	320	640	320	1280	1280	1280	1280	1280	
A/Cyprus/F56/2016	6B.1	2016-01-23	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Spain/9785/2016	6B.1	2016-01-26	MDCK3	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/Cyprus/F51/2016	6B.1	2016-01-26	MDCK1	640	640	320	320	640	320	1280	1280	640	1280	1280	
A/Kyrgyzstan/194/2016	6B.1	2016-01-29	MDCK2	640	640	320	320	1280	640	1280	1280	640	1280	1280	
A/Spain/11403/2016	6B.1	2016-02-02	MDCK2	640	640	320	320	640	320	1280	1280	640	1280	1280	
A/Spain/11878/2016	6B.1	2016-02-04	MDCK2	640	320	320	320	640	320	1280	1280	640	1280	1280	
A/Kyrgyzstan/209/2016	6B	2016-02-07	MDCK2	640	640	320	320	1280	320	1280	1280	640	1280	1280	
A/Kyrgyzstan/229/2016	6B.1	2016-02-08	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	2560	
A/Kyrgyzstan/212/2016	6B.1	2016-02-08	MDCK1	640	160	160	320	640	160	1280	1280	640	1280	640	
A/Bulgaria/857/2016	6B.1	2016-03-14	SIAT2/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280	
A/Bulgaria/855/2016	6B.1	2016-03-14	SIAT2/MDCK1	640	320	160	320	640	320	1280	640	640	640	640	
A/Bulgaria/865/2016	6B.1	2016-03-15	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Bulgaria/897/2016	6B.1	2016-03-21	SIAT1/MDCK1	1280	640	320	320	640	640	1280	1280	1280	1280	1280	

G155E

G155E>G, D222G

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

Sequences in phylogenetic trees

Vaccine

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-05-04)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Passage history			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
	Ferret number						4	5	6	7	6A	6B	6B.1	6B.2	
	Genetic group														
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	2560	1280	640	640	2560	640	2560	2560	1280	2560	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	320	80	40	40	80	40	80	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	640	320	160	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	2560	640	2560	1280	1280	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	1280	640	640	1280	640	2560	1280	640	1280	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	160	320	160	1280	640	320	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	320	320	640	640	320	1280	640	640	640	640	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	2560	640	640	1280	1280	640	5120	2560	1280	2560	1280	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
TEST VIRUSES															
A/Kyrgyzstan/118/2015	6B.1	2015-12-25	MDCK2	1280	640	320	640	1280	640	1280	1280	1280	2560	2560	
A/Bulgaria/404/2016	6B.1	2016-02-09	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Bulgaria/359/2016	6B.1	2016-02-09	SIAT1/MDCK1	1280	640	320	640	640	640	1280	1280	1280	2560	1280	
A/Bulgaria/333/2016	6B.1	2016-02-09	SIAT2/MDCK1	1280	640	320	640	640	640	1280	1280	1280	2560	2560	
A/Bulgaria/332/2016		2016-02-09	SIAT2/MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Bulgaria/381/2016	6B.1	2016-02-10	SIAT1/MDCK1	1280	640	320	640	640	320	1280	1280	1280	1280	1280	
A/Bulgaria/507/2016	6B.1	2016-02-13	SIAT1/MDCK1	1280	640	320	640	1280	320	1280	1280	1280	2560	2560	
A/Bulgaria/505/2016	6B.1	2016-02-13	SIAT1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560	2560	
A/Bulgaria/531/2016	6B.1	2016-02-16	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Bulgaria/559/2016	6B.1	2016-02-17	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Bulgaria/552/2016	6B.1	2016-02-17	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/Bulgaria/671/2016	6B.1	2016-02-19	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Bulgaria/670/2016	6B.1	2016-02-19	SIAT2/MDCK1	1280	640	320	640	640	640	2560	1280	1280	2560	1280	
A/Bulgaria/665/2016	6B.1	2016-02-19	SIAT2/MDCK1	1280	640	320	640	640	640	2560	1280	1280	1280	1280	
A/Bulgaria/694/2016	6B.1	2016-02-22	SIAT1/MDCK1	1280	640	320	640	1280	640	1280	2560	1280	2560	2560	
A/Bulgaria/691/2016	6B.1	2016-02-22	SIAT1/MDCK1	640	640	320	320	640	320	1280	1280	640	1280	1280	
A/Bulgaria/811/2016	6B.1	2016-03-04	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-05-10)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK
	Ferret number			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2
REFERENCE VIRUSES														
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	160	80	40	40	40	40	40	40	40
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	160	160	320	320	160	320	160
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	1280	5120	2560	2560	2560	2560
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	1280	640	2560	2560	1280	2560	1280
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	640	640	1280	640	2560	1280	1280	2560	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	640	640	640	1280	640	2560	1280	640	2560	1280
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	320	640	320	1280	1280	640	1280	640
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	640	640	640	1280	640	1280	1280	1280
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	1280	1280	640	1280	2560	1280	5120	2560	1280	2560	2560
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
TEST VIRUSES														
A/Bulgaria/265/2016	6B.1	2016-02-03	SIAT2/MDCK1	1280	640	320	640	640	320	2560	1280	1280	1280	1280
A/Bulgaria/301/2016	6B.1	2016-02-04	SIAT2/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560
A/Bulgaria/276/2016	6B.1	2016-02-04	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/314/2016	6B.1	2016-02-05	SIAT2/MDCK1	640	640	320	640	640	640	2560	1280	1280	1280	1280
A/Bulgaria/312/2016	6B.1	2016-02-05	SIAT1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/311/2016	6B.1	2016-02-05	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/306/2016	6B.1	2016-02-05	SIAT2/MDCK1	640	320	320	640	640	320	1280	1280	640	1280	1280
A/Bulgaria/349/2016	6B.1	2016-02-07	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Bulgaria/345/2016	6B.1	2016-02-07	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/378/2016	6B.1	2016-02-08	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/363/2016	6B.1	2016-02-08	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/353/2016	6B.1	2016-02-08	SIAT2/MDCK1	640	640	320	640	640	640	2560	1280	1280	1280	1280
A/Bulgaria/331/2016	6B.1	2016-02-08	SIAT2/MDCK1	1280	640	320	640	640	640	2560	1280	1280	1280	1280
A/Bulgaria/329/2016	6B.1	2016-02-08	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Ghana/FS-16-0249/2016	6B	2016-02-22	Cx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Ghana/FS-16-0268/2016	6B.1	2016-02-25	Cx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Montenegro/174/2016	6B.1	2016-02-26	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	640	1280	1280
A/Ghana/FS-16-0292/2016	6B.1	2016-03-02	Cx/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Montenegro/237/2016	6B.1	2016-03-06	MDCK1/MDCK1	640	640	320	640	640	320	1280	1280	640	1280	1280
A/Montenegro/258/2016	6B.1	2016-03-09	MDCK1/MDCK1	640	320	320	640	640	320	1280	1280	640	2560	1280
A/Ghana/FIP-16-1886/2016	6B.1	2016-03-10	Cx/MDCK1	2560	1280	640	1280	2560	640	2560	2560	1280	2560	2560
A/Ghana/FS-16-0320/2016	6B.1	2016-03-11	Cx/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Ghana/DILI-16-0280/2016	6B.1	2016-03-15	Cx/MDCK1	1280	640	320	640	640	320	2560	1280	640	1280	1280
A/Ghana/DILI-16-0284/2016	6B.1	2016-03-16	Cx/MDCK1	1280	640	320	640	1280	640	2560	1280	640	2560	1280
A/Montenegro/318/2016	6B.1	2016-03-17	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Montenegro/323/2016	6B.1	2016-03-18	MDCK1/MDCK1	640	320	160	320	640	320	1280	640	640	1280	640
A/Ghana/FS-16-0382/2016	6B.1	2016-03-21	Cx/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Ghana/DILI-16-0309/2016	6B.1	2016-03-22	Cx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Ghana/DILI-16-0302/2016	6B.1	2016-03-22	Cx/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560
A/Ghana/FS-16-0377/2016	6B.1	2016-03-23	Cx/MDCK1	1280	1280	640	640	1280	640	2560	1280	1280	2560	2560
A/Ghana/FIP-16-0011/2016	6B.1	2016-03-24	Cx/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560
A/Ghana/DILI-16-0317/2016	6B.1	2016-03-29	Cx/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Ghana/DILI-16-0321/2016		2016-03-29	Cx/MDCK1	1280	640	320	640	640	640	2560	1280	1280	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

G155E
G155E>G, D222G

Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-05-17)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK
	Ferret number			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2
REFERENCE VIRUSES														
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	640	640	1280	640	2560	1280	1280	1280	1280
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	160	80	40	40	40	40	80	80	40
A/Lviv/6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	160	160	160	320	160	320	160
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	2560	1280	5120	2560	1280	2560	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	1280	1280	5120	2560	1280	2560	1280	1280
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	640	640	1280	640	2560	1280	640	2560	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	640	640	640	640	1280	640	2560	1280	640	1280	1280
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	80	160	320	160	640	640	320	640	320
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	320	320	320	640	640	1280	640	640	640	640
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	1280	640	640	1280	1280	1280	5120	2560	1280	2560	1280
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	320	640	640	640	2560	1280	640	1280	1280
TEST VIRUSES														
A/Oman/7534/2015	6B.1	2015-12-22	SIAT2/SIAT1	1280	640	320	320	1280	640	2560	1280	1280	2560	1280
A/Oman/7542/2015	6B.1	2015-12-22	SIAT2/MDCK1	1280	640	160	320	1280	320	1280	640	640	640	640
A/Oman/292/2016	6B.1	2016-01-05	SIAT3/MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Oman/631/2016		2016-01-12	MDCK1/MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Egypt/1494/2016	6B.1	2016-01-22	MDCK1	1280	1280	640	1280	2560	1280	2560	2560	1280	2560	1280
A/Egypt/1492/2016	6B.1	2016-01-22	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Egypt/1572/2016	6B.1	2016-01-23	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Egypt/1571/2016	6B.1	2016-01-23	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Egypt/1570/2016	6B.1	2016-01-23	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Egypt/1518/2016	6B.1	2016-01-23	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Montenegro/45/2016	6B.1	2016-01-29	MDCK2/MDCK1	640	320	160	320	640	320	1280	640	640	1280	640
A/Montenegro/51/2016	6B.1	2016-02-01	MDCK2/MDCK1	640	320	320	320	640	640	1280	1280	640	1280	1280
A/Oman/1565/2016	6B.1	2016-02-04	MDCK1	1280	640	320	320	1280	640	2560	1280	640	1280	1280
A/Montenegro/64/2016	6B.1	2016-02-08	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	1280	1280
A/Montenegro/63/2016	6B.1	2016-02-08	MDCK1/MDCK1	1280	320	320	640	1280	640	1280	1280	1280	1280	1280
A/Montenegro/65/2016	6B.1	2016-02-09	MDCK1/MDCK1	640	320	320	320	640	320	1280	640	640	1280	1280
A/Oman/1726/2016	6B.1	2016-02-09	MDCK2/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280
A/Oman/1770/2016	6B.1	2016-02-09	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Montenegro/67/2016	6B.1	2016-02-10	MDCK1	1280	640	320	320	640	640	1280	1280	1280	1280	1280
A/Oman/1748/2016	6B.1	2016-02-10	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Oman/1910/2016	6B.1	2016-02-10	MDCK1	640	320	320	640	640	640	1280	1280	640	1280	1280
A/Montenegro/71/2016	6B.1	2016-02-11	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Ghana/FS-16-0237/2016	6B.1	2016-02-18	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280
A/Ghana/FS-16-0269/2016	6B.1	2016-02-18	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Montenegro/97/2016	6B.1	2016-02-18	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Montenegro/93/2016	6B.1	2016-02-18	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Montenegro/92/2016	6B.1	2016-02-18	MDCKx/MDCK1	640	640	320	320	640	640	1280	1280	640	1280	1280
A/Montenegro/110/2016	6B.1	2016-02-19	MDCK2	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Montenegro/129/2016	6B.1	2016-02-23	MDCK1/MDCK1	640	320	160	320	640	320	1280	640	640	1280	640
A/Oman/2477/2016	6B.1	2016-03-02	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280
A/Botosani/195345/2016	6B.1	2016-03-29	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	640
A/Iasi/195762/2016	6B.1	2016-04-06	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Iasi/196079/2016	6B.1	2016-04-11	MDCK1/MDCK1	640	640	320	640	640	320	2560	1280	640	1280	640
A/Dolj/196287/2016	6B.1	2016-04-11	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-14. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-05-24 + 2016-06-03 + 2016-06-08)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Slh Afr	A/Slow	A/Israel	A/Ukraine
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	6907/15
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	MDCK
	Passage history			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F28/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ¹	F08/16 ²	F17/16 ²
	Ferret number						4	5	6	7	6A	6B	6B.1	6B.2	6B.2
	Genetic group														New
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E3	1280	640	640	640	1280	640	2560	1280	1280	2560	2560	640
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	640	320	80	40	40	40	40	80	80	40	<
A/Lviv/6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	160	160	160	320	160	320	160	80
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	1280	5120	2560	2560	5120	2560	640
A/Astrakhan/12/2011	5	2011-02-28	MDCK1/MDCK5	1280	1280	640	1280	2560	1280	5120	2560	2560	2560	2560	640
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	640
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	1280	1280	640	2560	1280	5120	2560	1280	2560	2560	640
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	320	640	320	1280	1280	640	640	640	320
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	320	640	640	1280	640	1280	640	640	320
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	640
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	640	640	1280	1280	5120	2560	1280	2560	2560	640
A/Ukraine/6907/2015	6B.2	2015-12-16	C2/MDCK2	320	640	320	320	640	640	1280	640	1280	1280	2560	640
TEST VIRUSES															
A/Oman/7821/2015	6B.1	2015-12-29	SIAT2/MDCK1	2560	640	320	640	2560	640	5120	2560	2560	5120	5120	640
A/Jordan/20086/2016	6B.1	2016-01-10	MDCK1	1280	1280	640	1280	1280	640	5120	2560	2560	2560	2560	1280
A/Jordan/20112/2016	6B.1	2016-01-16	MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	1280
A/Jordan/20127/2016	6B.1	2016-01-18	MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560	1280
A/Jordan/30040/2016	6B.1	2016-01-19	MDCK1	1280	1280	640	1280	1280	640	5120	2560	2560	2560	2560	1280
A/Egypt/1573/2016	6B.1	2016-01-23	MDCK2	1280	1280	640	640	1280	1280	5120	2560	2560	5120	2560	1280
A/Egypt/1568/2016	6B.1	2016-01-23	MDCK2	1280	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	1280
A/Egypt/1564/2016	6B.1	2016-01-23	MDCK2	1280	1280	320	640	1280	640	5120	2560	1280	2560	2560	640
A/Jordan/30047/2016	6B.1	2016-01-25	MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	2560	1280
A/Jordan/10640/2016	6B.1	2016-01-26	MDCK1	1280	1280	640	1280	1280	1280	5120	2560	1280	2560	2560	1280
A/Jordan/10645/2016	6B.1	2016-01-26	MDCK1	1280	1280	640	1280	1280	640	5120	2560	1280	2560	2560	640
A/Bulgaria/133/2016	6B.1	2016-01-27	SIAT1/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	1280
A/Jordan/20186/2016	6B.1	2016-01-28	MDCK1	1280	1280	640	1280	1280	640	5120	2560	1280	2560	2560	640
A/Jordan/10647/2016	6B.1	2016-01-28	MDCK1	640	640	320	640	1280	640	2560	1280	1280	2560	1280	640
A/Jordan/10673/2016	6B.1	2016-01-28	MDCK1	1280	1280	640	1280	1280	640	5120	2560	1280	2560	2560	640
A/Bulgaria/178/2016	6B.1	2016-01-29	SIAT2/MDCK1	640	640	320	320	640	320	2560	1280	1280	1280	1280	320
A/Bulgaria/175/2016	6B.1	2016-01-29	SIAT2/MDCK1	640	320	160	320	640	320	1280	640	640	1280	640	320
A/Bulgaria/171/2016	6B.1	2016-01-29	SIAT2/MDCK1	1280	320	160	320	640	320	1280	1280	640	1280	640	320
A/Jordan/10689/2016	6B.1	2016-01-30	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	640
A/Jordan/10687/2016	6B.1	2016-01-30	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	640
A/Jordan/20220/2016	6B.1	2016-01-30	MDCK2	2560	1280	1280	1280	2560	1280	5120	2560	2560	5120	2560	ND
A/Jordan/10730/2016	6B.1	2016-01-31	MDCK1	1280	1280	320	1280	1280	640	5120	2560	1280	2560	2560	640
A/Bulgaria/213/2016	6B.1	2016-02-01	SIAT2/MDCK1	640	320	160	320	640	320	2560	1280	1280	1280	1280	320
A/Bulgaria/205/2016	6B.1	2016-02-01	SIAT2/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	640
A/Kaliningrad/967/2016		2016-02-01	E3/E1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560	ND
A/Bulgaria/242/2016	6B.1	2016-02-02	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	640
A/Sibiu/190931/2016	6B.1	2016-02-02	MDCK1/MDCK1	1280	640	320	640	640	320	1280	640	640	1280	1280	640
A/Jordan/10768/2016	6B.1	2016-02-03	MDCK1	1280	1280	640	1280	2560	640	5120	2560	1280	2560	2560	1280
A/Jordan/10816/2016	6B.1	2016-02-04	MDCK1	640	320	320	320	640	320	1280	1280	1280	1280	1280	320
A/Jordan/10820/2016	6B.1	2016-02-04	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	640
A/Bucuresti/191223/2016	6B.1	2016-02-06	MDCK1/MDCK1	1280	640	320	640	640	320	1280	640	640	1280	2560	640
A/Jordan/10862/2016	6B.1	2016-02-08	MDCK1	640	640	320	640	640	320	2560	1280	1280	2560	1280	640
A/Jordan/10875/2016	6B.1	2016-02-09	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	640
A/Jordan/10906/2016	6B.1	2016-02-11	MDCK1	1280	640	320	640	2560	640	5120	2560	2560	5120	2560	ND
A/Jordan/20255/2016	6B.1	2016-02-12	MDCK1	1280	640	320	640	1280	320	2560	1280	1280	2560	1280	640
A/Georgia/694/2016		2016-02-13	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	ND
A/Jordan/30073/2016	6B.1	2016-02-14	MDCK1	2560	1280	640	1280	1280	1280	5120	2560	2560	2560	2560	1280
A/Georgia/760/2016	6B.1	2016-02-16	C2/MDCK1	2560	1280	1280	1280	2560	1280	5120	2560	2560	2560	2560	1280
A/Georgia/774/2016	6B.1	2016-02-19	C3/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	640
A/Jordan/20303/2016	6B.1	2016-02-21	MDCK1	1280	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	1280
A/Georgia/791/2016	6B.1	2016-02-23	C3/MDCK1	80	640	640	320	640	160	640	320	320	1280	2560	640
A/Jordan/11141/2016	6B.1	2016-02-25	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	640
A/Oman/2427/2016	6B.1	2016-02-28	MDCK2	1280	640	320	640	1280	640	5120	2560	1280	2560	2560	1280
A/Jordan/11161/2016	6B.1	2016-02-29	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	640
A/Dolj/193185/2016	6B.1	2016-02-29	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	1280
A/Jordan/11177/2016	6B.1	2016-03-02	MDCK1	640	320	160	320	320	320	1280	640	640	640	640	320
A/Jordan/20310/2016	6B.1	2016-03-04	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	640
A/Jordan/20317/2016	6B.1	2016-03-07	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	640
A/Jordan/11218/2016	6B.1	2016-03-07	MDCK1	1280	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	1280
A/Isi/193746/2016	6B.2	2016-03-09	MDCK1/MDCK1	320	320	160	160	320	320	640	320	640	640	1280	640
A/Isi/193859/2016	6B.1	2016-03-10	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	320
A/Isi/193967/2016	6B.2	2016-03-11	MDCK1/MDCK1	640	320	320	320	640	640	1280	1280	1280	1280	1280	640
A/Georgia/914/2016	6B.1	2016-03-11	MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	ND
A/Georgia/916/2016	6B.1	2016-03-11	MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	ND
A/Jordan/11268/2016	6B.1	2016-03-13	MDCK1	1280	1280	640	1280	1280	640	5120	2560	2560	2560	2560	1280
A/Bacau/194305/2016	6B.1	2016-03-15	MDCK1/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	640
A/Suceava/194312/2016	6B.1	2016-03-16	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	1280	

Table 5-15. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-06-14)

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/Sl. P	A/Sl. P	A/HK	A/Sth Afr	A/Slov	A/Israel
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK
Passage history				F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ¹	F08/16 ²
Ferret number														
Genetic group							4	5	6	7	6A	6B	6B.1	6B.2
REFERENCE VIRUSES														
A/California/7/2009	clone 38-32	2009-04-09	E3/E3	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	160	80	40	40	40	40	80	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	1280	320	80	80	80	320	80	320	160
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	1280	5120	2560	640	5120	2560	2560	2560	2560
A/Astrakhan/1/2011		2011-02-28	MDCK1/MDCK5	1280	640	640	1280	2560	640	5120	2560	2560	2560	2560
A/Sl. Petersburg/27/2011		2011-02-14	E1/E4	1280	640	640	1280	1280	640	2560	2560	2560	2560	2560
A/Sl. Petersburg/100/2011	6	2011-02-14	E1/E4	1280	640	640	1280	1280	640	2560	2560	2560	2560	2560
A/Hong Kong/5659/2012		2011-03-14	E1/E4	1280	640	640	1280	1280	640	2560	2560	2560	2560	2560
A/South Africa/3626/2013		2012-05-21	MDCK4/MDCK2	320	160	80	160	320	160	1280	640	320	640	640
A/Slovenia/2903/2015	clone 37	2013-06-06	E1/E3	640	640	640	320	640	640	1280	1280	2560	2560	1280
A/Israel/Q-504/2015		2015-10-26	E4/E1	2560	1280	640	1280	2560	1280	5120	1280	2560	5120	5120
		2015-12-15	C1/MDCK2	1280	640	320	320	1280	320	2560	2560	2560	2560	2560
TEST VIRUSES														
A/Bulgaria/033/2016	6B.1	2016-01-14	SIAT1/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280
A/Bulgaria/031/2016	6B.1	2016-01-14	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Bulgaria/037/2016	6B.1	2016-01-15	SIAT1/MDCK1	640	1280	160	320	1280	320	1280	1280	1280	2560	1280
A/Bulgaria/043/2016		2016-01-18	SIAT1/MDCK1	320	320	160	320	640	320	1280	640	640	1280	640
A/Bulgaria/042/2016	6B.1	2016-01-18	SIAT1/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	640
A/Bulgaria/052/2016	6B.1	2016-01-19	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/099/2016	6B.1	2016-01-22	SIAT2/MDCK1	1280	1280	160	640	1280	640	2560	1280	1280	2560	1280
A/Bulgaria/093/2016	6B.1	2016-01-22	SIAT2/MDCK1	640	320	160	320	640	320	1280	1280	640	2560	1280
A/Bulgaria/129/2016	6B.1	2016-01-25	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280
A/Bulgaria/110/2016	6B.1	2016-01-25	SIAT2/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Bulgaria/107/2016	6B.1	2016-01-25	SIAT1/MDCK1	640	640	320	640	640	640	1280	1280	1280	2560	1280
A/Bulgaria/132/2016	6B.1	2016-01-27	SIAT2/MDCK1	640	640	160	320	640	320	1280	1280	1280	2560	1280
A/Bulgaria/127/2016	6B.1	2016-01-27	SIAT2/MDCK1	640	320	160	320	640	640	2560	1280	1280	2560	1280
A/Bulgaria/126/2016	6B.1	2016-01-27	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280
A/Ukraine/7207/2016	6B.1	2016-02-02	P2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Kyiv/295/2016	6B.1	2016-02-02	SIAT2/MDCK1	1280	320	320	640	1280	640	2560	1280	1280	2560	2560
A/Povazska Bystrica/101/2016	6B.1	2016-02-03	MDCK2/MDCK1	640	320	320	320	640	640	2560	1280	1280	2560	1280
A/Ukraine/7203/2016	6B.1	2016-02-03	P1/MDCK1	640	320	320	320	640	320	1280	1280	640	2560	1280
A/Kyiv/310/2016	6B.1	2016-02-03	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Odessa/685/2016	6B.1	2016-02-03	SIAT2/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280
A/Ukraine/7206/2016	6B.1	2016-02-05	P2/MDCK1	640	640	320	640	1280	320	1280	1280	640	2560	1280
A/Odessa/690/2016	6B.1	2016-02-05	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Kyiv/331/2016	6B.1	2016-02-06	SIAT1/MDCK1	1280	640	640	640	2560	640	2560	2560	2560	2560	5120
A/Odessa/693/2016	6B.1	2016-02-06	SIAT2/MDCK1	320	160	160	160	320	160	640	640	640	640	640
A/Kyiv/325/2016	6B.1	2016-02-07	SIAT1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560	2560
A/Ukraine/7293/2016	6B.1	2016-02-08	P0/MDCK1	640	640	320	320	1280	320	1280	1280	640	2560	1280
A/Odessa/699/2016	6B.1	2016-02-08	SIAT2/MDCK1	160	640	640	320	640	320	1280	640	640	1280	2560
A/Ukraine/7205/2016	6B.1	2016-02-09	P1/MDCK1	640	640	640	320	1280	320	1280	1280	1280	2560	2560
A/Ukraine/7160/2016	6B.1	2016-02-09	P0/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Ukraine/7209/2016	6B.1	2016-02-11	P1/MDCK1	1280	640	640	640	1280	320	2560	2560	1280	2560	2560
A/Ukraine/7162/2016	6B.1	2016-02-11	P0/MDCK1	1280	640	640	640	1280	320	2560	2560	1280	2560	2560
A/Ukraine/7314/2016	6B.1	2016-02-14	P1/MDCK1	1280	1280	640	1280	2560	640	2560	2560	1280	2560	2560
A/Trencin/142/2016	6B.1	2016-02-15	MDCK4/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Ukraine/7319/2016	6B.1	2016-02-15	P1/MDCK1	320	320	320	320	640	160	1280	640	640	1280	1280
A/Komarno/148/2016	6B.1	2016-02-16	MDCK4/MDCK1	640	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Ukraine/7315/2016	6B.1	2016-02-18	P1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Ukraine/7294/2016	6B.1	2016-02-18	P0/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560
A/Ukraine/7276/2016	6B.1	2016-02-18	P2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Ukraine/7277/2016	6B.1	2016-02-19	P2/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	1280
A/Zaporizka/632/2016	6B.1	2016-02-21	SIAT2/MDCK1	320	640	320	320	1280	320	1280	640	1280	2560	2560
A/Ukraine/7309/2016	6B.1	2016-02-23	P2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Ternopil/749/2016	6B.1	2016-02-23	SIAT2/MDCK1	640	640	640	640	1280	640	2560	2560	1280	5120	2560
A/Novosibirsk/56/2016		2016-02-24	MDCK1/MDCK1	320	320	320	320	640	320	2560	640	640	1280	1280
A/Ukraine/7288/2016	6B.1	2016-02-24	P0/MDCK1	640	640	320	320	640	320	1280	1280	640	1280	1280
A/Khmelnytsky/672/2016	6B.1	2016-02-24	SIAT2/MDCK1	640	640	320	320	640	320	1280	1280	1280	2560	1280
A/Ukraine/7316/2016	6B.1	2016-02-25	P1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	1280
A/Ukraine/7304/2016	6B.1	2016-02-26	P2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Ukraine/7295/2016	6B.1	2016-02-26	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Ukraine/7292/2016	6B.1	2016-02-26	P0/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	1280	1280
A/Ukraine/7290/2016	6B.1	2016-02-26	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Ukraine/7289/2016	6B.1	2016-02-26	P0/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560
A/Ukraine/7303/2016	6B.1	2016-02-29	P2/MDCK1	80	320	320	320	640	160	640	640	640	1280	1280
A/Khmelnytsky/663/2016	6B.1	2016-02-29	SIAT2/MDCK1	640	320	160	320	640	320	1280	1280	640	2560	1280
A/Ukraine/7317/2016	6B.1	2016-03-02	P1/MDCK1	640	320	320	320	640	320	1280	1			

Table 5-16. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-06-22)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	A/HK	
				7/09 Egg F06/16 ¹	6/09 MDCK F09/15 ¹	N6/09 MDCK F14/13 ¹	16/10 Egg F15/14 ¹	1/11 MDCK F22/13 ¹	27/11 Egg F26/14 ¹	100/11 Egg F24/11 ¹	5659/12 MDCK F30/12 ¹	3628/13 Egg F03/14 ¹	2903/2015 Egg F02/16 ³	Q-504/15 MDCK F08/16 ³	90/16 MDCK F22/16 ³	
							4	5	6	7	6A	6B	6B.1	6B.2	6B.2	New
REFERENCE VIRUSES																
A/California/7/2009	clone 38-32	2009-04-09	E3/E3	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	640	320	80	40	40	40	80	40	40	40	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	2560	1280	320	160	160	320	160	320	160	320	160	
A/Christchurch/16/2010	4	2010-07-12	E2/E2	2560	2560	1280	5120	2560	640	5120	5120	2560	2560	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	1280	640	1280	2560	640	2560	2560	1280	2560	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	1280	640	640	2560	1280	2560	2560	1280	2560	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	160	640	160	1280	1280	640	1280	640	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	640	640	1280	1280	1280	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	2560	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	2560	640	640	1280	250	640	5120	2560	1280	5120	2560	2560	
A/Hong Kong/90/2016	6B.2	2016-01-05	MDCK1/MDCK2	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	2560	
TEST VIRUSES																
A/Lithuania/3475/2016	6B.1	2016-02-01	MDCKx/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Czech Republic/21/2016	6B.1	2016-02-02	C3/MDCK1	1280	640	320	640	1280	640	1280	2560	1280	2560	1280	2560	
A/Padova/8/2016	6B.1	2016-02-03	MDCK3/MDCK1	320	640	320	320	640	320	1280	1280	640	2560	1280	1280	
A/Czech Republic/108/2016	6B.1	2016-02-03	C2/MDCK1	1280	640	160	640	640	320	2560	1280	1280	2560	1280	2560	
A/Lithuania/4555/2016	6B.1	2016-02-07	MDCKx/MDCK1	1280	640	320	1280	1280	640	2560	2560	2560	2560	2560	2560	
A/Lithuania/4551/2016	6B.1	2016-02-07	MDCKx/MDCK1	2560	1280	320	1280	2560	640	2560	2560	2560	2560	2560	2560	
A/Lithuania/4554/2016	6B.1	2016-02-07	MDCKx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	2560	
A/Czech Republic/102/2016	6B.1	2016-02-08	C2/MDCK1	1280	320	160	640	640	320	1280	1280	1280	1280	1280	1280	
A/Iceland/14/2016	6B.1	2016-02-08	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	2560	
A/Czech Republic/106/2016	6B.1	2016-02-09	C3/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Lisboa/105/2016	6B.1	2016-02-09	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	5120	
A/Roma/8/2016	6B.1	2016-02-10	Cx/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	2560	
A/Czech Republic/104/2016	6B.1	2016-02-10	C2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	2560	
A/Czech Republic/105/2016	6B.1	2016-02-10	C3/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560	1280	2560	
A/Ukraine/7161/2016		2016-02-11	P9/MDCK2	1280	1280	640	1280	2560	1280	2560	2560	1280	2560	2560	5120	
A/Bolzano/11/2016	6B.1	2016-02-16	MDCK3/MDCK1	1280	1280	320	640	1280	1280	2560	2560	1280	2560	2560	2560	
A/Czech Republic/103/2016	6B.1	2016-02-16	C2/MDCK1	1280	640	320	640	640	320	1280	1280	1280	2560	1280	1280	
A/Lithuania/6088/2016	6B.1	2016-02-16	MDCKx/MDCK1	1280	1280	320	1280	1280	640	2560	5120	1280	2560	2560	2560	
A/Lithuania/6081/2016	6B.1	2016-02-16	MDCKx/MDCK1	1280	640	160	640	640	640	1280	1280	1280	1280	1280	2560	
A/Lithuania/6071/2016	6B.1	2016-02-17	MDCKx/MDCK1	1280	640	160	640	1280	640	2560	1280	1280	2560	1280	2560	
A/Lithuania/6090/2016	6B.1	2016-02-17	MDCKx/MDCK1	1280	1280	160	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Iceland/21/2016	6B.1	2016-02-17	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560	2560	2560	
A/Lithuania/6173/2016	6B.1	2016-02-18	MDCKx/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	2560	5120	
A/Lithuania/6016/2016	6B.1	2016-02-18	MDCKx/MDCK1	1280	640	160	640	1280	640	1280	2560	1280	2560	1280	2560	
A/Milano/43/2016	6B.1	2016-02-22	MDCK1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560	1280	2560	
A/Panna/39/2016	6B.1	2016-02-24	MDCK1/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560	2560	
A/Czech Republic/107/2016	6B.1	2016-02-24	C2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	2560	
A/Palermo/8/2016	6B.1	2016-02-29	MDCK2/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Friuli-Venezia Giulia/15/2016	6B.1	2016-02-29	MDCK3/MDCK1	640	640	320	640	1280	640	1280	2560	1280	2560	2560	2560	
A/Lisboa/127/2016	6B.1	2016-03-01	SIAT2/MDCK1	1280	320	160	640	640	320	2560	1280	1280	2560	1280	2560	
A/Czech Republic/109/2016	6B.1	2016-03-06	C2/MDCK1	1280	1280	160	640	1280	640	1280	2560	1280	2560	1280	2560	
A/Lisboa/129/2016	6B.1	2016-03-06	SIAT1/MDCK1	1280	1280	320	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Czech Republic/82/2016	6B.1	2016-03-09	C4/MDCK1	1280	320	160	320	640	640	1280	1280	1280	1280	1280	2560	
A/Lisboa/123/2016	6B.1	2016-03-10	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	2560	
A/Palermo/10/2016	6B.1	2016-03-15	MDCK3/MDCK1	2560	640	320	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Lisboa/130/2016	6B.1	2016-03-16	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Lisboa/124/2016	6B.1	2016-03-18	SIAT1/MDCK1	2560	640	320	640	1280	640	2560	2560	1280	2560	2560	2560	
A/Lisboa/128/2016	6B.1	2016-03-18	SIAT2/MDCK1	1280	640	320	640	640	640	1280	2560	1280	2560	1280	2560	
A/Iceland/43/2016	6B.1	2016-03-22	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	2560	
A/Lisboa/125/2016	6B.1	2016-04-01	SIAT1/MDCK1	1280	640	320	640	1280	320	2560	2560	1280	2560	1280	2560	
A/Perugia/26/2016	6B.1	2016-04-09	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	1280	2560	
A/Iceland/51/2016	6B.1	2016-04-13	MDCK1/MDCK1	160	640	320	640	640	320	640	1280	640	2560	1280	2560	
A/Iceland/54/2016	6B.1	2016-04-22	MDCK1/MDCK1	160	640	320	640	640	320	1280	1280	640	2560	2560	2560	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Sequences in phylogenetic trees

Vaccine

G155E
G155E>G, D222G

Table 5-17. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-06-29)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Ferret number			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E3	1280	1280	640	640	1280	1280	2560	2560	1280	2560	1280	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	640	320	160	40	40	40	80	80	80	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	80	160	160	320	160	320	320	
A/Christchurch/16/2010	4	2010-07-12	E2/E2	2560	2560	2560	5120	2560	640	5120	2560	1280	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	1280	640	1280	2560	640	2560	2560	1280	2560	1280	
A/St. Petersburg/2/2011	6	2011-02-14	E1/E4	2560	1280	640	640	1280	1280	2560	2560	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	1280	640	640	1280	1280	2560	2560	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	320	640	320	1280	1280	640	640	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	2560	1280	1280	1280	1280	2560	2560	2560	2560	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	2560	1280	640	1280	1280	1280	5120	5120	1280	2560	2560	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	640	640	1280	640	5120	2560	1280	2560	2560	
TEST VIRUSES															
A/Finland/570/2015	6B.1	2015-12-21	MDCK1/MDCK1	2560	1280	640	2560	2560	1280	5120	5120	2560	5120	5120	
A/Ukraine/7029/2016	6B.2	2016-01-02	P2/MDCK1	320	320	160	320	320	320	640	640	640	1280	1280	
A/Ukraine/7154/2016	6B.1	2016-01-11	P0/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Ukraine/7153/2016	6B.1	2016-01-11	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Ukraine/7031/2016	6B.2	2016-01-12	P2/MDCK1	320	160	80	160	320	320	640	640	640	640	640	
A/Ukraine/7092/2016	6B.1	2016-01-18	P1/MDCK1	1280	320	160	320	640	640	1280	1280	1280	2560	1280	
A/Ukraine/7182/2016	6B.1	2016-01-22	P2/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	2560	
A/Ukraine/7156/2016	6B.1	2016-01-22	P0/MDCK1	2560	1280	640	1280	1280	1280	5120	5120	2560	5120	5120	
A/Ukraine/7157/2016	6B.2	2016-01-26	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Odessa/696/2016	6B.1	2016-01-28	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/Finland/575/2016	6B.1	2016-01-31	MDCK1/MDCK1	1280	1280	640	1280	2560	640	2560	2560	1280	2560	2560	
A/Slovenia/460/2016	6B.1	2016-02-01	MDCKx/MDCK1	160	320	320	320	640	320	640	640	640	1280	1280	
A/Stockholm/17/2016	6B.1	2016-02-01	MDCKx/MDCK1	1280	1280	320	1280	1280	640	2560	2560	1280	2560	2560	
A/IV-Ryazan/48/2016	6B.1	2016-02-01	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	2560	
A/Finland/602/2016	6B.1	2016-02-02	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	5120	
A/Slovenia/558/2016	6B.1	2016-02-03	MDCKx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/IV-Penza/110-L/2016	6B.1	2016-02-04	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	1280	2560	2560	
A/IV-Birobidzhan/133-B/2016	6B.1	2016-02-09	MDCK4/MDCK1	1280	320	160	320	640	320	1280	1280	640	1280	1280	
A/Slovenia/667/2016	6B.1	2016-02-10	MDCKx/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Krasnodar/32/2016	6B.1	2016-02-10	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Finland/580/2016	6B.1	2016-02-11	MDCK1/MDCK1	1280	1280	320	1280	1280	640	5120	2560	1280	2560	2560	
A/Slovenia/716/2016	6B.1	2016-02-12	MDCKx/MDCK1	1280	640	320	640	640	640	2560	1280	640	2560	1280	
A/Yaroslavl/14/2016	6B.1	2016-02-12	MDCK1/MDCK1	2560	640	320	1280	1280	640	2560	2560	1280	2560	2560	
A/Vladivostok/69/2016	6B.1	2016-02-13	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Slovenia/938/2016	6B.1	2016-02-20	MDCKx/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	5120	
A/Finland/620/2016	6B.1	2016-02-20	MDCK1/MDCK1	2560	1280	320	640	1280	640	2560	2560	1280	2560	2560	
A/IV-Orenburg/111/2016	6B.1	2016-02-23	MDCK1/MDCK1	1280	640	320	640	640	640	1280	2560	1280	2560	1280	
A/Finland/616/2016	6B.1	2016-02-24	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	2560	
A/Vladivostok/102/2016	6B.1	2016-02-29	MDCK1/MDCK1	640	320	160	320	320	320	640	640	640	1280	640	
A/IV-Cheboksary/129/2016	6B.1	2016-03-03	MDCK1/MDCK1	640	320	160	640	640	320	1280	1280	640	1280	1280	
A/Finland/619/2016	6B.1	2016-03-06	MDCK1/MDCK1	2560	1280	640	1280	1280	1280	2560	5120	1280	2560	2560	
A/Slovenia/1143/2016	6B.1	2016-03-07	MDCKx/MDCK1	2560	640	320	640	1280	640	5120	2560	1280	2560	2560	
A/Stockholm/27/2016	6B.1	2016-03-10	MDCKx/MDCK1	2560	1280	640	1280	1280	1280	5120	5120	2560	5120	2560	
A/Slovenia/1283/2016	6B.1	2016-03-18	MDCKx/MDCK1	2560	1280	640	1280	640	640	5120	2560	1280	2560	2560	
A/Finland/622/2016	6B.1	2016-03-21	MDCK1/MDCK1	2560	1280	1280	1280	2560	1280	2560	5120	1280	5120	2560	
A/Cote D'Ivoire/475/2016	6B.1	2016-03-23	P1/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	5120	
A/Cote D'Ivoire/471/2016	6B.1	2016-03-24	P1/MDCK1	1280	1280	640	1280	1280	1280	2560	2560	2560	5120	2560	
A/Slovenia/1377/2016	6B.1	2016-03-28	MDCKx/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Slovenia/1423/2016	6B.1	2016-04-01	MDCKx/MDCK1	2560	640	320	1280	1280	640	2560	2560	1280	2560	2560	
A/IV-Moscow/132/2016	6B.1	2016-04-07	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/Cote D'Ivoire/575/2016	6B.1	2016-04-11	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/Cote D'Ivoire/582/2016	6B.1	2016-04-12	P0/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	1280	2560	2560	
A/Cote D'Ivoire/596/2016	6B.1	2016-04-15	P0/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	5120	
A/Cote D'Ivoire/597/2016	6B.1	2016-04-15	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Cote D'Ivoire/598/2016	6B.1	2016-04-15	P0/MDCK1	1280	640	320	640	1280	640	1280	2560	1280	2560	2560	
A/Cote D'Ivoire/600/2016	6B.1	2016-04-18	P0/MDCK1	1280	1280	320	1280	1280	640	2560	2560	1280	2560	2560	
A/Cote D'Ivoire/602/2016	6B.1	2016-04-19	P0/MDCK1	2560	1280	320	1280	2560	1280	5120	5120	2560	5120	2560	
A/Cote D'Ivoire/603/2016	6B.1	2016-04-19	P0/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	
A/Cote D'Ivoire/604/2016	6B.1	2016-04-19	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Cote D'Ivoire/616/2016	6B.1	2016-04-20	P0/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	
A/Cote D'Ivoire/617/2016		2016-04-20	P0/MDCK1	2560	1280	320	640	1280	640	2560	2560	1280	2560	2560	
A/Cote D'Ivoire/618/2016	6B.1	2016-04-20	P0/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	5120	
A/Cote D'Ivoire/641/2016	6B.1	2016-04-25	P2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Cote D'Ivoire/645/2016	6B.1	2016-04-26	P2/MDCK1	1280	320	160	320	640	320	1280	1280	640	1280	1280	
A/Cote D'Ivoire/646/2016	6B.1	2016-04-26	P2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/Cote D'Ivoire/669/2016	6B.1	2016-04-27	P0/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	5120	
A/Cote D'Ivoire/680/2016	6B.1	2016-05-05	P2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

G155E
G155E>G, D222G

Table 5-18. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-07-05 + 2016-07-12)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Ferret number			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	2560	1280	640	1280	2560	1280	5120	2560	1280	5120	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	640	320	160	40	40	40	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	1280	320	80	160	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	2560	2560	2560	5120	2560	1280	5120	2560	2560	5120	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	1280	640	1280	2560	1280	5120	2560	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	5120	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	160	320	640	320	1280	1280	640	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	1280	640	640	1280	640	2560	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	5120	1280	640	2560	2560	2560	5120	5120	2560	5120	5120	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	2560	640	320	1280	1280	640	2560	2560	1280	2560	2560	
TEST VIRUSES															
A/Norway/2914/2015	6B.1	2015-12-14	MDCK1/MDCK1	2560	1280	640	2560	2560	1280	5120	5120	2560	5120	5120	
A/Albania/6376/2016	6B.1	2016-02-01	MDCK3/MDCK1	2560	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Albania/6378/2016	6B.1	2016-02-01	MDCK3/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	2560	
A/Albania/6425/2016	6B.1	2016-02-04	MDCK3/MDCK1	1280	640	320	1280	1280	640	2560	2560	1280	2560	2560	
A/Albania/6443/2016	6B.1	2016-02-05	MDCK3/MDCK1	1280	640	320	640	1280	1280	2560	2560	2560	2560	2560	
A/Albania/6485/2016	6B.1	2016-02-10	MDCK3/MDCK1	1280	320	320	640	640	640	1280	1280	1280	1280	1280	
A/Albania/6484/2016	6B.1	2016-02-10	MDCK3/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	2560	
A/Albania/6507/2016	6B.1	2016-02-15	MDCK3/MDCK1	2560	1280	640	1280	1280	1280	2560	2560	2560	2560	2560	
A/Kazakhstan/4481/2016	6B.1	2016-02-16	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	
A/Kazakhstan/4514/2016	6B.1	2016-02-16	MDCK1	2560	640	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Kazakhstan/4693/2016	6B.1	2016-02-16	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Kazakhstan/4762/2016	6B.1	2016-02-16	MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	2560	
A/Kazakhstan/4692/2016	6B.1	2016-02-17	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Kazakhstan/4694/2016	6B.1	2016-02-18	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/South Africa/R0736/2016	6B.1	2016-02-18	MDCK1	2560	640	640	1280	1280	1280	5120	2560	2560	2560	2560	
A/Kazakhstan/4695/2016	6B.1	2016-02-19	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Albania/6665/2016	6B.1	2016-02-19	MDCK2/MDCK1	2560	640	640	1280	2560	1280	5120	2560	2560	5120	2560	
A/Kazakhstan/4507/2016	6B.1	2016-02-22	MDCK1	640	1280	640	640	1280	640	2560	2560	1280	2560	1280	
A/Albania/6675/2016	6B.1	2016-02-27	MDCK3/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	5120	
A/Albania/6660/2016	6B.1	2016-02-28	MDCK3/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	5120	
A/Kazakhstan/4763/2016	6B.1	2016-03-01	MDCK1	2560	1280	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Kazakhstan/4764/2016	6B.1	2016-03-01	MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Albania/6690/2016	6B.1	2016-03-02	MDCK3/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Albania/6713/2016	6B.1	2016-03-03	MDCK3/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Norway/2036/2016	6B.1	2016-03-10	MDCK3/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	2560	2560	
A/Kazakhstan/4697/2016	6B.1	2016-03-14	MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	5120	
A/Kazakhstan/4702/2016	6B.1	2016-03-18	MDCK1	1280	1280	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Norway/2298/2016	6B.1	2016-03-21	MDCK2/MDCK1	1280	640	320	640	640	640	1280	1280	1280	1280	2560	
A/South Africa/R2487/2016	6B.1	2016-05-10	MDCK1/MDCK1	1280	640	320	1280	1280	1280	5120	2650	2560	2560	2560	
A/South Africa/R2539/2016	6B.1	2016-05-11	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	
A/Norway/3591/2016	6B.1	2016-05-21	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	2560	2560	1280	2560	2560	
A/South Africa/R3518/2016	6B.1	2016-06-07	MDCK1	2560	1280	640	1280	1280	1280	2560	2560	2560	5120	2560	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-19. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-07-26)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/CaI	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	G155E G155E>G, D222G
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Passage history			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
	Ferret number						4	5	6	7	6A	6B	6B.1	6B.2	
	Genetic group														
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	160	80	40	40	40	40	40	40	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	640	320	80	160	80	160	80	160	160	
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	1280	5120	2560	640	2560	2560	1280	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	320	1280	1280	640	2560	1280	640	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	2560	1280	640	1280	2560	1280	5120	2560	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	640	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	1260	80	160	320	320	640	640	320	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	1280	640	640	1280	1280	1280	1280	1280	1280	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	2560	640	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
TEST VIRUSES															
A/Czech Republic/11/2016	6B.1	2016-01-28	C3/MDCK1	2560	640	320	1280	1280	640	2560	2560	1280	2560	2560	
A/Estonia/98215/2016	6B.1	2016-02-04	MDCK1/MDCK1	2560	1280	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Estonia/98192/2016	6B.1	2016-02-04	MDCK2/MDCK1	2560	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Estonia/98290/2016	6B.1	2016-02-08	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Estonia/98475/2016	6B.1	2016-02-12	MDCK1/MDCK1	1280	640	320	1280	1280	1280	2560	2560	1280	2560	2560	
A/Mardin/3612/2016	6B.1	2016-02-14	SIAT1/MDCK1	2560	1280	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Antalya/1150/2016	6B.1	2016-02-16	SIAT1/MDCK1	1280	320	320	640	640	640	1280	1280	640	1280	1280	
A/Aydin/3677/2016	6B.1	2016-02-17	SIAT1/MDCK1	2560	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Izmir/3728/2016	6B.1	2016-02-17	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Antalya/3748/2016	6B.1	2016-02-17	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Estonia/98636/2016	6B.1	2016-02-19	MDCK1/MDCK1	1280	1280	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Karabuk/3926/2016	6B.1	2016-02-24	MDCK1/MDCK1	2560	640	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Estonia/98801/2016	6B.1	2016-02-26	MDCK1/MDCK1	2560	2560	640	2560	2560	2560	5120	5120	2560	5120	5120	
A/Estonia/98923/2016	6B.1	2016-03-04	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Estonia/99005/2016	6B.1	2016-03-08	MDCK2/MDCK1	640	640	640	640	640	640	1280	1280	1280	2560	2560	
A/Estonia/99218/2016	6B.1	2016-03-17	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Estonia/99242/2016	6B.1	2016-03-21	MDCK1/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Estonia/99288/2016	6B.1	2016-03-22	MDCK1/MDCK1	1280	1280	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Estonia/99350/2016	6B.1	2016-03-24	MDCK1/MDCK1	1280	640	320	640	640	640	1280	1280	1280	2560	2560	
A/Estonia/99352/2016	6B.1	2016-03-24	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	5120	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-20. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-08-02 + 2016-08-09 + 2016-08-16)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15
	Passage history			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK
	Ferret number			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²
	Genetic group						4	5	6	7	6A	6B	6B.1	6B.2
REFERENCE VIRUSES														
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	640	640	1280	1280	2560	2560	1280	2560	1280
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	80	80	40	<	<	40	40	40	<
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	160	1280	640	320	80	80	80	160	80	160	160
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	1280	5120	1280	640	2560	1280	640	2560	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	320	640	1280	640	2560	2560	1280	2560	1280
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	320	640	1280	640	2560	1280	640	2560	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	640	320	640	640	640	1280	1280	640	1280	640
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	80	160	320	160	640	640	320	640	320
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	320	320	320	320	640	640	640	640	640	320
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	1280	640	320	1280	1280	640	2560	1280	1280	2560	1280
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	640	320	640	640	640	2560	1280	1280	2560	1280
TEST VIRUSES														
A/Tajikistan/4/2016	6B.1	2016-01-18	MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560	5120	5120
A/Mardin/2840/2016	6B.1	2016-02-01	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Batman/2857/2016	6B.1	2016-02-01	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	2560	2560	2560
A/Diyarbakir/1006/2016	6B.1	2016-02-02	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Izmir/1016/2016	6B.1	2016-02-02	SIAT1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	2560	2560
A/Antalya/1038/2016	6B.1	2016-02-03	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280
A/Zonguldak/2996/2016	6B.1	2016-02-03	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Burdur/3015/2016	6B.1	2016-02-03	SIAT1/MDCK1	640	320	160	320	640	320	1280	640	640	1280	640
A/Antalya/1103/2016	6B.1	2016-02-05	SIAT1/MDCK1	1280	640	320	640	640	640	2560	1280	1280	2560	1280
A/Aydin/3180/2016	6B.1	2016-02-05	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Serbia/824/2016		2016-02-08	C1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Ankara/3266/2016	6B.1	2016-02-08	MDCK1/MDCK1	1280	640	320	640	640	320	1280	1280	1280	2560	1280
A/Manisa/3292/2016	6B.1	2016-02-09	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Samsun/3315/2016	6B.1	2016-02-09	SIAT1/MDCK1	1280	1280	320	1280	1280	640	2560	2560	1280	2560	2560
A/Eskisehir/3317/2016	6B.1	2016-02-09	SIAT1/MDCK1	1280	640	160	320	640	320	1280	1280	640	2560	1280
A/Denizli/3365/2016	6B.1	2016-02-10	SIAT1/MDCK1	2560	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Ankara/3467/2016	6B.1	2016-02-11	MDCK1/MDCK1	640	640	320	640	1280	640	2560	2560	1280	2560	2560
A/Borisov/1519/2016	6B.1	2016-02-22	MDCK2/MDCK1	1280	640	320	640	640	640	2560	1280	1280	2560	1280
A/Minsk/1495/2016	6B.1	2016-02-23	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Mogilev/1595/2016	6B.1	2016-02-29	MDCK2/MDCK1	1280	640	320	320	1280	640	2560	1280	1280	2560	2560
A/Ghana/FIP-16-2018/2016	6B.1	2016-04-25	Cx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560
A/Ghana/DILI-16-0649/2016	6B.1	2016-06-11	Cx/MDCK1	1280	640	640	1280	1280	640	5120	2560	1280	2560	2560
A/Ghana/DILI-16-0610/2016	6B.1	2016-06-13	Cx/MDCK1	1280	640	320	640	640	640	2560	1280	640	2560	2560
A/Ghana/DILI-16-0662/2016	6B.1	2016-06-27	Cx/MDCK1	1280	640	320	1280	1280	640	2560	1280	1280	2560	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Sequences in phylogenetic trees

Vaccine

Table 5-21. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-08-24 + 2016-08-31)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	G155E G155E>G, D222G
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Passage history			F06/16 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
	Ferret number						4	5	6	7	6A	6B	6B.1	6B.2	
	Genetic group														
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	160	80	40	40	40	40	40	40	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK2	160	640	1280	320	80	160	80	320	160	320	160	
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	640	1280	5120	1280	640	2560	1280	1280	1280	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	320	320	640	1280	640	2560	2560	1280	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	320	320	640	1280	640	2560	1280	640	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	1280	320	320	640	640	640	2560	1280	640	1280	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	160	80	160	320	160	640	640	320	320	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	320	320	320	320	320	640	640	640	640	640	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	320	160	320	640	320	1280	1280	640	1280	1280	
TEST VIRUSES															
A/Serbia/664/2016		2016-02-01	MDCK2	640	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Lorraine/2132/2016	6B.1	2016-03-29	MDCK1/MDCK1	640	320	320	640	640	640	2560	1280	640	1280	1280	
A/Paris/2257/2016	6B.1	2016-04-11	MDCK1/MDCK1	640	320	160	320	640	640	2560	1280	640	1280	1280	
A/Paris/2295/2016	6B.1	2016-04-11	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/England/932/2016		2016-04-12	SIAT1/MDCK1	1280	320	640	640	640	320	1280	1280	640	1280	1280	
A/Ghana/FS-16-0469/2016		2016-04-12	MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Alsace/2318/2016	6B.1	2016-04-12	MDCK1/MDCK1	640	320	160	320	640	320	2560	1280	1280	2560	1280	
A/Bretagne/2441/2016	6B.1	2016-04-12	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Franche Comte/2335/2016	6B.1	2016-04-14	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Nord Pas de Calais/2353/2016	6B.1	2016-04-18	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Pays de Loire/2364/2016	6B.1	2016-04-18	MDCK1/MDCK1	1280	640	320	640	1280	1280	5120	2560	1280	2560	2560	
A/England/950/2016		2016-04-19	SIAT2/MDCK1	1280	320	640	320	640	320	1280	640	640	1280	1280	
A/Centre/2373/2016	6B.1	2016-04-21	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	1280	1280	
A/Paris/2374/2016	6B.1	2016-04-23	MDCK1/MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	5120	2560	
A/Paris/2391/2016		2016-04-26	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	1280	
A/England/923/2016		2016-05-01	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Ghana/DILI-16-0686/2016		2016-07-04	SIAT1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560	
A/South Africa/R5236/2016		2016-07-25	MDCK1	1280	640	320	640	640	640	1280	1280	1280	2560	1280	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-22. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-09-13)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	G155E G155E>G, D222G
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Passage history			F06/16 ⁻¹	F09/15 ⁻¹	F14/13 ⁻¹	F15/14 ⁻¹	F22/13 ⁻¹	F26/14 ⁻¹	F24/11 ⁻¹	F30/12 ⁻¹	F03/14 ⁻¹	F02/16 ⁻²	F08/16 ⁻²	
	Ferret number						4	5	6	7	6A	6B	6B.1	6B.2	
	Genetic group														
REFERENCE VIRUSES															
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	320	160	40	40	40	80	80	80	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	640	320	80	160	80	320	80	10	160	
A/Christchurch/16/2010		2010-07-12	E2/E2	1280	1280	1280	5120	1280	640	2560	2560	1280	2560	1280	
A/Astrakhan/1/2011		2011-02-28	MDCK1/MDCK6	640	640	320	640	1280	640	2560	1280	1280	1280	1280	
A/St. Petersburg/27/2011		2011-02-14	E1/E4	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	
A/St. Petersburg/100/2011		2011-03-14	E1/E4	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Hong Kong/5659/2012		2012-05-21	MDCK4/MDCK1	320	320	160	160	320	320	1280	1280	320	320	320	
A/South Africa/3626/2013		2013-06-06	E1/E3	640	640	320	640	640	640	1280	640	1280	1280	640	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E1	1280	640	640	1280	2560	1280	5120	2560	1280	2560	2560	
A/Israel/Q-504/2015		2015-12-15	C1/MDCK2	1280	640	320	1280	1280	640	2560	2560	1280	2560	2560	
TEST VIRUSES															
A/Iceland/12/2016		2016-02-03	MDCK1/MDCK1	640	640	320	640	640	320	1280	1280	640	1280	1280	
A/Niger/3474/2016		2016-02-06	MDCK1	1280	640	320	640	640	640	2560	1280	1280	1280	1280	
A/Iceland/16/2016		2016-02-11	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	2560	
A/Niger/3439/2016		2016-02-15	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Niger/3427/2016		2016-02-16	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Niger/3456/2016		2016-02-17	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Poitiers/1851/2016		2016-02-17	MDCKx/MDCK1/MDCK1	640	640	320	640	640	640	1280	1280	1280	2560	1280	
A/Iceland/32/2016		2016-03-01	MDCK1/MDCK1	640	640	320	640	640	640	2560	1280	1280	2560	2560	
A/Iceland/35/2016		2016-03-04	MDCK1/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	2560	
A/Dakar/02/2016		2016-03-04	P0/MDCK1	1280	640	320	1280	1280	1280	2560	2560	1280	2560	2560	
A/Niger/3544/2016		2016-03-05	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	2560	
A/Jordan/20404/2016		2016-03-24	MDCK1	1280	640	320	1280	1280	640	2560	2560	1280	2560	2560	
A/Jordan/20421/2016		2016-04-05	MDCK1	320	640	320	640	640	320	1280	640	640	2560	2560	
A/Dakar/03/2016		2016-04-06	P0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	
A/Lyon/1644/2016		2016-04-19	MDCK2/MDCK1	2560	1280	320	1280	1280	1280	5120	2560	1280	5120	2560	
A/Dakar/04/2016		2016-05-04	P1/MDCK1	640	320	160	640	640	640	1280	1280	1280	1280	1280	
A/Dakar/05/2016		2016-05-09	P2/MDCK1	640	320	160	320	640	320	1280	1280	1280	1280	1280	
A/Dakar/06/2016		2016-05-11	P0/MDCK1	1280	640	320	1280	1280	640	2560	2560	640	5120	2560	
A/La Reunion/2020/2016		2016-07-01	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	
A/Hong Kong/2394/2016		2016-08-04	MDCK1/MDCK1	1280	640	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Hong Kong/2395/2016		2016-08-09	MDCK1/MDCK1	1280	640	640	1280	1280	1280	2560	2560	1280	2560	2560	
A/Hong Kong/2397/2016		2016-08-10	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	1280	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <80

Vaccine

Sequences in phylogenetic trees

Table 5-23. Antigenic analyses of influenza A(H1N1)pdm09 viruses with collection dates after 2016-01-31 - Summary

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre													
				Post-infection ferret antisera													
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	A/Ukraine	A/HK	
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	6907/15	90/16	
Passage history																	
Genetic group																	
REFERENCE VIRUSES																	
A/California/7/2009	clone 38-32	2009-04-09	E3/E2	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	640	2560	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	40	320	160	80	40	40	40	40	40	40	40	<	40	
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK2	160	640	1280	320	80	160	80	320	160	320	160	80	80	
A/Christchurch/16/2010		4	2010-07-12	E2/E2	1280	640	1280	5120	1280	640	2560	1280	1280	2560	640	2560	
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	320	320	640	1280	640	2560	2560	1280	1280	640	1280	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	1280	320	320	640	1280	640	2560	1280	640	2560	1280	640	2560
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	1280	320	320	320	640	640	2560	1280	640	1280	1280	640	2560
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	320	160	80	160	320	160	640	640	320	320	320	320	640
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	640	320	320	320	320	320	640	640	640	640	640	320	1280
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E1	1280	640	320	640	1280	640	2560	2560	1280	2560	2560	640	2560
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK2	1280	320	160	320	640	320	1280	1280	640	1280	1280	640	2560	
A/Ukraine/6907/2015	6B.2	2015-12-16	C2/MDCK2	320	640	320	320	640	640	1280	640	1280	1280	2560	640	ND	
A/Hong Kong/90/2016	6B.2	2016-01-05	MDCK1/MDCK2	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	2560	ND	2560	
TEST VIRUSES																	
Number of viruses tested*				413	413	413	413	413	413	413	413	413	413	413	30	43	
No with titre reduction ≤2-fold				393	413	172	1	361	384	396	411	380	383	405	30	43	
%				95.2	100.0	41.6	0.2	87.4	93.0	95.9	99.5	92.0	92.7	98.1	100.0	100.0	
No with titre reduction =4-fold				10		203	84	45	29	14	2	30	24	8			
%				2.4		49.2	20.4	10.9	7.0	3.4	0.5	7.3	5.8	1.9			
No with titre reduction ≥8-fold				10		38	328	7		3		3	6				
%				2.4		9.2	79.4	1.7		0.7		0.7	1.5				
* Of those with available HA sequence (374) clades represented: 6B (3 = 0.8%), 6B.1 (369 = 98.7%), 6B.2 (2 = 0.5%)				Vaccine													

Figure 1. Phylogenetic comparison of influenza A(H1N1)pdm09 HA genes

Vaccine virus

Reference viruses

Collection date

Feb-Mar 2016

Apr 2016

May 2016

Jun-Jul 2016

HA2 numbering

* indicates amino acid polymorphism

@ proposed serology antigens

HI result in table

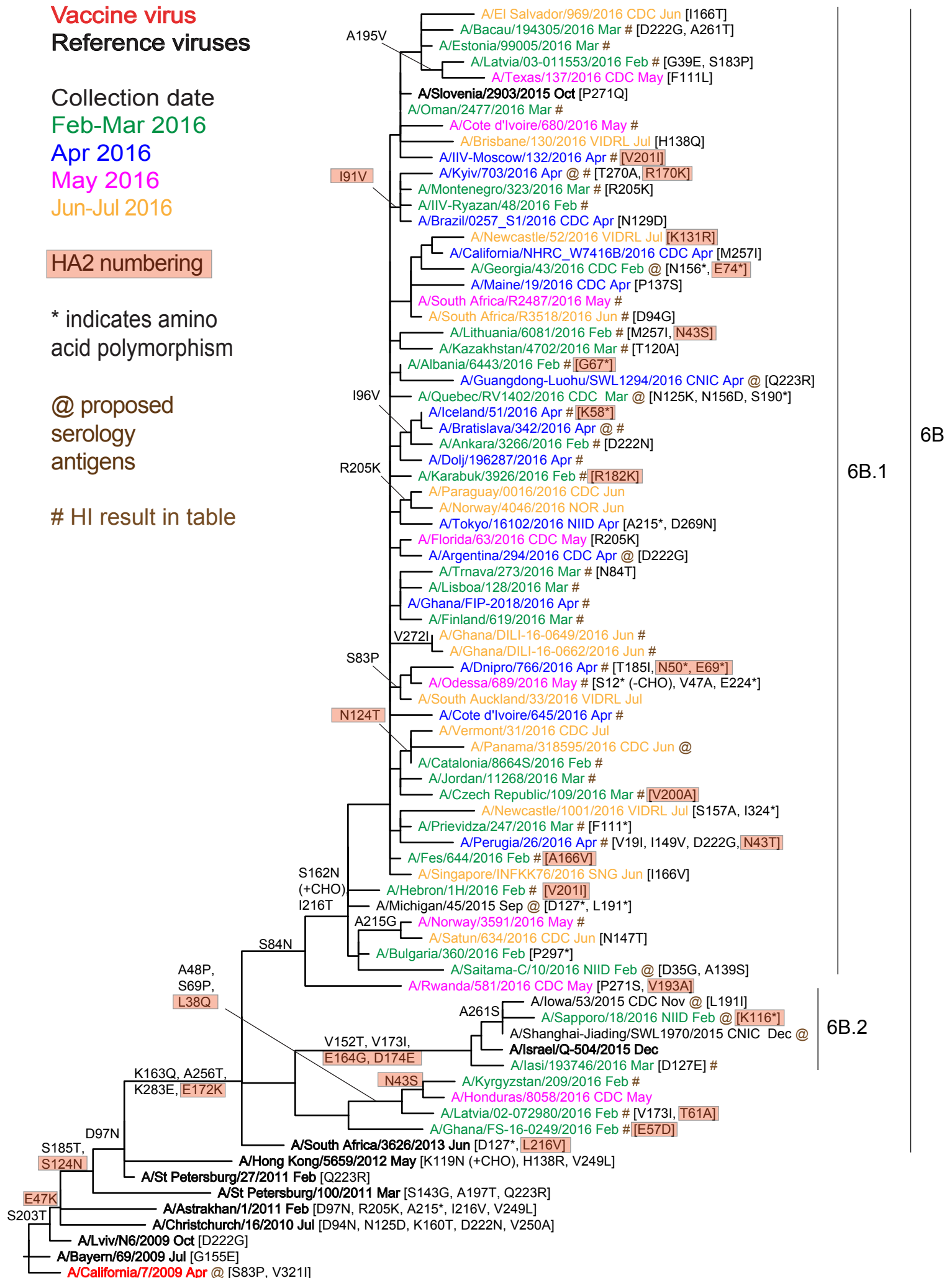
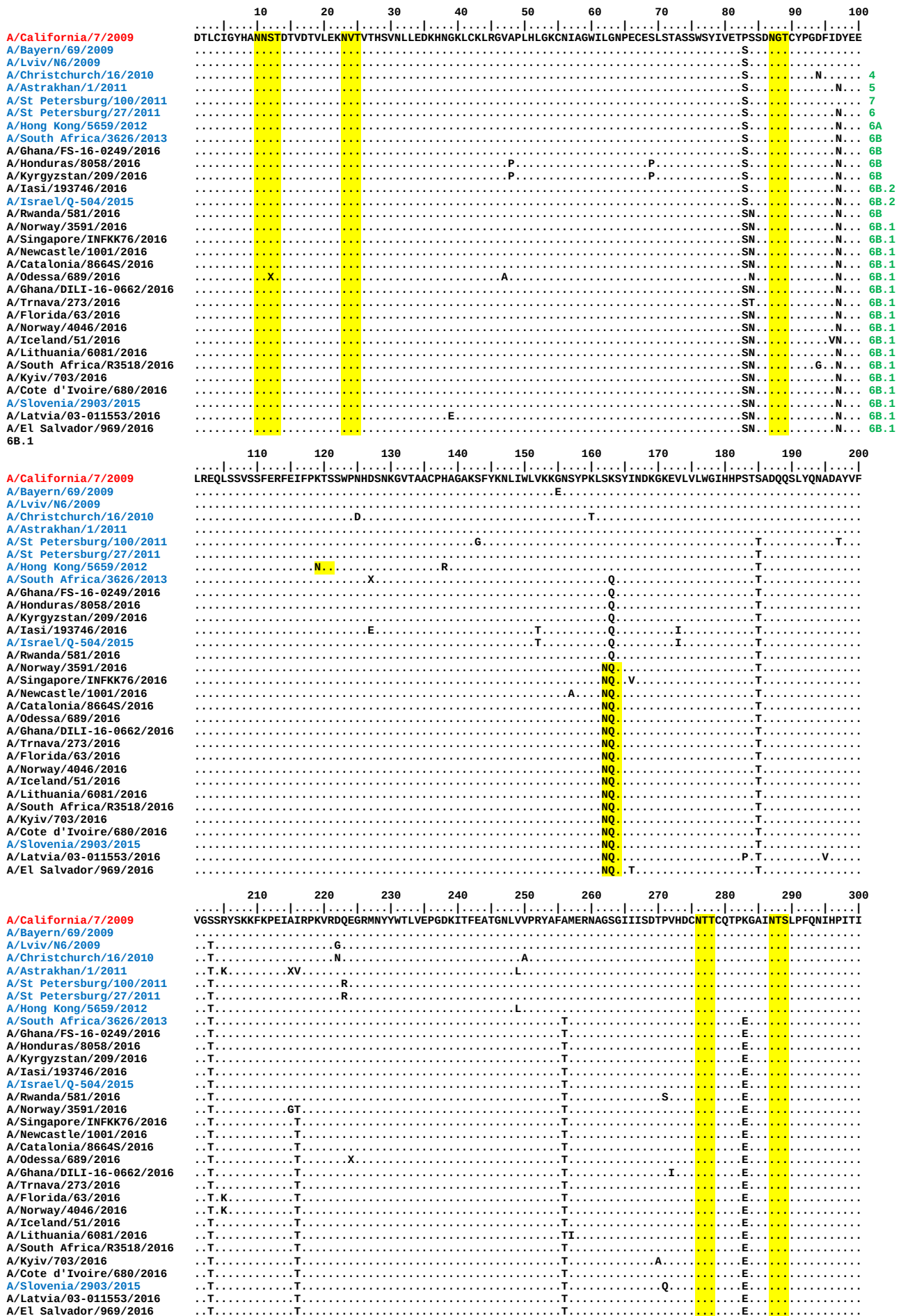


Figure 2. HA protein alignment for influenza A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



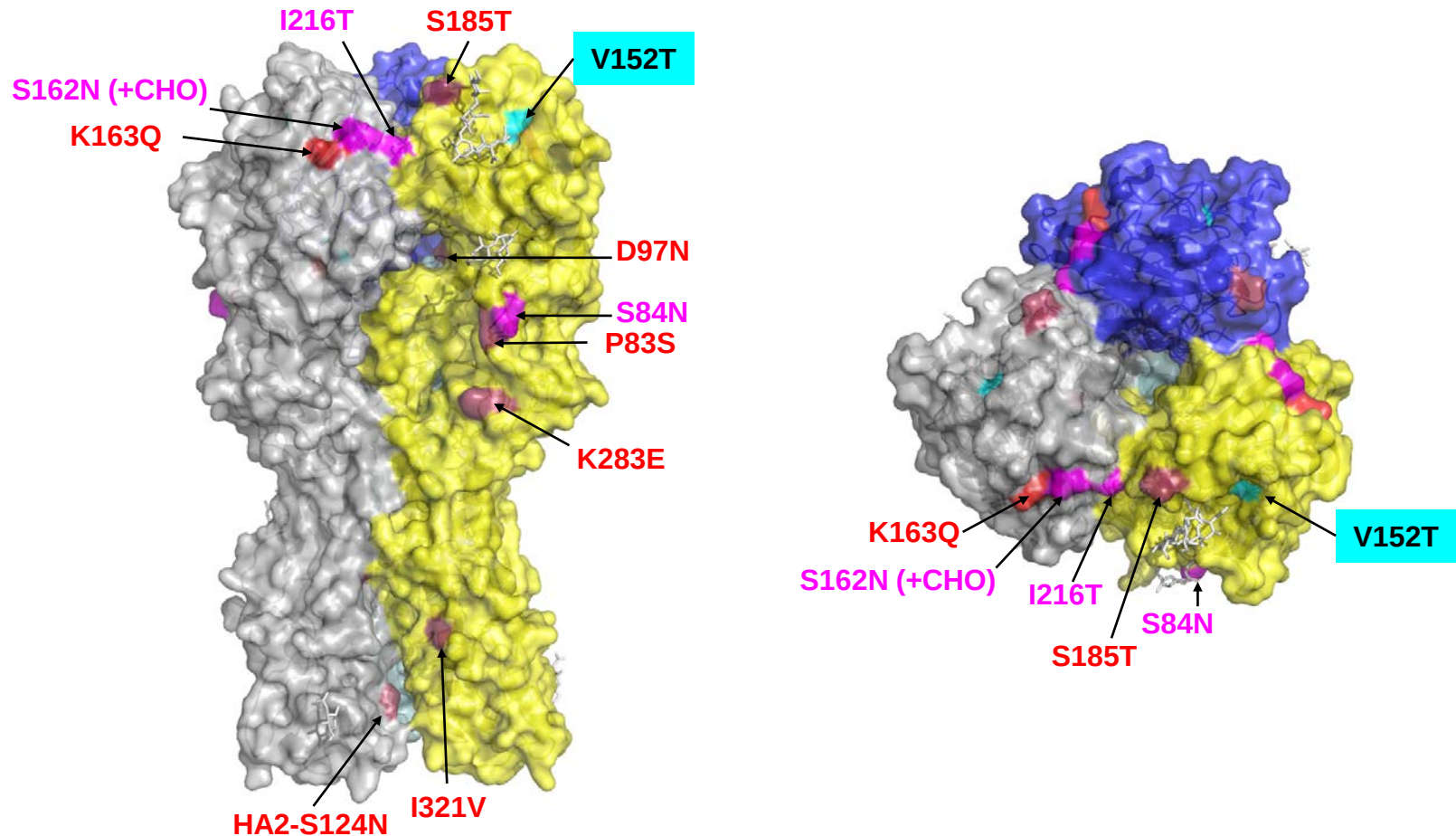
	310	320	3	13	23	33	43	53	63	73
A/California/7/2009	GKCPKYVKSTKLRLATGLRNIPSIQSRGLFGAIAGFIEGGWTGMVDGWYGVHHQNEQSGGYAADLKSTQNAIDEITNKVNSVIEKMNTQFTAVGKEFNHL									
A/Bayern/69/2009		V								
A/Lviv/N6/2009		V								
A/Christchurch/16/2010		V						K		
A/Astrakhan/1/2011		V						K		
A/St Petersburg/100/2011		V						K		
A/St Petersburg/27/2011		V						K		
A/Hong Kong/5659/2012		V						K		
A/South Africa/3626/2013		V						K		
A/Ghana/FS-16-0249/2016		V						K	D	
A/Honduras/8058/2016		V					Q	S	K	
A/Kyrgyzstan/209/2016		V					Q	S	K	
A/Iasi/193746/2016		V						K		
A/Israel/Q-504/2015		V						K		
A/Rwanda/581/2016		V						K		
A/Norway/3591/2016		V						K		
A/Singapore/INFKK76/2016		V						K		
A/Newcastle/1001/2016		V	X					K		
A/Catalonia/8664S/2016		V						K		
A/Odessa/689/2016		V						K		
A/Ghana/DILI-16-0662/2016		V						K		
A/Trnava/273/2016		V						K		
A/Florida/63/2016		V						K		
A/Norway/4046/2016		V						K		
A/Iceland/51/2016		V						K	X	
A/Lithuania/6081/2016		V					S	K		
A/South Africa/R3518/2016		V						K		
A/Kyiv/703/2016		V						K		
A/Cote d'Ivoire/680/2016		V						K		
A/Slovenia/2903/2015		V						K		
A/Latvia/03-011553/2016		V						K		
A/El Salvador/969/2016		V						K		

	83	93	103	113	123	133	143	153	163	173
A/California/7/2009	EKRIENLNKKVDDGFLDIWTYNALVLLENERLTLDYHDSNVKNLYEKVRSQKNNAKEIGNGCFFYHKCDNTCMESVKNGTYDYPKYSEEAKLNREEI									
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Christchurch/16/2010										
A/Astrakhan/1/2011										
A/St Petersburg/100/2011					N					
A/St Petersburg/27/2011					N					
A/Hong Kong/5659/2012					N					
A/South Africa/3626/2013					N					K
A/Ghana/FS-16-0249/2016					N					K
A/Honduras/8058/2016					N					K
A/Kyrgyzstan/209/2016					N					K
A/Iasi/193746/2016					N					K
A/Israel/Q-504/2015					N					K
A/Rwanda/581/2016					N					K
A/Norway/3591/2016					N					K
A/Singapore/INFKK76/2016					N					K
A/Newcastle/1001/2016					N					K
A/Catalonia/8664S/2016					N					K
A/Odessa/689/2016					N					K
A/Ghana/DILI-16-0662/2016					N					K
A/Trnava/273/2016					N					K
A/Florida/63/2016					N					K
A/Norway/4046/2016					N					K
A/Iceland/51/2016					N					K
A/Lithuania/6081/2016					N					K
A/South Africa/R3518/2016					N					K
A/Kyiv/703/2016		V			N					K
A/Cote d'Ivoire/680/2016					N					K
A/Slovenia/2903/2015					N					K
A/Latvia/03-011553/2016					N					K
A/El Salvador/969/2016					N					K

	183	193	203	213
A/California/7/2009	DGVKLESTRIYQILAIYSTVASSLLVLVSLGAISFWMNCSNGSLQCRICI			
A/Bayern/69/2009				
A/Lviv/N6/2009				
A/Christchurch/16/2010				
A/Astrakhan/1/2011				
A/St Petersburg/100/2011				
A/St Petersburg/27/2011				
A/Hong Kong/5659/2012				
A/South Africa/3626/2013				
A/Ghana/FS-16-0249/2016				
A/Honduras/8058/2016				
A/Kyrgyzstan/209/2016				
A/Iasi/193746/2016	E			
A/Israel/Q-504/2015	E			
A/Rwanda/581/2016		A		
A/Norway/3591/2016				
A/Singapore/INFKK76/2016				
A/Newcastle/1001/2016				
A/Catalonia/8664S/2016				
A/Odessa/689/2016				
A/Ghana/DILI-16-0662/2016				
A/Trnava/273/2016				
A/Florida/63/2016				
A/Norway/4046/2016				
A/Iceland/51/2016				
A/Lithuania/6081/2016				
A/South Africa/R3518/2016				
A/Kyiv/703/2016				
A/Cote d'Ivoire/680/2016				
A/Slovenia/2903/2015				
A/Latvia/03-011553/2016				
A/El Salvador/969/2016				

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 3. H1 HA locations of amino acids substitutions in genetic subgroups 6B.1 and 6B.2



Amino acid substitutions compared with A/California/7/2009 that define genetic group 6B are in **red** (S203T, A256T in HA1 with E47K, E172K in HA2 not visible on this structure), those that define subgroup 6B.1 in **magenta**, those that define subgroup 6B.2 in **cyan** (V173I in HA1 with E164G, D174E in HA2 not visible on this structure). A number of viruses in subgroup 6B.2 carry additional substitutions (e.g. R113K, D127E in HA1 with K47Q in HA2).

Figure 4. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

Vaccine virus
Reference viruses

Collection date
Feb-Mar 2016
Apr 2016
May 2016
Jun-Jul 2016

* indicates amino
acid polymorphism

@ proposed
serology
antigens

HI result in table

D151*

*when N (+CHO)

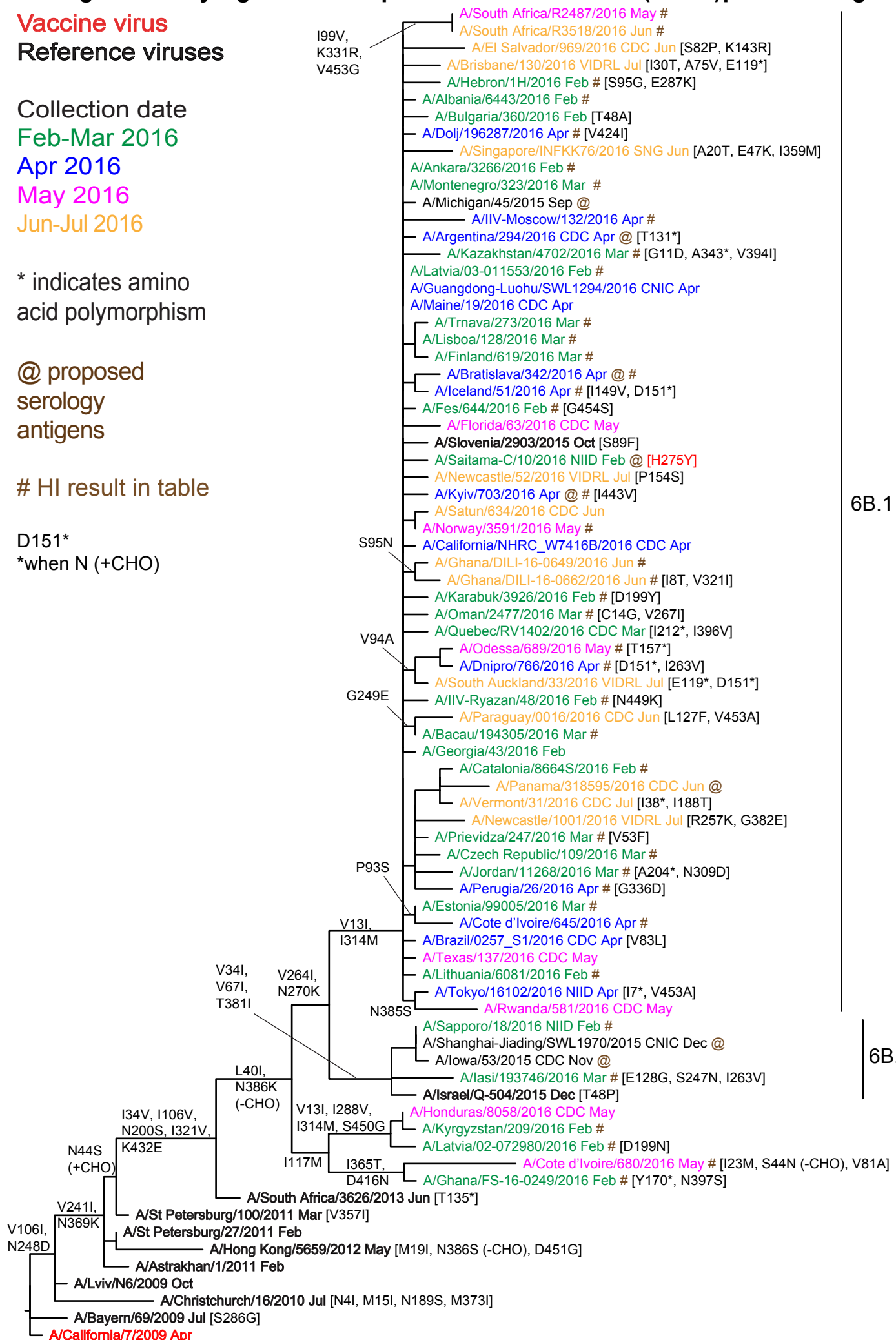
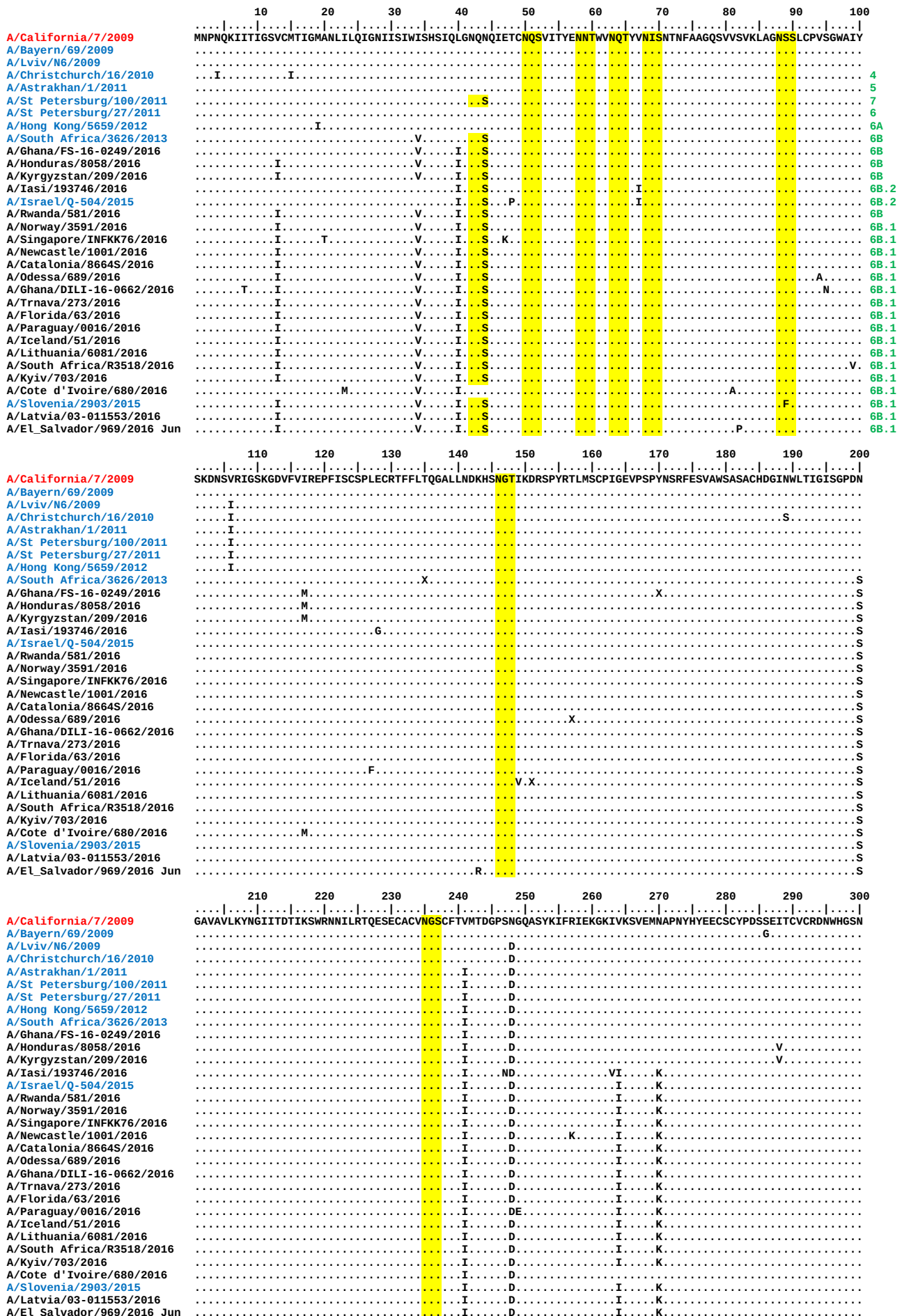


Figure 5. NA protein alignment for influenza A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400					
A/California/7/2009	RPWVSFNQ	LEYQIGY	ICSGIFG	DNPRND	KTGSCGP	VSSNGAN	GKGFSG	KYGNVWI	GRTKSI	SSRNGF	EMIWDPN	GWGTGTD	NFSSIKQ	DIVGIN	EW
A/Bayern/69/2009															
A/Lviv/N6/2009															
A/Christchurch/16/2010								I							
A/Astrakhan/1/2011							K								
A/St Petersburg/100/2011						I									
A/St Petersburg/27/2011							K								
A/Hong Kong/5659/2012							K						S		
A/South Africa/3626/2013			V				K								
A/Ghana/FS-16-0249/2016			V				T	K					K		S
A/Honduras/8058/2016		M	V					K					K		
A/Kyrgyzstan/209/2016		M	V					K					K		
A/Iasi/193746/2016			V					K					I	K	
A/Israel/Q-504/2015			V					K					I	K	
A/Rwanda/581/2016		M	V					K						SK	
A/Norway/3591/2016		M	V					K						K	
A/Singapore/INFKK76/2016		M	V				M	K						K	
A/Newcastle/1001/2016		M	V					K					E	K	
A/Catalonia/8664S/2016		M	V					K						K	
A/Odessa/689/2016		M	V					K						K	
A/Ghana/DILI-16-0662/2016		M						K						K	
A/Trnava/273/2016		M	V					K						K	
A/Florida/63/2016		M	V					K						K	
A/Paraguay/0016/2016		M	V					K						K	
A/Iceland/51/2016		M	V					K						K	
A/Lithuania/6081/2016		M	V					K						K	
A/South Africa/R3518/2016		M	V		R			K						K	
A/Kyiv/703/2016		M	V					K						K	
A/Cote d'Ivoire/680/2016			V				T	K						K	
A/Slovenia/2903/2015		M	V					K						K	
A/Latvia/03-011553/2016		M	V					K						K	
A/El Salvador/969/2016 Jun		M	V					K						K	

	410	420	430	440	450	460
A/California/7/2009	GYSGSFVQ	HPELTGL	DCIRPCF	WVLIIRG	RPKENTIW	TSGSSISF
A/Bayern/69/2009						
A/Lviv/N6/2009						
A/Christchurch/16/2010						
A/Astrakhan/1/2011						
A/St Petersburg/100/2011						
A/St Petersburg/27/2011						
A/Hong Kong/5659/2012					G	
A/South Africa/3626/2013					E	
A/Ghana/FS-16-0249/2016		N			E	
A/Honduras/8058/2016					E	G
A/Kyrgyzstan/209/2016					E	G
A/Iasi/193746/2016					E	
A/Israel/Q-504/2015					E	
A/Rwanda/581/2016					E	
A/Norway/3591/2016					E	
A/Singapore/INFKK76/2016					E	
A/Newcastle/1001/2016					E	
A/Catalonia/8664S/2016					E	
A/Odessa/689/2016					E	
A/Ghana/DILI-16-0662/2016					E	
A/Trnava/273/2016					E	
A/Florida/63/2016					E	
A/Paraguay/0016/2016					E	A
A/Iceland/51/2016					E	
A/Lithuania/6081/2016					E	
A/South Africa/R3518/2016					E	G
A/Kyiv/703/2016					E	V
A/Cote d'Ivoire/680/2016		N			E	
A/Slovenia/2903/2015					E	
A/Latvia/03-011553/2016					E	
A/El Salvador/969/2016 Jun					E	

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2016-01-31 have been received from NICs in 32 countries or administrative regions in four WHO Regions: African (7: Algeria, Cameroon, Cote d'Ivoire, Ghana, Madagascar, Niger and South Africa), Eastern Mediterranean (4: Egypt, Jordan, Morocco and Oman), European (24 countries) and Western Pacific (1: Hong Kong SAR). Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pig, turkey and humans or the loss of the ability of viruses to agglutinate any of these RBCs.

Of the successfully propagated viruses collected since 2015-08-31, less than 40% had sufficient HA titre to be analysed by HI assays conducted using guinea pig RBCs in the presence of 20nM oseltamivir, added to circumvent NA-mediated binding to the RBCs (**Tables 1, 2 & 7, Figure 6**). The remainder were unable to agglutinate guinea pig RBCs at all (**Tables 2 & 6**), or were unable to agglutinate RBCs in the presence of 20nM oseltamivir. Of the viruses in these last two categories that were sequenced, all but one belonged to subclade 3C.2a (**Table 6**). Of the 42 viruses in subclade 3C.2a with collection dates after 2016-01-31, that were able to bind guinea pig RBCs in the presence of oseltamivir, 14 (33%) showed either loss of or polymorphism at the HA1 158-160 glycosylation motif acquired by viruses in this subclade (**Tables 8-4 to 8-9**). Of the viruses able to agglutinate turkey RBCs, for which the addition of oseltamivir reduced their respective titres on guinea pig RBCs by two-fold or less and for which the respective HA sequences were known, ~80% (19 of 24) had HA sequences that fell into genetic clade 3C.3a (**Figure 6**).

HI analyses.

A total of 74 viruses with collection dates after 2016-01-31 could be analysed by HI assay and summaries of the HI results are shown in **Tables 2 and 7**. Individual HI results are shown in **Tables 8-1 to 8-9**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 8-4 to 8-9** are viruses with collection dates after 2016-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. HI results for egg-propagated cultivars of viruses from Australia, South Africa, Japan and Slovenia are shown in **Tables 8-3 to 8-5**.

A summary of the HI results for viruses collected since 2016-01-31 with antisera raised against the cell culture-propagated cultivars of A/Hong Kong/5738/2014 and A/Hong Kong/4801/2014 and the egg-propagated cultivar of A/Hong Kong/4801/2014 is shown in **Table 7**. A more detailed summary of the recognition of the viruses by the full panel of antisera is shown in **Table 8-10**, together with a breakdown based on gene sequencing for viruses in subclades 3C.2a and 3C.3a. The antiserum raised against A/Hong Kong/5738/2014 recognised all test

viruses at titres within 4-fold of the titre of the antiserum with the homologous virus, 88% within 2-fold of the homologous titre. Similarly, the antiserum raised against the cell culture-propagated A/Hong Kong/4801/2014 recognised 93% of test viruses with collection dates after 2016-01-31 at titres within 4-fold of the titre of the antiserum with the homologous virus and 70% at titres within 2-fold of the titre with the homologous virus. However, although the antiserum raised against egg-propagated A/Hong Kong/4801/2014 recognised 82% of test viruses with collection dates after 2016-01-31 at titres within 4-fold of the titre of the antiserum with the homologous virus only 35% of the test viruses were recognised at titres within 2-fold of the titre with the homologous virus.

For test viruses with collection dates after 2016-01-31 characterised genetically and antigenically by HI since the February 2016 report:

- an antiserum raised against the egg-propagated cultivar of A/Hong Kong/4801/2014 (3C.2a) recognised 69% of those viruses in subclade 3C.2a and all of the viruses in subclade 3C.3a at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/4801/2014 (3C.2a) recognised 90% of those viruses in subclade 3C.2a and 95% of the viruses in subclade 3C.3a at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/5738/2014 (3C.2a) recognised all test viruses in subclades 3C.2a and 3C.3a at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against a cell culture-propagated cultivar of A/Georgia/532/2015 (3C.2a), a cloned reference virus that carries the HA1 amino acid substitutions R142K, T160K (resulting in loss of the potential 158-160 glycosylation motif) and Q197R, recognised only 28.6% of those viruses in subclade 3C.2a and only 44% of the viruses in subclade 3C.3a at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the previously recommended vaccine virus, egg-propagated A/Switzerland/9715293/2013 (3C.3a), recognised only a single virus at a titre within 4-fold of the titre of the antiserum with the homologous virus;
- the antiserum raised against a cell culture-propagated cultivar of A/Switzerland/9715293/2013 (3C.3a) had a low homologous titre of 80 and, in contrast, recognised all the test viruses in both subclades at titres within 4-fold of the homologous titre;
- an antiserum raised against a cell culture-propagated cultivar of A/Stockholm/6/2014 (3C.3a) had a higher homologous titre of 640 and recognised 79% of those viruses in subclade 3C.2a and all of the test viruses in subclade 3C.3a at titres within 4-fold of the homologous titre of the antiserum;

- an antiserum raised against an egg-propagated cultivar of A/Stockholm/6/2014 (3C.3a) had an homologous titre of 320 and recognised only 36% of those viruses in subclade 3C.2a but all of the test viruses in subclade 3C.3a at titres within 4-fold of the homologous titre of the antiserum.

Antisera raised against viruses in other genetic groups generally recognised the test viruses poorly; these were raised against A/Texas/50/2012 (a former vaccine virus in genetic group 3C.1) and reference viruses in other genetic groups - A/Hong Kong/146/2013 (3C.2), A/Samara/73/2013 (3C.3) and A/Netherlands/525/2014 (3C.3b).

Virus Neutralisation.

Plaque reduction neutralisation assays (PRNA) were used to complement HI tests and allowed antigenic characterisation of viruses that could not be assayed by HI. The results of virus neutralisation assays are shown in **Tables 9-1 to 9-7**.

Selected antisera were raised against the currently recommended vaccine virus, egg-propagated A/Hong Kong/4801/2014, and four cell culture-propagated viruses from clade 3C.2a: A/Hong Kong/7295/2014, A/Slovenia/3188/2015 (3C.2a1), A/South Africa/R2665/2015 and A/Serbia-NS/1517/2016 - all four of which retained the HA1 158-160 glycosylation motif. A/Hong Kong/7295/2014 represented a non-polymorphic virus genetically very closely related to the prototype A/Hong Kong/4801/2014; A/Slovenia/3188/2015 represented a group (3C.2a1) that carries HA1 N171K and S312R substitutions together with I77V and G155E substitutions in HA2, this virus showed a mixed sequence for residue 275 of HA1; A/South Africa/R2665/2015 was chosen as a representative of viruses that carried HA1 R142K and Q197R substitutions; and A/Serbia-NS/1517/2016 was chosen as a representative of viruses that carried HA1 Q197R substitution. Antisera raised against two 3C.3a viruses, both propagated in cell culture, were used. The cell culture-propagated cultivar of A/Switzerland/9715293/2013 was chosen as the prototype for subclade 3C.3a and A/Bucuresti/193605/2016 was chosen as a representative of subclade 3C.3a with three HA1 substitutions: Y94H, S198P and S312N.

Test viruses were selected to represent both recently emerged genetic subclades. Of the 88 viruses examined 69 were from subclade 3C.2a (78%) and 19 were from subclade 3C.3a (22%).

The antiserum raised against egg-propagated A/Hong Kong/4801/2014 recognised 51 of 69 (74%) of the 3C.2a viruses tested at titres within 4-fold of the titre with the homologous virus, but none of the 19 subclade 3C.3a viruses was recognised at titres within 4-fold of the titre with the homologous virus.

Antisera raised against three of the four cell culture-propagated viruses, A/Hong Kong/7295/2014, A/Slovenia/3188/2015 (3C.2a1) and A/Serbia-NS/1517/2016,

recognised similar or higher proportions of the 3C.2a viruses at titres within 4-fold of the titres for their homologous viruses compared to the antiserum raised against egg-propagated A/Hong Kong/4801/2014: A/Hong Kong/7295/2014 (100%), A/Slovenia/3188/2015 (3C.2a1: 96%), A/Serbia-NS/1517/2016 (80%). The antiserum raised against A/South Africa/R2665/2015 had a high homologous titre and recognised only 13 (19%) of the test viruses, seven of which carried dual HA1 R142K and Q197R amino acid substitutions, at titres within 4-fold of the titre for the homologous virus. The seven test viruses carrying R142K and Q197R substitutions were also recognised well generally by the other antisera in the panel.

The 19 3C.3a test viruses were all recognised at titres within 2-fold of the homologous titre by the antiserum raised against the 3C.3a reference virus, cell culture-propagated A/Switzerland/9715293/2013, which had a titre of only 137 for the homologous virus. The antiserum raised against cell culture-propagated A/Bucuresti/193605/2016, another 3C.3a virus, had a titre of 377 for the homologous virus and recognised 15 of the 19 test viruses in subclade 3C.3a at titres within 2-fold of the homologous titre. However, the subclade 3C.3a test viruses were recognised less well by the antisera raised against the four cell culture-propagated subclade 3C.2a viruses with 89% of viruses being recognised at titres within 4-fold of the homologous titre by the antiserum raised against A/Hong Kong/7295/2014, 26% by the antiserum raised against A/Slovenia/3188/2015 (3C.2a1), while none was recognised at a titre within 4-fold of the homologous titre by the antisera raised against either A/South Africa/R2665/2015 or A/Serbia-NS/1517/2016.

Antisera raised against cell culture-propagated cultivars of subclade 3C.3a viruses, A/Switzerland/9715293/2013 and A/Bucuresti/193605/2016, recognised 100% and 84%, respectively, of test viruses in subclade 3C.2a at titres within 4-fold of those for their homologous viruses.

Viruses in some of the genetic clusters within subclade 3C.2a have warranted special attention. Those focussed on here are viruses in clusters with HA1 substitutions of: N171K alone; N121K combined with S144K or N171K; and those with substitutions at 197 (Q197R or Q197K) with Q197R sometimes combined with R142K.

The antiserum raised against egg-propagated A/Hong Kong/4801/2014 recognised at titres within 4-fold of its homologous titre (718):

- 15 of 19 viruses with N171K substitution,
 - o 11 of 13 carried the combined N121K and N171K substitutions,
- 21 of 28 viruses with the N121K substitution, of which
 - o 10 of 15 carried the combined N121K and S144K substitution,
 - o Again, 11 of 13 carried the combined N121K and N171K substitutions,

- 7 of 13 viruses with the Q197R substitution, of which
 - 5 of 8 carried the combined R142K and Q197R substitutions,
- All 7 of the viruses with the Q197K substitution.

The antiserum raised against cell culture-propagated A/Serbia-NS/1517/2014 gave a homologous titre of 1550 and recognised some test viruses of the 3C.2a subclade less well. Numbers recognised at titres within 4-fold of its homologous titre were:

- 14 of 19 viruses with N171K substitution,
 - 10 of 13 carried the combined N121K and N171K substitutions,
- 22 of 28 viruses with the N121K substitution, of which
 - 12 of 15 carried the combined N121K and S144K substitutions,
 - Again, 10 of 13 carried the combined N121K and N171K substitutions,
- 11 of 13 viruses with the Q197R substitution, of which
 - 7 of 8 carried the combined R142K and Q197R substitutions,
- 6 of 7 of the viruses with the Q197K substitution.

Only two of the 15 viruses examined with the combined substitutions N121K and S144K showed titres reduced ≥ 8 -fold over the homologous titre by the antiserum raised against A/Slovenia/3188/2015 (3C.2a1).

From 79% to 88% of the test viruses in subclade 3C.2a with the substitutions N171K, or N121K or Q197R were recognised at titres within 4-fold of the homologous titre for antiserum raised against cell culture-propagated A/Bucuresti/193605/2016 and all seven of the subclade 3C.2a viruses with Q197K substitution were recognised at titres within 4-fold of the homologous titre.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 7**. The HA genes fell within genetic group 3C. This group has three subdivisions: 3C.1, 3C.2 and 3C.3; subdivisions 3C.2 and 3C.3 have undergone further evolution to yield subclades designated as 3C.2a, 3C.3a and 3C.3b respectively (**Figure 7**).

The previously used vaccine virus A/Texas/50/2012 belongs to genetic subdivision 3C.1. Viruses received, collected during the course of the 2015-2016 season, that have been sequenced fell into subclades 3C.2a (267: 79%) and 3C.3a (70: 21%). Amino acid substitutions that define these subclades compared with A/Texas/50/2012 are:

(**3C.2a**) **L3I**, **N128T** (resulting in the gain of a potential glycosylation site), **N144S** (resulting in the loss of a potential glycosylation site), **N145S**, **F159Y**, **K160T** (resulting in the gain of a potential glycosylation site), **V186G**[@], **P198S**, **F219S**, **N225D** and **Q311H** in **HA1**, e.g. A/Hong Kong/5738/2014. Substitutions at residues **121**, **142** and **197** in **HA1** occur at several places in the phylogeny.

(**3C.3a**) **T128A** (resulting in the loss of a potential glycosylation site), **A138S**, **R142G**, **N145S**, **F159S**, **V186G**[@], **P198S**, **F219S** and **N225D** in **HA1**, e.g. A/Switzerland/9715293/2013. Some recent viruses also carry **Y94H** in **HA1** and others have the substitution **S198P** in **HA1** with several other substitutions defining distinct genetic clusters with the 3C.3a subclade.

Viruses of the **3C.2a** and **3C.3a** subclades have circulated recently and subclusters have emerged in both subclades with a **3C.2a** cluster designated as **3C.2a1** being detected in the greatest numbers (**Figure 7**) but retaining antigenic similarity to other **3C.2a** viruses. **3C.2a1** cluster viruses carry substitutions associated with **3C.2a** viruses together with **HA1 N171K** substitution, e.g. A/Bolzano/7/2016, often with **N121K**, e.g. A/Scotland/63440583/2016.

[@]Note: the **G186V** substitution in HA1 occurred during adaptation of A/Texas/50/2012 to propagation in hens' eggs.

Figure 8 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny (**Figure 7**). The vaccine virus A/Hong Kong/4801/2014 is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The group/subclade to which each of the sequences belongs is also shown.

Figure 9 shows the location on a 2005 H3 HA structure of amino acid substitutions that define the main subclades 3C.2a (**Figure 9A**) and 3C.3a (**Figure 9B**) each compared with the former vaccine virus A/Texas/50/2012.

Figure 10 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. **Figure 11** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine virus is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the group/subclade to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of recent viruses with HA genes in genetic group 3C generally cluster similarly to the HA sequences. However, the NA genes of some recently collected viruses with HA genes in subclade 3C.3a have NA genes seemingly evolved independently and have acquired an **I392T** substitution in their NA glycoproteins. Those in subclade 3C.2a carry **T267K** and **I380V** with a subcluster carrying **S245N** (resulting in the acquisition of a glycosylation motif) and

S247T. Other subclusters are apparent in the NA genes of viruses in genetic group 3C.2a: additional to the substitutions **S245N** and **T267T** others also have (i) **V149A** substitution or (ii) **P468H** with a significant number of viruses in this subcluster also carrying **D339N** substitution (the **3C.2a1** viruses). **Figure 12** indicates the positions of the amino acid substitutions on the structure of an N2 NA for four reference viruses in the three main genetic clusters.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 260 viruses, with 220 being recovered from samples with collection dates after 2016-01-31. All showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 14**).

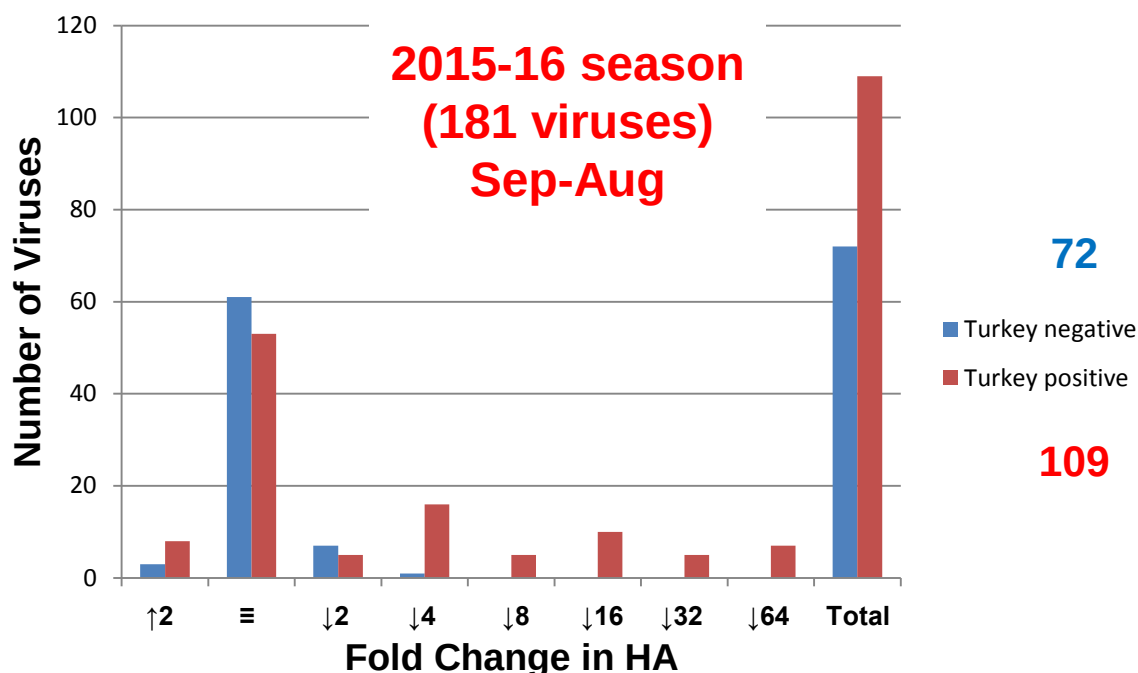
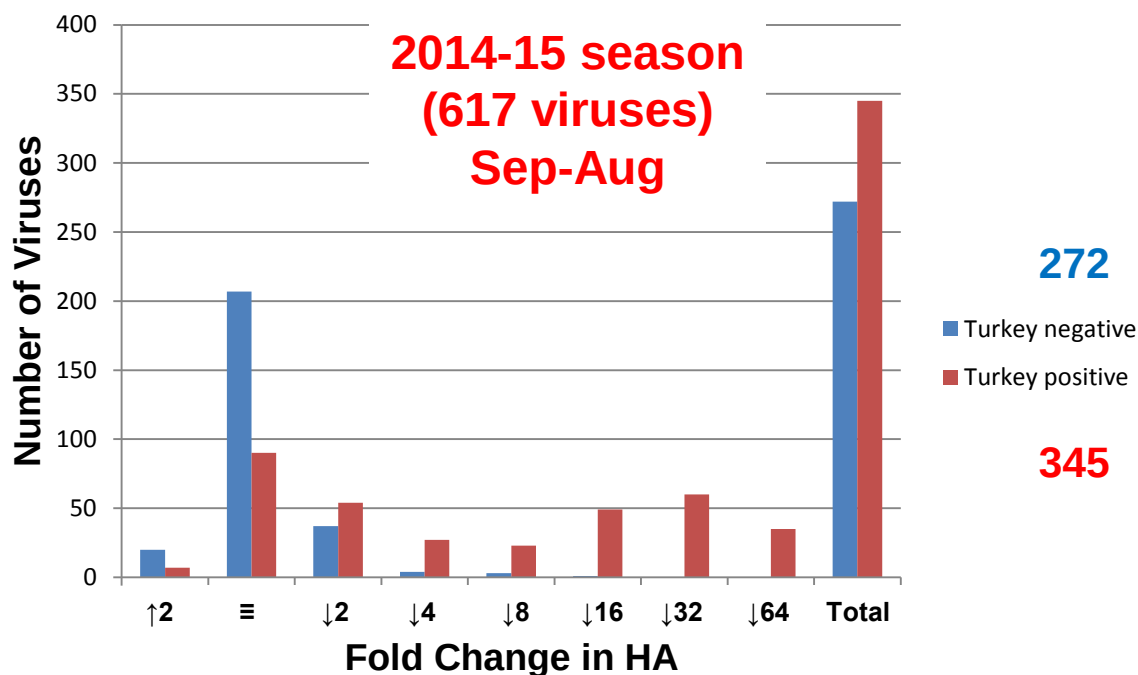
Conclusion.

Viruses in subclade 3C.2a were the most common but two-thirds of viruses in this subclade did not agglutinate RBCs and could not therefore be characterised by HI assay.

In HI assays the antiserum raised against a cell culture-propagated 3C.2a virus (polymorphic for the HA1 158-160 glycosylation motif) recognised viruses in subclades 3C.2a and 3C.3a similarly well. The antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 recognised the majority of viruses at titres within 4-fold of its homologous titre. The antiserum raised against the egg-propagated virus A/Stockholm/6/2014, a virus from subclade 3C.3a, recognised viruses in subclade 3C.3a better than those in subclade 3C.2a. The antiserum raised against egg-propagated A/Switzerland/9715293/2013, also a subclade 3C.3a virus, recognised only a small minority of the test viruses at titres within 4-fold of its homologous titre.

In PRNA, subclade 3C.2a viruses were recognised best by antisera raised against three cell culture-propagated viruses in genetic group 3C.2a and by antisera raised against cell culture-propagated viruses in 3C.3a but not by an antiserum raised against a 3C.2a virus with a **R142K** substitution in **HA1**. However, antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 (3C.2a) recognised a lower proportion of viruses in subclade 3C.2a and none of the test viruses in subclade 3C.3a.

Figure 6. Analysis of H3N2 Influenza Virus Agglutination of Guinea Pig RBCs and the Effect of 20nM Oseltamivir



**Fold shift in HA activity in the presence of 20nM oseltamivir:
[Enhancement (↑2) Equivalence (≡) Reduction (↓2-↓64)]**

Table 6. A(H3N2) viruses recovered from specimens collected during the 2015-2016 influenza season (after 2015-08-31), as assessed by NA activity (MUNANA-based assay)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3b	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3.3
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir							
+	+	+	Yes	88	84 (95%)	0	59	25	0
-	+	+	Yes	72	67 (93%)	0	10	56	1
-	-	+	Yes	0	0	0			
+	-	+	Yes	0	0	0			
			Totals	160	151 (94%)	0	69	81	1
+	+	-	No	21	15 (71%)	0	0	15	0
+	-	-	No	0	0	0	0	0	0
-	+	-	No	28	25 (89%)	0	0	25	0
-	-	-	No	187	147 (79%)	0	1	146	0
			Totals	236	187 (79%)	0	0	186	0
			Totals	396	338 (85%)	0	70	267	1

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2016-09-16: a number still in process

Situation at the time of the February 2016 VCM

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3b	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3.3
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir							
+	+	+	Yes	37	25 (68%)	0	17	8	0
-	+	+	Yes	9	9 (100%)	0	3	4	1
-	-	+	Yes	0	0	0			
+	-	+	Yes	0	0	0			
			Totals	46	34 (74%)	0	20	12	1
+	+	-	No	3	2 (67%)	0	0	2	0
+	-	-	No	0	0	0	0	0	0
-	+	-	No	9	5 (56%)	0	0	5	0
-	-	-	No	47	29 (62%)	0	0	29	0
			Totals	59	36 (61%)	0	0	36	0
			Totals	105	70 (67%)	0	20	48	1

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2016-02-15: a number still in process

Table 7. Summary of influenza A(H3N2) viruses analysed, collected after 2016-01-31

MONTH	H3N2										
Country	Number received	Number propagated ¹	≤2 fold ²	4 fold ²	≥8 fold ²	≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴
FEBRUARY											
Algeria	3	0									
Bosnia & Herzegovina	1	0									
Bulgaria	1	0									
Egypt	1	0									
Finland	1	1	1			1			1		
France	1	0									
Germany	2	2	2			2				1	1
Ghana	2	0									
Italy	10	7	7			5	2			5	2
Jordan	1	0									
Kazakhstan	2	1	1			1			1		
Macedonia	1	0									
Montenegro	1	0									
Morocco	1	0									
Oman	8	1	1				1			1	
Romania	2	2	2			1	1		1	1	
Russia	1	1	1			1					1
Serbia	9	1	1			1				1	
Slovakia	1	1	1			1			1		
Slovenia	9	4	2	2		2	1	1	1	1	2
Sweden	2	2	1	1		1		1	1	1	1
Tajikistan	1	0									
Turkey	21	11	11			10	1		10	1	
Ukraine	7	2	2			2				2	
MARCH											
Algeria	1	0									
Bulgaria	2	0									
Czech Republic	1	1		1				1			1
France	1	0									
Georgia	1	1	1			1				1	
Germany	1	0									
Ghana	2	0									
Iceland	1	0									
Italy	5	4	4			2	2			2	2
Jordan	7	3	3			3			1	2	
Kazakhstan	1	0									
Madagascar	1	0									
Montenegro	1	0									
Oman	2	0									
Romania	1	1	1				1			1	
Serbia	4	0									
Slovenia	6	3	3			2	1		1	2	
Sweden	1	1		1			1				1
Turkey	4	3	2	1		1	1	1	1	2	
Ukraine	7	0									
United Kingdom	1	0									
APRIL											
Cote d'Ivoire	8	4	3	1		1	2	1		3	1
France	1	0									
Georgia	5	1	1			1				1	
Iceland	2	0									
Italy	1	0									
Jordan	1	0									
Russia	3	0									
Sweden	1	0									
Tajikistan	1	0									
Ukraine	1	0									
United Kingdom	1	1	1			1			1		
MAY											
Cameroon	3	0									
Cote d'Ivoire	3	1	1				1			1	
Ghana	2	0									
Iceland	2	1	1			1					1
Jordan	1	0									
Madagascar	7	1	1			1			1		
Norway	1	1	1			1			1		
South Africa	2	2	1	1		1	1		1	1	
JUNE											
Cameroon	7	0									
Egypt	1	0									
Ghana	9	0									
Niger	2	0									
Norway	3	1	1			1			1		
South Africa	11	4	3	1		2	2		1	3	
United Kingdom	3	0									
JULY											
Cameroon	4	1	1			1			1		
Egypt	2	0									
France	1	in progress									
Ghana	5	0									
Hong Kong	2	2	2			2				2	
Niger	4	0									
South Africa	11	1	1			1			1		
United Kingdom	1	0									
AUGUST											
Hong Kong	1	0									
United Kingdom	6	in progress									
35 Countries											
	259	74	65	9	0	51	18	5	26	35	13
			87.8%	12.2%	0.0%	68.9%	24.3%	6.8%	35.1%	47.3%	17.6%

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Compared to homologous titres for ferret antisera raised against tissue culture-propagated virus A/Hong Kong/5738/2014 (3C.2a)

3. Compared to homologous titres for ferret antisera raised against tissue culture-propagated virus A/Hong Kong/4801/2014 (3C.2a)

4. Compared to homologous titres for ferret antisera raised against egg-propagated virus A/Hong Kong/4801/2014 (3C.2a)

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-02-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT			
	Ferret number			F36/12 ⁻¹	F35/15 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	F18/15 ⁻¹	F32/14 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F43/15 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹	
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	40	640	320	320	160	320	80	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	2560	1280	640	640	160	640	640	640	640	1280	320	640
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	160	80	640	320	320	160	80	160	160	320	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	160	160	320	80	640	80	80	160	320	80	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	80	40	320	160	80	80	40	40	80	80	80	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	160	320	80	640	80	80	320	320	80	160
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	640	160	320	160	80	160	640	160	80	160	80	160
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	640	320	640	160	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	80	<	160	80	40	40	<	40	80	160	80	80
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	40	160	160	80	40	40	80	160	320	160	80
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	80	80	160	80	80	40	80	40	320	640	320	640
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	40	160	160	80	80	<	80	160	320	160	320
TEST VIRUSES																
A/Antsirabe/3857/2015		3C.3a	2015-09-08	MDCK1/MDC1	80	40	160	320	80	80	40	40	80	80	80	80
A/Ghana/ARI-15-0132/2015		3C.2a	2015-09-22	Cx/SIAT2	80	40	80	160	40	40	<	40	80	160	80	80
A/Oman/4681/2015		3C.2a	2015-10-10	SIAT2/SIAT1	80	40	80	160	80	40	<	40	80	160	80	80
A/Istanbul/188/2015		3C.3a	2015-12-25	MDCK1/MDCK3	160	80	320	640	160	80	40	80	160	160	160	160
A/Turkey/TR-20/2016		3C.3a	2016-01-13	SIAT1	160	40	320	640	80	80	40	80	160	160	160	160
A/Turkey/TR-11/2016		3C.3a	2016-01-14	SIAT2	80	40	160	320	80	80	40	40	80	80	80	80
A/Turkey/TR-17/2016		3C.3a	2016-01-14	SIAT1	160	80	320	640	160	160	80	160	320	320	160	160

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
NH 2015-16Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-02-26

				Haemagglutination inhibition titre												
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	
				Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	
				F36/12 ⁻¹	F35/15 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	F18/15 ⁻¹	F32/14 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F43/15 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹	
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	2560	640	160	640	80	640	320	320	160	320	80	320
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	2560	640	640	640	160	640	320	640	640	640	320	320
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	160	80	320	160	160	80	40	80	80	160	80	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	80	160	320	80	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	80	40	320	160	80	80	40	40	80	80	80	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	160	320	80	640	80	80	320	320	80	160
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	640	160	160	160	40	80	640	160	80	160	80	160
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	80	640	320	640	320	640	160	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	40	160	160	80	80	40	80	160	320	160	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1/SIAT1	80	40	160	160	80	80	<	80	160	320	160	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	80	160	160	80	80	40	80	40	320	640	320	640
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	80	160	160	80	80	<	80	160	320	160	640
TEST VIRUSES																
A/Ghana/FS-15-0583/2015		3C.2a	2015-09-25	Cx/SIAT3	80	80	320	160	80	80	<	80	160	160	80	80
A/Oman/4926/2015		3C.2a	2015-09-27	SIAT2/SIAT2	40	<	80	40	40	<	<	<	80	80	40	40
A/Oman/5009/2015		3C.2a	2015-09-29	SIAT2/SIAT2	40	<	160	80	80	40	<	<	160	160	80	40
A/Oman/5011/2015		3C.2a	2015-09-30	SIAT2/SIAT2	80	40	160	80	80	40	<	80	320	160	80	160
A/Oman/5357/2015		3C.2a	2015-10-13	MDCK2/SIAT2	<	<	40	<	<	<	<	<	40	40	<	<
A/Konya/1926/2015		3C.2a	2015-10-15	MDCK1/SIAT1	80	<	160	80	<	40	<	40	160	160	80	80
A/Oman/5422/2015		3C.2a	2015-10-17	SIAT2/SIAT2	80	40	160	80	80	40	<	80	160	160	80	80
A/Oman/5691/2015		3C.2a	2015-10-27	SIAT2/SIAT2	40	40	160	80	40	40	<	40	80	160	80	40
A/Oman/4940/2015		3C.2a	2015-10-28	SIAT2/SIAT2	80	40	320	80	80	40	<	80	160	160	80	80
A/Italy-FVG/105/2015		3C.2a	2015-11-25	SIAT3/SIAT2	40	<	80	40	40	40	<	40	80	80	40	40
A/Switzerland/15772531/2015		3C.3a	2015-11-28	SIAT1	80	40	320	160	80	80	40	40	160	80	80	80
A/Malatya/5037/2015		3C.3a	2015-12-10	SIAT1/SIAT1	160	40	320	320	160	160	80	80	160	160	160	160
A/Ankara/5082/2015		3C.2a	2015-12-14	SIAT1/SIAT1	80	<	160	40	<	<	<	40	80	80	80	40
A/Ankara/5139/2015		3C.2a	2015-12-22	MDCK1/SIAT1	80	<	160	40	<	<	<	40	80	80	80	80
A/Turkey/TR-25/2016		3C.3a	2016-01-13	SIAT1	160	40	320	160	80	80	40	80	160	160	160	80
A/Turkey/TR-26/2016		3C.3a	2016-01-13	SIAT1	80	40	320	160	80	80	40	80	160	80	80	80
A/Berlin/7/2016		3C.3a	2016-01-18	C2/SIAT1	160	80	640	320	160	80	40	80	160	160	160	160
A/Sachsen/2/2016		3C.3a	2016-01-18	C2/SIAT1	160	80	640	320	160	160	40	160	160	160	160	160
A/Berlin/11/2016		3C.3a	2016-01-25	C2/SIAT1	160	40	640	320	80	80	40	80	160	160	160	160
A/Jordan/4054/2015		3C.2a		SIAT1	<	<	160	40	<	<	<	40	80	80	80	80

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
NH 2015-16Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-03-04 + 2016-03-11 + 2016-03-18

Haemagglutination inhibition titre																
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	
				Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	
				F36/12 ⁻¹	F35/15 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	F18/15 ⁻¹	F32/14 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F43/15 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹	
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	2560	640	160	640	80	640	320	320	160	320	80	320
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	2560	640	640	640	160	640	320	640	640	640	320	320
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	160	80	320	160	160	80	40	80	80	160	80	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	80	160	320	80	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	80	40	320	160	80	80	40	40	80	80	80	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	160	320	80	640	80	80	320	320	80	160
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	640	160	160	160	40	80	640	160	80	160	80	160
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	80	640	320	1280	320	640	160	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	40	160	160	80	80	40	80	80	320	160	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1/SIAT1	80	40	160	160	80	80	<	80	160	320	160	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	40	<	40	<	160	640	320	320
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	80	160	160	80	80	<	80	160	320	160	640
TEST VIRUSES																
A/South Australia/9/2015		3C.2a	2015-02-12	E5/E2	40	80	<	40	<	40	40	<	80	320	160	160
A/South Australia/9/2015	NYMC X-265	3C.2a	2015-02-12	Cx	<	40	<	<	<	<	<	<	80	320	160	80
A/Mie/21/2015		3C.2a	2015-07-07	MDCK3/SIAT2	40	<	40	40	<	<	<	40	40	80	40	40
A/Laos/1810/2015		3C.2a	2015-08-03	MDCK2/SIAT2	80	40	80	80	40	40	<	40	80	160	80	160
A/Shizuoka/34/2015		3C.2a	2015-11-06	MDCK1/SIAT2	80	40	80	80	40	40	<	40	80	80	80	80
A/Catalonia/8396S/2015		3C.2a	2015-11-25	C0/MDCK3	40	<	160	40	<	<	<	<	40	80	40	40
A/Cyprus/F16/2016		3C.2a	2016-01-15	SIAT1	40	<	160	40	40	<	<	40	80	160	80	80

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-04-08 + 2016-04-15 + 2016-04-22

Haemagglutination inhibition titre																		
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera														
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	A/Sth Africa	A/Saitama	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	2825/15	103/14	
				Passage history	Egg	SIAT	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	Egg	Egg		
				Ferret number	F36/12 ⁻¹	F35/15 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	F18/15 ⁻¹	F32/14 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F43/15 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹	F12/16 ⁻¹	JPN Ferret
Genetic group	3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a	New	
REFERENCE VIRUSES																		
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	2560	640	160	640	40	640	320	320	160	320	80	160	160	640	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	640	640	320	320	80	320	320	320	640	160	320	320	1280		
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	40	40	320	160	160	80	40	80	80	160	80	80	320	320	
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	320	80	160	320	80	640	80	160	320	40	80	80	320	
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	<	<	160	80	80	40	40	80	80	40	80	80	160	
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	160	160	160	320	40	640	40	80	160	80	160	80	320	
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	320	160	160	160	80	80	1280	160	80	160	80	160	640	
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	1280	320	80	320	80	320	160	640	160	320	80	320	2560	
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	40	40	160	80	40	40	<	40	160	80	80	160	640	
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1/SIAT1	40	40	160	80	80	40	<	40	80	160	80	160	640	
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	40	40	40	<	160	320	320	640	2560	
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	80	40	160	80	80	40	<	80	160	160	80	320	640	
TEST VIRUSES																		
A/South Africa/R2825/2015	3C.2a	2015-06-01	E6	<	<	40	40	40	<	40	<	640	640	80	320	1280	ND	
A/Saitama/103/2014	3C.2a	2014-11-17	NC3/E20/E1	640	640	320	640	40	320	160	320	640	2560	640	1280	ND	2560	
A/Cameroon/15V-6741/2015	3C.2a	2015-09-21	SIAT1	<	<	160	40	<	<	<	40	80	80	40	40	80	ND	
A/Cameroon/15V-7134/2015	3C.2a	2015-10-02	SIAT1	<	<	160	40	<	<	<	40	80	80	40	40	80	ND	
A/Cameroon/15V-8163/2015	3C.2a	2015-11-05	SIAT1	<	<	160	40	<	<	<	40	80	80	40	40	80	ND	
A/Cameroon/15V-8793/2015	3C.3a	2015-11-26	SIAT1	<	<	320	160	40	40	<	40	80	80	80	80	160	ND	
A/Berlin/27/2016	3C.3a	2016-01-23	C3/SIAT1	<	40	320	160	80	80	40	40	80	80	80	80	160	ND	
A/Berlin/28/2016	3C.3a	2016-01-28	C2/SIAT1	40	40	320	160	80	80	40	80	160	160	80	160	160	ND	
A/Baden-Wuerttemberg/36/2016		2016-02-08	C2/SIAT1	<	<	320	160	40	40	<	40	80	80	80	80	160	ND	
A/Berlin/52/2016	3C.2a	2016-02-10	C3/SIAT1	<	<	160	40	<	<	<	40	80	80	40	40	80	ND	
A/Serbia-NS/1565/2016	3C.3a	2016-02-22	SIAT1	40	80	320	160	80	160	40	80	160	160	80	80	ND	ND	

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ <= <40; ND = Not Done

Vaccine
NH 2015-16

Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-05-13 + 2016-05-20

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre															
				Post-infection ferret antisera															
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/HK	A/Georgia	A/Slov	A/Slov	A/S Africa
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	3188/15	129/16	129/16	R2665/15
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	SIAT	Egg	Egg	
	Ferret number			F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	F18/16 ¹	F19/16 ¹	F20/16 ¹	
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a	3C.2a	
																New	New	New	
REFERENCE VIRUSES																			
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	1280	640	80	640	40	320	320	320	160	320	80	160	160	160	80
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	640	640	640	320	80	320	320	320	320	640	160	320	320	320	160
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	80	80	320	320	160	160	80	160	160	320	160	160	640	320	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	320	80	160	320	80	640	80	40	160	320	40	80	320	80	ND
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	<	40	320	160	80	80	40	40	80	80	80	80	160	80	ND
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	160	160	160	320	80	640	40	80	160	160	80	160	320	80	80
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	320	160	160	160	40	80	640	160	80	160	80	160	80	<	40
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	1280	640	80	320	40	640	320	640	320	640	160	320	320	80	80
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	40	80	160	80	40	80	40	80	160	320	160	160	160	80	80
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	40	40	160	80	40	40	<	80	160	320	160	160	160	80	80
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	<	80	80	40	40	40	<	160	320	320	320	320	320	320	320
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	80	80	160	80	40	80	<	80	160	320	160	640	160	320	320
TEST VIRUSES																			
A/Slovenia/129/2016	clone 38	3C.2a	2016-01-12	E6	<	80	160	40	40	40	40	320	640	40	2560	320	2560	2560	
A/South Africa/R2665/2015	clone 67	3C.2a	2015-05-25	E5	<	40	40	40	40	<	<	320	320	80	640	320	1280	1280	
A/South Africa/R2665/2015	clone 29	3C.2a	2015-05-25	E7	<	<	<	<	<	<	<	320	160	40	640	160	1280	1280	
A/Berlin/28/2016		3C.3a	2016-01-28	C2/SIAT1	40	40	320	160	80	80	40	80	160	160	80	80	160	160	80
A/Berlin/27/2016		3C.3a	2016-01-23	C3/SIAT1	40	40	320	160	80	80	40	80	160	160	80	80	160	80	80
A/Bucuresti/191366/2016		3C.3a	2016-02-10	SIAT1/SIAT1	<	<	160	160	40	40	40	80	80	80	80	ND	ND	ND	ND
A/Arges/1480-3059/2016		3C.3a	2016-02-10	SIAT1/SIAT1	40	40	320	160	80	80	40	80	160	160	160	160	ND	ND	ND
A/Oman/1819/2016		3C.3a	2016-02-13	SIAT2/SIAT1	<	40	320	160	40	80	40	80	80	80	80	ND	ND	ND	ND
A/Georgia/864/2016		3C.3a	2016-03-02	SIAT1	<	<	160	40	<	<	<	40	80	160	80	40	160	80	40
A/Bucuresti/193605/2016		3C.3a	2016-03-08	SIAT1/SIAT1	<	40	320	160	80	80	40	80	160	80	80	80	ND	ND	ND
A/Jordan/4470/2016		3C.3a	2016-03-27	SIAT1	40	40	320	160	80	80	40	80	160	160	80	80	160	160	80
A/Georgia/1041/2016		3C.2a	2016-04-07	SIAT1	<	40	320	160	80	80	40	80	160	160	80	80	160	80	40

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Vaccine
NH 2015-16

Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-06-10 + 2016-06-17 + 2016-06-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	
	Ferret number			F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	2560	640	80	640	40	320	320	320	80	320	40	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	320	320	640	640	160	320	320	320	320	1280	160	320
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	40	80	640	320	320	160	80	160	160	320	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	160	80	80	320	80	320	40	40	80	320	40	40
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	<	<	320	160	80	80	<	40	80	80	80	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	320	320	320	640	80	640	80	160	320	640	80	320
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	320	320	160	160	40	160	1280	160	80	320	80	160
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	40	640	40	640	320	1280	160	640	80	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	80	80	320	160	40	80	40	80	160	320	160	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1/SIAT1	80	80	320	160	40	80	40	80	160	320	160	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	160	160	80	80	80	80	40	320	640	320	640
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	80	80	160	160	40	80	40	80	320	320	160	640
TEST VIRUSES																
A/Bari/11/2016		3C.2a	2016-01-26	SIAT3/SIAT2	80	<	160	40	40	<	<	40	80	160	80	80
A/Palermo/1/2016		3C.2a	2016-02-01	SIAT3/SIAT2	80	40	320	80	40	40	<	80	80	160	80	80
A/Milano/23/2016		3C.2a	2016-02-01	MDCK1/SIAT2	40	<	160	<	<	<	<	40	80	80	40	40
A/St. Petersburg/210/2016		3C.2a	2016-02-06	MDCK1/SIAT1	<	<	80	<	<	<	<	80	160	<	<	40
A/Odessa/700/2016		3C.2a	2016-02-07	SIAT2/SIAT2	80	80	320	80	40	40	40	80	80	320	80	160
A/Odessa/695/2016		3C.2a	2016-02-13	SIAT2/SIAT2	80	40	160	80	40	40	<	80	160	160	80	160
A/Roma/12/2016		3C.2a	2016-02-15	SIAT2/SIAT2	40	<	160	40	<	<	<	40	80	160	80	80
A/Padova/17/2016		3C.2a	2016-02-15	SIAT3/SIAT2	40	<	160	40	40	40	<	40	80	160	80	80
A/Slovenia/852/2016		3C.2a	2016-02-18	MDCK1/SIAT2	80	40	160	80	40	40	<	40	80	320	80	160
A/Slovenia/920/2016		3C.2a	2016-02-19	MDCK1/SIAT2	40	<	160	40	40	40	<	40	40	80	40	160
A/Palermo/6/2016		3C.2a	2016-02-22	SIAT2/SIAT2	40	<	160	40	40	40	<	40	80	320	80	40
A/Bratislava/184/2016		3C.2a	2016-02-22	MDCKx/SIAT1	80	40	160	80	40	40	40	80	160	320	160	320
A/Parma/38/2016		3C.2a	2016-02-23	MDCK2/SIAT1	80	40	160	80	40	40	<	80	160	320	80	80
A/Slovenia/972/2016		3C.2a	2016-02-24	MDCK1/SIAT2	40	<	80	40	<	<	<	40	40	40	40	80
A/Friuli-Venezia Giulia/10/2016		3C.2a	2016-02-25	SIAT1/SIAT2	40	<	160	40	<	<	<	40	80	80	40	40
A/Friuli-Venezia Giulia/14/2016		3C.2a1	2016-03-01	SIAT1/SIAT2	40	<	160	40	<	<	<	40	80	80	40	40
A/Slovenia/1070/2016		3C.2a	2016-03-01	MDCK1/SIAT2	40	<	160	40	<	<	<	40	80	80	80	160
A/Slovenia/1172/2016		3C.2a	2016-03-09	MDCK1/SIAT2	160	80	320	160	80	40	40	80	160	640	160	160
A/Bolzano/7/2016		3C.2a	2016-03-15	SIAT3/SIAT2	80	40	160	80	80	<	<	40	80	160	80	80
A/Czech Republic/122/2016		3C.2a	2016-03-21	C2/SIAT1	40	<	80	40	<	<	<	40	40	40	<	40
A/Slovenia/1307/2016		3C.2a	2016-03-21	MDCK1/SIAT1	80	40	320	80	40	40	40	80	160	320	80	160
A/Perugia/23/2016		3C.2a	2016-03-23	SIAT3/SIAT2	80	40	160	80	80	40	<	40	160	160	80	160
A/Bolzano/9/2016		3C.2a	2016-03-27	SIAT3/SIAT2	40	<	160	40	<	<	<	40	80	80	40	40
A/Iceland/59/2016		3C.2a	2016-05-08	MDCK1/SIAT1	80	<	80	40	<	40	<	40	80	160	40	80

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
NH 2015-16Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-06-24 + 2016-07-01 + 2016-07-08

				Haemagglutination inhibition titre												
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	
				Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	
	Passage history			F36/12 ⁻¹	F35/15 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	F18/15 ⁻¹	F32/14 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F43/15 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹	
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	40	640	320	320	160	320	80	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	2560	640	640	640	160	640	320	640	320	640	160	320
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	160	40	640	160	160	160	80	80	160	160	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	1280	160	160	320	80	640	80	80	160	640	40	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	40	40	320	160	80	80	40	80	80	80	40	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	320	320	80	1280	40	80	320	320	80	160
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT4	640	160	320	160	40	160	1280	160	80	160	80	160
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	5120	1280	160	640	40	640	320	1280	320	640	160	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	40	160	80	40	40	<	80	160	320	160	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1/SIAT1	80	40	160	80	40	80	40	80	160	320	160	640
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	80	80	80	80	40	40	40	<	320	640	160	640
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	40	160	80	40	80	40	80	160	320	160	640
TEST VIRUSES																
A/Ukraine/7030/2016		3C.2a	2016-01-04	P2/MDCK1	80	<	160	40	<	40	<	40	80	160	80	80
A/Slovenia/486/2016		3C.2a	2016-02-01	MDCK1/SIAT2	80	<	320	80	40	40	<	40	160	320	80	320
A/Finland/586/2016		3C.3a	2016-02-15	SIAT1/SIAT1	80	<	320	160	80	80	<	40	80	160	80	80
A/Stockholm/24/2016		3C.2a	2016-02-22	MDCK3/SIAT1	80	<	320	80	40	<	<	80	160	160	40	40
A/Stockholm/25/2016		3C.2a	2016-02-24	MDCK2/SIAT1	<	<	40	<	<	<	<	<	40	40	<	<
A/Kazakhstan/4766/2016		3C.3a	2016-02-29	SIAT1	80	<	320	160	80	40	<	40	80	160	80	80
A/Istanbul/1946182/2016		3C.3a	2016-03-03	MDCK1/SIAT1	40	<	320	80	<	80	<	<	40	40	40	40
A/Stockholm/28/2016		3C.2a1	2016-03-10	MDCK2/SIAT1	<	<	160	40	<	<	<	40	40	80	<	40
A/Cote D'Ivoire/544/2016		3C.2a	2016-04-06	P1/SIAT1	<	<	80	40	<	<	<	<	80	40	<	<
A/Cote D'Ivoire/595/2016		3C.2a	2016-04-15	P1/SIAT1	40	<	160	40	<	<	<	40	80	80	40	40
A/Cote D'Ivoire/599/2016		3C.2a	2016-04-15	P1/SIAT1	40	<	160	80	<	<	<	80	80	160	40	40
A/Cote D'Ivoire/651/2016		3C.2a	2016-04-25	P2/SIAT1	<	<	160	40	<	<	<	<	40	80	40	40
A/Cote D'Ivoire/695/2016		3C.2a	2016-05-06	P1/SIAT1	40	<	160	40	<	<	<	<	80	80	40	40
A/South Africa/R3026/2016		3C.2a	2016-05-26	MDCK1/SIAT1	<	<	80	<	<	<	<	<	40	80	40	80
A/South Africa/R3142/2016		3C.2a	2016-05-30	MDCK1/SIAT1	80	40	160	40	40	40	<	40	160	160	80	160
A/South Africa/R3328/2016		3C.2a	2016-06-02	MDCK1/SIAT1	40	<	160	40	40	40	<	40	160	160	80	80
A/South Africa/VW0036/2016		3C.2a1	2016-06-04	MDCK1/SIAT1	40	<	80	40	<	<	<	<	80	80	40	40

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-07-22 + 2016-07-29

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Bucuresti	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	193605/16	525/14	146/13	5738/14	4801/14	4801/14	532/15
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	SIAT	Egg	MDCK	MDCK	Egg	SIAT
	Ferret number			F36/12 ⁻¹	F35/15 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	F18/15 ⁻¹	F32/14 ⁻¹	F26/16 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F43/15 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a
										New						
REFERENCE VIRUSES																
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	5120	1280	160	1280	40	640	80	640	640	160	640	80	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	1280	640	320	640	80	640	160	320	640	320	640	160	320
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	80	40	320	160	160	80	320	40	40	160	160	80	160
A/Stockholm/6/2014	isolate 2 3C.3a	2014-02-06	E4/E1	640	160	160	320	80	640	160	80	40	160	640	80	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	40	40	320	160	80	80	160	40	40	80	80	40	80
A/Switzerland/9715293/2013	clone 123 3C.3a	2013-12-06	E4/E1	320	160	320	320	80	1280	160	40	80	320	320	40	320
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT2	40	40	320	160	80	80	160	40	40	160	160	40	80
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT4	320	160	160	160	40	80	80	640	80	80	160	80	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	2560	1280	160	1280	40	1280	160	640	1280	640	1280	80	640
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	80	40	160	160	80	40	80	<	80	160	320	80	160
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK2	80	40	160	80	40	40	40	<	80	160	320	160	160
A/Hong Kong/4801/2014	isolate 1 3C.2a	2014-02-26	E6/E2	40	80	80	80	40	40	80	80	<	320	640	160	640
A/Georgia/532/2015	plaq 20 3C.2a	2015-03-09	SIAT1/SIAT3	160	80	320	160	80	80	80	40	80	160	320	160	640
TEST VIRUSES																
A/Mardin/2834/2016	3C.3a	2016-02-01	MDCK1/SIAT1	80	40	640	160	160	80	ND	40	80	160	160	80	160
A/Izmir/1020/2016	3C.3a	2016-02-02	SIAT1/SIAT1	80	40	320	160	160	80	ND	40	80	160	160	80	80
A/Aydin/3060/2016	3C.3a	2016-02-03	SIAT1/SIAT1	80	40	320	160	160	160	ND	80	80	160	160	80	160
A/Izmir/1089/2016	3C.3a	2016-02-05	SIAT1/SIAT1	80	80	640	320	160	160	ND	80	160	320	320	160	160
A/Zonguldak/3161/2016	3C.3a	2016-02-05	MDCK1/SIAT1	80	80	640	320	160	160	ND	80	160	320	320	160	160
A/Aydin/3446/2016	3C.3a	2016-02-11	SIAT1/SIAT1	80	40	640	160	160	80	ND	40	80	160	160	80	80
A/Kutahya/3503/2016	3C.3a	2016-02-12	MDCK1/SIAT1	80	40	320	160	80	80	ND	40	80	160	160	80	80
A/Antalya/3562/2016	3C.3a	2016-02-12	MDCK1/SIAT1	80	40	320	160	160	80	ND	40	80	160	160	80	160
A/Izmir/3575/2016	3C.3a	2016-02-13	MDCK1/SIAT1	80	40	640	320	160	80	ND	40	80	160	160	80	160
A/Denizli/3801/2016	3C.3a	2016-02-19	MDCK1/SIAT1	80	40	640	160	160	80	ND	40	80	160	160	80	160
A/Bolu/3994/2016	3C.3a	2016-02-26	MDCK1/SIAT1	40	<	320	160	40	80	80	<	40	160	80	40	80
A/Aksaray/4048/2016	3C.3a	2016-03-01	MDCK1/SIAT1	40	<	320	160	40	40	80	<	40	80	80	40	80
A/Bitlis/4351/2016	3C.3a	2016-03-18	SIAT1/SIAT1	80	40	640	320	80	160	160	40	80	160	160	80	160
A/Norway/3545/2016	3C.3a	2016-05-13	SIAT1	80	40	640	320	160	80	ND	80	80	160	160	80	160
A/Antsirabe/2267/16	3C.2a	2016-05-23	SIAT1	80	<	160	40	<	<	<	<	40	80	160	80	40
A/South Africa/R3620/2016	3C.2a1	2016-06-08	SIAT1	40	<	160	40	<	<	<	<	40	80	160	40	40
A/Norway/3798/2016	3C.3a	2016-06-09	SIAT1	80	40	640	320	160	80	ND	80	80	160	160	80	80

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used)⁻¹ < = <40; ND = Not done

Vaccine
NH 2015-16

Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-08-26 + 2016-09-02 + 2016-09-09 + 2016-09-16

				Haemagglutination inhibition titre											
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15
				Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT
				Ferret number	F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹
Genetic group	3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a			
REFERENCE VIRUSES															
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	5120	640	320	640	80	640	320	640	160	640	80	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	1280	640	640	640	160	640	320	640	640	1280	160	640
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	160	80	640	320	320	160	80	160	160	320	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	160	160	320	80	640	80	160	640	40	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40	40	320	160	80	80	40	40	80	160	40	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	160	640	40	1280	80	80	320	320	40
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT4	320	160	320	160	80	80	640	80	80	160	40	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	2560	640	160	640	80	640	320	1280	640	1280	160	640
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	80	40	160	80	80	80	<	80	160	320	80	160
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK2	80	80	320	160	80	80	40	80	160	320	160	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	80	40	80	40	320	640	640
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	160	320	160	80	80	40	160	320	640	1280
TEST VIRUSES															
A/Marseille/2588/2015		2015-11-30	MDCK3/SIAT1	80	<	160	40	40	<	<	40	160	160	40	40
A/Jordan/30124/2016		2016-03-25	SIAT2	80	80	320	80	80	40	<	80	320	320	80	640
A/Jordan/IIIj-15-4470-01NT/2016		2016-03-27	SIAT1	80	40	320	160	80	80	<	40	160	160	40	80
A/England/929/2016		2016-04-22	SIAT1/SIAT2	80	40	160	80	80	40	40	80	320	80	80	80
A/South Africa/R4140/2016	3C.2a	2016-06-27	MDCK2/SIAT1	<	<	40	40	<	<	<	<	40	80	40	40
A/South Africa/R4626/2016	3C.2a	2016-07-06	MDCK1/SIAT1	80	40	320	160	80	40	<	80	160	320	160	320
A/Cameroon/6284/2016		2016-07-16	SIAT2	40	40	160	40	40	<	<	40	160	160	80	80
A/Hong Kong/2262/2016		2016-07-28	MDCK1/SIAT1	80	<	160	40	<	<	<	<	80	160	40	160
A/Hong Kong/2271/2016		2016-07-28	MDCK1/SIAT1	80	<	160	80	40	<	<	40	80	160	40	160

* Superscripts refer to antiserum properties (<relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016
NH 2016-17

Sequences in phylogenetic trees

Table 8-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) with collection dates after 2016-01-31 - Summary

				Haemagglutination inhibition titre												
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Bucuresti	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	193605/16	525/14	146/13	5738/14	4801/14	4801/14	532/15
				Passage history	Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	SIAT	Egg	MDCK	MDCK	Egg
Genetic group	3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a		
REFERENCE VIRUSES																
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	5120	640	320	640	80	640	80	320	640	160	640	80	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	1280	640	640	640	160	640	160	320	640	640	1280	160	640
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	160	80	640	320	320	160	320	80	160	160	320	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	E4/E1	640	160	160	320	80	640	160	80	80	160	640	40	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40	40	320	160	80	80	160	40	40	80	160	40	80
A/Switzerland/9715293/2013	clone 123	3C.3a	E4/E1	640	160	160	640	40	1280	160	80	80	320	320	40	160
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT2	40	40	320	160	80	80	160	40	40	160	160	40	40
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT4	320	160	320	160	80	80	80	640	80	80	160	40	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	2560	640	160	640	80	640	160	320	1280	640	1280	160	640
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	80	40	160	80	80	80	80	<	80	160	320	80	160
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK2	80	80	320	160	80	80	40	40	80	160	320	160	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	E6/E2	40	80	80	80	80	40	80	80	40	320	640	160	640
A/Georgia/532/2015	plaq 20	3C.2a	SIAT1/SIAT3	160	160	320	160	80	80	80	40	160	320	640	160	1280
TEST VIRUSES																
Number of viruses tested				74	74	74	74	74	74	5	74	74	74	74	74	74
No with titre reduction ≤2-fold						39	28	48		3		65	51	26	3	
%						52.7	37.8	64.9		60.0		87.8	68.9	35.1	4.1	
No with titre reduction =4-fold					2	26	16	26*	1			9	18	35	22	
%					2.7	35.1	21.7	35.1	1.4			12.2	24.3	47.3	29.7	
No with titre reduction ≥8-fold				74	72	9	30		73	2	74	74		5	13	49
%				100.0	97.3	12.2	40.5		98.6	40.0	100.0	100.0		6.8	17.6	66.2
Number of clade 3C.2a viruses tested				42	42	42	42	42	42	2	42	42	42	42	42	42
No with titre reduction ≤2-fold						11	2	19				34	25	8	2	
%						26.2	4.7	45.2				81.0	59.5	19.0	4.8	
No with titre reduction =4-fold					2	22	13	23*				8	13	21	10	
%					4.8	52.4	31.0	54.8				19.0	31.0	50.0	23.8	
No with titre reduction ≥8-fold				42	40	9	27		42	2	42	42		4	13	30
%				100.0	95.2	21.4	64.3		100.0	100.0	100.0	100.0		9.5	31.0	71.4
Number of clade 3C.3a viruses tested				25	25	25	25	25	25	3	25	25	25	25	25	25
No with titre reduction ≤2-fold						25	23	23		3		24	19	16		
%						100.0	92.0	92.0		100.0		96.0	76.0	64.0		
No with titre reduction =4-fold							1	2*	1			1	5	9	11	
%							4.0	8.0	4.0			4.0	20.0	36.0	44.0	
No with titre reduction ≥8-fold				25	25		1		24		25	25		1		14
%				100.0	100.0		4.0		96.0		100.0	100.0		4.0		56.0

* Assigned as 4-fold as homologous titre was 80

Vaccine
NH 2015-16Vaccine
SH 2016
NH 2016-17

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Table 9-1 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-07-20

Viruses	Collection Date	Passage History	Neutralisation titre ¹																HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013*
			Post-infection ferret antisera																
			A/HK 4801/14		A/HK 7295/14		A/Slove 3188/15		A/Sth Afr R2665/15		A/Ser-NS 1517/16		A/Switz 9715923/13		A/Bucu 193605/16				
			Egg		MDCK		SIAT		SIAT		SIAT		SIAT		SIAT				
			F12/15		F02/15		F18/16		F50/15		F28/16		NIB F18/15		F26/16				
Passage history	Ferret number	Genetic group	3C.2a		3C.2a		3C.2a1		3C.2a		3C.2a		3C.3a		3C.3a				
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																			
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none	
A/Slovenia/3188/2015	3C.2a1	2015-12-26	MDCKx/SIAT3	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E	
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K	
A/Serbia-NS/1517/2016	3C.2a	2016-02-10	SIAT1	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	160	160	153	160	130	160	137	320	320		
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT1	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N	
TEST VIRUSES																		Additional information	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	C1/SIAT1	320	440	1280	1408	1280	1813	5120	3982	1280	1800	320	320	640	786	N121K, S144K, S198P	
A/Moscow/136/2016	3C.2a	2016-04-06	MDCK1/SIAT1	160	170	640	738	1280	1129	1280	1506	640	938	160	202	160	160	I48T, I300V	
A/Moscow/137/2016	3C.2a1	2016-04-12	MDCK1/SIAT1	320	341	640	823	1280	1894	2560	2148	1280	1430	160	192	320	248	N121S, N171K, S113A	
A/Moscow/135/2016	3C.2a	2016-04-13	MDCK1/SIAT1	320	315	1280	1176	2560	2316	>5120	>5120	5120	4066	320	306	320	315	E62K, R142K, M168V, Q197R	
A/Cote d'Ivoire/591/2016	3C.2a	2016-04-15	C1/SIAT1	640	488	1280	1262	1280	1415	5120	4655	2560	2033	320	427	320	320	N121K, S144K	
A/Cote d'Ivoire/595/2016	3C.2a	2016-04-15	C1/SIAT1	160	141	960	960	1280	985	1280	1870	2560	2133	160	236	640	507	N121K, S144K	
A/Cote d'Ivoire/599/2016	3C.2a	2016-04-15	C1/SIAT1	160	135	640	597	640	640	1280	1138	1280	1034	160	176	480	480	N121K, S144K, F193S,S198P	
A/Cote d'Ivoire/607/2016	3C.2a	2016-04-19	P2/SIAT1	320	342	1280	1189	640	868	1280	1341	2560	2048	240	240	320	414	N121K, S144K, N28X	
A/Cote d'Ivoire/651/2016	3C.2a	2016-04-25	P2/SIAT1	320	457	640	731	1280	1280	1280	1746	1280	1227	320	338	320	445	N121K, S144K	
A/Cote d'Ivoire/670/2016	3C.2a	2016-04-27	P2/SIAT1	80	80	240	240	320	267	320	409	160	236	80	107	80	86	N121K, S144K, S198P	
A/Cote d'Ivoire/679/2016	3C.2a	2016-05-03	C1/SIAT1	40	50	160	150	160	204	160	204	80	109	80	63	40	57	V102M, N121K, S144K, K207R	
A/Cote d'Ivoire/697/2016	3C.2a	2016-05-04	C1/SIAT1	320	248	1280	1135	1280	1031	2560	2163	1280	1768	320	307	320	273	N121K, S144K	
			Vaccine SH 2016 NH 2016-17																

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Table 9-2 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-07-25

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹														HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013 ²
				Post-infection ferret antisera														
				A/HK 4801/14 Egg	A/HK 7295/14 MDCK	A/Slove 3188/15 SIAT	A/Sth Afr R2665/15 SIAT	A/Ser-NS 1517/16 SIAT	A/Switz 9715923/13 SIAT	A/Bucu 193605/16 SIAT								
				F12/15 3C.2a	F02/15 3C.2a	F18/16 3C.2a1	F50/15 3C.2a	F28/16 3C.2a	NIB F18/15 3C.3a	F26/16 3C.3a								
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																		
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none
A/Slovenia/3188/2015	3C.2a1	2015-12-26	MDCKx/SIAT3	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K
A/Serbia-NS/1517/2016	3C.2a	2016-02-10	SIAT1	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	160	160	153	160	130	160	137	320	320	
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT1	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N
TEST VIRUSES																		Additional information
A/Slovenia/665/2016	3C.2a	2016-02-07	MDCK1/SIAT1	160	220	640	498	1280	1193	2560	2970	640	876	120	120	160	129	A163S, Q197K, K174R
A/Slovenia/704/2016	3C.2a	2016-02-10	MDCK1/SIAT1	80	112	320	320	640	693	20480	18286	1280	1034	80	76	80	118	R142K, M168V, Q197R
A/Slovenia/815/2016	3C.2a	2016-02-16	MDCK1/SIAT1	160	235	640	782	1280	1550	2560	2987	1280	1393	80	92	160	142	A163S, Q197K, K174R
A/Slovenia/1108/2016	3C.2a	2016-03-03	MDCK1/SIAT1	80	90	160	235	640	487	5120	4640	640	505	80	83	80	76	V88I, R142K, M168V, Q197R
A/Slovenia/1172/2016	3C.2a	2016-03-09	MDCK1/SIAT1	320	293	320	298	320	431	1280	1351	320	384	80	88	160	134	T160E, A163S, Q197K
A/Stockholm/28/2016	3C.2a1	2016-03-10	MDCK2/SIAT1	320	269	640	603	1280	971	320	436	320	268	160	203	160	133	N121K, T160X, N171K, P221X, I177V, G155E
A/Slovenia/1282/2016	3C.2a	2016-03-18	MDCK1/SIAT1	160	218	640	530	1280	1053	1280	1882	1280	970	80	99	80	98	A163S, Q197K
A/Slovenia/1381/2016	3C.2a	2016-03-25	MDCKx/SIAT1	320	390	1280	1054	2560	2068	5120	5120	1280	1858	160	124	160	171	R142K, M168V, Q197R, A212T
A/Iceland/6202/2016	3C.2a	2016-03-30	SIAT1	160	154	320	460	640	739	2560	3243	640	747	80	84	160	129	L197M
A/Iceland/44/2016	3C.2a1	2016-04-01	MDCK1/SIAT1	40	52	160	149	320	356	320	253	80	117	40	50	40	56	N171K, 216X, P221X, I77V, G155E
A/Stockholm/26/2016	3C.2a1	2016-04-02	MDCK2/SIAT1	80	72	160	137	320	350	320	244	80	112	80	63	80	69	I67M, N121K, S124X, N171K, I77V, G155E
A/Cote d'Ivoire/639/2016	3C.2a	2016-04-25	C2/SIAT1	40	42	160	130	160	196	160	160	80	77	40	40	40	40	A93X, N121K, S144K, S198P, R127X
A/Iceland/59/2016	3C.2a	2016-05-08	MDCK1/SIAT1	80	113	120	120	160	222	160	177	80	72	40	50	40	56	T160K, P221X, L197M
				Vaccine SH 2016 NH 2016-17														

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Table 9-3 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-07-27

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹														HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013*	
				Post-infection ferret antisera															
				A/HK 4801/14	A/HK 7295/14	A/Slove 3188/15	A/Sth Afr R2665/15	A/Ser-NS 1517/16	A/Switz 9715923/13	A/Bucu 193605/16									
				Egg	MDCK	SIAT	SIAT	SIAT	SIAT	SIAT									
				F12/15 3C.2a	F02/15 3C.2a	F18/16 3C.2a1	F50/15 3C.2a	F28/16 3C.2a	NIB F18/15 3C.3a	F26/16 3C.3a									
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																			
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none	
A/Slovenia/3188/2015	3C.2a1	2015-12-26	MDCKx/SIAT3	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E	
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K	
A/Serbia-NS/1517/2016	3C.2a	2016-02-10	SIAT1	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	160	160	153	160	130	160	137	320	320		
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT1	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N	
TEST VIRUSES																Additional information			
A/Bucuresti/191366/2016	3C.3a	2016-02-10	SIAT1/SIAT1	80	86	160	137	320	320	320	347	320	273	160	157	320	287	Y94H, R326K	
A/Arges/1480-3059/2016	3C.3a	2016-02-10	SIAT1/SIAT1	80	107	80	108	320	333	320	267	320	249	160	132	320	373	Y94H, R326K	
A/Khmelnitsky/659/2016	3C.2a1	2016-02-28	SIAT2/SIAT1	80	93	160	127	320	294	160	168	80	113	40	58	80	70	160X, N171K, I77V, G155E	
A/Kyiv/571/2016	3C.2a	2016-03-01	SIAT1/SIAT1	640	676	1280	1092	2560	2115	2560	2440	1280	1256	320	297	480	480	G275S	
A/Jordan/20312/2016	3C.2a	2016-03-03	SIAT1	640	533	1280	1031	2560	2804	2560	2988	1280	1638	320	367	640	931	none	
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT1	320	360	640	914	1280	1520	2560	2560	2560	1997	320	328	320	335	N121K, N171K, I77V, G155E	
A/Jordan/20345/2016	3C.2a1	2016-03-08	SIAT1	320	397	640	827	1280	1877	5120	4680	1280	1877	320	287	640	640	N121K, T135A, N171K, V18M, I77V, G155E	
A/Khmelnitsky/668/2016	3C.2a1	2016-03-11	SIAT2/SIAT1	640	489	640	929	1280	1211	1280	1772	1280	1455	160	226	640	520	N171K, I77V, G155E	
A/Oman/2758/2016	3C.2a1	2016-03-12	SIAT1	640	540	1280	1038	1280	1124	2560	3491	1280	1415	240	240	320	457	N121K, S124N, T160X, N171K, I77V, G155E	
A/Khmelnitsky/728/2016	3C.2a	2016-03-19	SIAT2/SIAT1	160	215	640	671	1280	1408	640	914	640	814	160	144	160	260	none	
A/Khmelnitsky/722/2016	3C.2a	2016-03-23	SIAT2/SIAT1	320	302	640	788	1280	1156	640	864	1280	1248	160	196	320	422	none	
A/Khmelnitsky/719/2016	3C.2a	2016-03-25	SIAT2/SIAT1	320	253	640	625	1920	1920	640	606	320	292	320	310	320	256	V176L	
A/Bolzano/9/2016	3C.2a	2016-03-27	SIAT3/SIAT1	160	227	320	265	640	520	320	320	320	259	160	188	160	200	D73X	
A/Georgia/1139/2016	3C.2a1	2016-04-25	SIAT2	640	493	1280	1016	2560	2033	5120	3907	1280	1280	320	320	640	560	S46Y, N171K, I77V, G155E	
				Vaccine SH 2016 NH 2016-17															

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Table 9-4 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-07-14

Viruses	Collection Date	Passage History	Neutralisation titre ¹														HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013*	
			Post-infection ferret antisera															
			A/HK 4801/14	A/HK 7295/14	A/Slove 3188/15	A/Sth Afr R2665/15	A/Ser-NS 1517/16	A/Switz 9715923/13	A/Bucu 193605/16									
			Egg F12/15	MDCK F02/15	SIAT F18/16	SIAT F50/15	SIAT F28/16	SIAT NIB F18/15	SIAT F26/16									
			3C.2a	3C.2a	3C.2a1	3C.2a	3C.2a	3C.2a	3C.3a	3C.3a								
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																		
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none
A/Slovenia/3188/2015	3C.2a1	2015-12-26	MDCKx/SIAT3	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K
A/Serbia-NS/1517/2016	3C.2a	2016-02-10	SIAT1	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	160	160	153	160	130	160	137	320	320	
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT1	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N
TEST VIRUSES																		
A/Istanbul/1946182/2016	3c.3a	2016-03-03	MDCK1/SIAT1	40	53	80	85	80	117	80	57	40	55	80	80	160	234	E50K, Y89H, R326K
A/Jordan/4470/2016	3c.3a	2016-03-27	SIAT1	40	47	80	60	80	80	80	65	40	57	80	69	160	143	Y94H, I214T, R326K
A/Friuli-Venezia Giulia/14/2016	3c.2a1	2016-03-01	SIAT1/SIAT1	160	235	640	640	1280	980	1280	1505	640	760	160	218	160	227	N121K, N171K, I77V, G150E, G155E
A/Perugia/25/2016	3c.2a	2016-04-01	SIAT3/SIAT1	80	86	160	152	320	310	640	544	320	314	80	112	160	120	A138X, R142E, M168V, Q197R
A/South Africa/R3026/2016	3c.2a	2016-05-26	MDCK1/SIAT1	80	100	320	270	640	549	10240	9846	640	606	80	64	80	69	R142K, N158X, M168V, Q197R
A/South Africa/R3142/2016	3c.2a	2016-05-30	MDCK1/SIAT1	160	128	320	245	320	408	320	446	320	296	40	52	80	74	N158X, Q197R, V18M, D164N
A/South Africa/R3328/2016	3c.2a	2016-06-02	MDCK1/SIAT1	80	87	320	459	640	896	1280	1280	640	843	80	70	160	122	Q197R, V18M, D164N
A/South Africa/VW0036/2016	3c.2a1	2016-06-04	MDCK1/SIAT1	80	72	320	255	640	557	640	693	320	278	80	64	80	78	N121K, N171K, I77V, G155E
			Vaccine SH 2016 NH 2016-17															

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Table 9-5 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-08-03

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013*		
				Post-infection ferret antisera														
				A/HK 4801/14 Egg F12/15 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Slove 3188/15 SIAT F18/16 3C.2a1	A/Sth Afr R2665/15 SIAT F50/15 3C.2a	A/Ser-NS 1517/16 SIAT F28/16 3C.2a	A/Switz 9715923/13 SIAT NIB F18/15 3C.3a	A/Bucu 193605/16 SIAT F26/16 3C.3a								
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		2-fold	Read
				REFERENCE VIRUSES														
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none
A/Slovenia/3188/2015	3C.2a1	2015-12-26	MDCKx/SIAT3	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K
A/Serbia-NS/1517/2016	3C.2a	2016-02-10	SIAT1	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	160	160	153	160	130	160	137	320	320	
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT1	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N
TEST VIRUSES																	Additional information	
A/Bitlis/4351/2016	3C.3a	2016-03-18	SIAT1/SIAT1	160	150	160	184	640	543	160	139	160	157	160	169	320	384	Y94H, R326K, T328X, L202S
A/Antsirabe/2047/16	3C.2a	2016-05-02	SIAT1	640	602	1280	1261	2560	2560	5120	6095	2560	1954	640	696	1280	1067	V18M
A/Tsaralalana/2054/2016	3C.2a	2016-05-02	SIAT1	640	514	1280	1127	5120	4325	5120	4779	1280	1828	320	247	1280	1625	no changes
A/Antsirabe/2093/16	3C.2a	2016-05-10	SIAT1	640	597	1280	1792	2560	2170	2560	3480	2560	2462	640	716	640	909	K93R, N216S, N290S
A/Antananarivo/2156/16	3C.2a	2016-05-12	SIAT1	640	507	1280	1112	1280	1707	2560	2462	1280	1760	320	466	640	781	no changes
A/Antananarivo/2157/16	3C.2a	2016-05-12	SIAT1	640	600	1280	1080	2560	3109	2560	2360	1280	1668	640	674	320	471	no changes
A/Antsirabe/2173/16	3C.2a1	2016-05-17	SIAT1	320	436	640	835	2560	2960	2560	3108	1280	1536	320	269	320	394	N121K, N171K, I77V, G155E
A/Antsirabe/2267/16	3C.2a	2016-05-23	SIAT1	1280	1177	2560	2560	5120	5120	5120	4297	2560	3777	1280	1871	2560	1920	no changes
A/South Africa/R3412/2016	3C.2a	2016-06-06	SIAT1	320	247	640	735	2560	2880	5120	5447	2560	3151	320	286	320	307	Q197R, V213A, P289S, D291E
A/South Africa/R3620/2016	3C.2a1	2016-06-08	SIAT1	320	462	640	640	1280	1148	1280	1664	640	580	160	183	320	306	R142G, N171K, I77V, G155E
A/South Africa/R3703/2016	3C.2a1	2016-06-13	SIAT1	640	508	1280	1326	2560	1986	5120	5224	1280	1237	640	498	640	530	N121K, N171K, L38F, I77V, G155E
A/South Africa/R3742/2016	3C.2a	2016-06-13	SIAT1	320	304	1280	1071	1280	1146	1280	1706	1280	996	160	157	320	400	NO SEQUENCE
A/South Africa/R3978/2016	3C.2a	2016-06-14	SIAT1	320	408	640	953	2560	3277	5120	5818	2560	2844	320	434	320	299	Q197R, V213A, P289S, D291E
A/South Africa/R3915/2016	3C.2a	2016-06-17	SIAT1	320	360	640	939	1280	1028	>5120	20067	1280	1768	160	157	160	150	R142K, N158X, M168V, Q197R, D37X
				Vaccine SH 2016 NH 2016-17														

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Table 9-6 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-08-10

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹														HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013*
				Post-infection ferret antisera														
				A/HK 4801/14		A/HK 7295/14		A/Slove 3188/15		A/Sth Afr R2665/15		A/Ser-NS 1517/16		A/Switz 9715923/13		A/Bucu 193605/16		
				Egg		MDCK		SIAT		SIAT		SIAT		SIAT		SIAT		
				F12/15		F02/15		F18/16		F50/15		F28/16		NIB F18/15		F26/16		
				3C.2a		3C.2a		3C.2a1		3C.2a		3C.2a		3C.3a		3C.3a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none
A/Slovenia/3188/2015	3C.2a1	2015-12-26	MDCKx/SIAT3	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K
A/Serbia-NS/1517/2016	3C.2a	2016-02-10	SIAT1	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	160	160	153	160	130	160	137	320	320	
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT1	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N
TEST VIRUSES																		Additional information
A/Izmir/1089/2016	3C.3a	2016-02-05	SIAT1/SIAT1	40	47	80	75	160	138	160	124	80	109	80	84	160	186	Y94H, R326K
A/Zonguldak/3161/2016	3c.3a	2016-02-05	MDCK1/SIAT1	160	130	160	143	320	242	160	218	160	217	160	210	320	295	Y94H, R326K
A/Aydin/3446/2016	3C.3a	2016-02-11	SIAT1/SIAT1	80	67	80	93	160	231	80	95	80	63	160	138	320	334	Y94H, R326K
A/Kutahya/3503/2016	3C.3a	2016-02-12	MDCK1/SIAT1	80	60	80	103	160	146	80	119	160	129	160	117	160	237	Y94H, R326K
A/Antalya/3562/2016	3C.3a	2016-02-12	MDCK1/SIAT1	40	53	80	77	160	182	80	96	80	80	80	96	160	200	Y94H, I214T, R326K
A/Izmir/3575/2016	3c.3a	2016-02-13	MDCK1/SIAT1	80	50	80	78	160	135	80	80	40	57	80	94	160	166	Y94H, M268I, R326K
A/Denizli/3801/2016	3c.3a	2016-02-19	MDCK1/SIAT1	80	80	80	102	160	143	80	117	80	114	160	122	160	210	Y94H, R326K
A/Bolu/3994/2016	3C.3a	2016-02-26	MDCK1/SIAT1	80	73	80	106	320	273	80	116	80	75	80	114	160	221	Y94H, N144S, R326K
A/Aksaray/4048/2016	3C.3a	2016-03-01	MDCK1/SIAT1	40	57	80	77	160	145	160	126	80	80	80	118	160	197	Y94H, I214T, R326K, A147V
A/Norway/3545/2016	3C.3a	2016-05-13	SIAT1	80	76	160	140	160	152	80	103	80	73	160	131	320	250	K92R, Q197K, S198P, S312N
A/Norway/3798/2016	3C.3a	2016-06-09	SIAT1	40	40	80	74	160	122	80	72	80	65	80	93	160	203	K92R, Q197K, S198P, S312N
A/Norway/3806/2016	3C.2a1	2016-06-13	SIAT1	320	314	640	662	1280	1067	1280	960	640	629	320	288	320	278	N121K, I140M, N171K, I77V, G150E, G155E
A/Norway/3813/2016	3C.2a1	2016-06-16	SIAT1	320	259	640	614	1280	1182	1280	1280	640	907	160	170	320	393	N121K, I140M, N171K, I77V, G150E, G155E
				Vaccine SH 2016 NH 2016-17														

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Table 9-7 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-09-01

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013*		
				Post-infection ferret antisera														
				A/HK 4801/14 Egg F12/15 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Slove 3188/15 SIAT F18/16 3C.2a1	A/Sth Afr R2665/15 SIAT F50/15 3C.2a	A/Ser-NS 1517/16 SIAT F28/16 3C.2a	A/Switz 9715923/13 SIAT NIB F18/15 3C.3a	A/Bucu 193605/16 SIAT F26/16 3C.3a								
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																		
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none
A/Slovenia/3188/2015	3C.2a1	2015-12-26	MDCKx/SIAT3	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K
A/Serbia-NS/1517/2016	3C.2a	2016-02-10	SIAT1	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	160	160	153	160	130	160	137	320	320	
A/Bucuresti/193605/2016	3C.3a	2016-03-08	SIAT1/SIAT1	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N
TEST VIRUSES																		Additional information
A/Mardin/2834/2016	3C.3a	2016-02-01	MDCK1/SIAT1	160	122	80	86	160	209	320	258	80	111	80	108	160	228	T10X, Y94H, R326K
A/Izmir/1020/2016	3C.3a	2016-02-02	SIAT1/SIAT1	80	68	80	71	160	134	80	94	80	115	80	93	160	154	F79L, Y94H, I214T, R326K
A/Aydin/3060/2016	3C.3a	2016-02-03	SIAT1/SIAT1	80	61	40	59	80	93	80	107	80	68	80	69	160	127	Y94H, M168I, I214T, R326K
A/Tajikistan/15/2016	3C.2a	2016-04-12	SIAT1	320	373	640	885	1280	1513	2560	2133	1280	1056	160	168	320	270	no changes
A/Ghana/FS-16-0695/2016	3C.2a	2016-05-23	Cx/SIAT1	320	244	640	664	640	823	1280	1102	640	576	160	150	160	156	N121K, S144K,
A/Ghana/FS-16-0889/2016	3C.2a	2016-06-26	Cx/SIAT1	160	237	320	320	640	669	1280	1280	640	891	160	181	160	160	N121K, S144K, F193S, I89V
A/Ghana/DILI-16-0661/2016	3C.2a	2016-06-27	Cx/SIAT1	160	224	640	604	640	693	640	904	640	592	80	114	160	171	D32G, N121K, S144K
A/South Africa/VW0048/2016	3C.2a	2016-06-30	MDCK2/SIAT1	320	269	640	601	1280	1192	>5120	>5120	1280	1220	160	175	320	284	R142K, M168V, Q197R
A/South Africa/VW0050/2016	3C.2a	2016-07-04	MDCK2/SIAT1	160	202	640	731	640	853	640	725	640	727	160	207	320	269	N121K, S144K, Q197K
A/South Africa/R4598/2016	3C.2a1	2016-07-05	MDCK1/SIAT1	160	197	640	497	1280	1006	1280	1209	640	585	160	122	160	154	N121K, N171K, K189R, I77V, G155E, A201V
A/South Africa/VW0060/2016	3C.2a1	2016-07-16	MDCK1/SIAT1	160	233	640	557	1280	1112	1280	1575	640	674	160	136	160	165	N121K, N171K, I77V, G155E
A/South Africa/VW0067/2016	3C.2a	2016-07-20	MDCK1/SIAT1	320	273	640	587	1280	999	>5120	>5120	1280	1203	160	202	160	224	R142K, M168V, Q197R
A/South Africa/VW0071/2016	3C.2a	2016-07-21	MDCK1/SIAT1	320	282	640	795	1280	1035	1280	1280	640	935	160	173	160	216	N121K, S144K, Q197K
A/South Africa/VW0073/2016	3C.2a	2016-07-21	MDCK1/SIAT1	320	411	640	907	2560	2095	5120	6919	2560	2048	320	254	320	420	I25V, R33Q, Y94H, Q197K, R261Q
				Vaccine SH 2016 NH 2016-17														

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Table 9-8. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary

Viruses		Neutralisation titre ¹														HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate: HA1 substitutions for 3C.3a viruses compared to A/Switzerland/9715293/2013*	
		Post-infection ferret antisera															
		A/HK 4801/14	A/HK 7295/14		A/Slove 3188/15		A/Sth Afr R2665/15		A/Ser-NS 1517/16		A/Switz 9715923/13		A/Bucu 193605/16				
		Passage history Egg	MDCK		SIAT		SIAT		SIAT		SIAT		SIAT				
		Ferret number F12/15	F02/15		F18/16		F50/15		F28/16		NIB F18/15		F26/16				
Genetic group		3C.2a		3C.2a		3C.2a1		3C.2a		3C.2a		3C.3a		3C.3a			
		2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																	
A/Hong Kong/4801/2014	3C.2a	640	718	1280	1193	320	309	5120	4495	640	900	80	72	80	107	T160K, L194P	
A/Hong Kong/7295/2014	3C.2a	80	95	320	283	320	296	320	282	320	330	80	67	160	145	none	
A/Slovenia/3188/2015	3C.2a1	160	141	640	527	960	960	2560	2469	640	933	80	96	160	122	N171K, S312R, I77V, G155E	
A/South Africa/R2665/2015	3C.2a	160	191	1280	988	1280	1129	20480	19017	1280	1396	160	186	160	217	R142K, Q197R, R216K	
A/Serbia-NS/1517/2016	3C.2a	320	452	640	810	1280	1688	1280	1800	1280	1550	160	207	320	257	Q197K, R261Q	
A/Switzerland/ 9715293/2013	3C.3a	80	78	160	208	160	160	160	153	160	130	160	137	320	320		
A/Bucuresti/193605/2016	3C.3a	80	72	160	237	320	273	320	427	160	143	160	149	320	377	S198P, K278X, S312N	
TEST VIRUS SUBCLADE	Number tested	Number and % of viruses reacting ≤4-fold reduced compared to the homologous titre															
3C.3a	19	0		17	89.5	5	26.3	0		0		19	100.0	19	100.0		
3C.2a	69	51	73.9	69	100.0	66	95.7	13	18.8	55	79.7	69	100.0	58	84.1		
N171K	19	15	78.9	19	100.0	19	100.0	1	5.3	14	93.3	19	100.0	15	78.9	Viruses in subclade 3C.2a carrying the HA1 amino acid substitutions indicated	
N121K	28	21	75.0	28	100.0	26	92.9	1	3.6	22	78.6	28	100.0	23	82.1		
N121K + S144K	15	10	66.7	15	100.0	13	86.7	0		12	80.0	15	100.0	12	80.0		
N121K + N171K	13	11	84.6	13	100.0	13	100.0	3	23.1	10	76.9	13	100.0	11	84.6		
Q197R	13	7	53.8	13	100.0	13	100.0	8	61.5	11	84.6	13	100.0	11	84.6		
R142K + Q197R	8	5	62.5	8	100.0	8	100.0	6	75.0	7	87.5	8	100.0	7	87.5		
Q197K	7	7	100.0	7	100.0	7	100.0	1	14.3	6	85.7	7	100.0	7	100.0		
		Vaccine SH 2016 NH 2016-17															

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* HA2 substitutions

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 7. Phylogenetic comparison of influenza A(H3N2) HA genes

Vaccine virus

Reference viruses

Collection date

Feb-Mar 2016

Apr 2016

May 2016

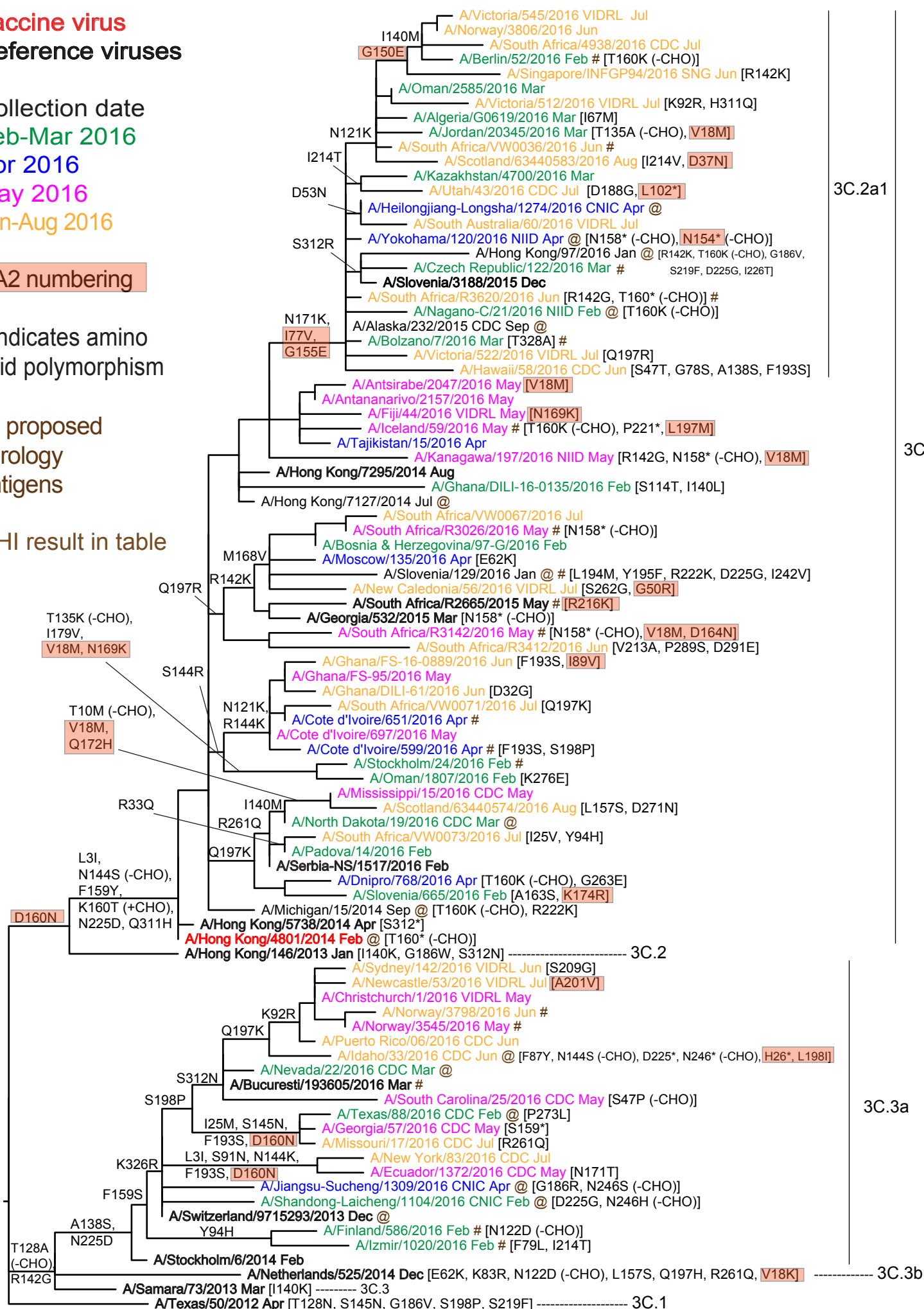
Jun-Aug 2016

HA2 numbering

* indicates amino acid polymorphism

@ proposed serology antigens

HI result in table



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**

	310	320	1	11	21	31	41	51	61	71
A/Texas/50/2012	TYGACPRYVKQSTLKLATGMRNVPEKQTRGIFGAIA	GIENGWEGMVDGWYGRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNKHFQIEKEFS								
A/Samara/73/2013										
A/Netherlands/525/2014										
A/Stockholm/6/2014										
A/Izmir/1020/2016										
A/Switzerland/9715293/2013			R							
A/New_York/83/2016			R							
A/Missouri/17/2016			R							
A/Bucuresti/193605/2016		N	R							
A/Idaho/33/2016		N	R				X			
A/Norway/3798/2016		N	R							
A/Sydney/142/2016		N	R							
A/Hong_Kong/146/2013		N								
A/Hong_Kong/4801/2014		H								
A/Hong_Kong/5738/2014		HX								
A/Dnipro/768/2016		H								
A/Serbia-NS/1517/2016		H								
A/Scotland/63440574/2016		H								
A/Oman/1807/2016		H								
A/Ghana/DILI-61/2016		H								
A/Georgia/532/2015		H								
A/South_Africa/R2665/2015		H								
A/South_Africa/R3026/2016		H								
A/Hong_Kong/7295/2014		H								
A/Iceland/59/2016		H								
A/Bolzano/7/2016		H	A							
A/Slovenia/3188/2015		HR								
A/Czech_Republic/122/2016		HR								
A/South_Australia/60/2016		H								
A/Kazakhstan/4700/2016		H								
A/South_Africa/VW0036/2016		H								
A/Norway/3806/2016 Jun		H								

	81	91	101	111	121	131	141	151	161	171
A/Texas/50/2012	EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTDLDSEMNKLF	EKTQQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF								
A/Samara/73/2013										
A/Netherlands/525/2014										
A/Stockholm/6/2014										
A/Izmir/1020/2016										
A/Switzerland/9715293/2013										
A/New_York/83/2016									N	
A/Missouri/17/2016									N	
A/Bucuresti/193605/2016									N	
A/Idaho/33/2016										
A/Norway/3798/2016										
A/Sydney/142/2016										
A/Hong_Kong/146/2013									N	
A/Hong_Kong/4801/2014									N	
A/Hong_Kong/5738/2014									N	
A/Dnipro/768/2016									N	
A/Serbia-NS/1517/2016									N	
A/Scotland/63440574/2016									N	
A/Oman/1807/2016									N	
A/Ghana/DILI-61/2016									N	
A/Georgia/532/2015									N	
A/South_Africa/R2665/2015									N	
A/South_Africa/R3026/2016									N	
A/Hong_Kong/7295/2014									N	
A/Iceland/59/2016									N	
A/Bolzano/7/2016		V							E	N
A/Slovenia/3188/2015		V							E	N
A/Czech_Republic/122/2016		V							E	N
A/South_Australia/60/2016		V							E	N
A/Kazakhstan/4700/2016		V							E	N
A/South_Africa/VW0036/2016		V							E	N
A/Norway/3806/2016 Jun		V							E	N

	181	191	201	211	221
A/Texas/50/2012	QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI				
A/Samara/73/2013					
A/Netherlands/525/2014					
A/Stockholm/6/2014					
A/Izmir/1020/2016					
A/Switzerland/9715293/2013					
A/New_York/83/2016					
A/Missouri/17/2016					
A/Bucuresti/193605/2016					
A/Idaho/33/2016			I		
A/Norway/3798/2016					
A/Sydney/142/2016					
A/Hong_Kong/146/2013					
A/Hong_Kong/4801/2014					
A/Hong_Kong/5738/2014					
A/Dnipro/768/2016					
A/Serbia-NS/1517/2016					
A/Scotland/63440574/2016	H				
A/Oman/1807/2016					
A/Ghana/DILI-61/2016					
A/Georgia/532/2015					
A/South_Africa/R2665/2015					K
A/South_Africa/R3026/2016					
A/Hong_Kong/7295/2014					
A/Iceland/59/2016					M
A/Bolzano/7/2016					
A/Slovenia/3188/2015					
A/Czech_Republic/122/2016					
A/South_Australia/60/2016					
A/Kazakhstan/4700/2016					
A/South_Africa/VW0036/2016					
A/Norway/3806/2016 Jun					

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 9A. Amino acid substitutions associated with the HA of subclade 3C.2a viruses relative to A/Texas/50/2012

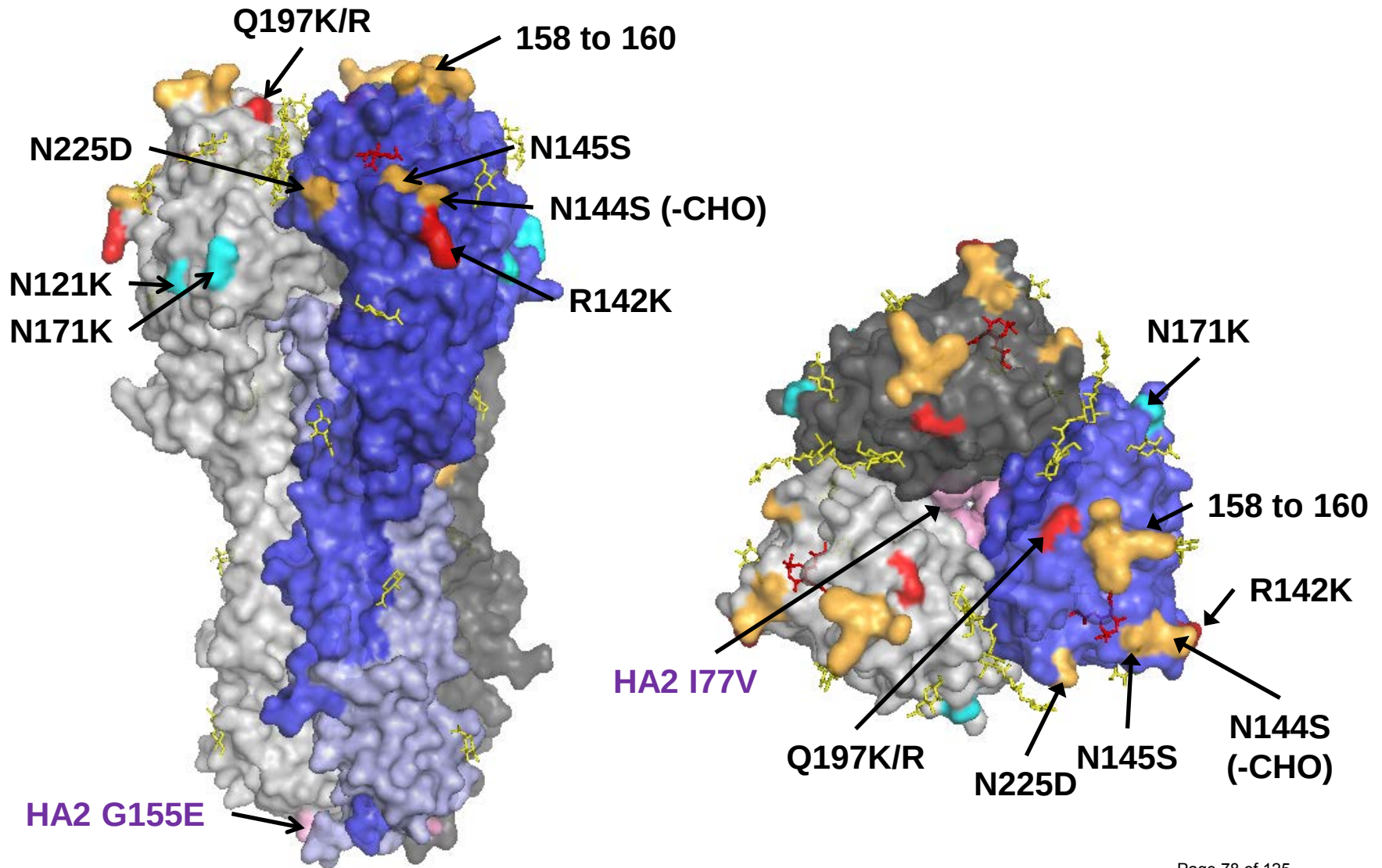


Figure 9B. Amino acid substitutions associated with the HA of subclade 3C.3a viruses relative to A/Texas/50/2012

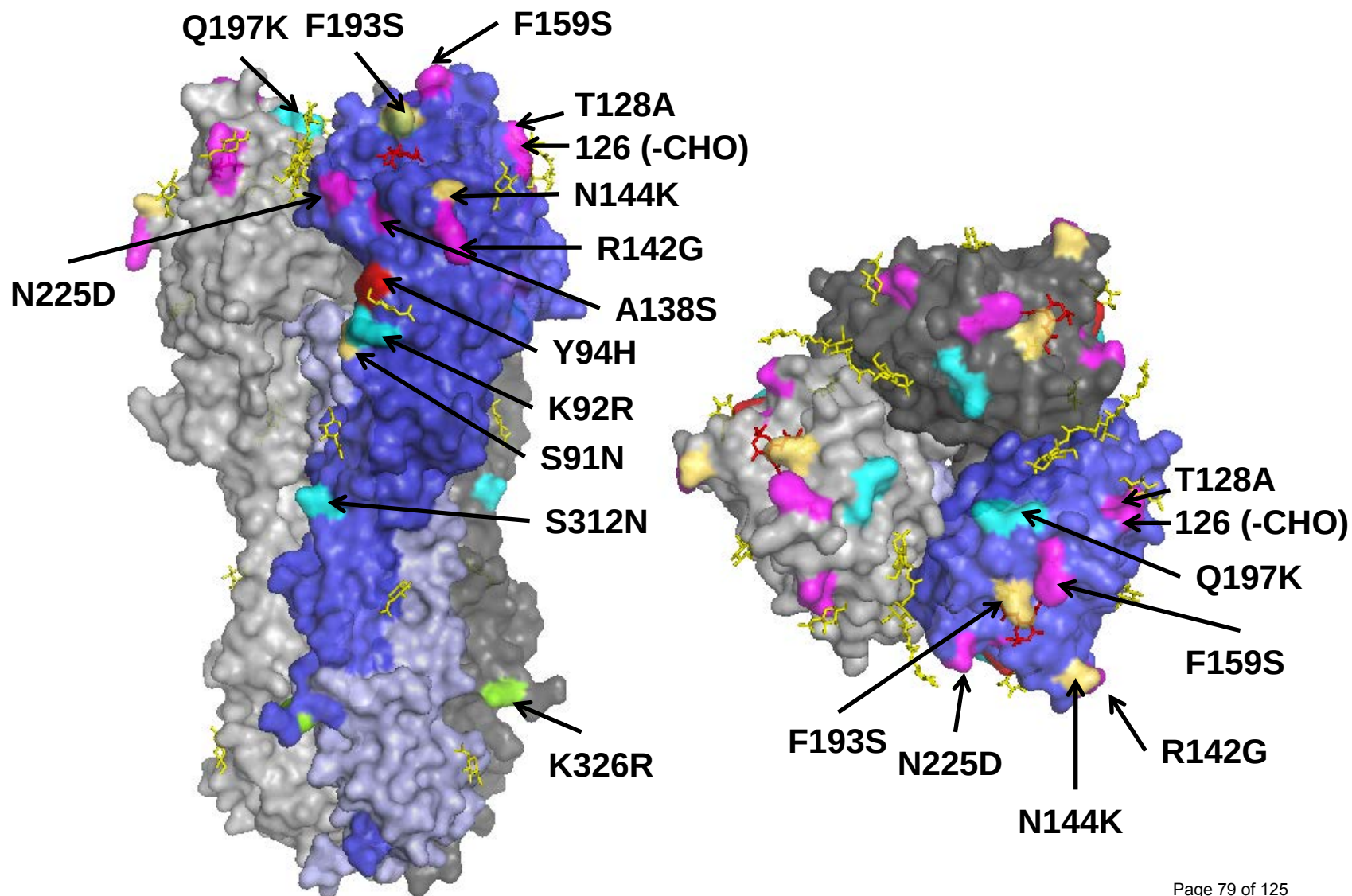
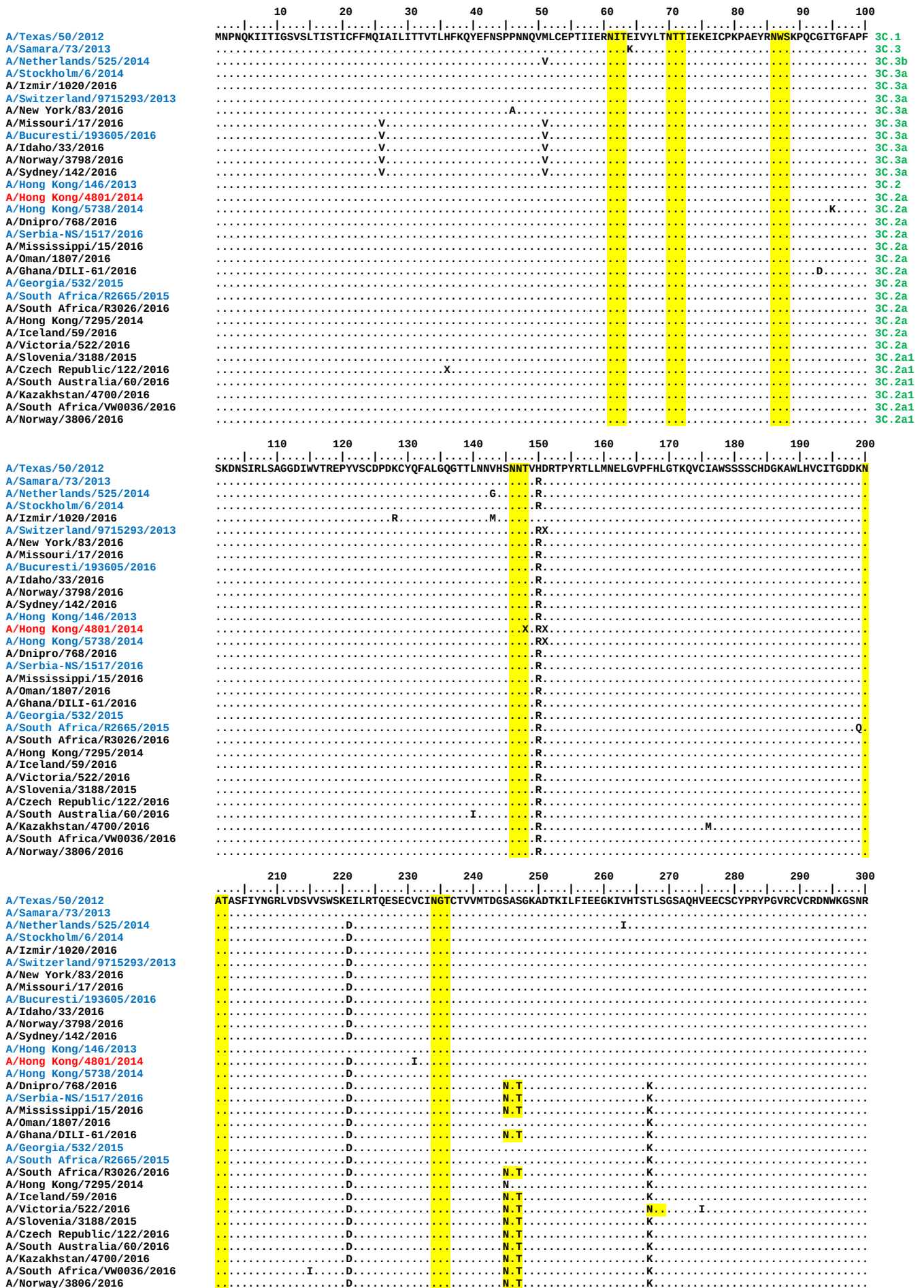


Figure 11. NA protein alignment for influenza A(H3N2) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
A/Texas/50/2012	PIVDINIKDHS	IVSSYVCSGL	VGDTPRKND	SSSSSHCLDPN	NEEGGHGVKG	WAFDDGNDV	WMGRITINET	SRLGYETFFK	VIEGWSNP	SKSLQINRQVIVDR
A/Samara/73/2013										
A/Netherlands/525/2014										
A/Stockholm/6/2014						G				T
A/Izmir/1020/2016		R								T.L.K
A/Switzerland/9715293/2013										T
A/New York/83/2016				K				R		T
A/Missouri/17/2016		R	S							
A/Bucuresti/193605/2016		R	S							
A/Idaho/33/2016		R	S							
A/Norway/3798/2016		R	S							
A/Sydney/142/2016		R	S							
A/Hong Kong/146/2013										
A/Hong Kong/4801/2014										T
A/Hong Kong/5738/2014										T
A/Dnipro/768/2016			G						V	
A/Serbia-NS/1517/2016			G						V	
A/Mississippi/15/2016			G						V	G
A/Oman/1807/2016									V	
A/Ghana/DILI-61/2016				N					V	
A/Georgia/532/2015									V	
A/South Africa/R2665/2015									V	
A/South Africa/R3026/2016		S							V	
A/Hong Kong/7295/2014									V	
A/Iceland/59/2016			G						V	
A/Victoria/522/2016				N					V	
A/Slovenia/3188/2015				N					V	
A/Czech Republic/122/2016				N					V	
A/South Australia/60/2016				N					V	
A/Kazakhstan/4700/2016				N					V	
A/South Africa/VW0036/2016				N					V	
A/Norway/3806/2016			S	N					V	
	410	420	430	440	450	460				
A/Texas/50/2012	GDRSGYSGIF	SV	EGKSCIN	RCFYVELI	RGRKEETE	VLWTSNS	IVVFCG	TS	SGTYGTG	SWPDGADLNLMP
A/Samara/73/2013										
A/Netherlands/525/2014										
A/Stockholm/6/2014										
A/Izmir/1020/2016										
A/Switzerland/9715293/2013										
A/New York/83/2016										
A/Missouri/17/2016										
A/Bucuresti/193605/2016										
A/Idaho/33/2016										
A/Norway/3798/2016										
A/Sydney/142/2016										
A/Hong Kong/146/2013										
A/Hong Kong/4801/2014										
A/Hong Kong/5738/2014										
A/Dnipro/768/2016							N			
A/Serbia-NS/1517/2016										
A/Mississippi/15/2016										
A/Oman/1807/2016									H	
A/Ghana/DILI-61/2016									L	
A/Georgia/532/2015										
A/South Africa/R2665/2015										
A/South Africa/R3026/2016									H	
A/Hong Kong/7295/2014										
A/Iceland/59/2016						N		H		
A/Victoria/522/2016							V			
A/Slovenia/3188/2015									H	
A/Czech Republic/122/2016									H	
A/South Australia/60/2016							A		V	
A/Kazakhstan/4700/2016									H	
A/South Africa/VW0036/2016									H	
A/Norway/3806/2016									H	

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 12A. Amino acid substitutions associated with the NA of subclade 3C.2a viruses relative to A/Texas/50/2012

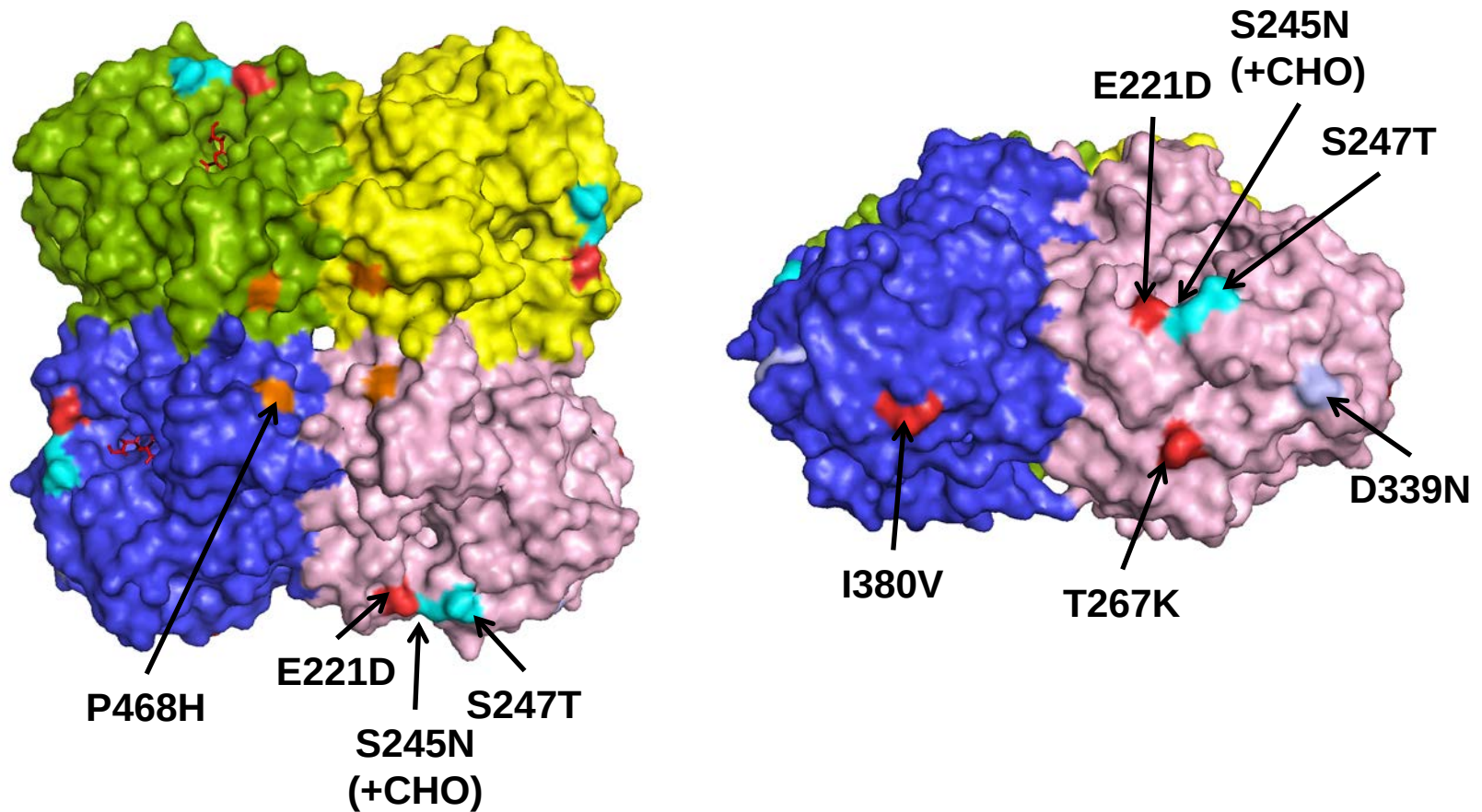
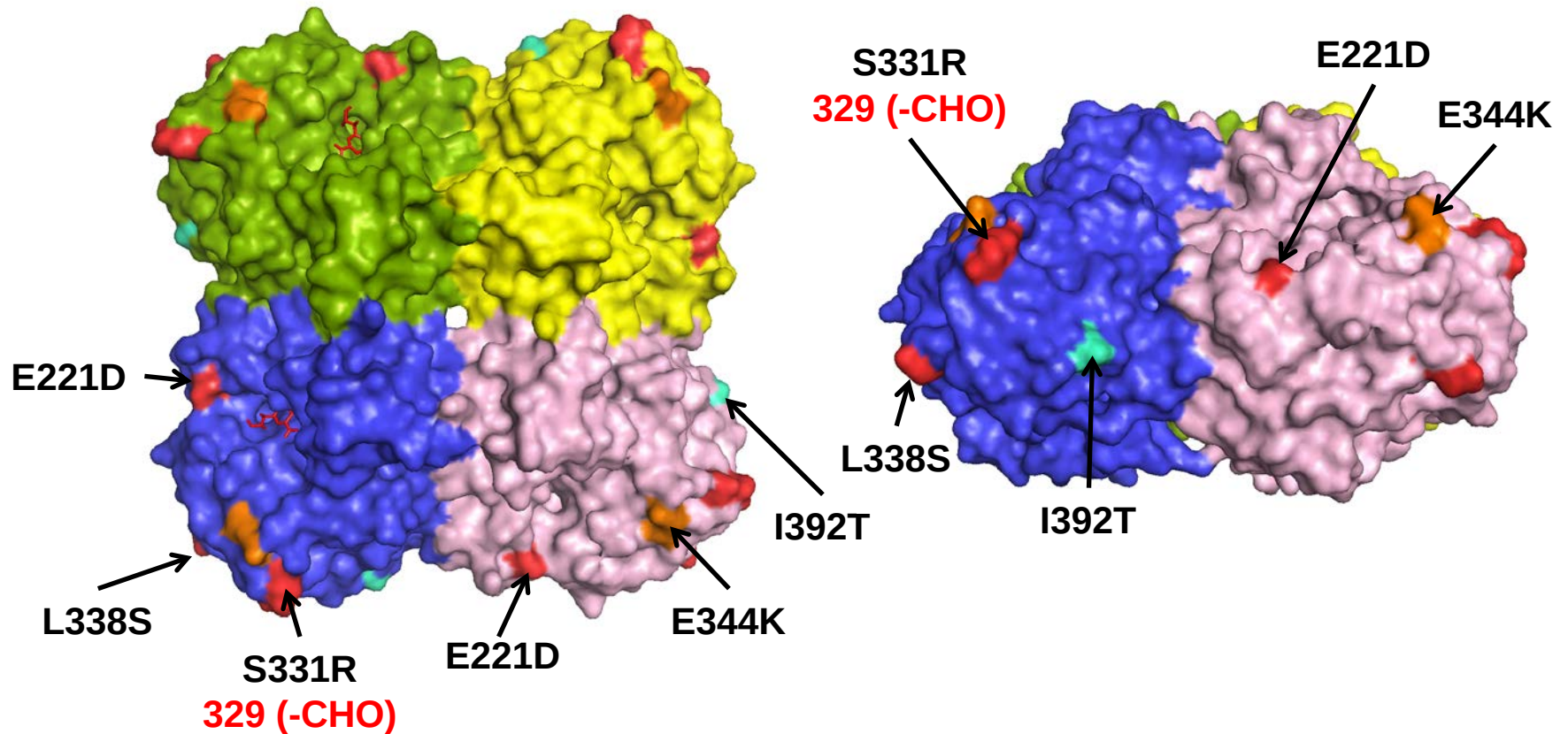


Figure 12B. Amino acid substitutions associated with the NA of subclade 3C.3a viruses relative to A/Texas/50/2012



Influenza B virus analyses.

Influenza B viruses represented 29.5% of the samples received with collection dates after 2016-01-31 (**Table 1**). B/Victoria lineage viruses predominated over those of the B/Yamagata lineage at a ratio of approximately 6.1:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from 33 countries or administrative regions in four WHO Regions (**Table 1**): African (3: Ghana, Madagascar and South Africa), European (25: Albania, Bulgaria, Cyprus, Czech Republic, Estonia, Finland, France, Georgia, Germany, Iceland, Italy, Kazakhstan, Lithuania, Montenegro, Norway, Portugal, Romania, Russia, Serbia, Slovakia, Slovenia, Spain, Sweden, Turkey, United Kingdom), Eastern Mediterranean (4: Egypt, Jordan, Oman and Palestine) and Western Pacific (1: Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 10**.

HI results for the propagated viruses are shown in **Tables 11-1 to 11-14**: test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are those with collection dates after 2016-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. The results for an egg-propagated virus received from the USA are shown in **Table 11-14**. A summary of the recognition of the test viruses collected after 2016-01-31 is shown in **Table 11-15**. Only six (2.6%) of the 234 cell-propagated test viruses were recognised at titres within 2-fold of the homologous titres with post-infection ferret antisera raised against the vaccine virus B/Brisbane/60/2008, and a total of 59 (25%) test viruses were recognised at titres within 4-fold of the homologous titres. Similarly, the cell culture-propagated test viruses were poorly recognised by post-infection ferret antisera raised against other egg-propagated reference viruses: B/Malta/636714/2011, B/South Australia/81/2012 and B/Johannesburg/3964/2012, with less than 7% of the tested viruses being recognised at titres within 4-fold of the homologous titres of these antisera. In contrast, over 90% of the test viruses were recognised at titres within 2-fold of the titres with the corresponding homologous viruses for antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in tissue culture; such viruses are considered to be surrogate cell-propagated antigens representing the egg-propagated B/Brisbane/60/2008 prototype strain. These antisera, used in the majority of HI assays, were raised against B/Iceland/56/2015, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016. Antisera raised against older cell culture-propagated viruses (B/Paris/1762/2009, B/Hong Kong/514/2009 and B/Odessa/3886/2010) were used in some assays and they recognised 99%, 90% and 85%, respectively, of test viruses at titres within 2-fold of the titres with the corresponding homologous viruses.

An antiserum raised against cell-propagated B/Formosa/V2367/2012 recognised the cell culture-propagated test viruses less well. Only 45 of 234 test viruses (19%) were recognised at titres within 2-fold of the titres of the antiserum for the homologous virus

and 168 viruses (72%) were recognised at titres within 4-fold of the homologous virus, but 66 viruses (28% of the total) were recognised at titres ≥ 8 -fold lower than the homologous titre of the antiserum. This antiserum was raised against B/Formosa/V2367/2012 which had become polymorphic for the egg-adaptive substitution at the HA1 197-199 glycosylation site despite isolation and propagation exclusively in tissue culture.

Phylogenetic analysis of the HA genes of representative B/Victoria lineage viruses is shown in **Figure 13** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 14**. All recently circulating viruses analysed carried HA genes that fell into genetic group 1A, the B/Brisbane/60/2008 genetic group, the vast majority falling into a genetic group defined by two HA1 amino acid substitutions compared with B/Brisbane/60/2008: I117V and N129D. **Figure 15** shows a phylogenetic tree for the NA gene and **Figure 16** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins. In the NA gene phylogeny, a small group of reassortant viruses from Africa and Europe with HA genes from the B/Yamagata lineage but NA genes from the B/Victoria lineage is shown (**Figure 15**).

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from 17 countries or administrative regions in four WHO Regions (**Table 1**): African (2: Ghana and Niger), Eastern Mediterranean (3: Egypt, Jordan and Oman), European (11: France, Germany, Iceland, Italy, Norway, Romania, Russia, Slovakia, Slovenia, Sweden and United Kingdom) and Western Pacific (1: Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 12**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 13-1 to 13-7**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 13-3 to 13-7**, 37 in total, are viruses with collection dates after 2016-01-31. The clade into which the HA gene falls is shown where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. A summary of the results for all of the post-infection ferret antisera is shown in **Table 13-8**.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in the Northern Hemisphere for 2016/2017, recognised all of the test viruses collected after 2016-01-31 at titres within 4-fold of the titre with the homologous virus, 86% within 2-fold. The ferret antiserum raised against a cell-propagated cultivar of B/Phuket/3073/2013 recognised 81% of the recently collected test viruses at titres within 4-fold, but only 43% within 2-fold of its titre with the homologous virus.

Antisera raised against other viruses from the B/Phuket/3073/2013 clade (clade 3), egg-propagated B/Wisconsin/1/2010, B/Stockholm/12/2011 and B/Hong Kong/3417/2014 and tissue culture-propagated B/Hamburg/1/2016, also recognised test viruses collected since 2016-01-31 well, recognising 100%, 92%, 100% and

100%, respectively, at titres within 4-fold of the titres of the antisera for their homologous viruses, and 92%, 35%, 100% and 83%, respectively, at titres within 2-fold of the homologous titres of the antisera.

An antiserum raised against egg-propagated B/Massachusetts/02/2012 recognised only 38% of recently collected test viruses at titres within 4-fold of the titre of the antiserum for the homologous virus, and only two viruses were recognised at titres within 2-fold of the homologous titre. Antiserum raised against the cell culture-propagated cultivar of B/Massachusetts/02/2012 recognised a third of recently collected test viruses at a titre within 4-fold of its homologous titre. Test viruses were recognised somewhat better by an antiserum raised against cell culture-propagated B/Estonia/55669/2011, another virus from the B/Massachusetts/02/2012 clade (clade 2), with all of the 37 test viruses collected after 2016-01-31 being recognised at titres within 4-fold of the titre of the antiserum for the homologous virus (78% were within 2-fold of the homologous titre). However, the homologous titre of this antiserum was low at 80 or 160 in the HI assays.

Figure 17 shows a phylogenetic analysis of the HA genes of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 18**. Reassortant viruses with HA genes from the B/Yamagata lineage but NA genes from the B/Victoria lineage are indicated. The vast majority of the viruses fall in clade 3, the B/Phuket/3073/2013 clade, with most falling into a genetic group defined by two HA1 amino acid substitutions compared with B/Phuket/3073/2013: L172Q and M251V.

Figure 19 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 20** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. Like for the HA gene, most NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3).

Antiviral Analysis.

Sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was assessed phenotypically, backed up by full NA gene sequencing, for 308 influenza B viruses with 263 being collected after 2016-01-31 (**Table 14**). All viruses showed normal inhibition (no reduced susceptibility) by both antivirals.

Conclusion.

Viruses of the B/Victoria lineage have predominated in the samples received. Most viruses of the B/Victoria lineage were antigenically similar to viruses genetically closely related to B/Brisbane/60/2008. Viruses of the B/Yamagata lineage fell predominantly within genetic clade 3, the B/Phuket/3073/2013 clade, and were recognised well by antisera raised against egg-propagated clade 3 vaccine/reference viruses.

Table 10. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2016-01-31

MONTH	B Victoria lineage				
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³
FEBRUARY					
Albania	6	6	5	1	
Bulgaria	12	11	10	1	
Cyprus	1	1	1		
Estonia	2	2	2		
Italy	4	4	3	1	
Lithuania	2	2	1	1	
Montenegro	2	2	1	1	
Oman	5	5	5		
Romania	1	1	1		
Russia	5	5	5		
Serbia	9	9	7	2	
Slovakia	5	5	5		
Slovenia	3	3	3		
Spain	2	2	2		
Sweden	2	2	1	1	
Turkey	9	9	8	1	
MARCH					
Bulgaria	8	8	8		
Czech Republic	8	8	7	1	
Egypt	2	2	1	1	
Estonia	3	3	3		
Finland	3	3	2	1	
Georgia	3	3	3		
Germany	7	7	6	1	
Iceland	1	1		1	
Italy	4	4	4		
Jordan	6	6	6		
Kazakhstan	3	3	3		
Palestine	1	1	1		
Portugal	4	3	2	1	
Romania	1	1	1		
Russia	3	3	1	2	
Slovakia	5	5	5		
Slovenia	5	5	5		
Sweden	2	2	2		
Turkey	3	3	3		
APRIL					
France	12	12	12		
Georgia	3	2	2		
Iceland	3	3	2	1	
Italy	1	1	1		
Jordan	13	13	13		
Kazakhstan	2	2	2		
Portugal	5	5	1	4	
Romania	5	5	5		
Russia	2	2	2		
Slovenia	2	2	2		
Turkey	1	1	1		
United Kingdom	3	3	3		
MAY					
France	2	2	2		
Iceland	2	2	2		
Jordan	7	7	7		
Madagascar	1	1	1		
Norway	2	2	2		
Slovenia	2	2	2		
South Africa	5	5	1		4
United Kingdom	3	3	3		
JUNE					
Hong Kong	1	1			1
Iceland	1	1	1		
Madagascar	1	1	1		
South Africa	7	7	3		4
JULY					
Egypt	1	1	1		
Ghana	2	2	2		
Hong Kong	1	1	1		
South Africa	5	4	4		
	237	233	202	22	9
33 Countries			86.7%	9.4%	3.9%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016)

Table 11-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-02-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Paris 1762/09	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Odessa 3886/10
				Egg	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK
	Passage history			Sh 539/540 /543/544 ^{1,3}	F37/11 ²	F22/12 ²	F07/11 ²	F29/13 ²	F01/13 ³	F04/13 ³	F41/13 ²	F09/13 ²	F19/11 ²
	Ferret number												
	Genetic group			1A		1A	1A	1A	1A	1A	1A	1B	1B
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	320	20	<	80	160	80	160	10	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	640	40	160	40	320	320	320	640	40	40
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	640	<	<	80	40	80	80	80	40	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	40	160	40	320	320	320	640	40	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	320	640	160	1280	1280	1280	1280	160	80
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	80	40	160	320	320	320	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E2	640	40	80	40	160	320	160	640	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	640	<	<	80	40	40	40	40	40	80
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	640	<	<	80	80	80	80	80	40	80
TEST VIRUSES													
B/Oman/6383/2015	1A	2015-11-18	SIAT1/MDCK1	1280	<	<	80	40	80	80	80	40	40
B/Rennes/2483/2015	1A	2015-11-26	MDCK1/MDCK1	640	<	<	80	40	40	80	80	40	40
B/Bretagne/2322/2015	1A	2015-11-30	MDCK1/MDCK1	1280	<	<	80	80	40	80	80	40	40
B/Champagne Ardenne/2335/2015	1A	2015-11-30	MDCK1/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Roma/11/2015	1A	2015-11-30	SIAT1/MDCK1	1280	<	<	80	80	80	80	80	40	40
B/Bretagne/2368/2015	1A	2015-12-07	MDCK1/MDCK1	640	<	<	80	40	40	80	80	40	40
B/Rennes/2548/2015	1A	2015-12-11	MDCK1/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Rennes/2547/2015	1A	2015-12-13	MDCK1/MDCK1	1280	<	<	80	40	80	80	80	40	40
B/Switzerland/15976202/2015	1A	2015-12-18	MDCK1	640	<	<	80	40	80	80	80	40	40
B/Switzerland/15983215/2015	1A	2015-12-22	MDCK1	640	<	<	80	80	40	80	80	40	40
B/Estonia/97370/2015	1A	2015-12-29	MDCK1	640	<	<	80	40	40	80	80	40	40
B/Croatia/131/2016	1A	2016-01-11	MDCK1/MDCK1	640	80	40	20	80	160	80	320	20	10
B/Turkey/TR-19/2016	1A	2016-01-15	MDCK1/MDCK1	640	<	<	80	40	40	80	80	40	40
B/Turkey/TR-30/2016	1A	2016-01-15	MDCK1/MDCK1	640	10	<	80	40	40	40	80	40	40
B/Turkey/TR-35/2016	1A	2016-01-15	MDCK1/MDCK1	640	<	<	80	40	40	80	80	40	40
B/Moldova/46.03/2016	1A	2016-01-16	MDCK1	640	<	<	80	40	40	80	80	40	40
B/Hungary/19/2016	1A	2016-01-18	MDCK2/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Nordrhein-Westfalen/2/2016	1A	2016-01-18	C2/MDCK1	640	<	<	80	40	40	80	80	40	40
B/Hungary/16/2016	1A	2016-01-19	MDCK1/MDCK1	1280	40	<	80	80	80	160	160	40	40
B/Hungary/15/2016	1A	2016-01-19	MDCK1/MDCK1	1280	40	<	40	80	80	160	160	40	40
B/Berlin/2/2016	1A	2016-01-25	C2/MDCK1	640	40	<	80	40	80	80	80	40	40
B/Thuringen/8/2016	1A	2016-01-25	C2/MDCK1	1280	40	<	80	40	80	80	80	40	40
B/Bayern/1/2016	1A	2016-01-25	C2/MDCK1	640	<	<	40	20	<	40	40	20	40
B/Bremen/1/2016	1A	2016-01-25	C2/MDCK1	640	<	<	80	40	40	80	80	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 11-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-03-02 + 2016-03-18 + 2016-03-22 + 2016-03-29 + 2016-04-13

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				B/Bris	B/Mal	B/Bris	B/Paris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Odessa	B/For
				60/08	2506/04	60/08	1762/09	636714/11	3964/12	V2367/12	81/12	514/09	3886/10	V2367/12
				Egg	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
				Ferret number	Sh 630 ^{1,3}	F37/11 ²	F22/12 ²	F07/11 ²	F29/13 ²	F01/13 ¹	F04/13 ⁴	F41/13 ²	F09/13 ²	F19/11 ²
Genetic group	1A		1A	1A	1A	1A	1A	1A	1B	1B	1A	New		
REFERENCE VIRUSES														
B/Malaysia/2506/2004	Sh = NIB-58	2004-12-06	E3/E7	1280	160	<	10	80	160	40	80	10	<	ND
B/Brisbane/60/2008		2008-08-04	E4/E5	1280	40	160	40	320	320	320	640	40	40	320
B/Paris/1762/2009		2009-02-09	C2/MDCK2	160	<	<	40	40	40	40	40	20	40	ND
B/Malta/636714/2011		2011-03-07	E4/E1	1280	40	160	40	320	320	320	640	40	40	ND
B/Johannesburg/3964/2012		2012-08-03	E1/E2	5120	320	640	160	1280	1280	1280	1280	160	80	ND
B/Formosa/V2367/2012		2012-08-06	MDCK1/MDCK3	1280	20	80	40	160	160	160	320	20	20	320
B/South Australia/81/2012		2012-11-28	E4/E2	2560	40	160	40	160	320	320	640	40	40	ND
B/Hong Kong/514/2009		2009-10-11	MDCK3	80	<	<	40	40	<	40	40	40	40	ND
B/Odessa/3886/2010		2010-03-19	C2/MDCK2	320	<	<	40	40	40	80	80	40	40	ND
TEST VIRUSES														
B/Egypt/1978/2015		2015-02-11	MDCK3	160	<	<	40	20	<	40	40	20	20	ND
B/Egypt/5366/2015		2015-07-31	MDCK3	160	<	<	40	40	40	40	80	20	20	ND
B/Sapporo/23/2015	1A	2015-09-25	MDCK1/MDCK2/MDCK1	1280	20	40	20	160	80	160	640	40	20	ND
B/Belgium/2015G0749/2015	1A	2015-11-03	MDCK3	160	<	<	40	20	<	40	40	20	20	ND
B/Oman/6335/2015	1A	2015-11-17	SIAT2/MDCK1	160	<	<	40	40	<	40	40	20	20	ND
B/Netherlands/2928/2015	1A	2015-11-23	SIAT3/MDCK1	160	<	<	40	20	<	40	40	20	20	ND
B/Estonia/97326/2015	1A	2015-12-23	MDCK2	160	<	<	40	40	<	40	40	20	20	ND
B/Latvia/01-050352/2016	1A	2015-12-29	C2/MDCK1	160	<	<	40	40	40	40	80	40	40	160
B/Catalonia/8515S/2016	1A	2016-01-07	C0/MDCK2	80	<	<	40	20	10	20	40	20	20	ND
B/Catalonia/8514S/2016	1A	2016-01-07	C0/MDCK3	80	<	<	20	10	<	20	20	10	10	ND
B/Croatia/84/2016	1A	2016-01-08	MDCK2	160	<	<	40	40	<	40	40	20	20	ND
B/Bosnia & Herzegovina/12-G/2016	1A	2016-01-11	MDCK1	640	<	<	80	40	80	80	80	80	80	ND
B/Latvia/01-062963/2016	1A	2016-01-25	C1/MDCK1	320	<	<	80	40	40	40	80	40	40	160
B/Latvia/01-074325/2016	1A	2016-01-30	C1/MDCK1	320	<	<	40	40	40	40	40	40	40	160
B/Catalonia/162038880NS/2016	1A	2016-02-03	C0/MDCK1	320	<	<	80	20	20	40	40	40	40	ND
B/Catalonia/8693S/2016	1A	2016-02-04	C0/MDCK1	320	<	<	80	80	80	160	160	80	80	ND
B/Salfeet/2S/2016	1A	2016-03-20	MDCK1	160	<	<	40	40	40	40	40	40	40	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-04-22 + 2016-04-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				B/Bris	B/Mal	B/Bris	B/Paris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Odessa	B/Iceland	B/Ireland	B/Nord-West
				60/08	2506/04	60/08	1762/09	636714/11	3964/12	V2367/12	81/12	514/09	3886/10	56/15	3154/16	1/16
				Egg	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK	MDCK	MDCK
	Passage history			Sh 630 ^{1,3}	F37/11 ²	F22/12 ²	F07/11 ²	F29/13 ²	F01/13 ⁴	F04/13 ⁴	F41/13 ²	F09/13 ²	F19/11 ²	F14/16 ²	F15/16 ²	F16/16 ²
	Ferret number			1A		1A	1A	1A	1A	1A	1A	1B	1B	1A	1A	1A
	Genetic group													New	New	New
REFERENCE VIRUSES																
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	320	20	<	80	160	40	160	40	<	<	<	<
B/Brisbane/60/2008	Sh = NIB-58	2008-08-04	E4/E4	2560	40	160	40	320	320	160	640	80	40	80	40	40
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	160	<	<	80	40	40	40	80	80	80	80	40	80
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	40	80	40	320	320	160	640	80	20	80	40	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	160	320	80	640	640	640	1280	160	40	80	80	80
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	40	80	40	160	160	160	320	80	20	80	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	40	80	40	160	320	160	640	80	40	80	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	40	<	<	20	10	<	10	20	40	40	40	<	20
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK1	320	<	<	40	40	40	40	80	80	40	40	20	40
TEST VIRUSES																
B/Iceland/56/2015	1A	2015-12-30	MDCK0/MDCK1	160	<	<	40	40	40	40	80	80	40	80	40	80
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	320	<	<	80	40	40	80	80	80	40	160	40	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	320	<	<	80	80	80	80	160	160	80	160	40	80
B/Cyprus/F82/2016	1A	2016-01-29	MDCK2	160	<	<	80	40	40	40	80	80	40	40	80	80
B/Cyprus/F85/2016	1A	2016-01-28	MDCK1	2560	40	<	40	80	160	80	320	80	20	20	40	40
B/Cyprus/F84/2016	1A	2016-01-28	MDCK1	640	10	<	40	80	80	80	160	80	40	40	80	40
B/Serbia-NS/1475/2016	1A	2016-02-04	MDCK1	320	<	<	80	40	40	40	80	80	40	40	160	80
B/Cyprus/F174/2016	1A	2016-02-05	MDCK2	320	<	<	80	40	40	40	80	80	40	40	80	80
B/Serbia-NS/1499/2016	1A	2016-02-08	MDCK1	320	<	<	40	40	40	80	80	80	40	40	80	80
B/Serbia-NS/1500/2016	1A	2016-02-08	MDCK2	160	<	<	40	40	40	40	80	80	40	40	80	40
B/Serbia-NS/1515/2016	1A	2016-02-10	MDCK1	320	<	<	40	40	40	80	80	80	40	40	80	80
B/Serbia-NS/1527/2016	1A	2016-02-11	MDCK2	160	<	<	40	40	40	40	80	80	40	40	80	80
B/Serbia-NS/1540/2016	1A	2016-02-17	MDCK1	320	<	<	40	40	40	40	80	80	40	40	80	80
B/Serbia-NS/1548/2016	1A	2016-02-17	MDCK1	80	<	<	20	10	20	20	40	20	10	<	40	20
B/Serbia-NS/1568/2016	1A	2016-02-22	MDCK1	320	<	<	40	40	40	40	80	80	40	40	80	40
B/Serbia-NS/1569/2016	1A	2016-02-22	MDCK1	160	<	<	80	40	40	40	80	80	40	40	80	80
B/Sachsen/26/2016	1A	2016-03-07	C1/MDCK1	40	<	<	20	10	<	20	20	40	10	<	40	40
B/Mecklenburg-Vorpommern/4/2016	1A	2016-03-07	C1/MDCK1	160	<	<	40	40	20	40	80	80	40	40	80	40
B/Bayern/14/2016	1A	2016-03-09	C1/MDCK1	160	<	<	40	40	40	40	80	80	40	40	80	80
B/Baden-Wuerttemberg/56/2016	1A	2016-03-10	C1/MDCK1	160	<	<	40	40	20	40	80	80	40	40	80	40
B/Bulgaria/849/2016	1A	2016-03-12	SIAT2/SIAT1	160	<	<	ND	40	40	40	80	80	ND	40	80	80
B/Berlin/54/2016	1A	2016-03-14	C1/MDCK1	320	<	<	80	40	40	40	80	80	40	40	80	80
B/Hessen/16/2016	1A	2016-03-14	C1/MDCK1	320	<	<	80	40	40	40	80	80	40	40	80	80
B/Thuringen/62/2016	1A	2016-03-14	C1/MDCK1	80	<	<	20	10	<	40	10	40	40	20	40	20
B/Bulgaria/881/2016	1A	2016-03-16	SIAT2/SIAT1	160	<	<	ND	20	20	40	80	80	ND	40	80	80
B/Bulgaria/905/2016	1A	2016-03-18	SIAT1/SIAT1	160	<	<	ND	20	20	40	40	40	ND	20	40	40
B/Bulgaria/896/2016	1A	2016-03-21	SIAT1/SIAT1	320	<	<	ND	40	20	40	80	80	ND	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
NH 2015-16^a
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 11-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-05-05 + 2016-05-10 + 2016-05-17

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
	Passage history			Ferret number									
	Genetic group			Sh 630 ^{1,3}	F37/11 ²	F22/12 ²	F29/13 ²	F01/13 ⁴	F04/13 ⁴	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
				1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	320	20	80	160	40	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	80	160	320	320	320	640	160	80	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	40	80	320	320	160	640	80	40	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	640	1280	1280	1280	1280	320	320	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	20	40	160	160	160	320	80	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	40	80	160	320	320	640	80	80	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	160	<	<	40	20	40	40	80	80	80
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	160	<	<	20	<	40	40	80	80	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	320	<	<	40	40	80	80	80	160	80
TEST VIRUSES													
B/Oman/7822/2015	1A	2015-12-29	SIAT2/MDCK1	160	<	<	10	<	20	20	40	20	40
B/Egypt/1764/2016	1A	2016-01-24	MDCK1	320	<	<	40	40	80	80	80	80	40
B/Niger/2492/2014		2014-01-28	MDCK1	160	<	<	20	<	40	40	80	80	40
B/Bulgaria/238/2016	1A	2016-02-03	SIAT1/MDCK1	320	<	<	40	40	80	80	80	80	80
B/Bulgaria/261/2016	1A	2016-02-03	SIAT1/MDCK1	160	<	<	20	20	40	80	80	80	80
B/Bulgaria/343/2016	1A	2016-02-06	SIAT1/MDCK1	320	<	<	40	40	80	80	80	80	80
B/Oman/1737/2016		2016-02-09	MDCK2/MDCK1	160	<	<	10	<	20	20	40	80	40
B/Bulgaria/350/2016	1A	2016-02-09	SIAT2/MDCK1	320	<	<	40	40	80	80	80	160	80
B/Oman/1907/2016		2016-02-11	MDCK2/MDCK1	160	<	<	20	20	40	40	80	80	40
B/Bulgaria/489/2016	1A	2016-02-12	SIAT2/MDCK1	160	<	<	20	<	40	40	80	160	80
B/Oman/1830/2016	1A	2016-02-13	MDCK2/MDCK1	160	<	<	20	20	40	80	80	80	40
B/Oman/1950/2016	1A	2016-02-16	MDCK2/MDCK1	320	<	<	10	<	20	40	80	40	40
B/Bulgaria/669/2016	1A	2016-02-19	SIAT2/MDCK1	320	<	<	40	40	80	80	80	160	80
B/Bulgaria/692/2016	1A	2016-02-22	SIAT2/MDCK1	320	<	<	40	40	40	80	80	80	80
B/Montenegro/141/2016	1A	2016-02-23	MDCK1/MDCK1	160	<	<	20	20	40	80	80	80	40
B/Montenegro/149/2016	1A	2016-02-24	MDCK1/MDCK1	320	<	<	20	20	40	80	80	40	20
B/Bulgaria/757/2016	1A	2016-02-25	SIAT2/MDCK1	320	<	<	40	80	80	80	160	160	80
B/Bulgaria/790/2016	1A	2016-02-29	SIAT2/MDCK1	160	<	<	20	20	40	80	80	80	40
B/Bulgaria/774/2016	1A	2016-02-29	SIAT2/MDCK1	160	<	<	20	40	40	80	80	160	80
B/Bulgaria/792/2016	1A	2016-03-01	SIAT2/MDCK1	160	<	<	20	<	40	40	40	80	40
B/Bulgaria/830/2016	1A	2016-03-09	SIAT2/MDCK1	160	<	<	10	<	40	40	80	80	40
B/Bulgaria/831/2016	1A	2016-03-10	SIAT2/MDCK1	320	<	<	40	80	80	80	80	160	80
B/Bulgaria/841/2016	1A	2016-03-10	SIAT2/MDCK1	320	<	<	10	<	40	40	80	80	40
B/Bucuresti/195675/2016	1A	2016-04-05	MDCK1/MDCK1	160	<	<	20	20	40	80	40	40	40
B/Mures/195946/2016	1A	2016-04-07	MDCK1/MDCK1	160	<	<	20	<	20	40	80	40	40
B/Alba/195948/2016	1A	2016-04-07	MDCK1/MDCK1	160	<	<	20	20	40	40	80	40	40
B/Bucuresti/196001/2016	1A	2016-04-08	MDCK1/MDCK1	160	<	<	20	20	40	40	80	40	40
B/Mures/196205/2016	1A	2016-04-10	MDCK1/MDCK1	160	<	<	20	20	20	40	80	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 11-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-05-25 + 2016-06-03

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West
				60/08	2506/04	60/08	636714/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
				Sh 630 ^{*1,3}	F37/11 ^{*2}	F22/12 ^{*2}	F29/13 ^{*2}	F01/13 ^{*2}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}
				1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	320	10	40	80	40	80	10	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	40	80	160	160	160	640	40	20	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	80	640	320	320	640	80	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	320	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	40	80	160	160	320	320	80	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	40	80	160	160	160	640	80	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	80	<	<	40	20	40	40	80	80	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	160	<	<	40	20	40	80	40	80	40
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	160	<	<	40	40	80	80	80	80	40
TEST VIRUSES													
B/Oman/7822/2015	1A	2015-12-29	SIAT2/MDCK2	160	<	<	40	20	80	80	80	80	80
B/Bulgaria/177/2016	1A	2016-02-01	SIAT1/MDCK1	160	<	<	40	40	80	80	40	80	40
B/Bucuresti/679-C7895/2016	1A	2016-02-11	MDCK1/MDCK1	160	<	<	40	20	80	80	40	40	40
B/Oman/2069/2016	1A	2016-02-16	MDCK2	160	<	<	40	40	40	80	40	40	40
B/Jordan/4338/2016	1A	2016-03-02	MDCK1	160	<	<	20	10	40	40	40	40	40
B/Georgia/915/2016	1A	2016-03-11	C2/MDCK1	160	<	<	20	40	80	80	80	80	80
B/Jordan/4380/2016	1A	2016-03-13	MDCK1	160	<	<	40	20	80	80	80	80	40
B/Georgia/933/2016	1A	2016-03-17	C2/MDCK1	160	<	<	20	20	80	40	40	40	40
B/Jordan/4441/2016	1A	2016-03-22	MDCK1	160	<	<	40	20	80	80	40	80	40
B/Jordan/20358/2016	1A	2016-03-23	MDCK1	160	<	<	40	20	80	80	40	80	40
B/Bucuresti/194978/2016	1A	2016-03-24	MDCK1/MDCK1	80	<	<	20	10	40	40	40	40	20
B/Georgia/1017/2016	1A	2016-03-28	C2/MDCK1	160	<	<	20	20	40	40	40	40	40
B/Jordan/30105/2016	1A	2016-04-02	MDCK1	160	<	<	40	20	40	80	40	40	40
B/Jordan/4203/2016	1A	2016-04-04	MDCK1	160	<	<	40	20	80	80	80	80	40
B/Georgia/1089/2016	1A	2016-04-19	MDCK1	160	<	<	40	40	80	80	160	80	80
B/Georgia/1095/2016	1A	2016-04-21	MDCK1	640	<	<	80	80	160	160	80	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
 NH 2015-16[#]
 SH 2016
 NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-06-15

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West
				60/08	2506/04	60/08	636714/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
				Ferret number	Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F37/11 ²	F22/12 ²	F29/13 ²	F01/13 ²	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²
Genetic group	1A	1A	1A	1A	1A	1A	1A	1B	1A	1A			
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	640	20	80	160	80	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	5120	80	160	640	640	320	1280	160	40	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	80	640	320	320	640	80	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	640	1280	1280	1280	1280	320	320	320
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	80	320	320	320	640	80	80	40
B/South Australia/81/2012	1A	2012-11-28	E4/E2	5120	80	160	320	320	320	1280	80	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	<	40	20	40	80	80	80	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	5120	<	<	40	20	80	80	80	80	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	5120	<	<	40	40	80	80	80	80	80
TEST VIRUSES													
B/St. Petersburg/89/2016	1A	2016-02-01	MDCK3/MDCK1	5120	<	<	20	20	40	80	80	80	40
B/St. Petersburg/171/2016	1A	2016-02-01	MDCK3/MDCK1	5120	<	<	40	40	80	80	80	80	80
B/Nitra/125/2016	1A	2016-02-09	MDCK1/MDCK1	5120	<	<	40	40	80	80	80	80	80
B/Levice/134/2016	1A	2016-02-11	MDCKx/MDCK1	5120	<	<	40	40	80	80	80	80	40
B/Kosice/150/2016	1A	2016-02-11	MDCKx/MDCK1	5120	<	<	40	20	40	80	80	80	40
B/St. Petersburg/71/2016	1A	2016-02-15	MDCK2/MDCK1	2560	<	<	20	20	80	80	80	80	40
B/St. Petersburg/69/2016	1A	2016-02-15	MDCK2/MDCK1	5120	<	<	20	40	80	80	80	80	40
B/Novosibirsk/50/2016	1A	2016-02-20	MDCK1/MDCK2	5120	<	<	20	20	80	80	80	80	80
B/Trnava/192/2016	1A	2016-02-22	MDCKx/MDCK1	5120	<	<	40	40	80	80	80	80	80
B/Bratislava/226/2016	1A	2016-02-29	MDCKx/MDCK1	2560	<	<	20	20	40	80	40	40	40
B/Egypt/98/2016	1A	2016-03-01	MDCK1	5120	<	<	40	40	80	80	160	80	80
B/Topolcany/240/2016	1A	2016-03-02	MDCKx/MDCK1	5120	<	<	40	40	80	80	80	80	40
B/Banska Bystrica/306/2016	1A	2016-03-04	MDCKx/MDCK1	5120	<	<	40	40	80	80	80	80	80
B/Egypt/105/2016	1A	2016-03-06	E2/E1	2560	80	40	320	320	160	640	80	40	20
B/Trencin/266/2016	1A	2016-03-10	MDCKx/MDCK1	5120	<	<	40	40	80	80	80	80	40
B/Kaliningrad/83/2016	1A	2016-03-14	MDCK1/MDCK1	5120	40	80	320	320	320	640	80	40	40
B/Piestany/296/2016	1A	2016-03-14	MDCKx/MDCK1	5120	<	<	40	40	80	80	80	80	80
B/Banska Bystrica/336/2016	1A	2016-03-15	MDCKx/MDCK1	5120	<	<	20	20	80	80	80	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17[#] B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 11-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-06-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Mal 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
	Passage history												
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F37/11 ²	F26/13 ²	F29/13 ²	F04/16 ²	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	640	80	80	40	80	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	640	320	320	320	1280	80	40	40
B/Mal/636714/2011	1A	2011-03-07	E4/E1	2560	80	640	640	320	320	1280	80	40	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	640	160	160	320	640	80	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E2	5120	80	80	320	320	320	1280	80	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	40	40	40	80	80	80	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	5120	10	80	40	40	40	80	80	80	40
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	2560	<	80	40	80	80	80	80	80	80
TEST VIRUSES													
B/Padova/2/2016	1A	2016-02-02	MDCK3/MDCK1	2560	10	160	20	80	40	80	40	80	40
B/Firenze/2/2016	1A	2016-02-02	MDCK2/MDCK1	2560	<	40	10	40	20	40	40	40	20
B/Lithuania/3663/2016	1A	2016-02-02	MDCKx/MDCK1	2560	<	160	20	40	40	80	40	80	40
B/Lithuania/4516/2016	1A	2016-02-07	MDCKx/MDCK1	2560	<	80	20	40	20	40	40	40	20
B/Roma/6/2016	1A	2016-02-08	Cx/MDCK1	5120	20	320	40	80	80	160	80	80	40
B/Parma/114/2016	1A	2016-02-24	MDCK1/MDCK1	2560	10	160	20	80	40	80	40	80	40
B/Palermo/4/2016	1A	2016-03-02	MDCK3/MDCK1	2560	10	160	20	80	40	80	40	80	40
B/Milano/145/2016	1A	2016-03-07	MDCK1/MDCK1	5120	10	160	20	80	40	80	40	40	40
B/Friuli-Venezia Giulia/22/2016	1A	2016-03-11	MDCK1/MDCK1	5120	10	80	40	80	40	80	80	80	40
B/Palermo/6/2016	1A	2016-03-11	MDCK3/MDCK1	5120	10	80	40	80	80	80	80	80	40
B/Czech Republic/111/2016	1A	2016-03-15	C2/MDCK1	5120	20	80	20	80	40	80	40	80	40
B/Czech Republic/116/2016	1A	2016-03-15	C2/MDCK1	2560	10	80	<	20	20	20	40	40	20
B/Czech Republic/115/2016	1A	2016-03-17	C2/MDCK1	5120	10	160	40	80	80	160	80	80	40
B/Iceland/39/2016	1A	2016-03-17	MDCK1/MDCK1	2560	10	40	20	40	40	40	40	40	20
B/Czech Republic/117/2016	1A	2016-03-18	C2/MDCK1	2560	<	80	20	40	40	80	40	40	40
B/Czech Republic/118/2016	1A	2016-03-21	C3/MDCK1	2560	40	80	40	80	80	160	80	80	40
B/Czech Republic/119/2016	1A	2016-03-21	C2/MDCK1	2560	<	80	20	40	40	80	40	80	40
B/Lisboa/6/2016	1A	2016-03-22	SIAT1/MDCK1	2560	10	160	40	40	40	80	40	80	40
B/Lisboa/8/2016	1A	2016-03-23	SIAT1/MDCK1	2560	<	160	20	40	40	80	40	40	20
B/Lisboa/9/2016	1A	2016-03-23	SIAT1/MDCK1	2560	<	160	40	40	40	80	40	80	40
B/Czech Republic/120/2016	1A	2016-03-24	C2/MDCK1	2560	20	80	20	80	40	80	40	80	40
B/Czech Republic/121/2016	1A	2016-03-29	C2/MDCK1	2560	10	80	20	80	40	80	40	80	40
B/Lisboa/14/2016	1A	2016-04-06	SIAT2/MDCK1	1280	<	80	20	40	20	80	40	40	20
B/Perugia/84/2016	1A	2016-04-11	MDCK1/MDCK1	2560	10	80	40	80	80	80	80	80	40
B/Lisboa/16/2016	1A	2016-04-11	SIAT2/MDCK1	2560	10	160	20	40	40	80	40	40	20
B/Lisboa/15/2016	1A	2016-04-11	SIAT2/MDCK1	2560	<	160	20	40	40	80	40	40	20
B/Iceland/50/2016	1A	2016-04-12	MDCK1/MDCK1	2560	<	80	40	80	40	80	40	80	40
B/Iceland/52/2016	1A	2016-04-15	MDCK1/MDCK1	2560	10	80	20	40	40	40	40	80	40
B/Lisboa/18/2016	1A	2016-04-18	SIAT1/MDCK1	2560	<	40	10	40	20	40	40	40	20
B/Iceland/55/2016	1A	2016-04-26	MDCK1/MDCK1	2560	<	80	20	40	20	40	40	40	20
B/Iceland/57/2016	1A	2016-05-02	MDCK1/MDCK1	2560	10	160	40	80	40	80	40	80	40
B/Iceland/60/2016	1A	2016-05-11	MDCK1/MDCK1	2560	10	80	20	40	40	40	40	80	40
B/Iceland/10292/2016	1A	2016-06-05	MDCK1	2560	<	160	20	40	40	80	40	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20Vaccine
NH 2015-16^{*}
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 11-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-06-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West
				60/08	2506/04	60/08	636714/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
	Passage history												
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F37/11 ²	F26/13 ²	F29/13 ²	F04/16 ¹	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	640	40	80	80	40	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	640	320	320	160	1280	80	40	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	640	640	320	320	1280	80	40	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	640	320	160	320	640	80	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	80	160	320	320	320	640	80	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	40	40	40	80	80	40	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	5120	<	80	40	40	40	80	40	80	40
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	5120	<	80	40	40	80	80	80	80	40
TEST VIRUSES													
B/Stockholm/11/2016	1A	2016-02-08	MDCK1/MDCK1	5120	10	160	80	80	80	160	40	20	40
B/Stockholm/12/2016	1A	2016-02-08	MDCK0/MDCK1	5120	10	40	20	20	40	40	40	40	40
B/Slovenia/701/2016	1A	2016-02-09	MDCK1/MDCK1	5120	<	40	40	40	40	80	40	40	40
B/Slovenia/914/2016	1A	2016-02-21	MDCKx/MDCK1	5120	<	40	40	40	40	80	80	80	40
B/Slovenia/1036/2016	1A	2016-02-26	MDCK1/MDCK1	5120	<	40	40	40	80	80	80	80	40
B/Stockholm/15/2016	1A	2016-03-01	MDCK0/MDCK1	5120	<	40	40	40	40	80	40	40	80
B/Finland/588/2016	1A	2016-03-01	MDCK1/MDCK1	5120	<	80	40	40	40	80	40	20	40
B/Slovenia/1186/2016	1A	2016-03-09	MDCK1/MDCK1	5120	<	80	80	40	80	80	80	80	40
B/Slovenia/1171/2016	1A	2016-03-09	MDCKx/MDCK1	5120	<	80	40	40	40	80	40	40	40
B/Istanbul/1913250/2016	1A	2016-03-10	MDCK 1/MDCK1	5120	10	40	20	<	40	40	40	40	40
B/Finland/592/2016	1A	2016-03-11	MDCK1/MDCK1	5120	10	80	40	40	80	80	80	40	40
B/Slovenia/1228/2016	1A	2016-03-14	MDCK1/MDCK1	5120	<	40	40	40	40	80	80	40	40
B/Slovenia/1237/2016	1A	2016-03-15	MDCKx/MDCK1	5120	<	40	40	40	40	80	40	40	40
B/Moscow/7/2016	1A	2016-03-15	MDCK1/MDCK1	5120	<	40	20	20	40	40	40	20	40
B/Stockholm/16/2016	1A	2016-03-16	MDCK0/MDCK2	5120	10	80	40	40	40	80	40	40	80
B/Slovenia/1311/2016	1A	2016-03-21	MDCKx/MDCK1	5120	<	80	80	80	80	160	80	80	40
B/Finland/623/2016	1A	2016-03-21	MDCK1/MDCK1	5120	10	80	40	80	80	80	80	40	80
B/Moscow/16/2016	1A	2016-03-30	MDCK1/MDCK1	5120	<	40	20	20	40	40	40	20	40
B/Lisboa/13/2016	1A	2016-04-06	SIAT1/MDCK2	5120	<	80	80	40	40	80	80	80	40
B/Slovenia/1459/2016	1A	2016-04-06	MDCKx/MDCK1	5120	<	80	80	80	80	160	80	80	40
B/Moscow/48/2016	1A	2016-04-14	MDCK1/MDCK1	5120	10	40	40	20	40	40	40	40	40
B/Slovenia/1523/2016	1A	2016-04-16	MDCK1/MDCK1	5120	<	80	80	80	80	160	80	80	40
B/Moscow/51/2016	1A	2016-04-18	MDCK1/MDCK1	5120	10	40	20	<	40	40	40	40	40
B/Slovenia/1604/2016	1A	2016-05-01	MDCK1/MDCK1	2560	<	80	40	40	40	80	80	40	40
B/Slovenia/1601/2016	1A	2016-05-01	MDCK1/MDCK1	5120	<	80	80	40	80	80	80	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
 NH 2015-16[#]
 SH 2016
 NH 2016-17

B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 11-9. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-07-05 + 2016-07-12 + 2016-07-19

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West	
				60/08	2506/04	60/08	636714/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16	
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F37/11 ²	F26/13 ²	F29/13 ²	F01/13 ¹	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²	
Genetic group	1A	1A	1A	1A	1A	1A	1B	1A	1A					
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	640	40	80	160	40	160	20	<	<	
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	640	320	320	1280	160	80	40	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	640	320	320	640	80	40	20	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	1280	1280	320	320	320	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	640	320	160	160	640	80	40	20	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	80	320	320	320	320	1280	80	80	40	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	40	40	40	80	160	80	40	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	5120	10	80	40	40	80	80	80	80	40	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	5120	10	80	80	40	80	160	160	80	80	
TEST VIRUSES														
B/Albania/6382/2016	1A	2016-02-01	MDCK2/MDCK1	5120	40	160	80	80	80	160	80	40	40	
B/Albania/6384/2016	1A	2016-02-02	MDCK1/MDCK1	5120	20	160	80	80	80	160	160	160	80	
B/Albania/6419/2016	1A	2016-02-03	MDCK1/MDCK1	5120	20	160	80	160	160	160	160	160	80	
B/Albania/6423/2016	1A	2016-02-03	MDCK1/MDCK1	5120	40	320	160	160	160	640	80	40	20	
B/Albania/6470/2016	1A	2016-02-08	MDCK2/MDCK1	5120	10	160	80	80	80	160	80	80	40	
B/Albania/6483/2016	1A	2016-02-10	MDCK2/MDCK1	2560	10	80	80	40	80	160	80	80	40	
B/Kazakhstan/4682/2016	1A	2016-03-22	MDCK2	5120	<	80	40	40	80	160	80	80	40	
B/Kazakhstan/4767/2016	1A	2016-03-23	MDCK1	5120	10	160	80	80	80	160	160	160	80	
B/Kazakhstan/4720/2016	1A	2016-03-29	MDCK1	5120	10	80	80	80	80	160	80	80	40	
B/Kazakhstan/4768/2016	1A	2016-04-04	MDCK1	5120	10	160	80	80	80	160	80	80	40	
B/Kazakhstan/4815/2016	1A	2016-04-21	MDCK1/MDCK1	5120	10	80	80	40	80	160	80	80	40	
B/South Africa/R2537/2016	1A	2016-05-11	MDCK1/MDCK1	1280	<	80	40	40	40	80	80	40	<	
B/South Africa/R2895/2016	1A	2016-05-23	MDCK1/MDCK1	2560	10	80	40	40	40	160	80	40	40	
B/South Africa/R2943/2016	1A	2016-05-23	MDCK1/MDCK1	5120	<	80	40	40	40	80	80	40	<	
B/Norway/3678/2016	1A	2016-05-24	MDCK1/MDCK1	5120	10	80	80	80	80	160	160	160	80	
B/Norway/3701/2016	1A	2016-05-26	MDCK1/MDCK1	5120	20	80	80	40	80	160	80	80	80	
B/South Africa/R3037/2016	1A	2016-05-26	MDCK1/MDCK1	5120	<	80	80	40	80	160	80	80	<	
B/South Africa/R3073/2016	1A	2016-05-26	MDCK1/MDCK1	5120	<	80	40	40	80	160	80	80	<	
B/Tsaralalana/2412/2016	1A	2016-05-30	MDCK1	2560	<	80	40	40	40	80	80	40	40	
B/South Africa/R3279/2016	1A	2016-06-01	MDCK1/MDCK1	5120	<	80	80	80	80	160	80	80	<	
B/Manjakaray/2477/2016	1A	2016-06-01	MDCK1	2560	20	80	40	40	80	80	80	80	40	
B/South Africa/R3225/2016	1A	2016-06-02	MDCK1/MDCK1	5120	<	80	40	40	80	160	80	80	<	
B/South Africa/R3310/2016	1A	2016-06-02	MDCK1/MDCK1	5120	<	80	80	40	80	160	80	80	<	
B/South Africa/R3352/2016	1A	2016-06-04	MDCK1/MDCK1	5120	<	40	<	<	40	80	80	40	<	
B/South Africa/R3406/2016	1A	2016-06-06	MDCK1/MDCK1	5120	<	80	80	80	80	160	160	160	80	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-10. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-07-27

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West
				60/08	2506/04	60/08	636714/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
Passage history													
Ferret number				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F37/11 ²	F26/13 ²	F29/13 ²	F01/13 ²	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
Genetic group				1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	640	80	160	160	80	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	320	320	160	1280	80	40	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	320	320	160	640	80	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	20	640	320	320	160	640	80	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	5120	80	640	320	640	320	1280	160	80	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	40	40	20	80	80	80	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	80	20	20	40	40	40	40	20
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	2560	<	80	40	20	40	80	40	80	40
TEST VIRUSES													
B/Estonia/98464/2016	1A	2016-02-12	MDCK1/MDCK1	5120	40	80	80	80	80	160	80	80	40
B/Estonia/98655/2016	1A	2016-02-19	MDCK2/MDCK1	2560	<	80	80	40	40	80	80	80	40
B/Konya/1183/2016	1A	2016-02-26	MDCK1/MDCK1	2560	<	40	40	40	40	80	40	80	40
B/Izmir/1197/2016	1A	2016-02-26	MDCK1/MDCK1	2560	<	40	40	40	40	80	40	80	40
B/Antalya/1212/2016	1A	2016-03-03	MDCK1/MDCK1	5120	<	80	40	40	40	80	40	80	40
B/Estonia/98924/2016	1A	2016-03-04	MDCK1/MDCK1	2560	<	80	40	40	40	80	40	80	40
B/Estonia/99079/2016	1A	2016-03-11	MDCK1/MDCK1	5120	<	80	40	40	40	80	40	80	40
B/Estonia/99128/2016	1A	2016-03-14	MDCK1/MDCK1	5120	<	80	40	40	40	80	80	80	40
B/Ankara/4326/2016	1A	2016-03-16	SIAT1/MDCK1	2560	<	40	40	40	40	80	40	40	40
B/Denizli/4616/2016	1A	2016-04-13	SIAT1/MDCK2	2560	<	40	40	40	20	80	40	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-11. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-08-03 + 2016-08-16

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West
				60/08	2506/04	60/08	636714/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
Passage history													
Ferret number				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F41/14 ²	F26/13 ²	F29/13 ²	F01/13 ²	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
Genetic group				1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	320	80	80	160	40	160	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	320	320	160	160	640	80	20	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	320	320	160	640	80	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	640	1280	320	80	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	640	320	160	160	640	80	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	80	320	320	320	160	640	80	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	40	40	40	20	80	80	40	20
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	40	20	20	20	40	40	40	20
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	2560	<	40	20	20	20	40	40	40	40
TEST VIRUSES													
B/Ankara/3103/2016	1A	2016-02-04	SIAT1/MDCK1	2560	<	40	20	40	40	80	40	40	40
B/Izmir/1091/2016	1A	2016-02-05	SIAT1/MDCK1	1280	<	20	10	<	10	20	10	20	20
B/Ankara/1101/2016	1A	2016-02-05	SIAT1/MDCK1	5120	10	40	40	40	40	80	40	40	40
B/Denizli/3151/2016	1A	2016-02-05	MDCK1/MDCK1	2560	<	40	20	20	20	40	40	40	20
B/Ankara/3346/2016	1A	2016-02-09	SIAT1/MDCK1	2560	<	80	40	40	40	80	80	80	40
B/Isparta/3518/2016	1A	2016-02-12	MDCK1/MDCK1	2560	<	40	40	40	40	80	80	40	20
B/Ankara/3635/2016	1A	2016-02-15	SIAT1/MDCK2	2560	10	40	40	20	20	80	40	40	20
B/Ghana/DILI-16-0739/2016	1A	2016-07-25	Cx/MDCK1	5120	10	80	40	40	40	80	80	80	40
B/Ghana/DILI-16-0740/2016	1A	2016-07-25	Cx/MDCK1	5120	20	80	40	40	40	80	80	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-12. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-08-24

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West	
				60/08	2506/04	60/08	636714/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16	
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F41/14 ²	F26/13 ²	F29/13 ²	F01/13 ²	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²	
Genetic group	1A	1A	1A	1A	1A	1A	1B	1A	1A					
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E6	5120	320	80	160	160	40	160	20	<	<	
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	5120	160	640	640	320	320	1280	80	40	20	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	640	320	160	640	80	40	20	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	640	1280	320	160	160	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	640	320	160	160	320	80	40	20	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	5120	80	640	320	320	160	1280	80	40	20	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	40	20	20	80	80	40	40	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	80	20	20	20	40	80	40	40	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	5120	<	40	20	20	80	80	80	80	40	
TEST VIRUSES														
B/England/925/2016		2016-04-18	SIAT1/MDCK1	5120	<	80	20	40	40	80	80	80	40	
B/England/934/2016		2016-04-18	SIAT1/MDCK1	2560	<	40	20	20	40	40	80	40	40	
B/England/953/2016		2016-04-24	SIAT2/MDCK1	2560	<	40	20	20	20	40	80	40	40	
B/England/926/2016		2016-05-02	SIAT1/MDCK1	5120	<	80	40	40	40	80	80	80	40	
B/England/939/2016		2016-05-12	SIAT1/MDCK1	2560	<	80	40	40	40	80	80	80	40	
B/England/942/2016		2016-05-29	SIAT2/MDCK1	2560	<	80	40	40	40	80	80	40	40	
B/South Africa/R4546/2016	1A	2016-07-05	MDCK1/MDCK1	2560	<	80	40	40	40	80	80	80	40	
B/South Africa/R4658/2016	1A	2016-07-05	MDCK1/MDCK1	2560	<	80	40	40	40	80	80	80	40	
B/South Africa/R4607/2016	1A	2016-07-07	MDCK1/MDCK1	2560	<	80	40	40	40	80	80	80	40	
B/South Africa/R4917/2016		2016-07-14	MDCK1	2560	<	80	40	40	40	80	80	80	40	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-13. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-08-31

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK
	Passage history												
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F41/14 ²	F26/13 ²	F29/13 ²	F01/13 ²	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	160	40	80	160	40	160	10	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	320	160	160	80	1280	80	20	10
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	640	320	160	640	80	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	160	1280	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	320	160	160	160	320	80	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	80	320	160	160	160	1280	80	20	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	10	40	40	20	40	80	80	40	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	5120	40	80	40	40	80	40	80	80	40
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	2560	20	40	20	20	40	80	80	40	40
TEST VIRUSES													
B/Paris/2301/2016		2016-04-11	MDCK1/MDCK1	5120	20	80	40	40	40	20	80	80	40
B/Paris/2303/2016		2016-04-11	MDCK1/MDCK1	5120	20	80	40	40	40	20	80	80	40
B/Pays de Loire/2307/2016		2016-04-11	MDCK1/MDCK1	2560	20	80	40	40	40	20	80	40	40
B/Bretagne/2334/2016		2016-04-12	MDCK1/MDCK1	2560	<	40	20	20	40	10	80	80	40
B/Lorraine/2330/2016		2016-04-13	MDCK1/MDCK1	2560	20	80	40	40	40	20	80	80	40
B/Basse Normandie/2333/2016		2016-04-14	MDCK1/MDCK1	5120	20	80	80	40	40	20	80	80	80
B/Paris/2351/2016		2016-04-16	MDCK1/MDCK1	5120	10	80	40	40	40	20	80	80	40
B/Paris/2361/2016		2016-04-19	MDCK1/MDCK1	5120	<	80	40	40	40	20	80	80	40
B/Paris/2379/2016		2016-04-25	MDCK1/MDCK1	5120	20	80	40	40	40	20	80	80	40
B/South Africa/R3988/2016		2016-06-20	MDCK2	2560	10	80	80	40	40	20	80	40	40
B/South Africa/R4256/2016		2016-06-27	MDCK2	2560	<	80	20	20	40	10	80	40	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-14. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-09-13

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				B/Bris	B/Mal	B/Bris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Ireland	B/Nord-West	
				60/08	2506/04	60/08	6367/14/11	3964/12	V2367/12	81/12	514/09	3154/16	1/16	
				Egg	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ⁴¹	F41/14 ²	F26/13 ²	F29/13 ²	F01/13 ²	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²	
Passage history														
Ferret number														
Genetic group				1A		1A	1A	1A	1A	1A	1B	1A	1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	320	80	80	80	40	80	10	<	<	
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	320	320	160	640	80	40	20	
B/Malta/6367/14/2011	1A	2011-03-07	E4/E1	2560	80	640	640	320	160	320	40	40	20	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	1280	640	160	160	160	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	40	640	160	160	160	80	40	40	20	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	80	640	320	160	160	640	40	40	20	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	10	80	40	20	40	40	40	80	40	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	20	80	40	20	40	40	40	80	40	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK5	5120	20	160	40	40	40	40	40	80	80	
TEST VIRUSES														
B/Florida/78/2015		2015-11-09	C1/MDCK1	5120	40	160	40	40	80	20	40	80	80	
B/Florida/78/2015		2015-11-09	E4/E1	5120	20	320	80	80	40	40	80	80	80	
B/Jordan/20407/2016		2016-03-27	SIAT1	5120	20	160	20	20	40	20	40	80	40	
B/Jordan/IIIj-15-4472-01NT/2016		2016-03-27	MDCK1	2560	10	160	40	40	40	20	40	80	40	
B/Poitiers/1845/2016		2016-04-01	MDCKx/MDCK1/M	5120	20	320	160	160	80	80	20	40	40	
B/Clermont Ferrand/1473/2016		2016-04-01	MDCK2/MDCK1	2560	10	80	20	20	40	10	20	40	40	
B/Jordan/30131/2016		2016-04-02	MDCK2	2560	10	80	20	20	40	20	40	80	40	
B/Jordan/20423/2016		2016-04-04	MDCK2	5120	20	80	40	40	80	20	40	80	80	
B/Jordan/IIIj-15-4203-01NT/2016		2016-04-04	MDCK1	5120	10	80	40	40	80	20	40	160	80	
B/Jordan/20364/2016		2016-04-05	MDCK1	5120	20	160	40	80	80	20	40	80	40	
B/Jordan/20390/2016		2016-04-05	MDCK1	5120	20	160	40	80	80	20	40	80	40	
B/Jordan/IIIj-15-4521-01NT/2016		2016-04-07	MDCK1	5120	10	160	40	80	80	20	40	80	80	
B/Jordan/IIIj-15-4530-01NT/2016		2016-04-10	MDCK1	5120	20	160	40	40	40	20	40	80	40	
B/Jordan/20450/2016		2016-04-12	MDCK1	2560	10	160	40	40	40	20	20	40	40	
B/Jordan/IIIj-15-4527-01NT/2016		2016-04-12	MDCK1	2560	10	160	40	40	40	20	20	40	40	
B/Lyon/1620/2016		2016-04-15	MDCK2/MDCK1	5120	80	320	320	320	160	160	40	80	40	
B/Jordan/IIIj-15-4606-01NT/2016		2016-04-26	MDCK1	5120	20	160	40	40	40	20	40	80	40	
B/Jordan/20464/2016		2016-04-26	MDCK1	5120	20	160	40	40	40	20	40	80	40	
B/Jordan/20466/2016		2016-05-04	MDCK1	5120	20	160	40	40	40	20	40	80	40	
B/Jordan/20499/2016		2016-05-07	MDCK1	2560	20	160	40	40	40	20	20	40	40	
B/La Reunion/2003/2016		2016-05-11	MDCK2/MDCK1	5120	20	160	40	40	80	20	40	80	40	
B/Jordan/20474/2016		2016-05-14	MDCK1	5120	20	160	40	80	80	20	40	80	80	
B/Jordan/30155/2016		2016-05-15	MDCK1	2560	20	160	40	40	40	20	40	80	40	
B/Jordan/30159/2016		2016-05-17	MDCK2	5120	20	160	40	40	40	20	40	80	40	
B/Jordan/20485/2016		2016-05-18	MDCK1	5120	80	1280	640	640	320	320	80	80	40	
B/Jordan/20508/2016		2016-05-19	MDCK1	2560	20	160	40	40	40	20	40	80	40	
B/La Reunion/2027/2016		2016-05-30	MDCK2/MDCK1	5120	40	160	80	80	80	40	40	80	40	
B/Hong Kong/1347/2016		2016-06-03	MDCK1/MDCK1	160	10	<	<	<	<	<	<	<	<	
B/Egypt/6082/2016		2016-07-17	MDCK1	5120	10	80	10	<	20	10	20	80	40	
B/Hong Kong/1544/2016		2016-07-30	MDCK1/MDCK1	5120	20	160	40	41	80	20	40	80	40	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 11-15. Antigenic analyses of influenza B viruses (Victoria lineage) with collection dates after 2016-01-31 - Summary

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Mal 2506/04	B/Bris 60/08	B/Malta 636714/11	B/Jhb 3964/12	B/Sth Aus 81/12	B/HK 514/09	B/Odessa 3886/10	B/Paris 1762/09	B/For V2367/12	B/Iceland 56/15	B/Ireland 3154/16	B/Nord-West 1/16
				Egg	Egg	Egg	Egg	Egg	MDCK	MDCK	MDCK	MDCK	MDCK	MDCK	MDCK
	Passage history														
	Genetic group				1A	1A	1A	1A	1B	1B	1A	1A	1A	1A	1A
REFERENCE VIRUSES															
B/Malaysia/2506/2004		2004-12-06	E3/E7	160	40	80	160	160	10	<	<	40	<	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	80	320	160	160	1280	80	40	40	80	80	20	10
B/Malta/636714/2011	1A	2011-03-07	E4/E1	80	320	640	320	640	80	20	40	160	80	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	160	1280	1280	1280	1280	320	40	80	1280	80	160	160
B/South Australia/81/2012	1A	2012-11-28	E4/E2	80	320	160	160	1280	80	40	40	160	80	20	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	10	40	40	20	80	80	40	20	40	40	40	40
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK1	<	<	40	40	80	80	40	40	40	40	20	40
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	<	<	40	40	80	80	80	80	40	80	40	80
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	40	320	160	160	320	80	20	40	160	80	40	20
B/Iceland/56/2015	1A	2015-12-30	MDCK0/MDCK1	<	<	40	40	80	80	40	40	40	80	40	80
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	40	80	40	40	40	80	40	80	80	160	80	40
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	20	40	20	20	80	80	80	80	40	160	40	40
TEST VIRUSES															
Number of viruses tested*				233	233	233	233	233	233	20	20	233	21	230	230
No with titre reduction ≤2-fold					6	4	1	4	231	18	17	45	17	225	206
%					2.6	1.7	0.4	1.7	99.1	90.0	85.0	19.3	90.0	97.8	89.6
No with titre reduction =4-fold				2	53	11	3	7		2	3	123	2	4	15
%				0.9	22.7	4.7	1.3	3.0		10.0	15.0	52.8	5.0	1.8	6.5
No with titre reduction ≥8-fold				231	174	218	229	222	2			65	2	1	9
%				99.1	74.7	93.6	98.3	95.3	0.9			27.9	5.0	0.4	3.9

* All viruses sequenced carried HA genes that fell in the B/Brisbane/60/2008 clade (clade 1A)

Vaccine
NH 2015-16[#]
SH 2016
NH 2016-17

[#] B/Victoria-lineage virus recommended for use in quadravalent vaccines

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 13. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Feb-Mar 2016

Apr 2016

May 2016

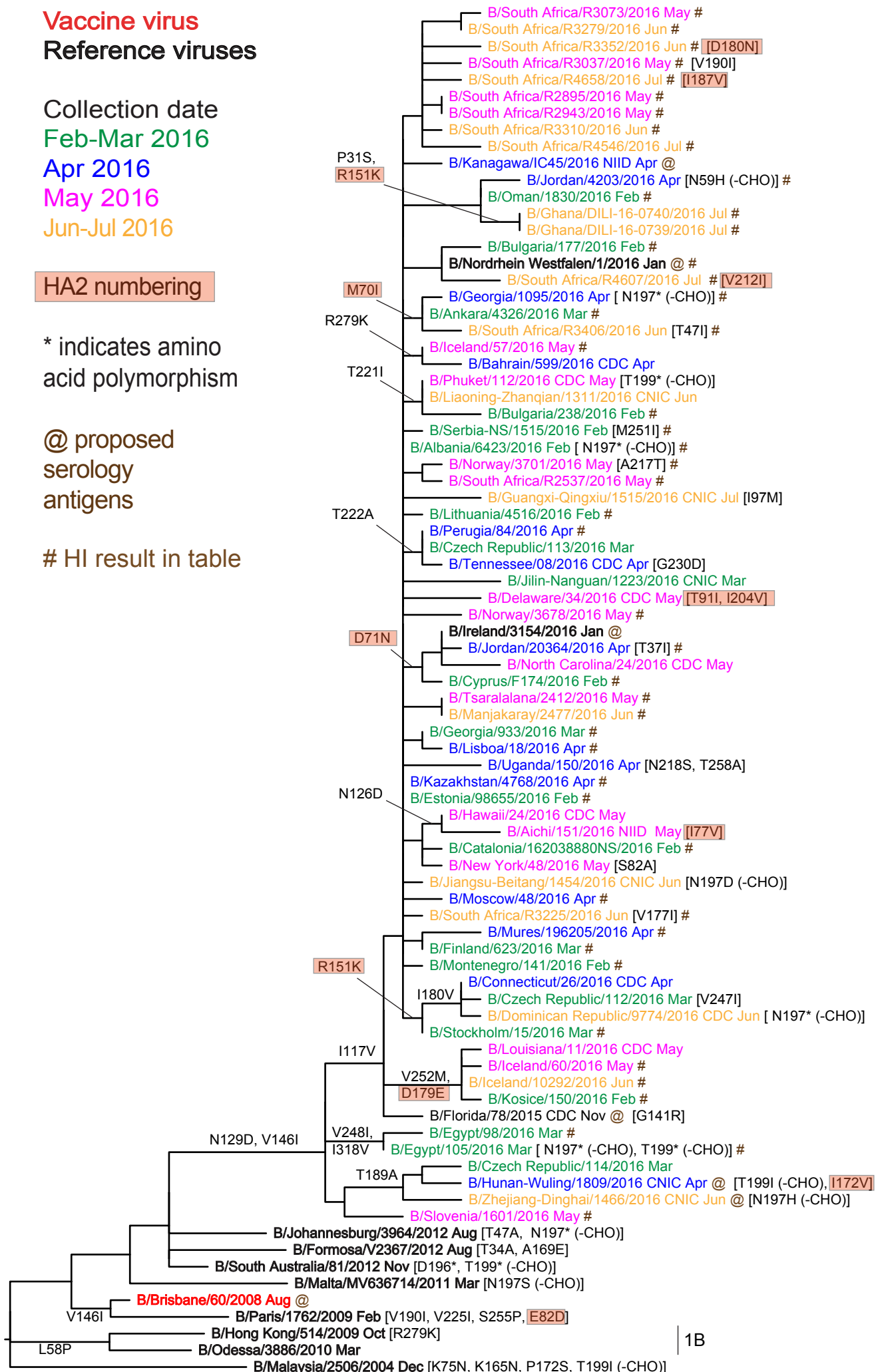
Jun-Jul 2016

HA2 numbering

* indicates amino acid polymorphism

@ proposed serology antigens

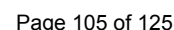
HI result in table



1A

1B

Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



	310	320	330	340	3	13	23	33	43	53
B/Malaysia/2506/2004	G	G	L	N	K	S	K	P	Y	T
B/Odessa/3886/2010	G	G	L	N	K	S	K	P	Y	T
B/Hong Kong/514/2009	G	G	L	N	K	S	K	P	Y	T
B/Paris/1762/2009	G	G	L	N	K	S	K	P	Y	T
B/Brisbane/60/2008	G	G	L	N	K	S	K	P	Y	T
B/Malta/MV636714/2011	G	G	L	N	K	S	K	P	Y	T
B/South Australia/81/2012	G	G	L	N	K	S	K	P	Y	T
B/Formosa/V2367/2012	G	G	L	N	K	S	K	P	Y	T
B/Johannesburg/3964/2012	G	G	L	N	K	S	K	P	Y	T
B/Slovenia/1601/2016	G	G	L	N	K	S	K	P	Y	T
B/Hunan-Wuling/1809/2016	G	G	L	N	K	S	K	P	Y	T
B/Egypt/98/2016	G	G	L	N	K	S	K	P	Y	T
B/Iceland/10292/2016	G	G	L	N	K	S	K	P	Y	T
B/Stockholm/15/2016	G	G	L	N	K	S	K	P	Y	T
B/Czech Republic/112/2016	G	G	L	N	K	S	K	P	Y	T
B/Catalonia/162038880NS/2016	G	G	L	N	K	S	K	P	Y	T
B/Aichi/151/2016	G	G	L	N	K	S	K	P	Y	T
B/Jordan/20364/2016	G	G	L	N	K	S	K	P	Y	T
B/Ireland/3154/2016	G	G	L	N	K	S	K	P	Y	T
B/Norway/3678/2016	G	G	L	N	K	S	K	P	Y	T
B/Perugia/84/2016	G	G	L	N	K	S	K	P	Y	T
B/Norway/3701/2016	G	G	L	N	K	S	K	P	Y	T
B/Bulgaria/238/2016	G	G	L	N	K	S	K	P	Y	T
B/Iceland/57/2016	G	G	L	N	K	S	K	P	Y	T
B/Georgia/1095/2016	G	G	L	N	K	S	K	P	Y	T
B/Nordrhein-Westfalen/1/2016	G	G	L	N	K	S	K	P	Y	T
B/Ghana/DILI-16-0740/2016	G	G	L	N	K	S	K	P	Y	T
B/South Africa/R4658/2016	G	G	L	N	K	S	K	P	Y	T
B/South Africa/R3352/2016	G	G	L	N	K	S	K	P	Y	T
B/South Africa/R3279/2016	G	G	L	N	K	S	K	P	Y	T

	63	73	83	93	103	113	123	133	143	153
B/Malaysia/2506/2004	S	L	S	E	L	V	K	N	L	Q
B/Odessa/3886/2010	S	L	S	E	L	V	K	N	L	Q
B/Hong Kong/514/2009	S	L	S	E	L	V	K	N	L	Q
B/Paris/1762/2009	S	L	S	E	L	V	K	N	L	Q
B/Brisbane/60/2008	S	L	S	E	L	V	K	N	L	Q
B/Malta/MV636714/2011	S	L	S	E	L	V	K	N	L	Q
B/South Australia/81/2012	S	L	S	E	L	V	K	N	L	Q
B/Formosa/V2367/2012	S	L	S	E	L	V	K	N	L	Q
B/Johannesburg/3964/2012	S	L	S	E	L	V	K	N	L	Q
B/Slovenia/1601/2016	S	L	S	E	L	V	K	N	L	Q
B/Hunan-Wuling/1809/2016	S	L	S	E	L	V	K	N	L	Q
B/Egypt/98/2016	S	L	S	E	L	V	K	N	L	Q
B/Iceland/10292/2016	S	L	S	E	L	V	K	N	L	Q
B/Stockholm/15/2016	S	L	S	E	L	V	K	N	L	Q
B/Czech Republic/112/2016	S	L	S	E	L	V	K	N	L	Q
B/Catalonia/162038880NS/2016	S	L	S	E	L	V	K	N	L	Q
B/Aichi/151/2016	S	L	S	E	L	V	K	N	L	Q
B/Jordan/20364/2016	S	L	S	E	L	V	K	N	L	Q
B/Ireland/3154/2016	S	L	S	E	L	V	K	N	L	Q
B/Norway/3678/2016	S	L	S	E	L	V	K	N	L	Q
B/Perugia/84/2016	S	L	S	E	L	V	K	N	L	Q
B/Norway/3701/2016	S	L	S	E	L	V	K	N	L	Q
B/Bulgaria/238/2016	S	L	S	E	L	V	K	N	L	Q
B/Iceland/57/2016	S	L	S	E	L	V	K	N	L	Q
B/Georgia/1095/2016	S	L	S	E	L	V	K	N	L	Q
B/Nordrhein-Westfalen/1/2016	S	L	S	E	L	V	K	N	L	Q
B/Ghana/DILI-16-0740/2016	S	L	S	E	L	V	K	N	L	Q
B/South Africa/R4658/2016	S	L	S	E	L	V	K	N	L	Q
B/South Africa/R3352/2016	S	L	S	E	L	V	K	N	L	Q
B/South Africa/R3279/2016	S	L	S	E	L	V	K	N	L	Q

	163	173	183	193	203	213	223
B/Malaysia/2506/2004	A	G	T	F	D	A	G
B/Odessa/3886/2010	A	G	T	F	D	A	G
B/Hong Kong/514/2009	A	G	T	F	D	A	G
B/Paris/1762/2009	A	G	T	F	D	A	G
B/Brisbane/60/2008	A	G	T	F	D	A	G
B/Malta/MV636714/2011	A	G	T	F	D	A	G
B/South Australia/81/2012	A	G	T	F	D	A	G
B/Formosa/V2367/2012	A	G	T	F	D	A	G
B/Johannesburg/3964/2012	A	G	T	F	D	A	G
B/Slovenia/1601/2016	A	G	T	F	D	A	G
B/Hunan-Wuling/1809/2016	A	G	T	F	D	A	G
B/Egypt/98/2016	A	G	T	F	D	A	G
B/Iceland/10292/2016	A	G	T	F	D	A	G
B/Stockholm/15/2016	A	G	T	F	D	A	G
B/Czech Republic/112/2016	A	G	T	F	D	A	G
B/Catalonia/162038880NS/2016	A	G	T	F	D	A	G
B/Aichi/151/2016	A	G	T	F	D	A	G
B/Jordan/20364/2016	A	G	T	F	D	A	G
B/Ireland/3154/2016	A	G	T	F	D	A	G
B/Norway/3678/2016	A	G	T	F	D	A	G
B/Perugia/84/2016	A	G	T	F	D	A	G
B/Norway/3701/2016	A	G	T	F	D	A	G
B/Bulgaria/238/2016	A	G	T	F	D	A	G
B/Iceland/57/2016	A	G	T	F	D	A	G
B/Georgia/1095/2016	A	G	T	F	D	A	G
B/Nordrhein-Westfalen/1/2016	A	G	T	F	D	A	G
B/Ghana/DILI-16-0740/2016	A	G	T	F	D	A	G
B/South Africa/R4658/2016	A	G	T	F	D	A	G
B/South Africa/R3352/2016	A	G	T	F	D	A	G
B/South Africa/R3279/2016	A	G	T	F	D	A	G

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 15. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine virus

Reference viruses

Collection date

Feb-Mar 2016

Apr 2016

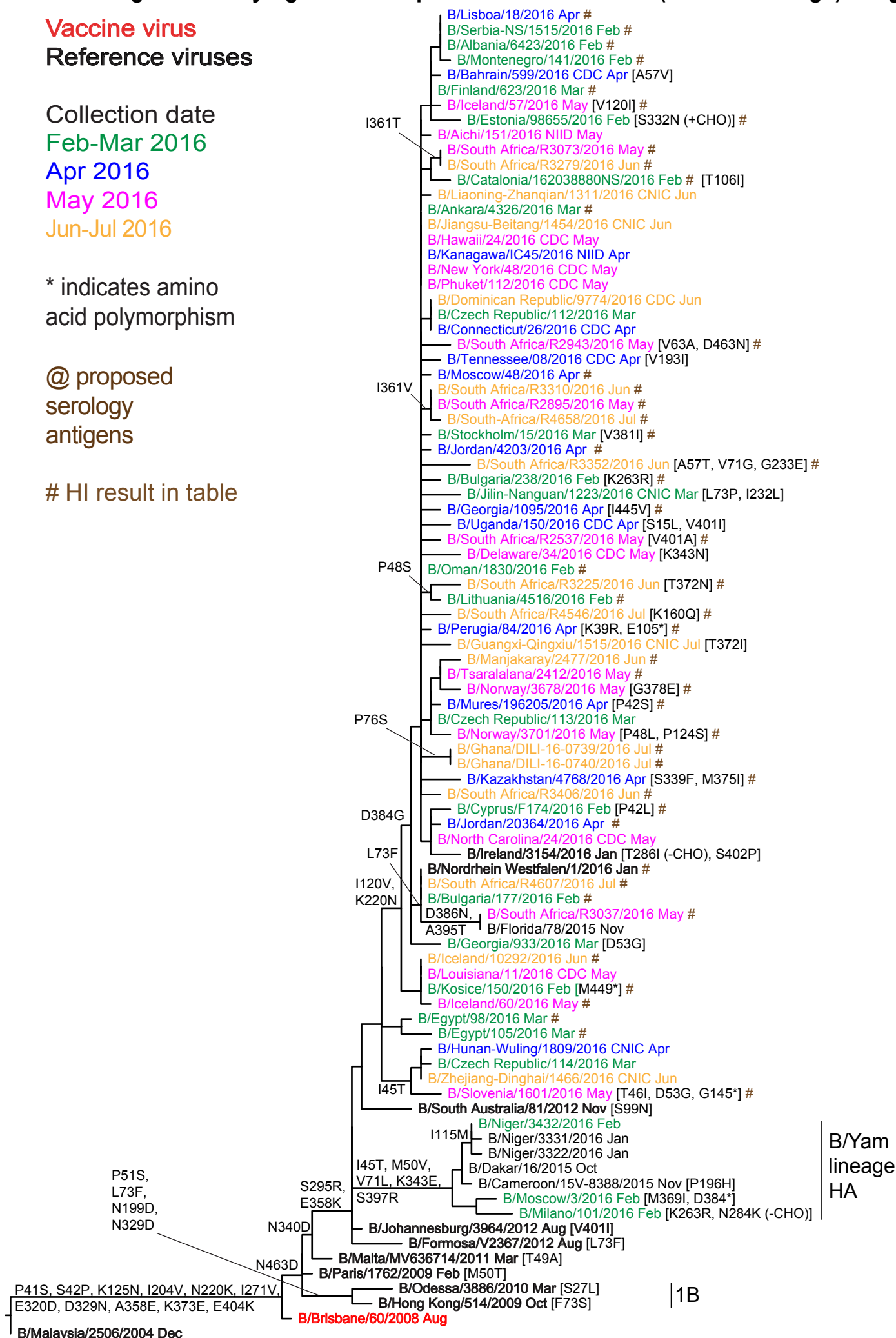
May 2016

Jun-Jul 2016

* indicates amino acid polymorphism

@ proposed serology antigens

HI result in table



1A

B/Yam
lineage
HA

1B

Figure 16. NA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
B/Malaysia/2506/2004	M	L	P	S	T	I	Q	T	L	T	L	F	L	T	S	G	G	V	L	L	S	L	Y	S	A	S	L	S	Y	L	L	Y	S	D	I	L	L	K	F	P	S	T	E	I	T	A	P	T	M	L	D	C	A	N	A	S	N	V	Q	A	V	N	R	S	A	T	K	G	V	T	L	L	L	P	E	P	E	W	T	Y	P	R	L	S	C	P	G	S	T	F	Q	K	A	L	L	I	S	P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
B/Odessa/3886/2010																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	</

	310	320	330	340	350	360	370	380	390	400														
B/Malaysia/2506/2004	PFVKLN	VETD	TAEIR	LMCT	ETYL	DT	PRPD	DSIT	GPCES	NGDK	SGG	IKG	GFVH	QRMASK	IGRW	YSRT	MSKT	KRMG	MGLY	VVKY	DGDP	WADSD	ALAF	SGVM
B/Odessa/3886/2010		D												E			E							
B/Hong Kong/514/2009		D												E			E							
B/Paris/1762/2009		D		N										E			E							
B/Brisbane/60/2008		D		N										E			E							
B/Malta/MV636714/2011		D		N		D								E			E							
B/South Australia/81/2012		D		N		D								K			E							
B/Formosa/V2367/2012		D		N		D								K			E							
B/Johannesburg/3964/2012		D		N		D								K			E							
B/Slovenia/1601/2016		D		N		D								K			E							
B/Hunan-Wuling/1809/2016		D		N		D								K			E							
B/Egypt/98/2016		D		N		D								K			E							
B/Iceland/10292/2016		D		N		D								K			E							
B/Stockholm/15/2016		D		N		D								K			E		I	G				
B/Czech Republic/112/2016		D		N		D								K			E			G				
B/Catalonia/162038880NS/2016		D		N		D								K			E			G				
B/Aichi/151/2016		D		N		D								K			E			G				
B/Jordan/20364/2016		D		N		D								K			E			G				
B/Ireland/3154/2016		D		N		D								K			E			G				
B/Norway/3678/2016		D		N		D								K			E		E	G				
B/Perugia/84/2016		D		N		D								K			E			G				
B/Norway/3701/2016		D		N		D								K			E			G				
B/Bulgaria/238/2016		D		N		D								K			E			G				
B/Iceland/57/2016		D		N		D								K			E			G				
B/Georgia/1095/2016		D		N		D								K			E			G				
B/Nordrhein-Westfalen/1/2016		D		N		D								K			E			G				
B/Ghana/DILI-16-0740/2016		D		N		D								K			E			G				
B/South Africa/R4658/2016		D		N		D								K	V		E			G				
B/South Africa/R3352/2016		D		N		D								K			E			G				
B/South Africa/R3279/2016		D		N		D								K	T		E			G				

	410	420	430	440	450	460
B/Malaysia/2506/2004	VSMEEP	GWYS	FGFE	IKDK	KCDV	PCIG
B/Odessa/3886/2010						
B/Hong Kong/514/2009						
B/Paris/1762/2009						
B/Brisbane/60/2008						
B/Malta/MV636714/2011						
B/South Australia/81/2012						
B/Formosa/V2367/2012						
B/Johannesburg/3964/2012	I					
B/Slovenia/1601/2016						
B/Hunan-Wuling/1809/2016						
B/Egypt/98/2016						
B/Iceland/10292/2016						
B/Stockholm/15/2016						
B/Czech Republic/112/2016						
B/Catalonia/162038880NS/2016						
B/Aichi/151/2016						
B/Jordan/20364/2016						
B/Ireland/3154/2016	P					
B/Norway/3678/2016						
B/Perugia/84/2016						
B/Norway/3701/2016						
B/Bulgaria/238/2016						
B/Iceland/57/2016						
B/Georgia/1095/2016				V		
B/Nordrhein-Westfalen/1/2016						
B/Ghana/DILI-16-0740/2016						
B/South Africa/R4658/2016						
B/South Africa/R3352/2016						
B/South Africa/R3279/2016						

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 12. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2016-01-31

MONTH	B Yamagata lineage				
Country	Number received¹	Number propagated²	≤2 fold³	4 fold³	≥8 fold³
FEBRUARY					
Germany	5	5	3	2	
Iceland	1	1	1		
Italy	1	1		1	
Niger	6	in progress	4		
Oman	1	1	1		
Russia	1	1	1		
United Kingdom	1	1	1		
MARCH					
Egypt	2	2	2		
France	1	1	1		
Germany	1	1	1		
Jordan	1	1	1		
Niger	1	1	1		
Romania	1	1	1		
Slovakia	2	2		2	
Sweden	1	1	1		
APRIL					
Iceland	1	1	1		
Jordan	2	2	2		
Niger	1	1		1	
Slovakia	1	1		1	
Slovenia	1	1	1		
United Kingdom	2	2	2		
MAY					
Norway	1	1	1		
JUNE					
Ghana	2	2	2		
JULY					
Hong Kong	1	1	1		
AUGUST					
Hong Kong	1	1	1		
	39	33	30	7	0
17 Countries			81.1%	18.9%	0.0%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 13-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-02-23 + 2016-03-02 + 2016-03-08

Haemagglutination inhibition titre															
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK	
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14	
				Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg	
				SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F05/15 ¹	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}	
Genetic Group	3	1	2	2	2	2	3	3	3	3	3	3			
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	640	320	80	160	640	160	160	40	320	160	
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	40	80	320	80	160	20	160	160	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	80	160	320	80	40	20	40	80	160	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	320	640	320	20	80	320	320	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	320	320	80	40	640	160	80	20	160	160	
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	20	<	160	160	80	40	160	160	
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	160	160	20	<	160	80	160	40	160	160	
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	160	160	320	320	320	40	640	320	320	
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	320	160	20	<	160	160	160	80	320	160	
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	40	<	<	40	80	20	20	40	160	
TEST VIRUSES															
B/Ghana/FS-15-0239/2015		2015-04-23	Cx/MDCK1	5120	160	160	160	320	320	320	20	640	320	160	
B/Ghana/DILI-15-0538/2015	3	2015-06-09	Cx/MDCK1	5120	320	320	320	1280	320	320	<	1280	640	320	
B/Ghana/DILI-15-0570/2015	3	2015-06-15	Cx/MDCK1	5120	160	80	160	40	80	160	<	640	320	20	
B/Zambia/04-00314/2015	3	2015-07-23	MDCK2/MDCK1	640	40	40	20	20	40	80	20	40	40	80	
B/Dakar/11/2015	3	2015-07-28	C1/MDCK1	5120	160	80	80	160	80	160	40	320	160	160	
B/Dakar/10/2015	3	2015-07-30	C0/MDCK1	5120	320	160	160	320	320	320	160	320	640	320	
B/Ghana/FS-15-0470/2015	3	2015-08-12	Cx/MDCK1	5120	320	160	320	640	320	320	<	1280	640	320	
B/Zambia/04-00354/2015	2	2015-08-21	MDCK2/MDCK1	640	80	80	160	160	160	40	<	40	40	80	
B/Ghana/FS-15-0487/2015	3	2015-08-25	cs/MDCK1	5120	160	160	160	320	160	160	160	640	320	160	
B/Zambia/04-00361/2015	3	2015-08-31	MDCK2/MDCK1	1280	80	80	40	40	80	80	40	160	160	80	
B/Zambia/04-00384/2015	3	2015-09-09	MDCK2/MDCK1	1280	40	80	40	40	40	80	40	80	80	80	
B/Dakar/13/2015	3	2015-09-10	C0/MDCK1	2560	80	80	40	40	160	80	40	160	160	160	
B/Zambia/04-00379/2015	3	2015-09-12	MDCK2/MDCK1	2560	40	80	40	<	80	80	40	80	80	160	
B/Dakar/14/2015	3	2015-09-12	C0/MDCK1	2560	160	80	40	40	160	160	40	160	160	160	
B/Antsirabe/3949/2015	3	2015-09-17	MDCK2/MDCK1	5120	160	160	320	640	320	320	80	1280	640	320	
B/Zambia/04-00397/2015	3	2015-09-22	MDCK2/MDCK1	2560	80	80	40	40	160	160	80	160	160	160	
B/Dakar/15/2015	3	2015-09-22	C0/MDCK1	2560	160	80	80	80	160	160	40	160	160	160	
B/Dakar/12/2015		2015-10-07	C0/MDCK1	5120	160	160	80	80	320	160	80	320	320	160	
B/Italy-FVG/19/2015	3	2015-10-12	SIAT2/MDCK1	5120	160	160	160	320	160	160	40	640	320	160	
B/Zambia/04-00413/2015	3	2015-10-12	MDCK2/MDCK1	2560	80	160	40	40	160	160	80	160	320	160	
B/Dakar/16/2015	3	2015-10-16	C0/MDCK1	5120	320	160	320	320	320	320	160	640	320	320	
B/Oman/6340/2015	3	2015-11-17	SIAT2/MDCK1	5120	80	160	160	160	320	160	80	320	320	160	
B/Oman/6341/2015	3	2015-11-17	SIAT2/MDCK1	5120	160	160	160	160	160	160	80	320	320	160	
B/Oman/6390/2015	3	2015-11-17	SIAT2/MDCK1	2560	80	160	80	80	160	160	80	160	320	160	
B/Oman/6392/2015	3	2015-11-18	SIAT2/MDCK1	5120	160	160	160	320	160	160	40	640	320	320	
B/Hamburg/1/2016	3	2016-01-18	C2/MDCK1	5120	160	80	40	80	160	160	40	160	160	320	
B/Rheinland-Pfalz/1/2016	3	2016-01-25	C2/MDCK1	2560	80	80	80	80	160	160	40	160	160	160	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ < + <40; ² < + <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

[#] B/Yamagata-lineage virus recommended for use in quadrivalent vaccines
Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Table 13-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-03-18 + 2016-04-13

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK	B/Calif	
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14	12/15	
				Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg	Egg	
Ferret number				SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F05/15 ¹	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}	CDC FS 2015-110	
Genetic Group				3	1	2	2	2	2	3	3	3	3	3		
REFERENCE VIRUSES																
B/Florida/4/2006		1	2006-12-15	E7/E1	2560	640	640	80	160	1280	320	160	40	320	160	320
B/Brisbane/3/2007		2	2007-09-03	E2/E3	1280	320	320	40	80	640	80	80	20	160	160	320
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	1280	80	80	160	320	80	40	40	40	80	160	160
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	1280	640	640	320	320	640	320	160	80	320	320	1280
B/Massachusetts/02/2012		2	2012-03-13	E3/E3	1280	320	640	80	80	640	160	160	20	160	160	320
B/Wisconsin/1/2010		3	2010-02-20	E3/E3	2560	320	320	20	40	320	320	160	40	160	160	320
B/Stockholm/12/2011		3	2011-03-28	E4/E1	1280	160	160	10	20	160	80	160	40	80	160	320
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	5120	320	320	160	320	320	320	160	640	320	320	640
B/Phuket/3073/2013		3	2013-11-21	E4/E3	2560	160	160	20	20	320	320	160	80	160	160	320
B/Hong Kong/3417/2014		3	2014-06-04	E4/E1	1280	80	80	10	10	80	80	40	20	40	160	320
TEST VIRUSES																
B/California/12/2015		3	2015-02-04	E4/E1	640	40	40	<	<	40	40	20	10	40	80	320
B/California/12/2015	NYMC BX-59A	3	2015-02-04	Ex	2560	80	40	20	40	20	80	80	160	160	160	640
B/California/12/2015	NYMC BX-59B	3	2015-02-04	Ex	1280	40	20	10	20	20	40	40	80	80	160	640
B/Jordan/30349/2015		3	2015-05-19	Cx/MDCK1	5120	80	80	80	160	160	160	80	160	160	160	ND
B/Jordan/30345/2015		3	2015-05-31	Cx/MDCK1	2560	80	80	80	80	160	160	80	160	160	160	ND
B/Jordan/30356/2015		3	2015-06-15	Cx/MDCK1	2560	80	80	40	80	80	160	40	160	80	160	ND
B/Jordan/20606/2015		3	2015-07-06	MDCK1	5120	160	160	160	320	320	320	160	640	320	160	ND
B/Okinawa/8/2015		3	2015-09-01	MDCK1/MDCK2/MDCK1	5120	160	160	160	160	320	320	80	320	320	160	ND
B/Cameroon/GAKO 160/2015		3	2015-10-15	MDCK1	2560	80	40	20	20	40	80	40	80	80	80	ND
B/Cameroon/15V-8388/2015		3	2015-11-17	MDCK2	2560	80	40	20	20	40	80	40	80	80	80	ND

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < + <40; ² < + <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC; ND = Not Done

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Table 13-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-04-20

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket	B/Fl	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
	Passage history			Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
	Ferret number			SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F05/15 ¹	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}
	Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	640	320	320	40	80	640	80	80	20	80	80
B/Brisbane/3/2007	2	2007-09-03	E2/E3	640	160	320	40	40	320	40	80	10	80	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	40	40	80	80	40	40	20	20	40	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	640	320	320	160	320	320	160	80	80	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	640	320	320	40	80	640	80	80	10	80	80
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	80	80	10	10	160	160	80	20	80	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	10	<	80	40	80	20	40	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	80	160	80	160	160	160	80	320	160	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	160	320	20	10	320	320	160	40	160	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	640	40	40	<	<	40	40	20	10	20	80
TEST VIRUSES														
B/Rheinland-Pfalz/4/2016	3	2016-02-15	C2/MDCK1	1280	80	40	40	40	40	80	20	80	80	160
B/Berlin/16/2016	3	2016-02-22	C2/MDCK1	1280	80	40	40	20	80	80	20	80	80	80
B/Nordrhein-Westfalen/26/2016	3	2016-02-22	C1/MDCK1	1280	80	40	20	20	40	40	20	80	40	160
B/Berlin/25/2016	3	2016-02-22	C1/MDCK1	1280	80	40	20	20	40	40	20	80	40	160
B/Bayern/5/2016	3	2016-02-29	C2/MDCK1	1280	80	80	40	20	160	80	40	80	80	160
B/Brandenburg/17/2016	3	2016-03-08	C2/MDCK1	1280	80	80	40	40	80	80	40	160	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < + <40; ² < + <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Table 13-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-05-04 + 2016-05-10 + 2016-05-17 + 2016-05-25

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
				Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
	Passage history													
	Ferret number			SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F05/15 ¹	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}
	Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	320	640	80	160	640	160	160	20	160	160
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	40	80	640	80	80	20	160	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	160	160	80	40	<	40	80	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	640	640	320	320	640	160	80	80	320	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	320	640	80	80	640	160	160	20	160	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	20	10	160	160	80	40	160	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	20	<	160	80	160	40	80	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	5120	160	160	160	320	160	160	80	320	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	320	160	20	20	320	320	160	80	160	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	1280	160	40	10	<	80	80	40	20	80	160
TEST VIRUSES														
B/Niger/3015/2015	3	2015-02-24	MDCK3	1280	80	40	40	40	40	40	20	80	80	80
B/Niger/3134/2015		2015-11-02	MDCK1	2560	40	40	40	40	40	20	20	40	80	160
B/Niger/3179/2015		2015-12-07	MDCK1	2560	40	40	20	20	40	20	20	40	80	160
B/Niger/3203/2015		2015-12-15	MDCK1	2560	40	40	40	40	40	40	20	40	80	160
B/Niger/3251/2015		2015-12-29	MDCK1	2560	40	40	20	20	20	20	20	40	40	160
B/Niger/3295/2016	3	2016-01-04	MDCK2	2560	160	80	40	40	160	160	40	160	160	160
B/Niger/3276/2016	3	2016-01-06	MDCK1	2560	80	40	40	40	80	80	40	80	80	160
B/Niger/3331/2016	3	2016-01-18	MDCK1	2560	80	40	40	<	40	80	<	160	80	160
B/Niger/3322/2016	3	2016-01-20	MDCK1	2560	80	40	40	<	80	80	20	80	80	160
B/Niger/3324/2016	3	2016-01-20	MDCK1	2560	80	40	40	40	80	80	40	80	80	160
B/Niger/3340/2016	3	2016-01-20	MDCK1	2560	80	40	40	<	40	80	20	80	80	160
B/Niger/3369/2016	3	2016-01-27	MDCK1	2560	80	40	40	40	80	80	40	160	160	160
B/Egypt/1832/2016	3	2016-01-28	MDCK1	2560	80	40	40	20	40	80	40	80	160	160
B/Egypt/1920/2016	3	2016-01-28	MDCK1	2560	80	80	40	40	80	160	40	160	160	160
B/Niger/3426/2016		2016-02-12	MDCK1	1280	80	40	40	40	40	80	20	160	160	160
B/Oman/1836/2016	3	2016-02-14	MDCK2/MDCK1	2560	80	80	40	40	80	160	40	80	80	160
B/Niger/3432/2016	3	2016-02-15	MDCK1	2560	160	80	80	40	160	160	40	160	160	160
B/Niger/3557/2016	3	2016-02-24	MDCK1	2560	160	80	80	80	160	160	40	320	320	160
B/Niger/3489/2016	3	2016-02-25	MDCK1	2560	80	40	40	<	80	80	20	80	80	160
B/Bucaresti/194094/2016	3	2016-03-15	MDCK1/MDCK1	2560	40	40	40	40	40	40	40	40	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < + <40; ² < + <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Table 13-5. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-06-15

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket	B/Fl	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
				Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
	Passage history													
	Ferret number			SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F05/15 ¹	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Judes F715/14 ^{2,4}
	Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	320	640	80	160	640	160	160	20	160	160
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	40	80	640	80	80	20	160	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	160	160	80	40	<	40	80	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	640	640	320	320	640	160	80	80	320	80
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	1280	80	160	1280	160	160	20	160	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	20	10	160	160	80	40	160	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	20	<	160	80	160	40	80	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	160	320	160	160	80	320	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	5120	320	320	20	20	320	320	160	40	320	320
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	1280	160	40	10	<	80	80	40	20	80	160
TEST VIRUSES														
B/Egypt/96/2016	3	2016-03-01	MDCK1	5120	160	160	80	80	160	320	80	320	320	320
B/Partizanske/246/2016	3	2016-03-03	MDCKx/MDCK1	2560	80	40	40	20	80	80	40	80	80	160
B/Egypt/110/2016	3	2016-03-06	MDCK1	5120	80	80	80	40	80	160	40	160	160	160
B/Poprad/320/2016	2	2016-03-22	MDCKx/MDCK1	2560	80	160	320	320	160	80	40	160	80	320
B/Bratislava/351/2016	3	2016-04-08	MDCKx/MDCK1	2560	80	80	40	<	80	160	40	160	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < + <40; ² < + <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Table 13-6. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-06-22 + 2016-06-28 + 2016-07-05 + 2016-08-16

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK	B/H'burg
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14	1/16
				Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg	MDCK
	Passage history														
	Ferret number			SH614 ^{1,3}	F17/13 ¹	F38/14 ²	F32/12 ²	F05/15 ¹	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}	F21/16 ²
	Genetic Group			3	1	2	2	2	2	3	3	3	3	3	3
															New
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	1280	640	80	80	1280	160	160	20	160	160	20
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	640	320	40	80	640	80	80	10	80	160	10
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	40	80	160	80	40	10	40	40	80	40
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	640	320	320	320	640	160	80	80	160	160	80
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	1280	640	80	160	1280	160	160	20	160	160	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	320	160	10	10	160	160	80	20	160	160	20
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	10	<	80	80	80	20	80	80	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	160	320	320	160	80	320	320	160	640
B/Phuket/3073/2013	3	2013-11-21	E4/E5	2560	320	160	20	10	160	160	80	40	160	160	40
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	1280	80	40	<	<	40	40	20	10	40	160	<
B/Hamburg/1/2016	3	2016-01-18	C2/MDCK2	2560	160	80	80	80	320	160	40	320	640	160	640
TEST VIRUSES															
B/Iceland/25/2016	3	2016-02-19	MDCK1/MDCK1	2560	160	80	80	40	160	160	40	160	160	160	320
B/Moscow/3/2016	3	2016-02-19	MDCK1/MDCK1	5120	160	80	160	160	160	160	80	320	160	160	640
B/Milano/101/2016	3	2016-02-22	MDCK3/MDCK1	5120	320	160	320	640	320	320	160	1280	640	160	1280
B/Stockholm/14/2016	3	2016-03-31	MDCK2/MDCK1	5120	160	160	40	20	320	320	80	160	160	320	320
B/Iceland/49/2016	3	2016-04-10	MDCK1/MDCK1	2560	80	80	80	80	80	160	40	160	160	160	320
B/Slovenia/1471/2016	3	2016-04-11	MDCKx/MDCK1	5120	80	80	40	<	80	160	40	80	80	160	160
B/Norway/3694/2016	3	2016-05-26	MDCK1/MDCK1	2560	80	80	40	<	80	160	40	80	80	160	ND
B/Ghana/FS-16-0730/2016	3	2016-06-02	Cx/MDCK1	2560	80	40	40	40	80	160	40	80	80	160	ND

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < + <40; ² < + <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Table 13-7. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-08-24 + 2016-09-14

Haemagglutination inhibition titre															
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				B/Phuket	B/Fl	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK	
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14	
				Passage history	Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
				Ferret number	SH614 ^{1,3}	F17/13 ¹	F38/14 ²	F32/12 ²	F05/15 ²	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude F715/14 ^{2,4}
Genetic Group	3	1	2	2	2	2	3	3	3	3	3				
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	1280	640	80	80	640	160	160	20	160	160	
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	640	320	40	80	640	80	80	10	80	80	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	80	160	80	40	20	40	40	80	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	320	640	320	160	80	160	320	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	640	640	80	80	640	160	160	20	160	160	
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	320	320	20	20	320	320	160	40	160	160	
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	10	<	160	80	160	20	80	160	
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	160	320	640	320	320	160	640	640	320	
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	320	160	20	20	160	160	80	40	160	160	
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	1280	80	40	10	<	40	80	20	10	40	160	
TEST VIRUSES															
B/England/729/2016		2016-02-05	SIAT1/MDCK1	5120	80	80	80	80	160	160	40	160	160	160	
B/Niger/3726/2016		2016-03-19	MDCK1	5120	160	80	80	<	320	320	40	320	160	320	
B/Lyon/1375/2016		2016-03-24	MDCK2/MDCK1	2560	80	80	40	<	80	160	40	80	80	160	
B/Jordan/20415/2016		2016-03-27	MDCK1	2560	80	80	40	<	40	160	20	40	80	160	
B/England/913/2016		2016-04-15	SIAT1/MDCK1	5120	80	80	80	80	160	160	40	160	160	160	
B/Jordan/30133/2016		2016-04-19	MDCK2	5120	160	160	320	640	160	160	80	640	320	160	
B/Jordan/30147/2016		2016-04-23	MDCK1	2560	160	80	80	80	160	320	40	80	160	160	
B/Niger/3836/2016		2016-04-24	MDCK1	5120	640	320	640	1280	640	640	160	1280	640	320	
B/England/941/2016		2016-04-24	SIAT2/MDCK1	5120	160	160	160	160	160	160	80	160	320	160	
B/Ghana/FS-16-0860/2016		2016-06-23	MDCK1	2560	80	80	40	20	80	160	40	80	80	160	
B/Hong Kong/1543/2016		2016-07-25	MDCK1/MDCK1	5120	80	80	40	<	160	320	40	80	80	160	
B/Hong Kong/1549/2016		2016-08-03	MDCK1/MDCK1	5120	160	80	40	<	160	160	40	80	80	160	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < + <40; ² < + <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Table 13-8. Antigenic analyses of influenza B viruses (Yamagata lineage) with collection dates after 2016-01-31- Summary

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK	B/H'burg
				4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14	1/16
				Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg	MDCK
				1	2	2	2	2	3	3	3	3	3	3
New														
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	640	80	80	1280	160	160	20	160	160	20
B/Brisbane/3/2007	2	2007-09-03	E2/E3	640	320	40	80	640	80	80	10	80	160	10
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	80	40	80	160	80	40	10	40	40	80	40
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	640	320	320	320	640	160	80	80	160	160	80
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	640	80	160	1280	160	160	20	160	160	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	320	160	10	10	160	160	80	20	160	160	20
B/Stockholm/12/2011	3	2011-03-28	E4/E1	160	80	10	<	80	80	80	20	80	80	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	160	160	160	320	320	160	80	320	320	160	640
B/Phuket/3073/2013	3	2013-11-21	E4/E5	320	160	20	10	160	160	80	40	160	160	40
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	80	40	<	<	40	40	20	10	40	160	<
B/Hamburg/1/2016	3	2016-01-18	C2/MDCK2	160	80	80	80	320	160	40	320	640	160	640
TEST VIRUSES														
Number of viruses tested*				37	37	37	37	37	37	37	37	37	37	6
No with titre reduction ≤2-fold				4	7	29	6	2	34	13	16	32	37	5
%				10.8	18.9	78.4	16.2	5.4	91.9	35.1	43.2	86.5	100.0	83.3
No with titre reduction =4-fold				14	21	8	6	12	3	21	14	5		1
%				37.8	56.8	21.6	16.2	32.4	8.1	56.8	37.8	13.5		16.7
No with titre reduction ≥8-fold				19	9		25	23		3	7			
%				51.4	24.3		67.6	62.2		8.1	19.0			

* All but one virus (B/Poprad/320/2016, clade 2: not tested with the B/Hamburg/1/2016 ferret antiserum) fell in clade 3

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Reference virus results are taken from individual tables as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Vaccine
NH 2015-16
SH2016[#]
NH 2016-17[#]

Figure 17. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Feb-Mar 2016

Apr 2016

May 2016

Jun-Jul 2016

HA2 numbering

* indicates amino acid polymorphism

@ proposed serology antigens

HI result in table

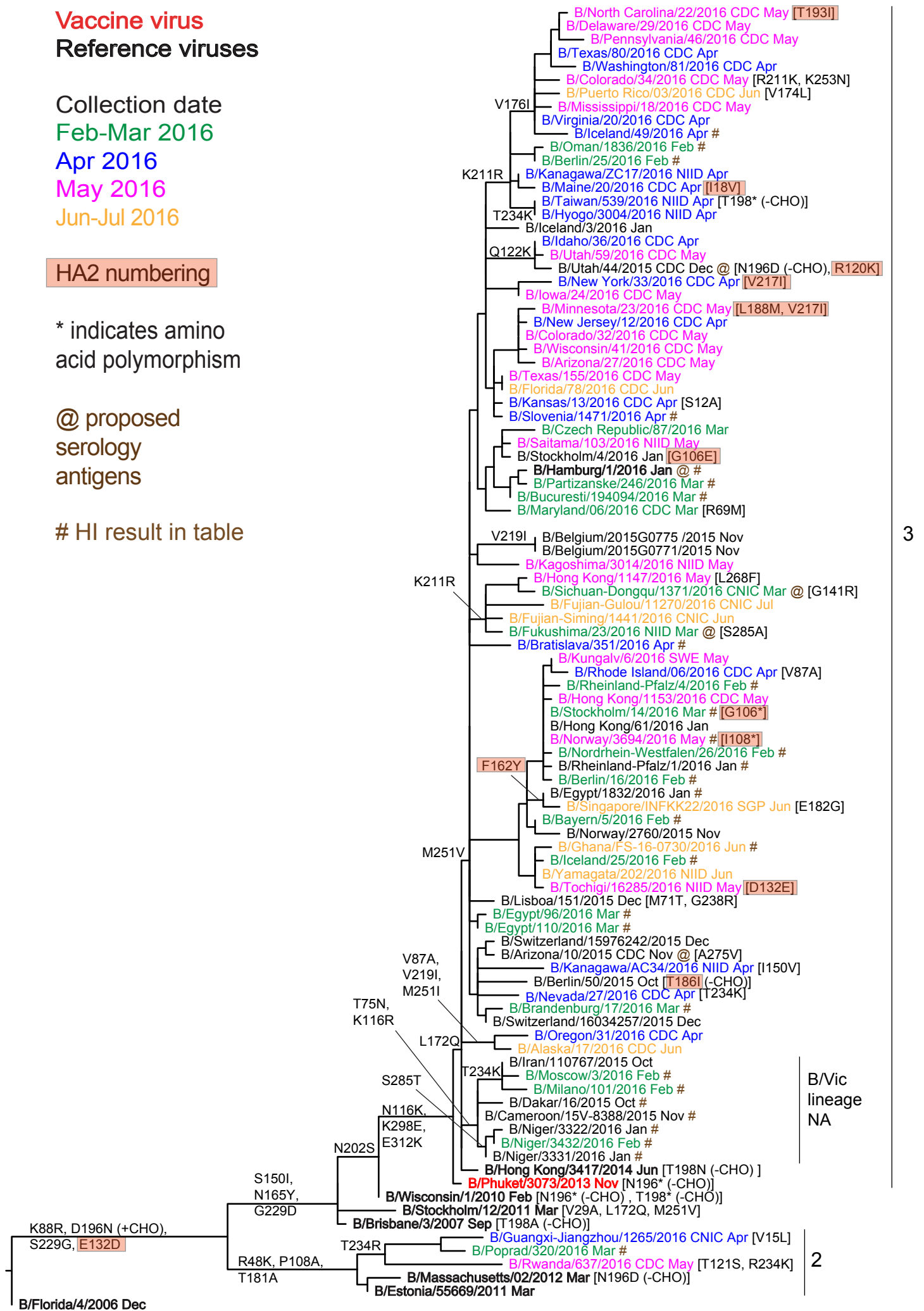


Figure 18. HA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Florida/4/2006	DR	I	C	T	G	I	T	S	S	N
B/Estonia/55669/2011										
B/Massachusetts/02/2012										
B/Rwanda/637/2016										
B/Poprad/320/2016										
B/Brisbane/3/2007										
B/Stockholm/12/2011										
B/Wisconsin/1/2010										
B/Phuket/3073/2013										
B/Hong Kong/3417/2014										
B/Niger/3432/2016										
B/Moscow/3/2016										
B/Alaska/17/2016										
B/Brandenburg/17/2016										
B/Kanagawa/AC34/2016										
B/Tochigi/16285/2016										
B/Singapore/INFKK22/2016										
B/Norway/3694/2016										
B/Kungälv/6/2016										
B/Fujian-Siming/1441/2016										
B/Hong Kong/1147/2016										
B/Hamburg/1/2016										
B/Saitama/103/2016										
B/Slovenia/1471/2016										
B/Minnesota/23/2016										
B/Utah/59/2016										
B/Hyogo/3004/2016										
B/Iceland/49/2016										
B/Puerto Rico/03/2016										
B/North Carolina/22/2016										
B/Florida/4/2006										
B/Estonia/55669/2011										
B/Massachusetts/02/2012										
B/Rwanda/637/2016										
B/Poprad/320/2016										
B/Brisbane/3/2007										
B/Stockholm/12/2011										
B/Wisconsin/1/2010										
B/Phuket/3073/2013										
B/Hong Kong/3417/2014										
B/Niger/3432/2016										
B/Moscow/3/2016										
B/Alaska/17/2016										
B/Brandenburg/17/2016										
B/Kanagawa/AC34/2016										
B/Tochigi/16285/2016										
B/Singapore/INFKK22/2016										
B/Norway/3694/2016										
B/Kungälv/6/2016										
B/Fujian-Siming/1441/2016										
B/Hong Kong/1147/2016										
B/Hamburg/1/2016										
B/Saitama/103/2016										
B/Slovenia/1471/2016										
B/Minnesota/23/2016										
B/Utah/59/2016										
B/Hyogo/3004/2016										
B/Iceland/49/2016										
B/Puerto Rico/03/2016										
B/North Carolina/22/2016										
B/Florida/4/2006										
B/Estonia/55669/2011										
B/Massachusetts/02/2012										
B/Rwanda/637/2016										
B/Poprad/320/2016										
B/Brisbane/3/2007										
B/Stockholm/12/2011										
B/Wisconsin/1/2010										
B/Phuket/3073/2013										
B/Hong Kong/3417/2014										
B/Niger/3432/2016										
B/Moscow/3/2016										
B/Alaska/17/2016										
B/Brandenburg/17/2016										
B/Kanagawa/AC34/2016										
B/Tochigi/16285/2016										
B/Singapore/INFKK22/2016										
B/Norway/3694/2016										
B/Kungälv/6/2016										
B/Fujian-Siming/1441/2016										
B/Hong Kong/1147/2016										
B/Hamburg/1/2016										
B/Saitama/103/2016										
B/Slovenia/1471/2016										
B/Minnesota/23/2016										
B/Utah/59/2016										
B/Hyogo/3004/2016										
B/Iceland/49/2016										
B/Puerto Rico/03/2016										
B/North Carolina/22/2016										

[illegible]

Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

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	310	320	330	340	350	360	370	380	390	400
B/Florida/4/2006	PFVKLN	VDTAE	IRLMCT	ETYLDP	TRPN	DGSGT	IPCESD	GGKSGG	IKGGFV	HQRMASK
B/Estonia/55669/2011										
B/Massachusetts/02/2012										
B/Rwanda/637/2016										
B/Poprad/320/2016									G	
B/Brisbane/3/2007										
B/Stockholm/12/2011				N	E					
B/Wisconsin/1/2010				N	E					
B/Phuket/3073/2013		K		N	E					
B/Hong Kong/3417/2014		K		E						
B/Kanagawa/ZC17/2016		K		NE						V
B/Ghana/FS-16-0730/2016		K		NE			Q			
B/Alaska/17/2016 Jun		K		E						
B/Brandenburg/17/2016		K		E						
B/Kanagawa/AC34/2016		K		E						
B/Tochigi/16285/2016		K		NE			Q			
B/Singapore/INFKK22/2016		K		NE			Q			
B/Norway/3694/2016		K		NE			Q			
B/Kungälv/6/2016		K		NE			Q			
B/Fujian-Siming/1441/2016		K		E	V					
B/Hong Kong/1147/2016		K		E						
B/Hamburg/1/2016		K		NE		T				
B/Saitama/103/2016		K		NE					S	
B/Slovenia/1471/2016		K		NE						
B/Minnesota/23/2016		K		NE						
B/Utah/59/2016		K		NE						
B/Hyogo/3004/2016	A	K		NE						
B/Iceland/49/2016		K		NE						
B/Puerto Rico/03/2016		K		NEX						
B/North Carolina/22/2016		K		NE						
	410	420	430	440	450	460				
B/Florida/4/2006	VSMEEP	GWYSF	GFEIK	DKKCD	VPCIG	IEMVHD	GKTTW	HSAA	AIYCL	MGSQ
B/Estonia/55669/2011										
B/Massachusetts/02/2012										
B/Rwanda/637/2016										
B/Poprad/320/2016										
B/Brisbane/3/2007										
B/Stockholm/12/2011									N	T
B/Wisconsin/1/2010									N	T
B/Phuket/3073/2013									N	T
B/Hong Kong/3417/2014						V			N	T
B/Kanagawa/ZC17/2016	I								N	T
B/Ghana/FS-16-0730/2016									N	T
B/Alaska/17/2016 Jun									N	T
B/Brandenburg/17/2016					V				N	T
B/Kanagawa/AC34/2016									N	T
B/Tochigi/16285/2016									N	T
B/Singapore/INFKK22/2016									N	T
B/Norway/3694/2016									N	T
B/Kungälv/6/2016									N	T
B/Fujian-Siming/1441/2016	I								N	T
B/Hong Kong/1147/2016	I								N	T
B/Hamburg/1/2016									N	T
B/Saitama/103/2016									N	T
B/Slovenia/1471/2016									N	T
B/Minnesota/23/2016									N	T
B/Utah/59/2016									N	T
B/Hyogo/3004/2016	I								N	T
B/Iceland/49/2016	I								N	T
B/Puerto Rico/03/2016	I								N	T
B/North Carolina/22/2016	I		X						N	T

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 14. Antiviral (NAI) testing of isolates with collection dates in 2016

Month of Collection	Number of viruses tested ¹					
	A(H1N1)pdm09		A(H3N2)		Influenza B	
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI
2016						
January	263	1 ^a	40		45	
February	267	2 ^{b,c}	78		83	
March	88	2 ^{d,e}	50		81	
April	43		23		55	
May	8		20		22	
June	4		26		12	
July	3		21		9	
August	2		2		1	
Totals	678	5	260		308	
Total No	1251					

¹ Numbers of influenza A subtype and influenza B viruses tested by phenotypic assay against oseltamivir and zanamivir. Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369–374.

a A/Czech Republic/11/2016 carries H275Y

b A/Oman/1726/2016 carries H275Y

c A/Odessa/699/2016 polymorphic*

d A/Norway/2298/2016 carries H275Y

e A/Norway/2036/2016 carries H275Y

Fold difference = os 392 zan 2

Fold difference = os 459 zan 1

Fold difference = os 1 zan 20

Fold difference = os 389 zan 1

Fold difference = os 353 zan 1

HRI oseltamivir

HRI oseltamivir

RI zanamivir

HRI oseltamivir

HRI oseltamivir

* Q136R(75%)/Q(25%), Y274D(25%)/Y(75%)